

SuperΣCMIV

INSTRUCTION MANUAL

MODEL

S-SIGMA-CM4-CTR-V5-N

S-SIGMA-CM4-CTR-V2-N

S-SIGMA-CM4-CTR-V5-P

S-SIGMA-CM4-CTR-V2-P

and customized products based on above products

Precautions and Requests When Using this Device

Description of symbol marks

	Always adhere to the following precautions, as failing to do so may cause serious injury or death.
	This symbol indicates that the product cannot be disposed of as household waste. For recycling, electric and electronic equipment and batteries must be handed over to an appropriate collection organization for disposal. Follow the laws and regulations of each country and region when disposing of the product and the supplied accessories. Recycling of the materials helps conserve natural resources and protect human health and environments. A coin-type lithium primary battery is used in this product.

Purpose of use and restrictions

This product is an air pulse type dispenser designed to dispense liquid material using compressed air. It allows liquid material in a container to be dispensed in accordance with a signal input from a switch on the device or an external device.

This product is an industrial monitoring and control device.

Ensure that it is used correctly in accordance with the instructions of the liquid material supervisor and product supervisor.

Read the instruction manual carefully before use to ensure correct operations.

	Warning	Always adhere to the following warnings, as failing to do so may cause serious injury or death. Do not perform any operations that are not outlined in this instruction manual.
When handling liquid material, ensure that it is used and managed appropriately in accordance with the Safety Data Sheet (SDS) under the supervision of qualified personnel. Some types of liquid material may have adverse effects on the human body. Some types of liquid material may react with parts of this device and cause an explosion or fire.		
Do not use the device near naked flames, explosives, or flammable or corrosive gas environments, or wash any parts that are not in direct contact (wetted parts) with liquid material. Doing so may cause an explosion, fire, earth leakage, or electric shock.		
Do not perform any unnecessary disassembly or modifications other than for maintenance purposes. Doing so may cause a fire, electric shock, or injury. * Removing any stickers or labels from the device, or performing any unnecessary disassembly or modifications other than for maintenance purposes will void the warranty. Contact our sales department if repairs are required.		
Use MUSASHI Engineering specified parts, accessories and cables, including the power cord. Use AC adapters and power cords that meet the standards of the country the device is being used. Incorrect specifications may cause a fire, electric shock, or injury. Parts that are not attached securely may fly off due to fast movement during dispensing operation. Sharp parts or liquid material may cause loss of vision or injury if they hit eyes or skin.		
Do not damage the power cord or cables. Do not allow liquid material to adhere to the connectors. Doing so may cause a fire, electric shock, or injury. Do not force the device between walls or shelves. Do not place heavy items on the device or subject it to loads. Regularly wipe off any dust. When doing so, always hold the plug to disconnect it.		
DC24V power should be supplied by using only the AC adapter or the DC IN/OUT connector. Mutual interference with the power supply may cause a malfunction or fire.		
When replacing parts or changing the layout of connected devices, always turn the power off, and remove cables, including the power cord. Moving the device with the power cord connected may cause a fire, electric shock, or injury.		
Wear protective goggles and gloves when using the device. Nozzles and other sharp parts or liquid material may cause loss of vision or injury if they hit eyes or skin.		
Attach securely any parts and accessories that are being used. Nozzles and other sharp parts or adapter tubes and other parts may fly off due to fast movement during dispensing operation. Sharp parts or liquid material may cause loss of vision or injury if they hit eyes or skin.		
When replacing parts or changing the layout of connected devices, always disconnect the air hoses. Doing so while pressurized may cause nozzles or liquid material to fly off. * Parts or liquid material flying off may cause loss of vision or injury if they hit eyes or skin.		

Do not look into the nozzle. Accidental dispense operation or leaking liquid material may cause loss of vision or injury if they come in contact with eyes or skin. Nozzles accidentally piercing eyes or hands may cause loss of vision or injury.
Avoid applying pressure quickly. Nozzles or parts detaching, resulting in liquid material flying off may cause loss of vision or injury if they hit eyes or skin. * Twist the filter/regulator knob fully counter-clockwise, and connect the pressure source after the pressure drops to 0 kPa. * After completing operations, turn the power off, twist the filter/regulator knob fully counter-clockwise to turn the pressure to 0 kPa.
Do not apply excessive pressure to the syringe. Doing so may cause fragments of nozzles or parts, or liquid material in syringes to fly off due to rupturing, which may result in loss of vision or injury if they hit eyes or skin. * The specifications of PSY series of syringes made by MUSASHI Engineering are designed for use at pressures of 0.5 MPa or less, and a temperature range of 10 to 60°C.
Do not use damaged parts. Doing so may mean the parts are not attached securely. This may cause liquid leaks or ruptures. * Fragments of nozzles or parts, or liquid material in syringes may cause loss of vision or injury if they hit eyes or skin. * Syringes made of plastic should be disposed of after use.

	Caution	Always adhere to the following precautions, as failing to do so may not only result in damage to the device, but may also cause injury or damage to surrounding items.
Take care as repeated dispense operations may cause the syringe to become hot. The higher the pressure and higher the dispensing frequency, the hotter the syringe becomes due to compression of air. There is a possibility the syringe may soften and rupture. Touching syringe that have become hot may cause burns.		
Do not operate the device with wet hands. Doing so may cause an electric shock.		
Do not touch the output terminal of the AC adapter. Doing so may cause electric shock. There may be residual voltage, even if the power cord is not connected.		
Place the device in a stable location. The device toppling over and falling down may cause an injury.		

Requests	Always adhere to the following requests, as failing to do so may not only result in poor performance of the device, but may also cause damage.
Use the device within the range of specifications specified for the device.	
Use USB cables and power cables connected to DC IN/OUT that are 3 m or less in length, and 30 m or less in length for other cables. Cables exceeding the above lengths do not comply with CE marking. This device and nearby devices may operate incorrectly if they are affected by noise.	
This product has built-in lithium-ion batteries (primary cells). When disposing of the batteries, do so in an appropriate manner in accordance with regulations.	
This device is a precision instrument, and data stored on the device may be lost due to static electricity, electrical noise, or vibrations. After purchasing the device, customers are advised to keep a second copy of data stored on the device in case of data loss.	
Note that no guarantee is provided for any data loss or details that could not be saved due to damage, incorrect operation or defects with the device.	
Do not use the same power supply line as air-conditioners or motors. Doing so may cause damage or incorrect operation.	
Please do not plug and unplug active line due to cause of malfunction. When you plug or unplug, please switch off all of related power beforehand, also use shielded wire to signal line to avoid a malfunction.	
Use MUSASHI Engineering specified parts and accessories. Select containers, nozzles and other parts that meet the specifications such as dispense amount and consumption.	
Allow the controller warm-up for 15 minutes or more after turning the power on. However, there are cases where the appropriate warm-up time differs, depending on dispensing conditions, so please adjust as appropriate.	
Ensure that the pressure supply is +0.100 [MPa] or more than the set dispensing pressure. To use this device, an "air supply" such as factory air or an air compressor is required. Always connect the separately sold regulator set to the pressure source, to supply clean, compressed air (dry air) to the device). Water, oil, dust or other impurities in the compressed air may cause damage to the compressor within the device. After completing operations, twist the knob of the regulator set connected to the pressure supply port fully counter-clockwise to turn the pressure to 0 [MPa].	
Do not allow liquid material to flow in reverse or adhere to the device. Doing so may cause a malfunction.	

Do not cut the power supply while in-dispensing. Depending on the conditions, doing so may cause dripping liquid.
Transfer any liquid material remaining in containers to other containers, or seal containers with dedicated caps (sold separately), and store them the specified location.
Do not use the device, parts or accessories in areas subjected to direct sunlight, areas with high temperatures, areas with sudden temperature changes, areas with high humidity or dust, or areas with poor ventilation. Doing so may cause a malfunction.
Wipe any liquid material or dust from the device, parts or accessories, and store them in a well-ventilated area away from direct sunlight.
Please wipe with IPA or ethanol content soft cloths when you clean products, also do not let the cleaning agent penetrate inside of the product to avoid a damage.
Purchase spares of consumables like syringes and nozzles in advance, and after use wipe clean adapter tubes and other parts that are reused.
* Nozzles cannot be reused if the holes become blocked with material.
Do not subject the device to strong loads like shocks or dropping the device. Doing so may cause a malfunction.
Do not place heavy items on the device. Doing so may result in the device tilting and cause a malfunction.

After-sales Service

The warranty of this device is valid for one year after it is delivered at a location specified by MUSASHI Engineering.

If malfunctions attributed to MUSASHI Engineering occur during the warranty period, parts shall be replaced or repaired by MUSASHI Engineering. Note that the warranty will be void in the following cases [1] through [6].

- [1] The malfunction is caused by a product that is not a MUSASHI Engineering product.
- [2] If the device cover is opened by persons other than MUSASHI Engineering, or the device has been modified or repaired.
- [3] The device is handled or used improperly by the customer.
- [4] Malfunctions caused during transportation that are not the responsibility of MUSASHI Engineering.
- [5] Malfunctions caused by other natural disasters or disasters that are not the responsibility of MUSASHI Engineering.
- [6] Parts that require replacement at specified timing.

Requesting repairs during the warranty period

Carefully read and check the instruction manual again, and if faults are present, contact our sales department with the serial number and product name (model name) included on the device. Depending on the warranty details, MUSASHI Engineering will perform repairs.

Requesting repairs after the warranty period

If repairs mean the device can continue functioning, repairs will be performed at the customer's expense.

Basic Operations

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1. Preparation

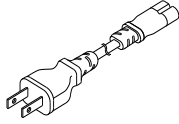
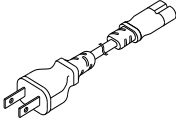
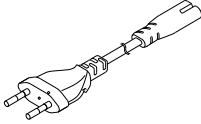
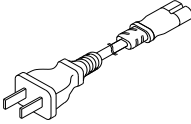
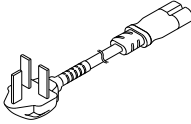
Confirmation of components

List of standard components

No.	Product Name	Model	Quantity
1	SUPER SIGMA CMIV DISPENSE CONTROLLER [V5 NPN] SUPER SIGMA CMIV DISPENSE CONTROLLER [V5 PNP] SUPER SIGMA CMIV DISPENSE CONTROLLER [V2 NPN] SUPER SIGMA CMIV DISPENSE CONTROLLER [V2 PNP]	S-SIGMA-CM4-CTR-V5-N S-SIGMA-CM4-CTR-V5-P S-SIGMA-CM4-CTR-V2-N S-SIGMA-CM4-CTR-V2-P	1 of either
2	POWER CORD	See power cord chart.	1 of either
3	AC ADAPTER [TYPE 4]	ACAT-4	1
4	MIST SEPARATOR REGULATOR SET [TYPE A]	F/R-MS-SETA	1
5	REGULATOR CONNECTION JOINT SET	F/R-MS-JSET	1
6	ADAPTER TUBE [WITH STOPPER Ø3ML 1.0M] ADAPTER TUBE [WITH STOPPER ØML SIGMA USE 1.0M] *Ø:05, 10, 20 to 70 ADAPTER TUBE [△ SIGMA USE 1.0M W-TYPE] *△:5, 10, 30/55	AT-3E-H-1.0M AT-○E-H-SIGMA-1.0M *○:5, 10, 50 AT-▲W-H-SIGMA-1.0M *▲:5, 10, 55	1 of either
7	CONNECTOR SET [DSUB25PIN]	MC-DSUB25P	1
8	CONNECTOR SET [DSUB9PIN]	MC-DSUB9	1
9	PC CONNECTION CABLE [DSUB 9PIN INTERLINK 1.8 m] PC CONNECTION CABLE [DSUB 9PIN INTERLINK 3.0 m]	MUCAD-C7-SMM-1.8M MUCAD-C8-SMM	1 of either
10	e-CON CONNECTOR SET [PLUG 3-pin]	MC-ECON3P	1
11	CONNECT CABLE [USB C-C TYPE 2.0M]	CB-USB-C-C-2.0M	1
12	MuCAD BOOK [FOR STARTER]	MUCADBOOK-1-ST	1
13	Instruction Manual	-	1
14	Warranty	-	1

This is a list of standard components required for using this product, and the chart also includes optional items. Select optional items for No. 2 through 11.
Regarding power supply without AC adapter, please refer to P.134"Product Specifications" and P.135"DC IN/OUT specifications".

Power cord chart

Model: PK2-1	Model: PK2-A	Model: PK2-C	Model: PK2-J	Model: PK2-BF
FOR JAPAN/TAIWAN	FOR NORTH AMERICA	FOR KOREA/EU	FOR CHINA	FOR SINGAPORE/UK
				

* Always use a cable that is suitable for the region in which you are using the product.

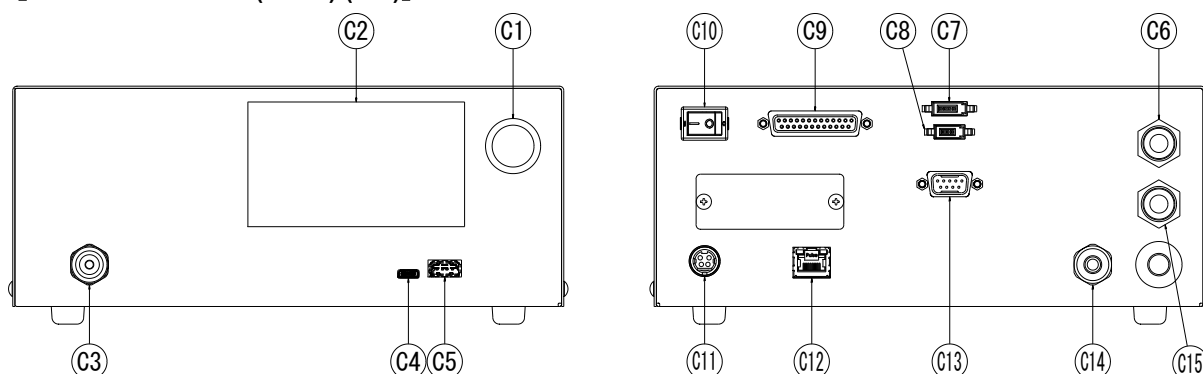
Models with specification differences are named according to the following chart.

Model	Name
S-SIGMA-CM4-CTR-V5-(N/P)	V5
S-SIGMA-CM4-CTR-V2-(N/P)	V2

Names of Parts

Controller

【S-SIGMA-CM4-CTR-(V5/V2)-(N/P)】

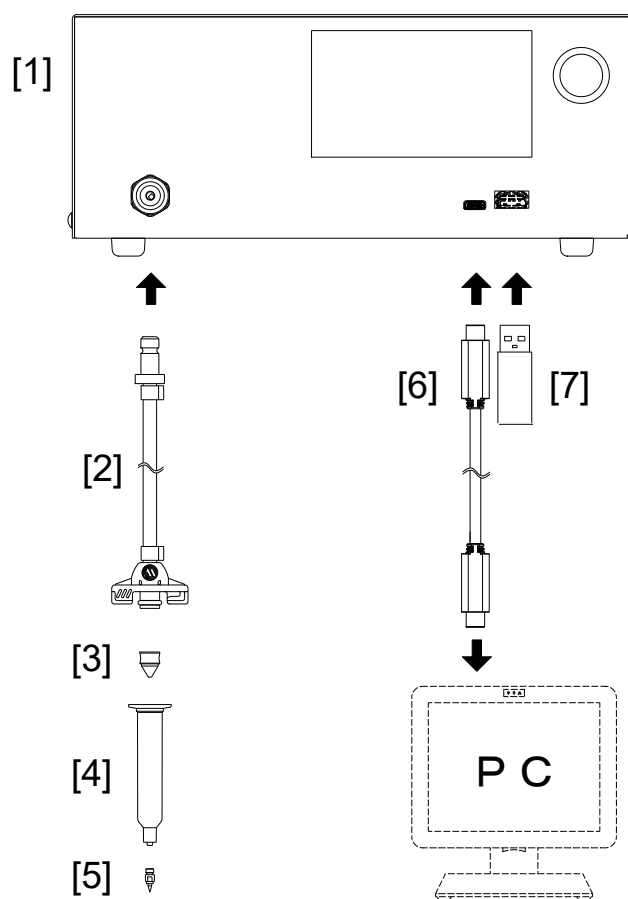


No.	Name	Function
C1	SHOT	Used for dispense request ON/OFF.
C2	LCD Touch Panel	Used for displaying various information and operation
C3	AIR OUT	Socket for supplying pressure.
C4	USB-1 connector	A USB Type-C connector. Used for communication with PC.
C5	USB-2 connector	A USB Type-A connector. Used for connecting with USB memory.
C6	Vacuum Exhaust Port	Discharges air for vacuum.
C7	COM	Used for connecting with temperature controller.
C8	DC IN/OUT	POWER SUPPLY section. (See P.135"DC IN/OUT specifications")
C9	I/O Connector	Used for input/output of various signals.
C10	POWER	Used for turning power ON/OFF.
C11	DC Jack	POWER SUPPLY section. (See P.134"Product Specifications"
C12	LAN Port	Used for Ethernet communications.
C13	RS-232C connector	Used for RS-232C communication.
C14	Air Input Port	Provides air supply pressure. V5: Dispensing pressure +100 kPa or more, up to 700 kPa V2: Dispensing pressure +100 kPa or more, up to 400 kPa
C15	Exhaust port	Discharges exhaust air.

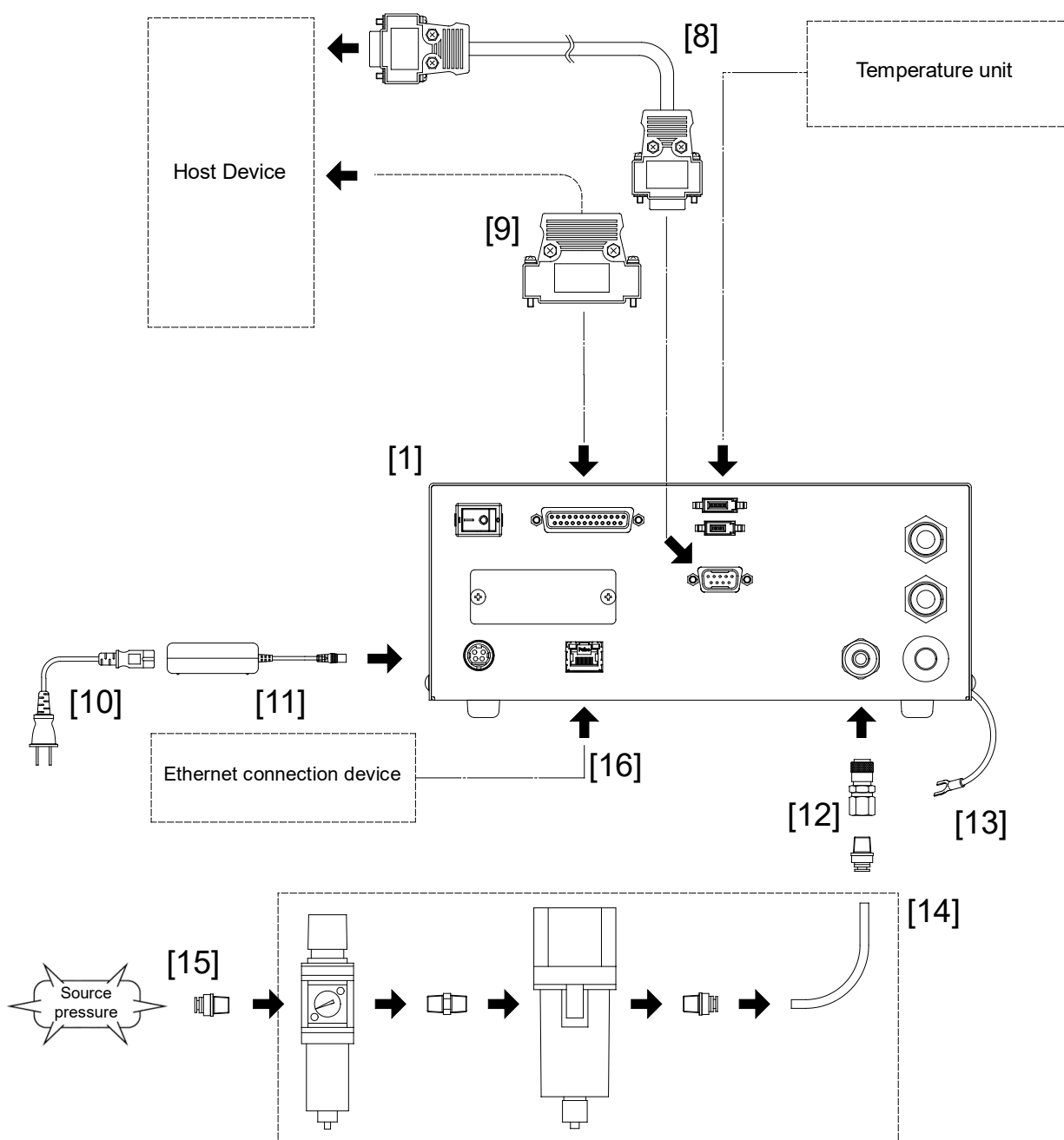
* Do not attach tubes to or block C6, C15. Doing so may impact dispensing performance.

Connections of Parts

Front connections

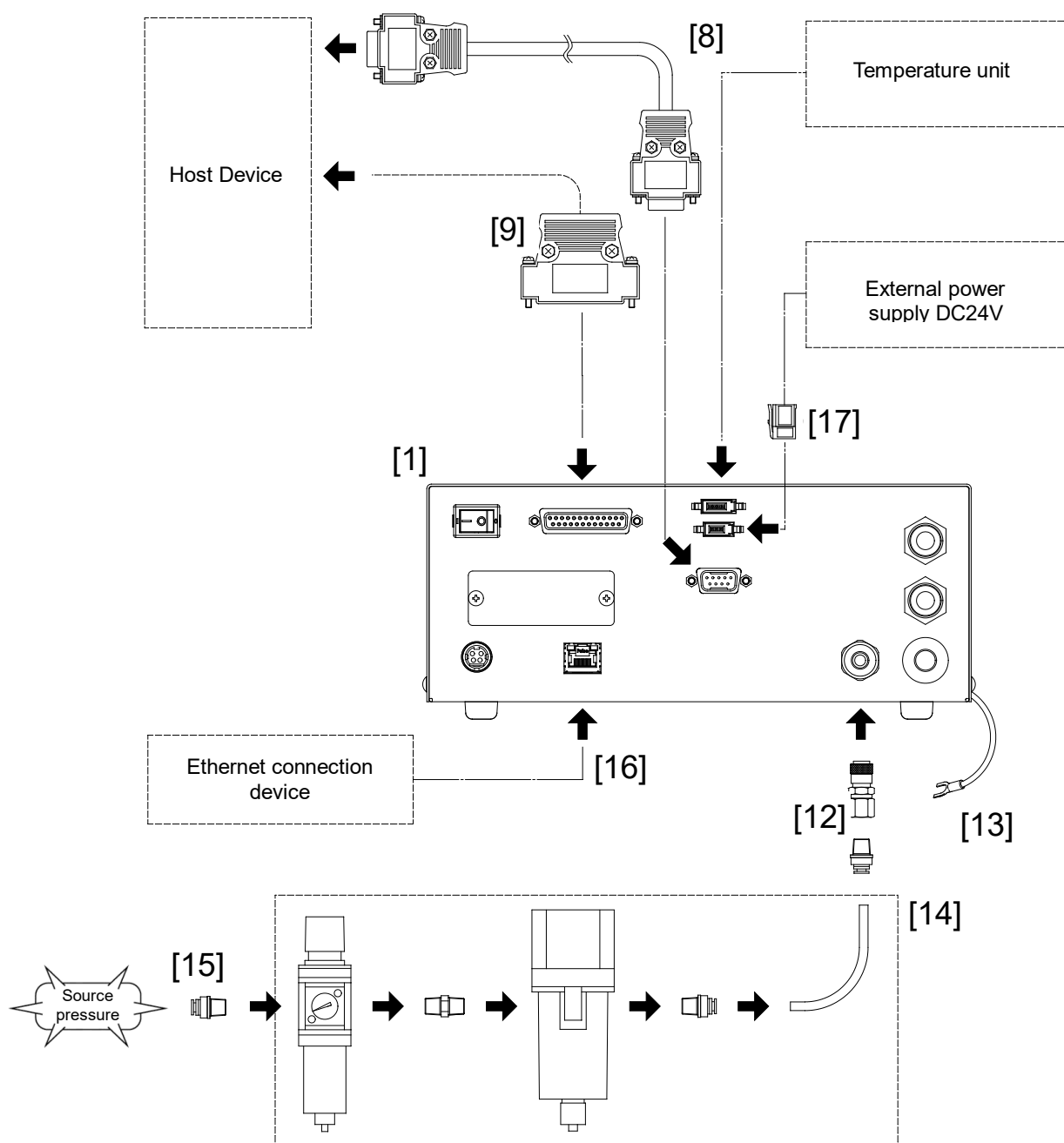


Examples of rear connections (from POWER SUPPLY DC jack)



- * Always connect a MIST SEPARATOR REGULATOR SET [TYPE A] in order to supply clean, dry air to the device.
- * If there is noise interference, use measures such as ferrite beads on cables for [8], [9], [10], and [11] to prevent noise.
- * The AC ADAPTER is not included with this product.
- * See P.134 "Product Specifications" for compatible POWER SUPPLIES and specifications.
- * Please fix grounding wire (provided by customer) on housing screw (M3 x 6L) of [1].
- * A common GND is used for the control board, POWER SUPPLIES, D-Sub25, and chassis.

Examples of rear connections (from POWER SUPPLY DC IN/OUT)



- * Always connect a MIST SEPARATOR REGULATOR SET [TYPE A] in order to supply clean, dry air to the device.
- * If there is noise interference, use measures such as ferrite beads on cables for [8] and [9] to prevent noise.
- * See P.134 "Product Specifications", P.135 "DC IN/OUT specifications" for compatible POWER SUPPLIES and specifications.
- * Please fix grounding wire (provided by customer) on housing screw (M3 x 6L) of [1].
- * A common GND is used for the control board, POWER SUPPLIES, D-Sub25, and chassis.

Components Chart

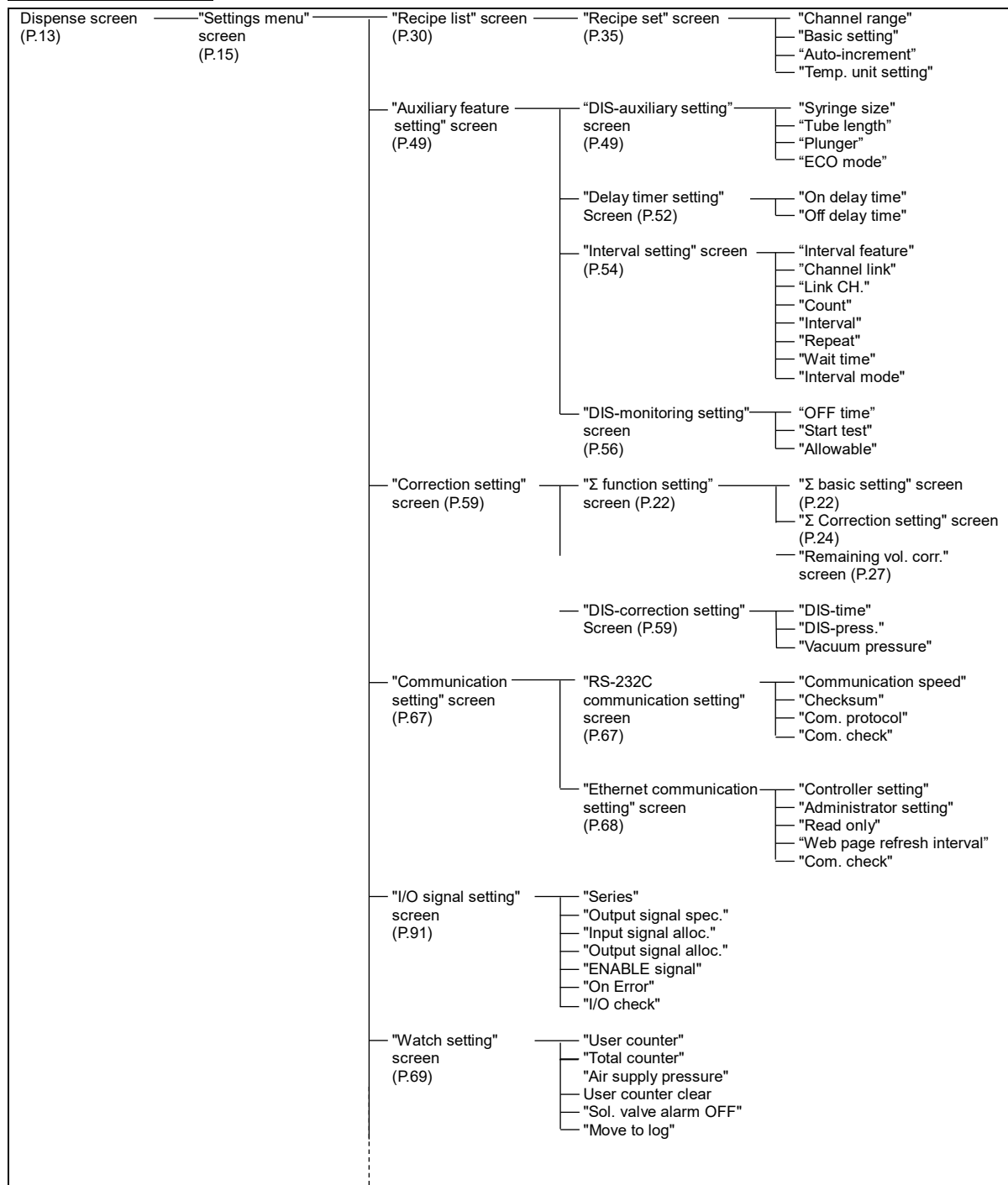
No.	Name	Model
[1]	SUPER SIGMA CMIV DISPENSE CONTROLLER [V5 NPN] SUPER SIGMA CMIV DISPENSE CONTROLLER [V5 PNP] SUPER SIGMA CMIV DISPENSE CONTROLLER [V2 NPN] SUPER SIGMA CMIV DISPENSE CONTROLLER [V2 PNP]	S-SIGMA-CM4-CTR-V5-N S-SIGMA-CM4-CTR-V5-P S-SIGMA-CM4-CTR-V2-N S-SIGMA-CM4-CTR-V2-P
[2]	ADAPTER TUBE [WITH STOPPER 03ML 1.0M] ADAPTER TUBE [WITH STOPPER ◎ SIGMA USE 1.0M] *◎:05, 10, 20 to 70 ADAPTER TUBE [△ SIGMA USE 1.0M W-TYPE] *△:5, 10, 30/55	AT-3E-H-1.0M AT-○E-H-SIGMA-1.0M *○:5, 10, 50 AT-▲W-H-SIGMA-1.0M *▲:5, 10, 55
[3]	Plunger	-
[4]	Liquid material container	-
[5]	Nozzle	-
[6]	CONNECT CABLE [USB C-C TYPE 2.0M]	CB-USB-C-C-2.0M
[7]	USB memory	-
[8]	PC CONNECTION CABLE [DSUB 9PIN INTERLINK 1.8 m] PC CONNECTION CABLE [DSUB 9PIN INTERLINK 3 m]	MUCAD-C7-SMM-1.8M MUCAD-C8-SMM
[9]	CONNECTOR SET [DSUB25PIN]	MC-DSUB25P
[10]	AC ADAPTER CORD [FOR JAPAN/TAIWAN 2PINS] AC ADAPTER CORD [FOR NORTH AMERICA 2PINS] AC ADAPTER CORD [KOREA/EU 2PINS] AC ADAPTER CORD [FOR CHINA 2PINS] AC ADAPTER CORD [SINGAPORE/UK 2PINS]	PK2-1 PK2-A PK2-C PK2-J PK2-BF
[11]	AC ADAPTER [TYPE 4]	ACAT-4
[12]	REGULATOR CONNECTION JOINT SET	F/R-MS-JSET
[13]	Grounding wire	-
[14]	MIST SEPARATOR REGULATOR SET [TYPE A]	F/R-MS-SETA
[15]	HALF UNION (CONNECTION DIAMETER: Rc 1/4)	J-U2B-8
[16]	LAN cable	-
[17]	e-CON CONNECTOR SET [PLUG 3-pin]	MC-ECON3P

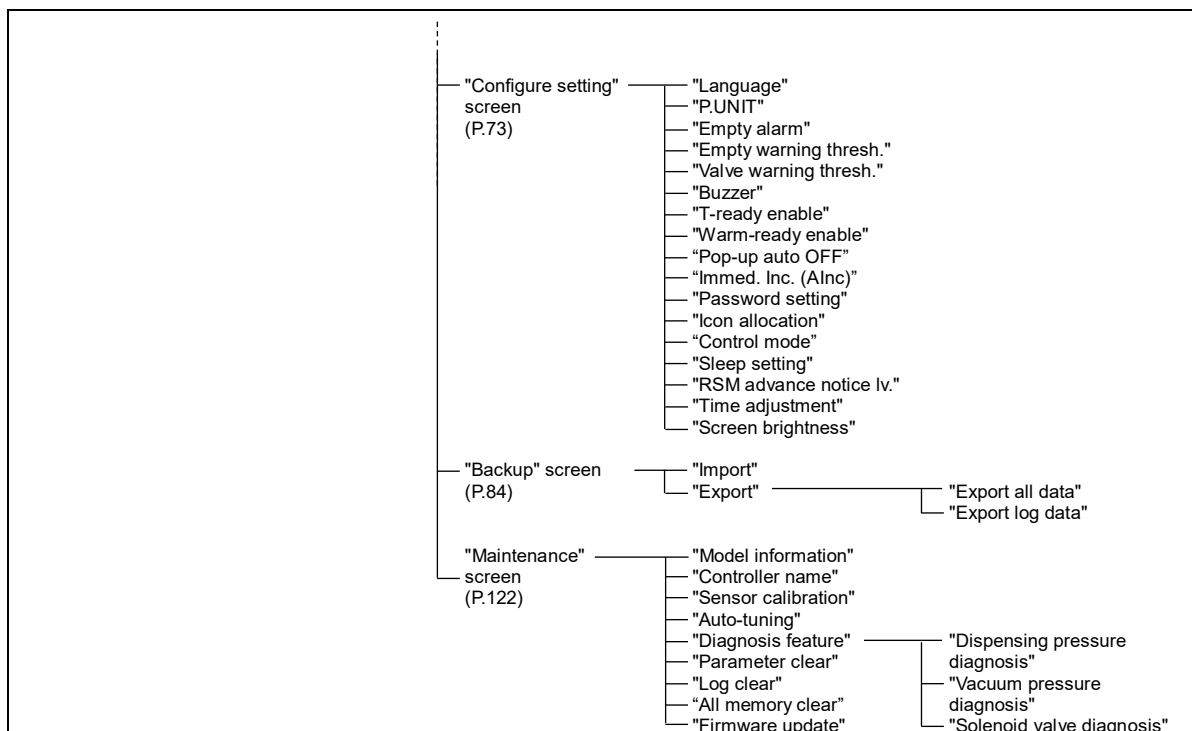
* Contact our sales department for various nozzles, plungers and liquid material containers.

2. Basic Operations

Description of Screens

Screen transitions





Screen operation

Navigate around the screen using the following methods.

[Tap]



Tap refers to the action of lightly touching LCD Touch Panel with your finger, and then releasing it quickly.

[Press and hold]



Press and hold refers to the action of lightly touching LCD Touch Panel for a certain period of time without moving your finger.

Screen Layout

[Dispense screen A]

In normal mode

Tapping this moves to the Number Input Screen or Character Input Screen.

Tapping this moves to the "Settings menu."

Main icons. Tapping these performs the actions associated with each icon. This can be customized in "Icon allocation" (P.78).

User counter. Also, the warm-up timer is displayed for 15 minutes after the power is turned on.

Sub icons. Tapping these performs the actions associated with each icon.

In Σ (sigma) mode

The dispensing time, dispensing pressure and vacuum pressure for the Σ mode CH cannot be edited. Σ mode must be canceled.

[Dispense screen B]

In normal mode

Tapping this moves to the Number Input Screen or Character Input Screen.

Tapping this moves to the "Settings menu."

Main icons. Tapping these performs the actions associated with each icon. This can be customized in "Icon allocation" (P.78).

User counter. Also, the warm-up timer is displayed for 15 minutes after the power is turned on.

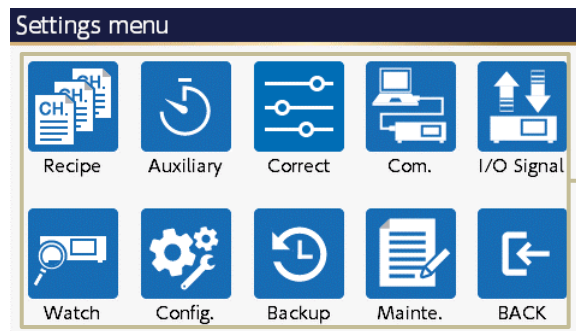
Sub icons. Tapping these performs the actions associated with each icon.

In Σ (sigma) mode

Displays remaining fluid volume in the syringe.

The dispensing time, dispensing pressure and vacuum pressure for the Σ mode CH cannot be edited. Σ mode must be canceled.

["Settings menu" screen]



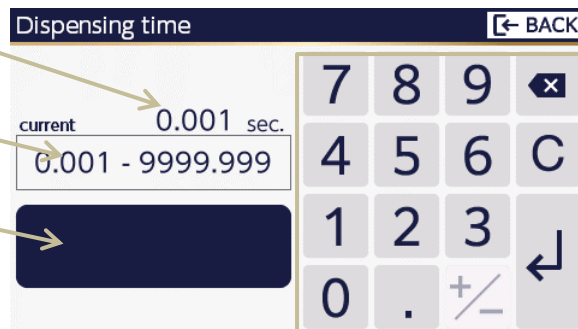
Tapping this moves to each setting.

[Number Input Screen]

The value currently set for the parameter is displayed.

The setting range of the parameter is displayed.

The current input value is displayed.

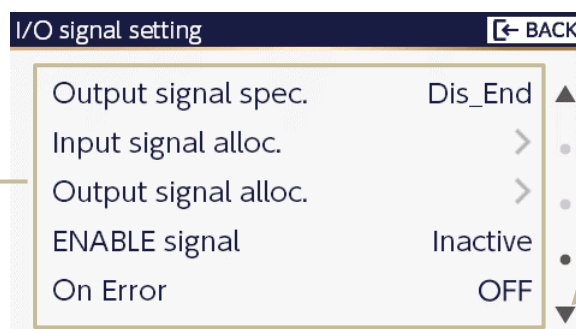


Returns to the previous screen.

Numeric keypad.

[Setting Screen]

Tapping this moves to the modify or number input screen.



Returns to the previous screen.

Tapping or pressing and holding this moves the displayed parameters up.

Tapping or pressing and holding this moves the displayed parameters down.

[Character Input Screen]

The entered number is displayed.

Tapping this can change the character type.



Returns to the previous screen.

Characters that can be entered are alphanumeric characters and symbols.

[Main Icon]

Main icons can be customized in "Icon allocation." See P.78 "Changing Icon Allocation (Icon Allocation)" for more details.

Main Icon	Function	Edit Allocation
	Moves to the "Settings menu."	×
 	Switches the Dispensing mode.  : Displayed if in TIMED mode.  : Displayed if in MANUAL mode.	○ (default value)
 	Moves to the screen for the Σ function.  : Displayed if in Normal mode. Tap to move to the "Σ basic setting" screen.  : Displayed if in Σ mode. Tap to move to the "Σ basic setting" screen. Press and hold to move to the "Σ mode setting" screen.	○ (default value)
	Moves to the "DIS-correction setting" screen.	○
	Moves to the "Channel list" screen or the "Channel graph" screen.	○ (default value)
	Moves up one channel. * Cannot move up beyond 401CH.	○
	If channel belongs to a recipe: Channel changes to Start CH. of the associated recipe. If channel does not belong to a recipe: Channel changes to 1CH.	○
	Moves down one channel. * Cannot move down below 0CH.	○
	If a channel belongs to a recipe and the Auto-increment function is ON, this changes the Start CH. and starts Auto-increment operation.	○
	Clears the User counter.	○
 	Switches the Interval feature between ON/OFF.  : Displayed if Interval feature is OFF.  : Displayed if Interval feature is ON.	○
	Moves to the "Select Start CH." screen. Selecting the recipe moves to the Start CH.	○
	No function available.	○

Display of   varies depending on the state of other features.

	- In Σ+TIMED mode - If Auto-increment function, where Dispensing mode is TIMED mode, has been turned ON - When the interval feature is ON
	- In Σ+MANUAL mode - If Auto-increment function, where Dispensing mode is MANUAL mode, has been turned ON
	- In dummy shot mode Tapping  in Σ+TIMED mode switches to dummy shot mode. Manual dispensing is performed in response to a dispense instruction. Tapping  switches to Σ+TIMED mode.

[Sub Icon]

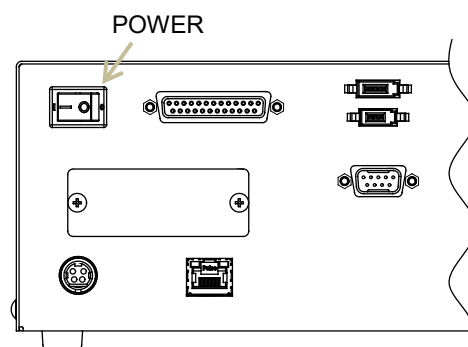
Sub Icon	Function
 	Moves to the password input screen. Lock/unlock the software by entering the password.  : Displayed if the password lock function is OFF.  : Displayed if the password lock function is ON.
  	Moves to the log screen.  : Displayed when conditions are normal.  : Displayed when an alarm occurs.  : Displayed when an error occurs.
	The following setting information is displayed with page feed. - Syringe size, adapter tube length, and plunger usage set for current CH. - ON/OFF state for each Σ function if current CH is in Σ mode - Channel range, liquid material name, and nozzle name if current CH belongs to a recipe
   	Displays the Temperature unit setting of the associated recipe.  : Displayed if channel does not belong to a recipe, or control of not even one temperature controller is set to ON.  : Displayed when the temperature controller is operating normally.  : Displayed when a temperature controller alarm occurs.  : Displayed when a temperature controller error occurs.
	Switches the Dispense screen display.

[Menu Icon]

Menu Icon	Function	Reference
	Moves to the "Recipe list" screen. Used to create or edit recipes.	P.30
	Moves to the "Auxiliary feature setting" screen. Used to configure the DIS-auxiliary parameters, delay timer, Interval feature and DIS-monitoring settings.	P.49
	Moves to the "Correction setting" screen. Used to configure " Σ mode" and "DIS-correction".	P.59
	Moves to the "Communication setting" screen. Used to change communication settings.	P.67
	Moves to the "I/O signal setting" screen. Used to configure I/O signal settings.	P.91
	Moves to the "Watch setting" screen. Used to check items like User counter and Source pressure.	P.69
	Moves to the "Configure setting" screen. Used to change Configure settings like alarm detection, password lock, language, and P.UNIT.	P.73
	Moves to the "Backup" screen. Used to Import and Export parameters saved on the device.	P.84
	Moves to the "Maintenance" screen. Used for sensor calibration or Parameter clear.	P.122
	Returns to the Dispense screen.	-

Turning the Power ON

- 1 Press POWER to turn the power ON.
* Do not input external signals until the Dispense screen is displayed. Doing so may cause incorrect operation.



- 2 The startup screen is displayed.



- 3 The "Select Language" screen is displayed. Select the required language.

* When starting up for the first time only.



- 4 After selecting the language, the "Time adjustment" screen is displayed. Adjust the current time.

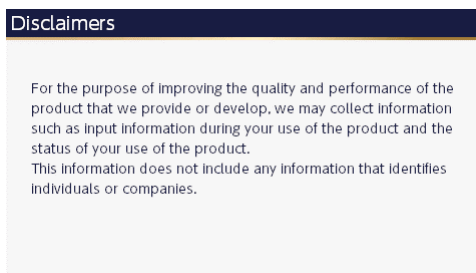
* When starting up for the first time only.

* For setting the time zone, refer to P.82.

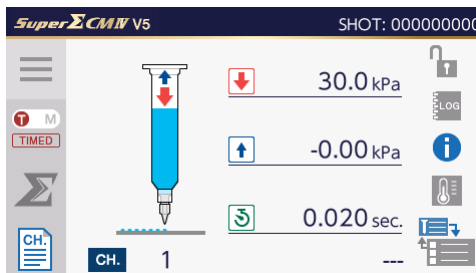


- 5 After completing Time adjustment, the "Disclaimers" screen is displayed.

* When starting up for the first time only.



- 6 Tapping the panel when the "Disclaimers" screen is displayed moves to Dispense screen B. Doing so allows dispense operations.



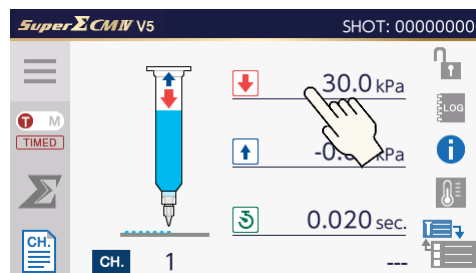
Dispensing Liquid Material

The following parameters related to dispensing are mainly set in Dispense screen A or B.

- Dispensing time
- Dispensing pressure
- Vacuum pressure
- Dispensing mode (TIMED mode/MANUAL mode)

Dispensing Pressure setting

- 1 Tap the displayed dispensing pressure.



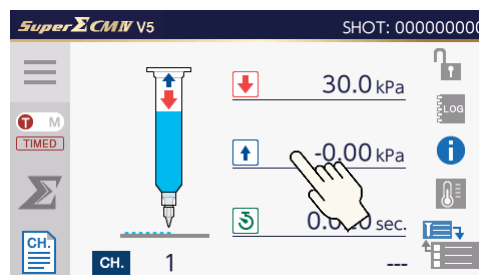
- 2 The "DIS-press." input screen is displayed. Set the dispensing pressure. When settings are completed, the screen returns to the previous screen.

* The image on the right is for V5.

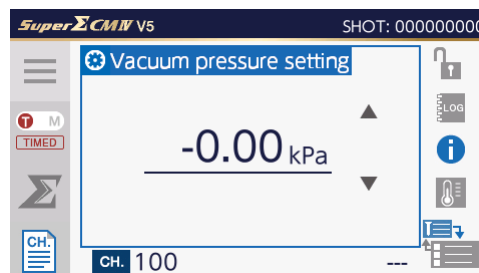


Vacuum Pressure setting

- 1 Tap the displayed vacuum pressure.



Tapping [▲] [▼] enables adjustment of Vacuum pressure while changing the setting.

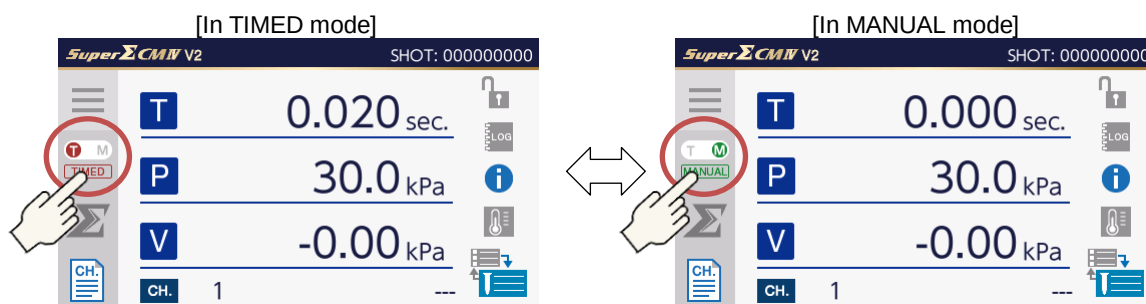


- * The "Vacuum pressure" input screen is displayed by pressing and holding the vacuum pressure value on the dispense screen, or tapping the vacuum value on the "Vacuum pressure setting" screen. Vacuum pressure can be input numerically. When settings are completed, the screen returns to the previous screen.



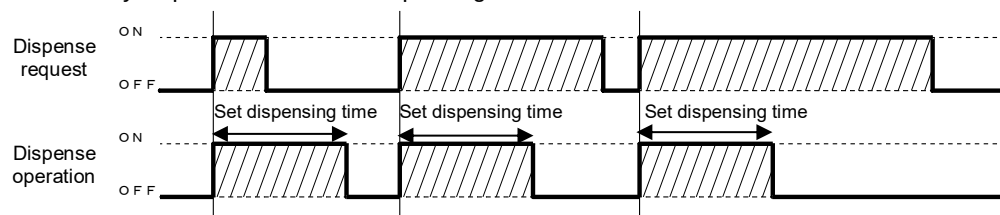
Dispensing Mode settings

Tapping [] or [] switches between TIMED mode and MANUAL mode.



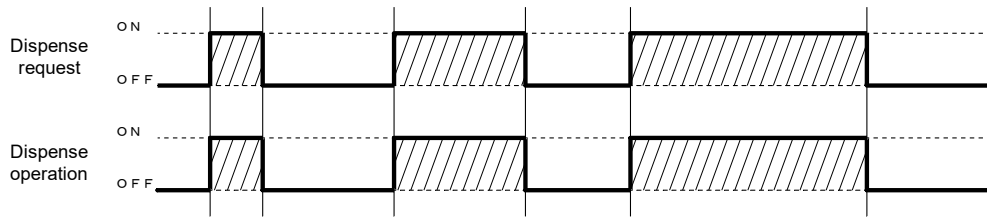
[TIMED mode]

Material is only dispensed for the set dispensing time.



[MANUAL mode]

Material is only dispensed while SHOT is pressed, or an external dispense signal is input.

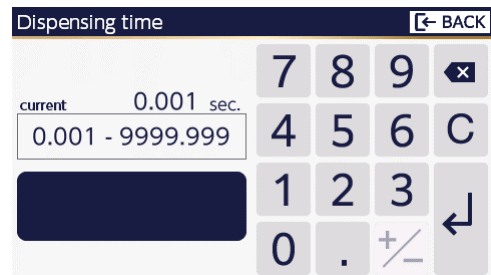


Dispensing Time setting

- 1 Tap the displayed dispensing time.



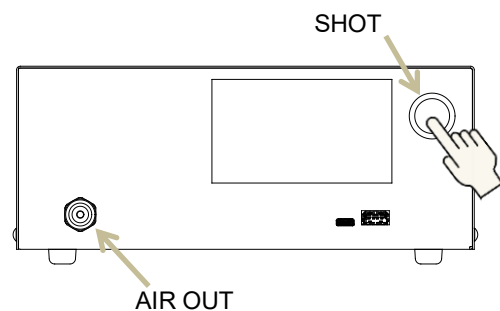
- 2 The "DIS-time" input screen is displayed. Set the dispensing time. When settings are completed, the screen returns to the previous screen.



Dispense

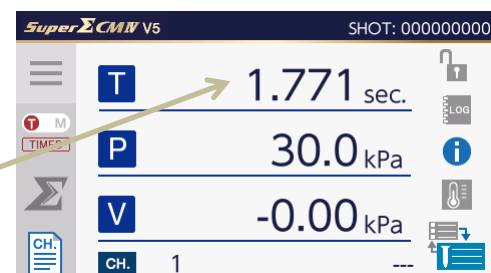
Dispensing is only valid with the Dispense screen.

- 1 Press SHOT. Air is output from AIR OUT in accordance with the Dispensing mode.



- 2 During In-dispensing, the dispensing time section displays the elapsed time from the start of dispensing.

Displays elapsed time



3 Sigma Function (Stable Dispensing Automation)

Basic Operation

Brief description of the sigma function

This device is equipped with the following functions to enable high-precision dispensing. These 3 functions are collectively referred to as the sigma function.

- | | |
|---|---|
| • Automatic correction of volume reduction function (*) | : Automatically corrects for variance in dispensing volume due to reductions in remaining fluid volume. |
| • Automatic fluid-drip prevention function | : Automatically adjusts vacuum pressure in accordance with the remaining fluid volume. |
| • Automatic remaining volume alert function | : An alarm (signal) is output when the remaining fluid amount reaches a set value. |

*Valid only in TIMED mode.

Sigma function use conditions

The sigma function can be used with the following combination of settings.

- | | |
|-------------------------|-------------------------------|
| • Adapter tube length | : 0.5, 1.0, 1.5, 2.0 m |
| • E/F-type syringe size | : 3, 5, 10, 20, 30, 50, 70 mL |
| • W-type syringe size | : 5, 10, 30, 55 mL |

*Contact our Sales Department for information on compatible models.

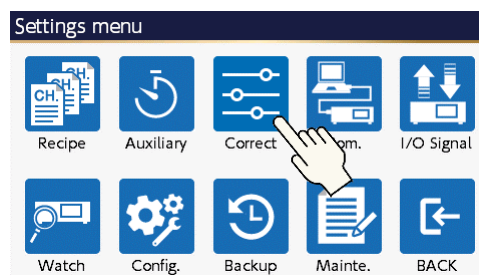
*Using air supply pressure values different from the recommended values (V5: 600 kPa, V2: 300 kPa) may result in incorrect operation.

Sigma function basic setting

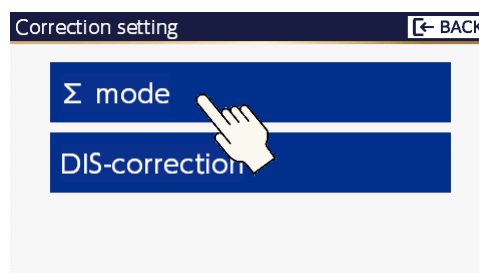
- 1 Tap the [≡] icon in Dispense screen A or B.



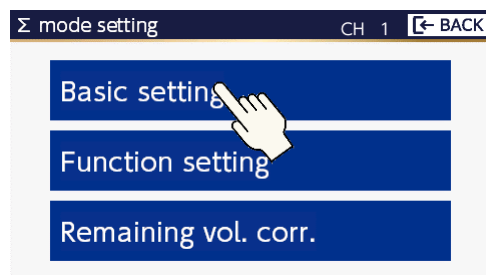
- 2 The "Settings menu" screen is displayed. Tap the [Settings] icon.



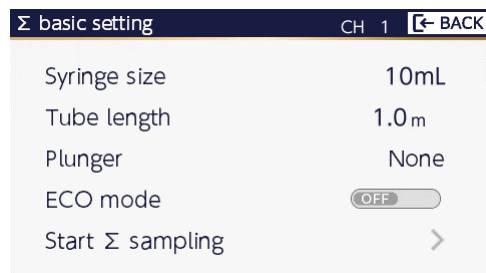
- 3 The correction setting screen is displayed. Tap "Σ mode."



- The Σ mode setting screen is displayed. Tap "Basic setting".



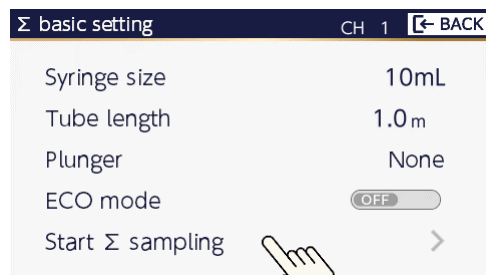
- The " Σ basic setting" screen is displayed. Set syringe size, adapter tube length, plunger usage, and ECO mode ON/OFF.



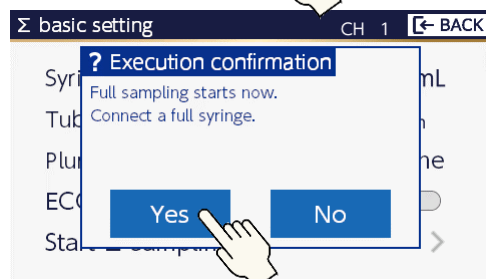
Enabling the Sigma Function

Use the following procedure to enable the sigma function. Perform after setting dispensing conditions.

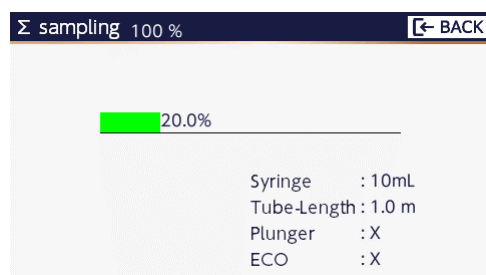
- Tap "Start Σ sampling" in the " Σ basic setting" screen.



- A confirmation screen appears. Connect a full syringe and tap "Yes". The sampling process starts.



- Returns to the Dispense screen when sampling is finished.



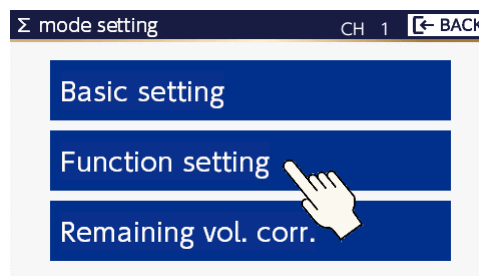
Sigma function tuning

Fine adjustment of the automatic correction of volume reduction function

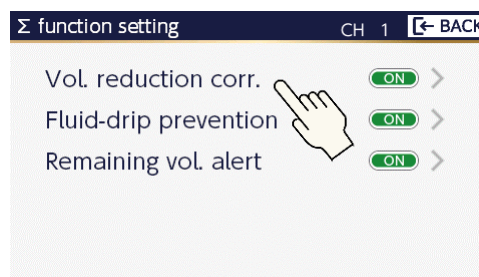
Alpha correction

This function operates in Σ mode and corrects for variance in the dispensing volume that occurs as the fluid volume in the syringe decreases.

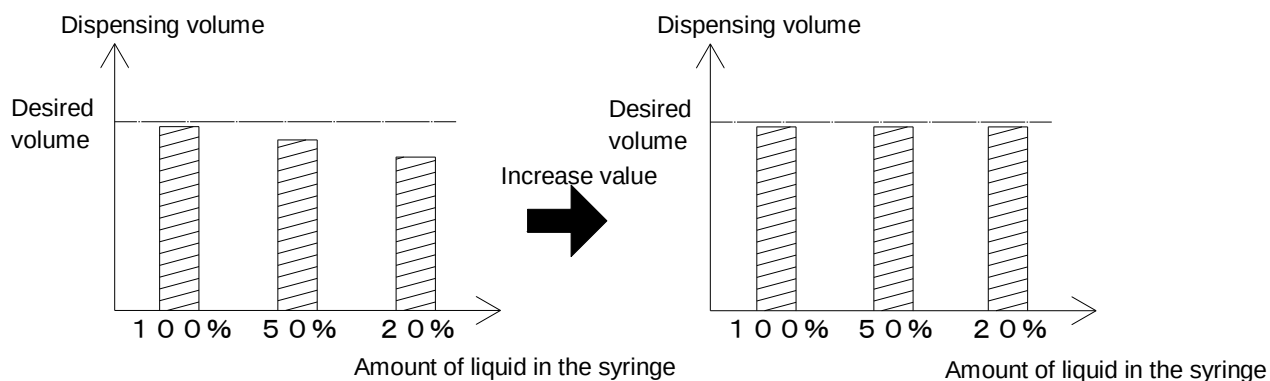
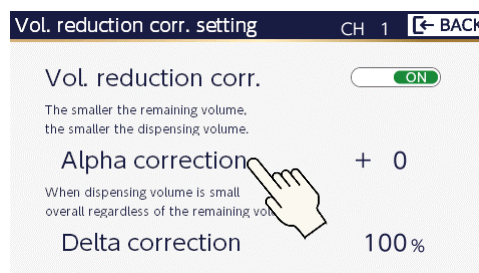
- 1 Tap "Function setting" in the " Σ mode setting" screen.



- 2 The " Σ function setting" screen is displayed." Tap "Vol. reduction corr."



- 3 The "Vol. reduction corr. setting" screen is displayed. Tap "Vol. reduction corr." and turn the function ON. Tap "Alpha correction" and move to the "Alpha correction" setting screen. Adjust the dispensing volume correction.

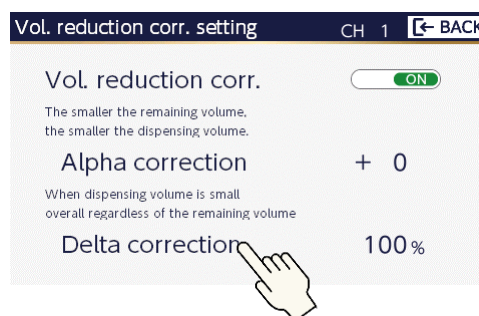


Delta correction

This function operates in TIMED mode only and corrects the dispensing volume when typically too much or too little from the time the syringe is completely filled with liquid until there is only a small amount remaining.

Tap "Vol. reduction corr." and turn the function ON on the "Vol. reduction corr. setting" screen.

Tap "Delta correction" and move to the "Delta correction" setting screen. Adjust the dispensing volume correction.



- * A larger correction value increases the dispensing volume per discharge. A lower correction value decreases the dispensing volume per discharge.
- * The corrected dispensing time will never be less than the minimum dispensing time. Delta correction may not function properly if the dispensing time is short. The following table lists the minimum dispensing time per adapter tube length.

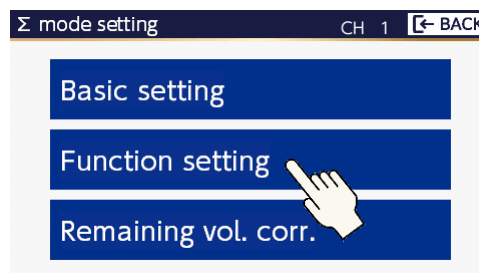
Adapter tube length	Minimum dispensing time
0.5 m	20 ms
1.0 m	20 ms
1.5 m	30 ms
2.0 m	40 ms

Fine adjustment of automatic fluid-drip prevention function

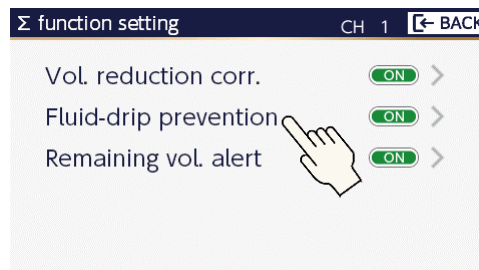
This function enables fine adjustments of vacuum pressure to stabilize the dispensing volume.

- * Liquid may leak if the vacuum pressure is too low.
- * If the vacuum pressure is too high, air bubbles may form in the syringe causing instability in dispensing volume. Note that liquid material may also reflux into the device, which will damage the device.

- 1 Tap "Function setting" in the "Σ mode setting" screen.



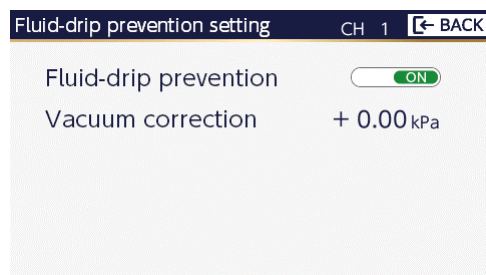
- 2 The "Σ function setting" screen is displayed." Tap "Fluid-drip prevention".



- 3 The "Fluid-drip prevention setting" screen is displayed. Tap "Fluid-drip prevention" and turn the function ON.

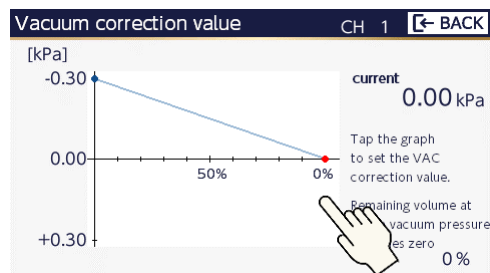
Tap "Vacuum correction".

*Cannot be changed if vacuum pressure set when Σ mode was enabled is zero.



- 4 "Vacuum correction value" setting screen is displayed.

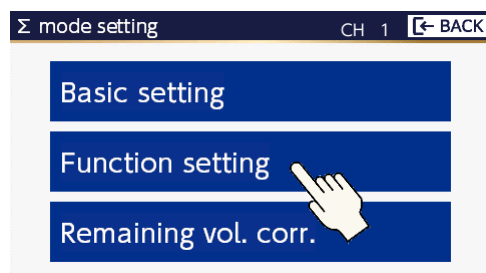
Tap the graph to set vacuum pressure at a remaining volume of 0%.



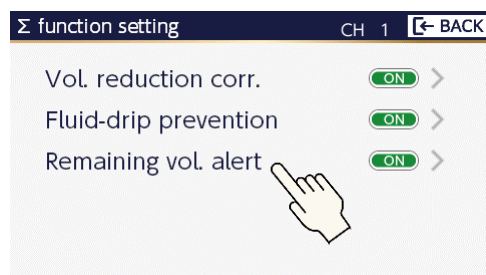
Remaining volume detection function setting

The automatic remaining amount alert is output when the liquid remaining in the syringe becomes low in Σ mode. In the default settings, "RSM level" is 10% and "RSM count" is 5 times. In this case, the alert is output if dispensing is done 5 times consecutively with the liquid at less than 10% of the full amount. The remaining volume detection level and the remaining volume detection count can be changed.

- 1 Tap "Function setting" in the " Σ mode setting" screen.

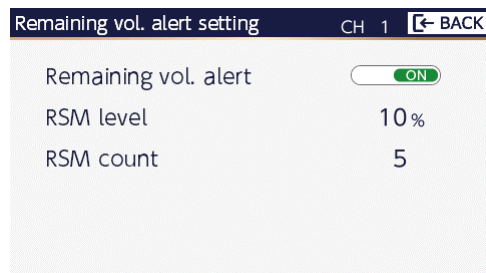


- 2 The " Σ function setting" screen is displayed." Tap "Remaining vol. alert".



- 3 The "Remaining vol. alert setting" screen is displayed. Tap "Remaining vol. alert" and turn the function ON.

Tap "RSM level" or "RSM count", move to the setting screen, and set the value.

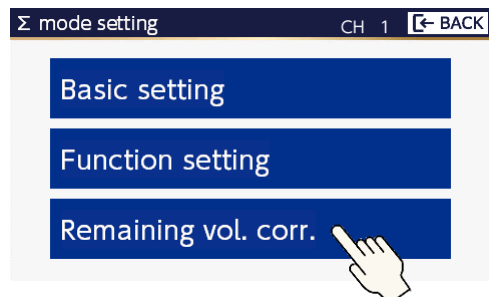


Fine adjustment of remaining volume display value

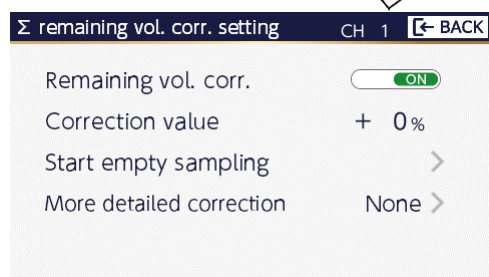
The displayed remaining volume display value can be tuned to more accurately represent the actual liquid level.

* The remaining volume display value is only an approximation. There will sometimes be variance with the actual amount of liquid in the syringe due to mechanical differences between devices, dispensing conditions, the air supply pressure, and the initial liquid amount.

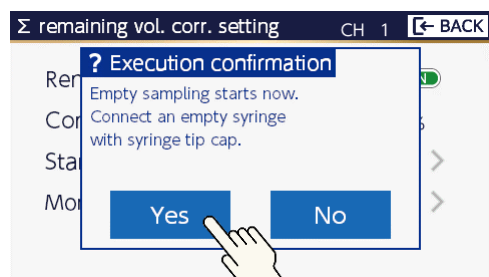
- 1 Tap "Remaining vol. corr." in the "Σ mode setting" screen.



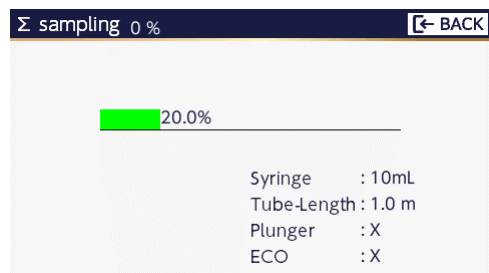
- 2 The "Σ remaining vol. corr. setting" screen is displayed. Tap "Remaining vol. corr." and turn the function ON.



- 3 Tap "Start empty sampling" to display the "Execution confirmation" screen. Connect an empty syringe with a nozzle cap, and tap "Yes".

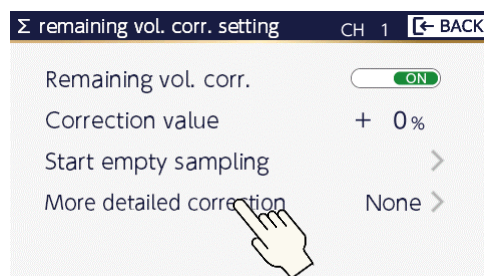


- 4 The empty sampling process starts.

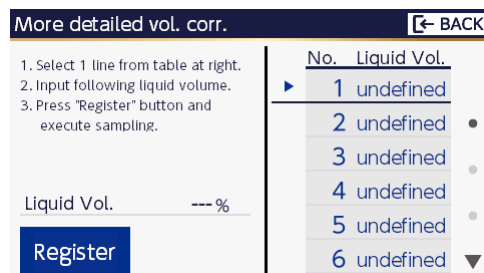


Also, more detailed adjustment of the remaining volume display value can be done through the following operation.

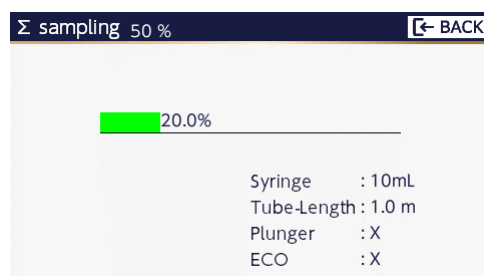
- 1 Turn ON the function on the "Remaining vol. corr." screen, and tap "More detailed correction".



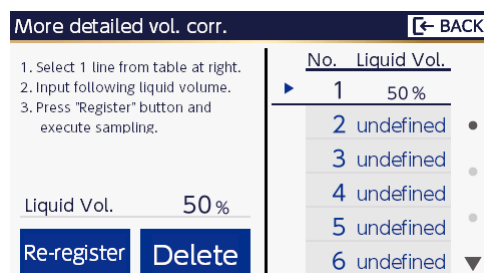
- 2 The "More detailed vol. corr." screen is displayed. Connect a syringe containing the liquid vol. set as the liquid vol. setting, and tap the "Register" button.



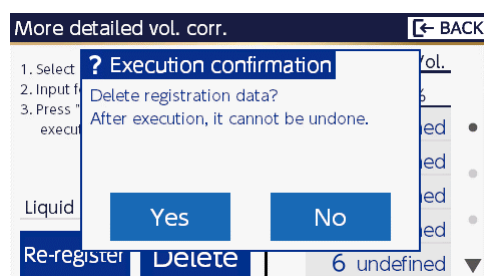
- 3 When "Yes" is tapped on the confirmation screen, the sampling process starts.



- 4 The registered data is added to the list on the right of the screen, and displayed.



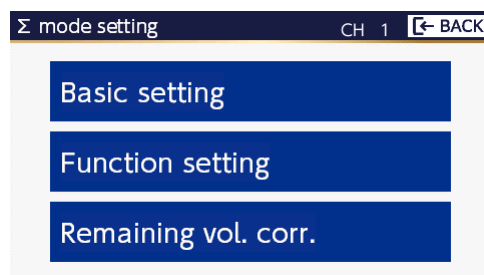
- * To delete registered data, tap the [Delete] button. A confirmation screen is displays, so select "Yes".



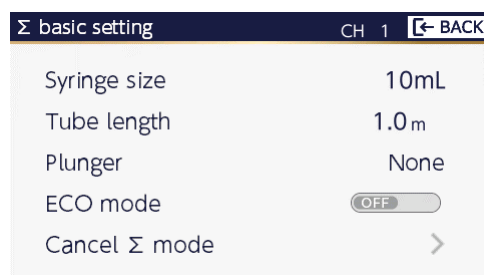
Disabling the Σ function

Perform the following procedure to disable the Σ function that has been used.

- 1 Tap "Basic setting" in the " Σ mode setting" screen.

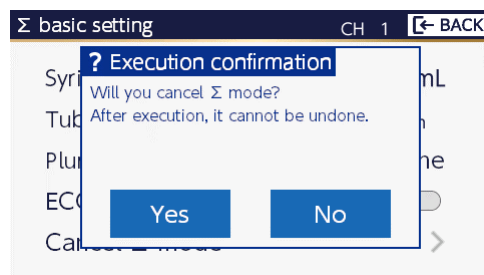


- 2 The " Σ basic setting" screen is displayed. Tap "Cancel Σ mode".



- 3 The "Execution confirmation" screen is displayed. Select "Yes" to cancel the Σ mode and enter the Normal mode.

* The sampling data used is cleared after the Σ mode is disabled. Cleared data cannot be recovered. Refer to P.23 "Enabling the Sigma Function" to reconfigure settings.



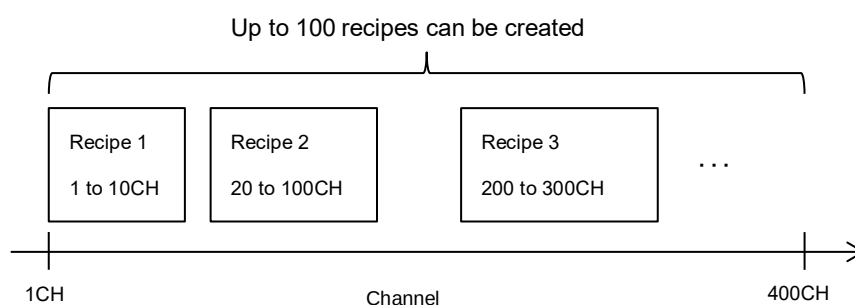
4 Various Functions

Managing Multiple Types

A single recipe can be created with multiple dispensing conditions, which is convenient when managing multiple types (100 recipes in total). Creating a recipe also allows the Auto-increment function (P.36), linkage with temperature unit (P.38) and other functions to be used.

Recipe and Channel

This device has 400 channels, and continuous channels can be created as a single recipe.



The following settings can be saved in one recipe.

Item	Description
"Start CH."	Sets the channel to start the recipe. Tapping the [CH] icon, or inputting a PRESET signal switches to the Start CH. of the currently associated recipe.
"End CH."	Sets the channel to end the recipe.
"Recipe name"	Sets up to 14 characters for the name for managing the recipe.
"Liquid material"	Sets up to 14 characters for the liquid material used with the recipe.
"Nozzle"	Sets up to 14 characters for the nozzle used with the recipe.
"Auto-increment"	Moves to the Auto-increment setting screen.
"Temperature controller"	Moves to the selection screen for the linked temperature controller.

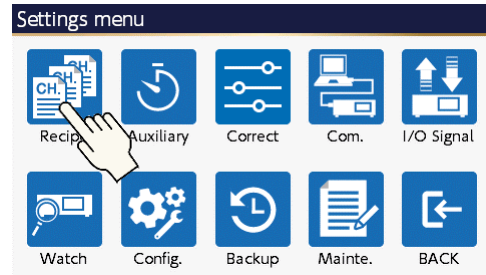
Recipe list

The recipe list displays a list of recipes on the device. These can be changed by tapping and selecting each recipe.

- 1 Tap the [≡] icon in Dispense screen A or B.



- The "Settings menu" screen is displayed.
Tap the  icon.



- The "Recipe list" screen is displayed.

Tapping undefined recipe creates a new recipe.

Tapping an existing recipe moves to the "Recipe set" screen. Pressing and holding displays recipe information.

Recipe list				← BACK
Recipe	CH.	Mode	Name	
2	undefined			
3	undefined			
4	undefined			
5	90 - 100	OFF	Work #001	
6	undefined			
7	undefined			

Tapping or pressing and holding this moves the displayed list up.

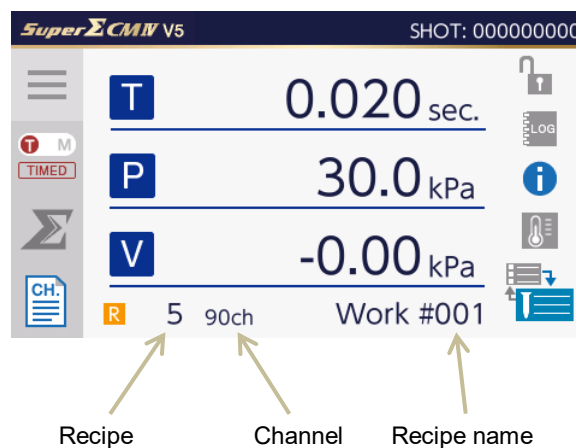
Tapping or pressing and holding this moves the displayed list down.

The meanings of the icons displayed in the [Mode] column are as follows.

Displayed icon	Description
[OFF]	Σ mode CH is not included when auto-increment function is OFF.
[A TIMED]	TIMED mode setting when auto-increment function is ON.
[A MANUAL]	MANUAL mode setting when auto-increment function is ON.
[Σ]	Σ mode CH is included in recipe.

• Displaying the Dispense screen if channel belongs to a recipe

If the channel belongs to a recipe, the Dispense screen is displayed as follows.



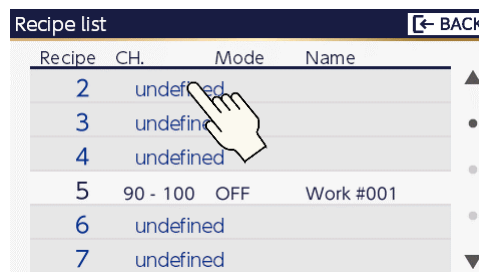
Creating a New Recipe

New recipes can be created with the following three methods.

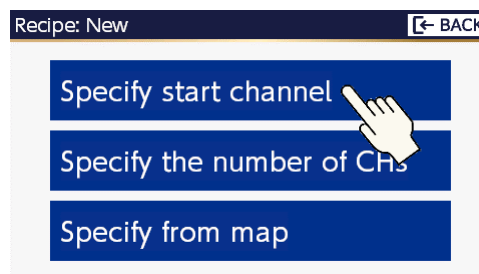
- Specify Start CH. and create
- Specify the number of CHs and create
- Specify from map and create

• To specify Start CH. and create

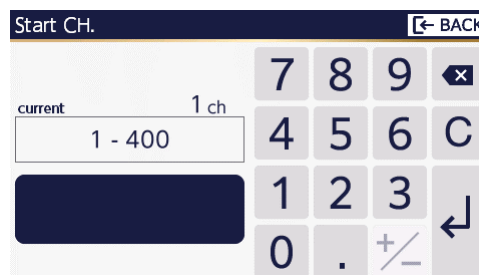
- 1 Tap the "undefined" recipe in the "Recipe list" screen.



- 2 The "Recipe: New" screen is displayed. Tap "Specify start channel."



- 3 The Start CH. input screen is displayed. Set the Start CH.



- 4 When the Start CH. settings are completed, the "Recipe set" screen is displayed. Configure each setting.

* End CH. is set with the maximum continuous channels from Start CH. that can be set for the recipe. If multiple recipes are used, set End CH. to suit the number of channels used in the recipe.

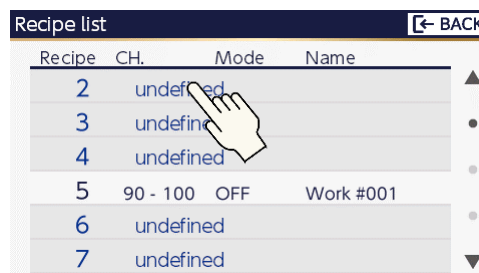


- 5 When all settings are completed, tap the [ SAVE] icon in the "Recipe set" screen. The settings are saved. The screen returns to the recipe list screen with the [ BACK] icon.

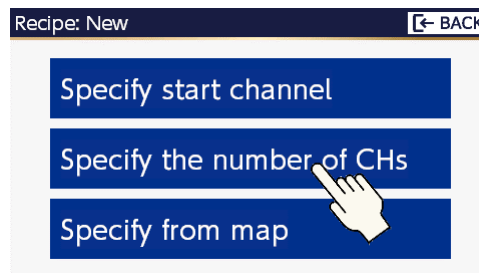
- * If the [ SAVE] icon is displayed, there is unsaved, changed data. If the [ BACK] icon is tapped in this state, the "Save confirmation" screen is displayed. Select "Yes" to save the settings and return to the Recipe list screen. Select "No" to not save the changed content and return to the Recipe list screen.

• Specify the number of CHs and create

- 1 Tap the "undefined" recipe in the "Recipe list" screen.



- 2 The "Recipe: New" screen is displayed. Tap "Specify the number of CHs."



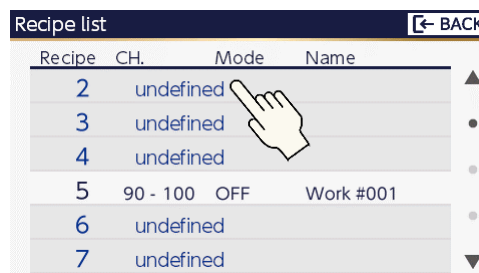
- 3 The "Num. of CHs." input screen is displayed. Set the Number of channels.



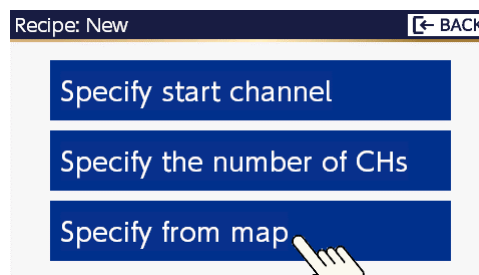
- 4 When the channel number settings are completed, the "Recipe set" screen is displayed. Configure each setting.
* Start CH. is set with the smallest channel that can be set for the recipe.
- 5 When all settings are completed, tap the [ SAVE] icon in the "Recipe set" screen. The settings are saved. The screen returns to the recipe list screen with the [ BACK] icon.

• Specify from map and create

- 1 Tap the "undefined" recipe in the "Recipe list" screen.



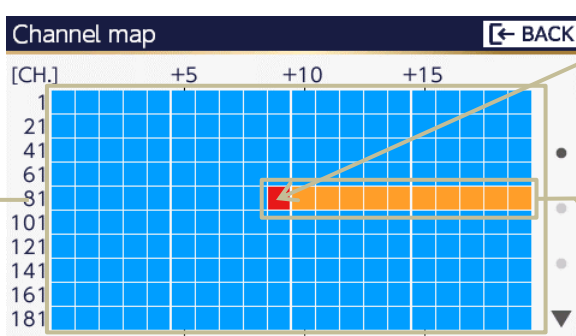
- 2 The Recipe: New screen is displayed. Tap "Specify from map."



- 3 The Channel map screen is displayed. Set the Start CH. by tapping a block in the map.

Blocks that represent channels.

The Σ mode CH is displayed with [Σ], [Σ], [Σ].



Indicates the Start CH. of recipes that have already been created. The Start CH. is displayed with [A] for recipes whose auto-increment function is ON.

Indicates channels used by recipes that have already been created.

- 4 When the Start CH. settings are completed, the "Recipe set" screen is displayed. Configure each setting.
* End CH. is set with the maximum continuous channels from Start CH. that can be set for the recipe.
If multiple recipes are used, set End Ch. to suit the number of channels used in the recipe.
- 5 When all settings are completed, tap the [Σ SAVE] icon in the "Recipe set" screen. The settings are saved. The screen returns to the recipe list screen with the [$\leftarrow$ BACK] icon.

Editing Recipes

Recipes can be edited with the following two methods by moving to the "Recipe set" screen.

- Tap an existing recipe in the "Recipe list" screen.
- Press and hold an existing recipe in the "Recipe list" screen, and tap "Edit."

Channel range

The Start CH. and End. CH. can be set when newly creating or reediting a recipe at the Channel range setting screen.

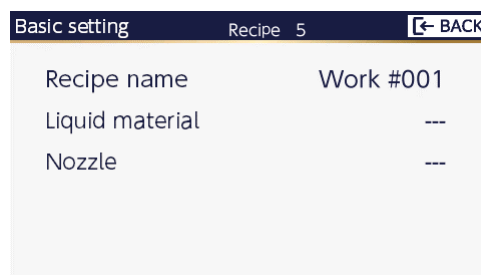


If you tap [← BACK] and return to the "Recipe set" screen, it is possible to determine whether the Σ mode CH is included in the specified CH range.



Basic setting

At the Basic setting screen, it is possible to set up to 14 characters for the recipe name, liquid material name, and nozzle name for managing the recipe.



Auto-increment

See P.36 "Auto-increment Function".

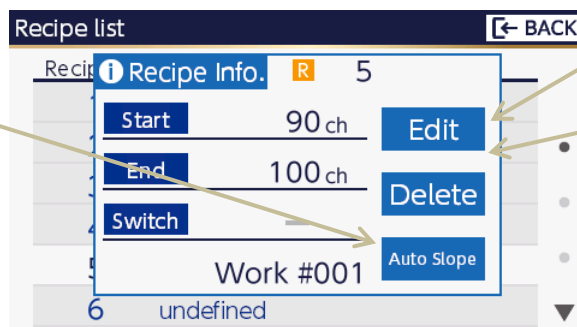
Temperature unit setting

See P.38 "Linkage with Temperature Controller".

Deleting Recipes

To delete a recipe, press and hold the existing recipe in the "Recipe list" screen, and tap "Delete".

Function for setting dispensing conditions as a batch. See P.61 "Setting Dispensing Conditions for a Specified Range as a Batch (Auto Slope function)" for more details.



Tapping "Edit" moves to the "Recipe set" screen.

Tapping "Delete" deletes the recipe.

* Cannot be restored to the original condition after deleting. Please take care with this function.

Auto-increment Function

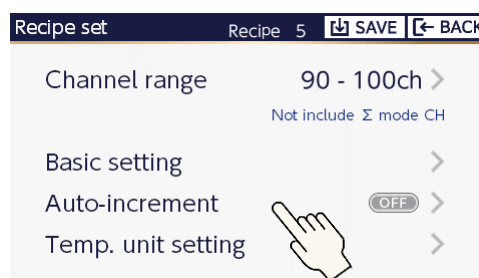
The Auto-increment function automatically switches channels with the set conditions. Automatically moves up the channel up from Start CH. to End CH. according to the set conditions. If a change is made to a channel belonging to a recipe whose auto-increment function is ON, the auto-increment function operation is started.

Operation of the Auto-increment function is set as follows.

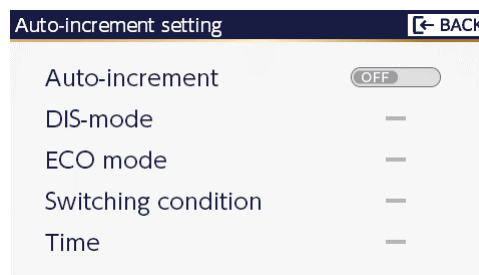
Item		Description
Auto-increment Function ON/OFF		Sets the Auto-increment function to ON/OFF.
Auto-increment Function dispensing mode		Dispensing mode when the Auto-increment Function is operating.
Auto-increment Function ECO mode		ECO mode ON/OFF when the Auto-increment Function is operating
Switching Condition		"Total time" : Switches channels at each set total dispensing time. Used for various dispensing times, or to change dispensing conditions to suit the remaining volume of liquid material in the syringe. "Total count" : Switches channels at each set total dispensing count. Used for a specific dispensing time, or to change dispensing conditions to suit the count. "Pot life" : Switches channels at the set elapsed time. Used to change dispensing conditions to suit the elapsed time regardless of the remaining volume of liquid material in the syringe.
Switching value	"Total time"	Set the value in 1 second increments.
	"Total count"	Set the value in 1shot increments.
	"Pot life"	Set the value in 1 second increments.
Immed. Inc. (Alnc)		OFF (default value) : Cannot move up the channel while In-dispensing, even if conditions are met. Moves up one channel according to the switched value after dispensing operations are complete. Multiple channels may move up at once. ON : If the conditions are met, the channel can be moved up even while In-dispensing.

•Auto-increment function settings

- 1 Tap "Auto-increment" in the "Recipe set" screen of the recipe for which to set the Auto-increment function.



- 2 The Auto-increment setting screen is displayed. Configure each setting.



- 3 When all settings are completed, tap the [SAVE] icon in the "Recipe set" screen. The settings are saved. The screen returns to the recipe list screen with the [BACK] icon.

•Restarts the Auto-increment function

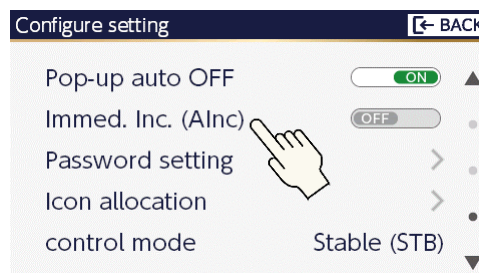
Automatic increment function will be restarted by the following operations.

- Tap on the [CH] icon when the channel belongs to a recipe in which the auto-increment function is ON.
- With the channel belonging to a recipe for which the Auto-increment function is ON, input the PRESET signal.
- Tap on the [CH] icon when the channel belongs to a recipe in which the auto-increment function is ON.
- Use the [RECIPE] icon to move to the Start CH. of the recipe for which the Auto-increment function is ON.

•CH. Immediately Increment settings

See P.73 "Changing Controller Configure Setting (Config.)" for moving to the "Configure setting" screen.

Tap "Immed. Inc. (AInc)" in the "Configure setting" screen. This allows switching between ON/OFF.



Linkage with Temperature Controller

Connecting a MUSASHI temperature controller allows the temperature controller settings and condition to be acquired (limited to compatible models). Up to five temperature controllers can be connected. A special connection cable is required to connect to a temperature controller. See the temperature controller instruction manual for details on the connection method.

Items that can be controlled with each temperature controller are as follows.

Item	Description
"Control"	Sets control of temperature controllers ON/OFF.
"Setting temperature"	Sets the target temperature. This overwrites the settings on the temperature controller.
"Upper limit"	Sets the upper limit.
"Lower limit"	Sets the lower limit.
Acquire alarm and error status	These are acquired automatically if control is ON.
"Auto-tuning"	Runs/cancels Auto-tuning.

• Preparation

Linkage with the temperature controller requires the setting the communication unit No. on the temperature controller side to 1 to 5. See the Temperature Controller instruction manual for details on the setting method.

* Linking multiple temperature controllers and setting overlapping communication unit No. will result in abnormal operation.

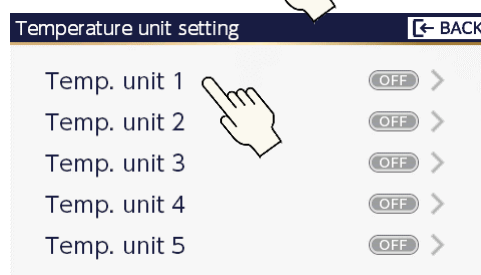
Please take care with this function.

• Temperature Controller settings

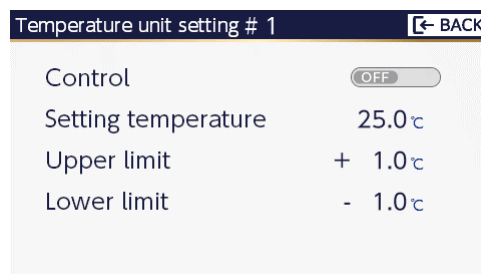
- 1 Tap "Temp. unit setting" in the "Recipe set" screen of the recipe for which to set the temperature controller.



- 2 The screen for selecting the controller to set is displayed. Tap the controller to change settings for.



- 3 The detailed setting screen is displayed. Configure each setting.

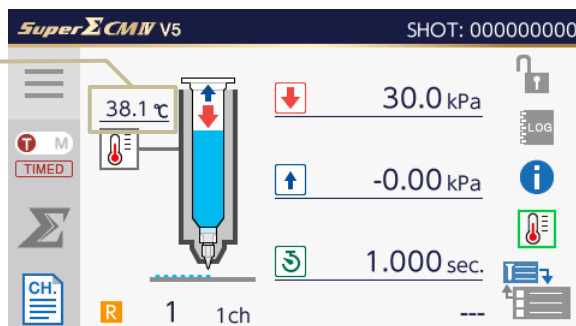


- 4 When all settings are completed, tap the [SAVE] icon in the "Recipe set" screen. The settings are saved. The screen returns to the recipe list screen with the [BACK] icon.

• Displaying Dispense Screen B by Linkage with Temperature Controller

If "control" of either of the temperature controllers is ON, Dispense screen B is displayed as follows.

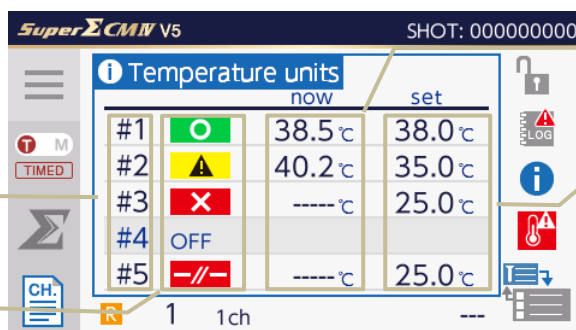
Displays the current temperature of the lowest communication unit No. when control of the temperature controller is ON.



• Check the Temperature Controller Status

The status of the temperature controller can be checked by tapping any of the [], [], [] or [] icons in the Dispense screen. Tapping each temperature controller moves to the detailed setting screen of each temperature controller, so values can be changed.

Temperature controller unit No.



Current temperature of each temperature controller.

Set temperature of each temperature controller.

Current status of each temperature controller.

Temperature Controller Status Icon	Description
	Displayed while the temperature controller is operating normally.
	Displayed while a temperature controller alarm occurs.
	Displayed while a temperature controller error occurs.
	Displayed if there is a communication problem with the temperature controller.

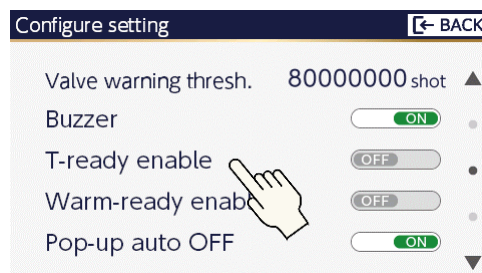
• T-ready Enable

This function restricts dispensing when temperature controller alarm or error occurs.

T-ready Enable	Operation
OFF (default value)	Dispense operations can be performed regardless of the temperature controller status.
ON	Dispensing cannot be performed if any alarm or error occur with temperature controllers for which control is ON.

See P.73 "Changing Controller Configure Setting (Config.)" for moving to the "Configure setting" screen.

Tap "T-ready enable" in the "Configure setting" screen. This allows switching between ON/OFF.

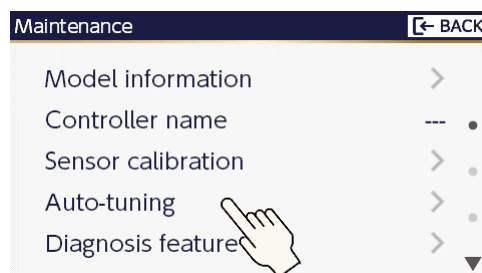


• Auto-tuning

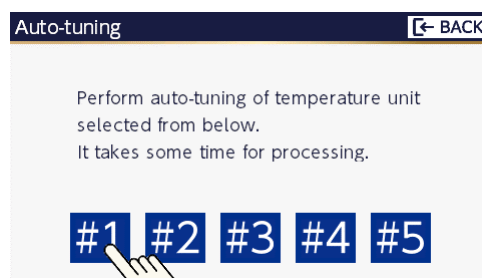
Auto-tuning is an adjustment function for appropriate control of temperature to suit the temperature control device being used, the heat characteristics of the liquid material, and the ambient temperature. Running auto-tuning is recommended for appropriate control of temperature when connecting to the temperature controller for the first time, changing the liquid material, or if there are changes in the ambient temperature.

See P.122 "Performing Controller Maintenance (Maintenance)" for moving to the "Maintenance" screen.

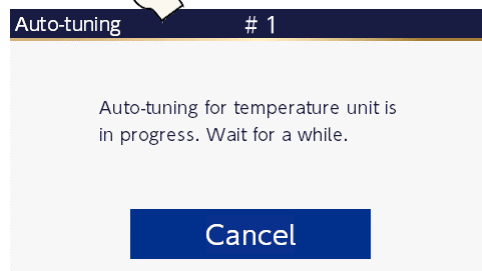
- 1 Tap "Auto-tuning" in the Maintenance screen.



- 2 The screen for selecting "Auto-tuning" is displayed. Tap the temperature controller unit No. for performing Auto-tuning.



- 3 Auto-tuning starts.
(Auto-tuning takes around 3 to 6 minutes)

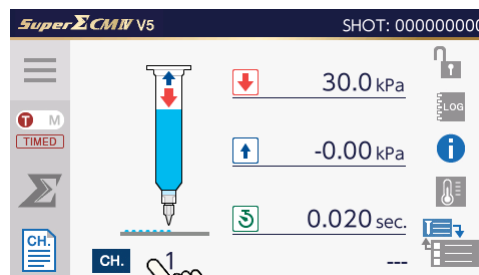


- 4 When Auto-tuning is completed, the screen returns to the screen for selecting "Auto-tuning."

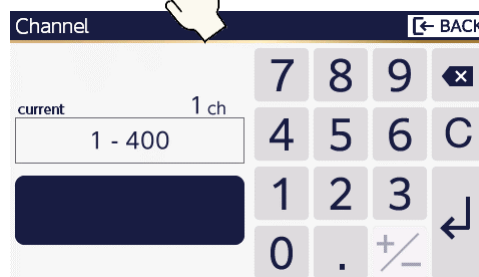
Switching Dispensing Conditions (Set Channel)

Multiple dispensing conditions can be saved with this device (400 CH in total).

- 1 Tap the displayed channel.



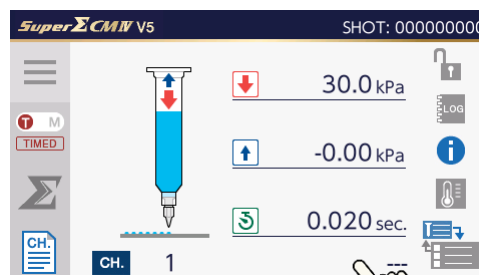
- 2 The "Channel" input screen is displayed. Set the channel.



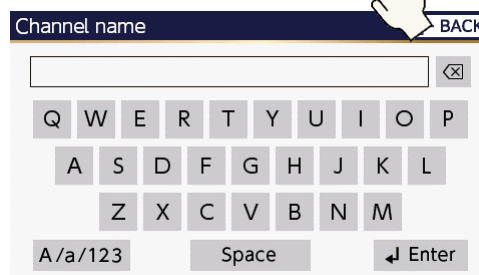
Add Name for Dispensing Conditions (Set Channel Name)

A name of up to 14 characters can be added to each channel.

- 1 Tap the displayed channel name.



- 2 The "Channel name" input screen is displayed. Set the channel name.



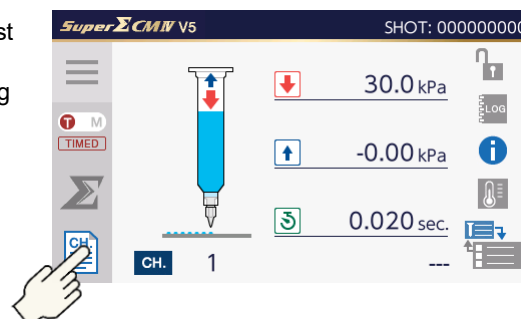
Setting Dispensing Conditions from List

Dispensing conditions can be displayed as a list with this device.

Displaying Channel List and Channel Graph

Tap the  icon in Dispense screen A or B. The Channel list screen or the Channel graph screen is displayed. The displayed screen can be changed with the CH. display setting (P.48).

* If the  icon is not displayed, see Icon allocation (P.78) and change the allocation.



Channel list

Changing dispensing parameter or copying channel with display of parameter list is available.

Displays the current channel.

CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

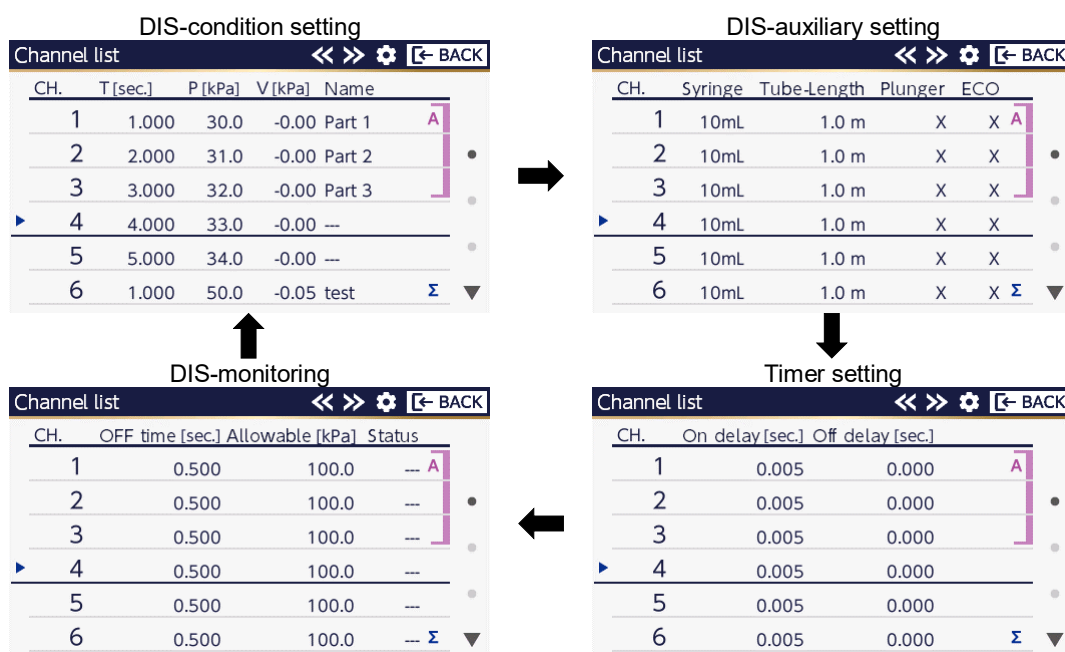
Moves to "CH. display setting." (P. 48)

Switches the page.

Displays whether used in recipe or is Σ mode.

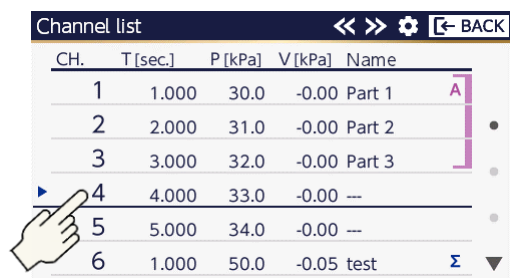
•Page switching

Tapping   switches the displayed parameters and displays them. The figure below shows page switching when  is tapped.



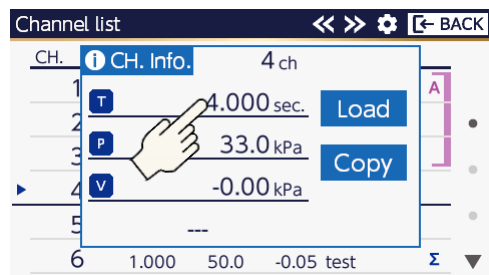
• **Setting Dispensing Conditions (Dispensing time, Dispensing pressure, Vacuum pressure, Channel name)**

- 1 Tap the channel to change parameters in the "Channel list" screen.



CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

- 2 "CH. Info." is displayed. Tap the parameters to change settings for.



CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

- 3 The tapped parameter input screen is displayed. Set the parameters. When settings are completed, the screen returns to the previous screen.

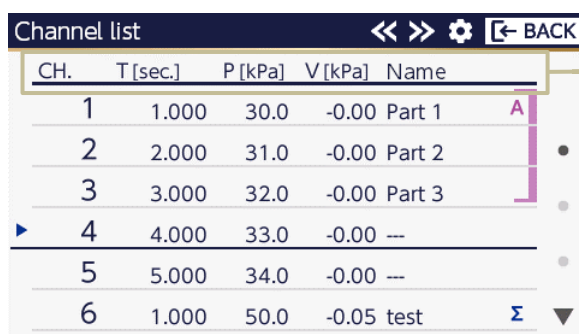


Dispensing time

current 0.001 sec.

0.001 - 9999.999

• **Auto Slope**



CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

Tapping here moves to the Auto slope setting screen.

See P.61 "Setting Dispensing Conditions for a Specified Range as a Batch information on how to use the Auto Slope function.

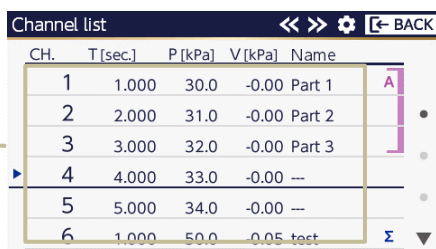
(Auto Slope function)" for

• Channel Load

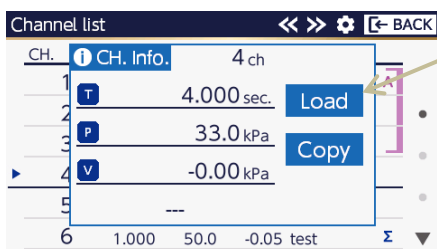
In the "Channel list" screen, there are the two following methods for loading channels.

- Tap "Load" from "CH. Info."
- Press and hold the channel in the Channel list

Press and hold channel



CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test



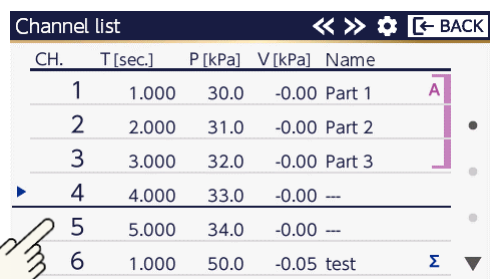
CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

Tap "Load"

• Channel copy/paste

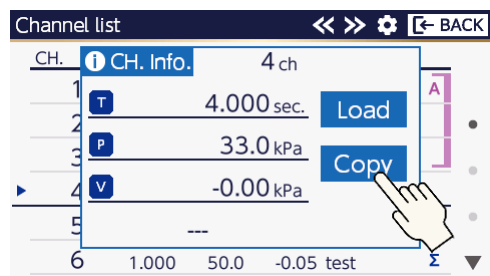
Dispensing conditions for the specified channel can be copied to other channels.

- 1 Tap the channel to be copied in the "Channel list" screen.



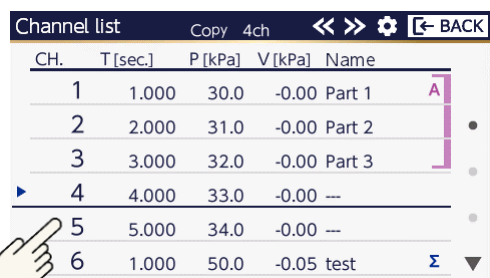
CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

- 2 "CH. Info." is displayed. Tap "Copy."



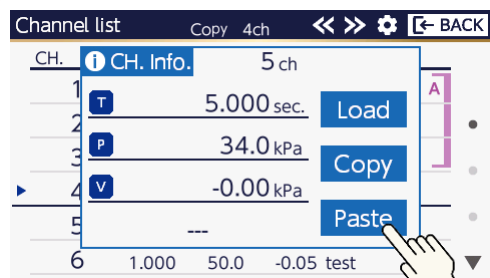
CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

- 3 The dispensing conditions are copied, and then returns to the Channel list screen. Tap the channel to copy to.



CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

- 4 "CH. Info." is displayed. Tap "Paste." The channel is copied.



CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	-0.00	Part 1
2	2.000	31.0	-0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

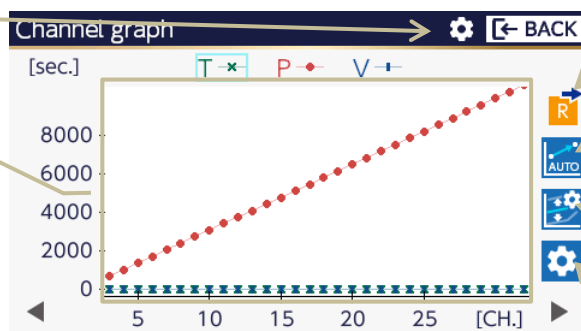
* Cannot be restored to the original condition after deleting.
Please take care with this function.

Channel graph

This displays the dispensing conditions as a graph, to change dispensing conditions, or perform Auto Slope.

Moves to "CH. display setting." (P.48)

Tapping within the graph displays "CH. Info." for the associated channel.



The specified recipe is displayed. (P.47)

Performs Auto Slope. (P.61)

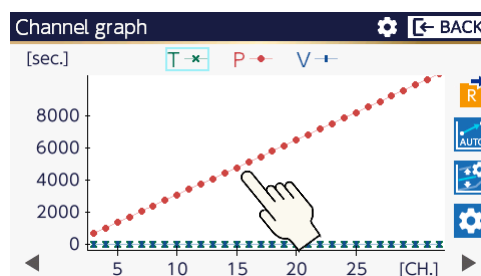
Performs DIS-correction. (P.59)

Sets the graph display options. (P.48)

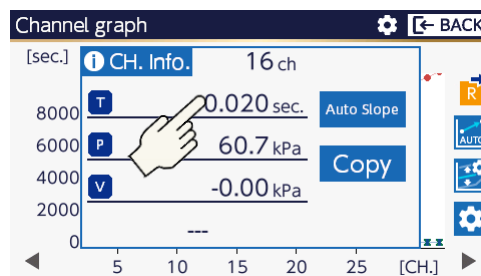
• Changing Dispensing Conditions (Dispensing time, Dispensing pressure, Vacuum pressure, Channel name)

- 1 Tap inside the graph in the Channel graph screen.

*The maximum number of channel displayed on the horizontal axis can be set with the graph display options.



- 2 "CH. Info." for the associated channel in the tapped position is displayed. Tap the parameters to change settings for.

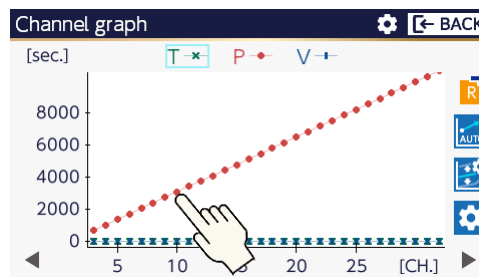


- 3 The tapped parameter input screen is displayed. Set the parameters. When settings are completed, the screen returns to the previous screen.

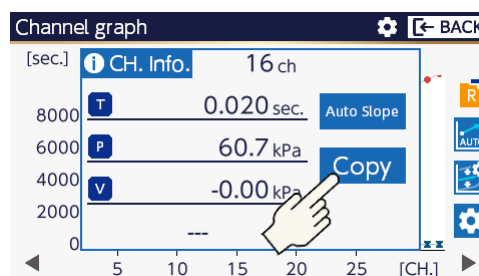
• Channel copy/paste

Dispensing conditions for the specified channel can be copied to other channels.

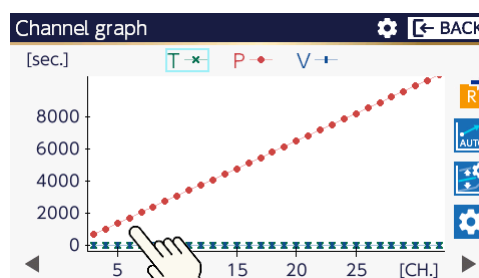
- 1 Tap inside the graph and select the channel to be copied in the "Channel graph" screen.



- 2 "CH. Info." is displayed. Tap "Copy." This registers it as the channel to be copied.

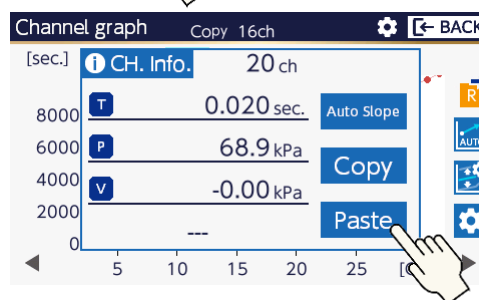


- 3 Tap inside the graph and select the channel to copy to.



- 4 "CH. Info." is displayed. Tap "Paste." The channel is copied.

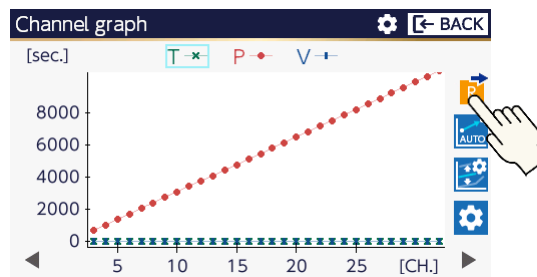
* Cannot be restored to the original condition after deleting.
Please take care with this function.



• Displaying Specified Recipe

The start channel displayed on the horizontal axis can be set as Start CH. of the specified recipe.

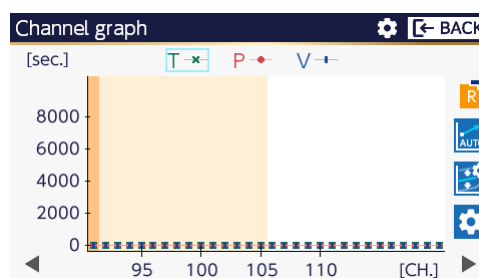
- 1 Tap the [R] icon in the Channel graph screen.



- 2 The "Select start CH." screen is displayed. This screen displays a list of recipes that have already been created. Tap the recipe to display from the list.

Select a recipe		BACK
Select recipe.	RecipeCH.	Name
	5	91 - 105 Part 1
Tap: Determine	8	201 - 205 Part 2
Hold: Show info.	17	351 - 400 Part 3

- 3 The graph starts being displayed from Start CH. of the specified recipe.



• Auto Slope

The Auto Slope function can be used from the Channel graph screen. See P.61 "Setting Dispensing Conditions for a Specified Range as a Batch (Auto Slope function)" for more details.

• DIS-correction

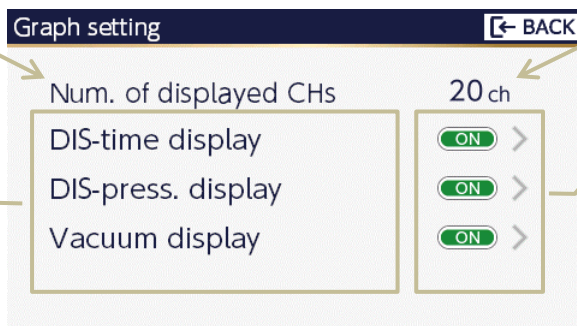
DIS-correction can be performed from the Channel graph screen. See P.59 "Making Minor Adjustments to Dispensing Conditions as a Batch (DIS-correction)" for more details.

•Graph Display Options

The display range of the horizontal and vertical axes and the types to display can be selected.

The maximum number of displayed channels of the horizontal axis can be set.

Moves to the display setting screen of each parameter.



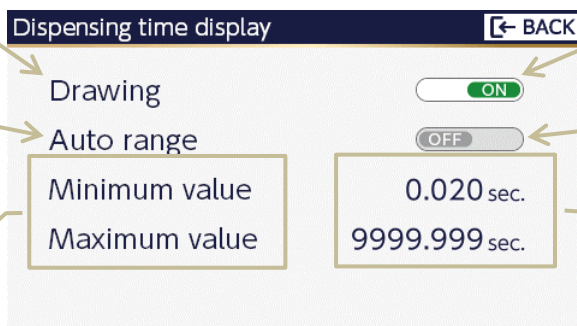
Displays the current maximum number of displayed channels.

Drawing currently ON/OFF.

Drawing can be set to on or off.

Sets the display range of the vertical axis to auto.

Sets the maximum and minimum values of the vertical axis. (Can be set when Auto range is OFF)



Drawing currently ON/OFF.

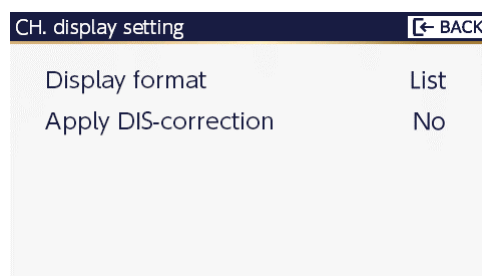
Auto range currently ON/OFF.

The current maximum and minimum values of the vertical axis. (Displayed when "Auto range" is OFF)

"CH. display setting"

Item	Description	Reference
"Display format"	List : Displays the Channel list screen. Graph : Displays the Channel graph screen.	-
"Apply DIS-correction"	Sets whether or not to apply DIS-correction for displaying the dispensing conditions.	P.59

- 1 Tap the [⚙️] icon in the title bar of the Channel list or Channel graph screen.
- 2 "CH. display setting" is displayed. Configure each setting.



Correcting Dispensing

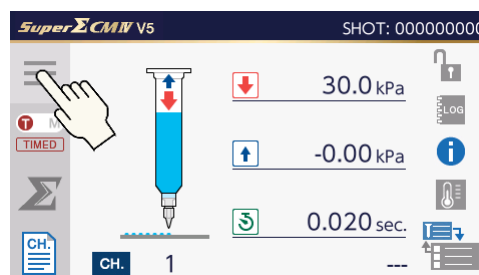
Setting dispensing environment (DIS-auxiliary setting)

Set the dispensing environment, such as the connected syringe and adapter tube, in the “DIS-auxiliary setting” screen.

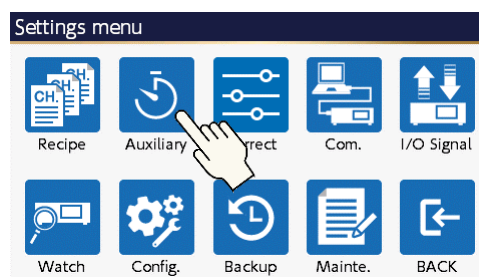
The syringe size, adapter tube length, and plunger must be configured correctly to use the Σ function. The sigma function will not operate correctly if these parameters are not configured correctly.

“DIS-auxiliary setting” screen

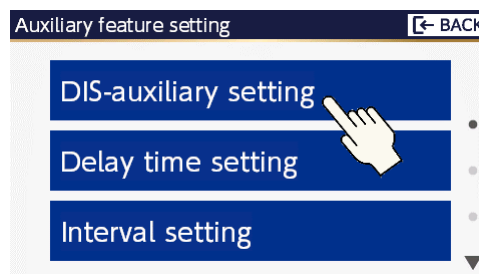
- 1 Tap the [≡] icon in Dispense screen A or B.



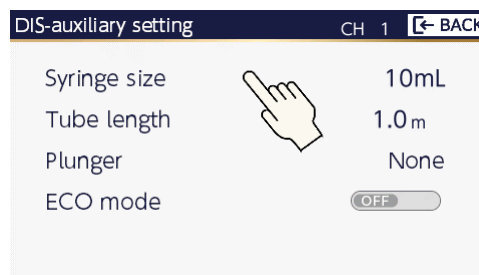
- 2 The "Settings menu" screen is displayed. Tap the [⚙️] icon.



- 3 The "Auxiliary feature setting" screen is displayed. Tap “DIS-auxiliary setting”.

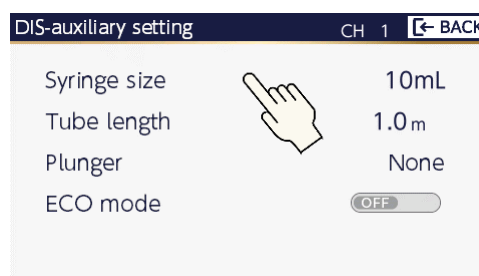


- 4 The “DIS-auxiliary setting” screen is displayed. Tap the parameters to change settings for.



Syringe size

- 1 Tap "Syringe size" in the "DIS-auxiliary setting" screen.



- 2 The "Select syringe size" screen is displayed. Tap the syringe size to be used.

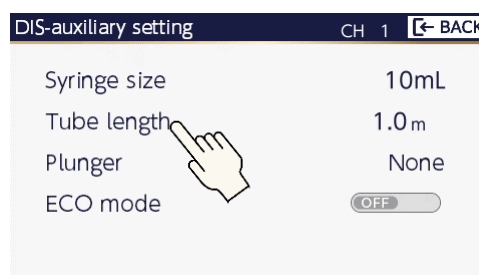
* The image on the right shows the case when "10 mL" is selected.



When using E/F-type syringes, select one of "3 mL", "5 mL", "10 mL", "20 mL", "30 mL", "50 mL", or "70 mL", and when using W-type syringes, select one of "W 5 mL", "W 10 mL", "W 30 mL", or "W 55 mL".

Adapter tube length

- 1 Tap "Tube length" in the "DIS-auxiliary setting" screen.



- 2 The "Adapter tube length" selection screen is displayed. Tap the tube length to be used.

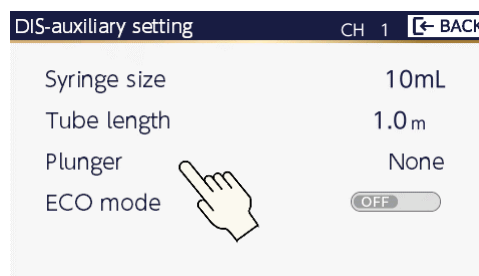
* The image on the right shows the case when "1.0 m" is selected.



Plunger enabled/disabled

- 1 Tapping "Plunger" in the "DIS-auxiliary setting" screen allows switching the enabled/disabled state of the plunger.

Each tap changes the setting in the following order.
"Disabled" → "Enabled" → "Disabled" → ...



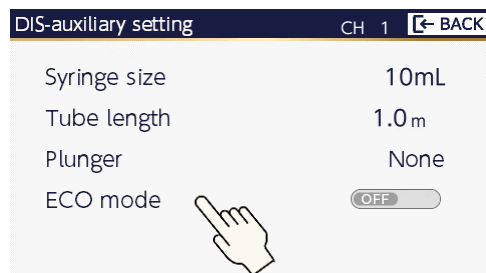
ECO mode

Turning ECO mode ON allows reduction of air consumption.

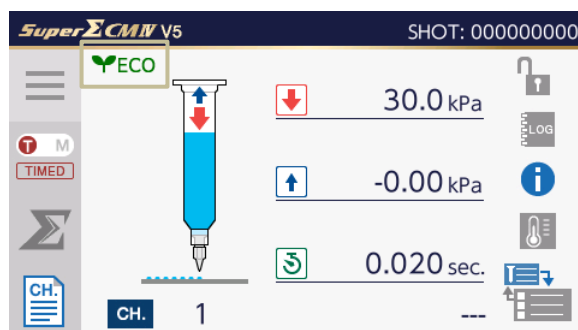
- * There is a risk of losing dispensing stability, depending on the dispensing conditions. To perform more stable dispensing, use with ECO mode turned OFF.

- 1 Tapping "ECO mode" in the "DIS-auxiliary setting" screen allows ON/OFF switching of the ECO mode function.

Each tap changes the setting in the following order.
"OFF" → "ON" → "OFF" → ...



When ECO mode is ON, [🌱ECO] is displayed at the upper left of the dispense screen.



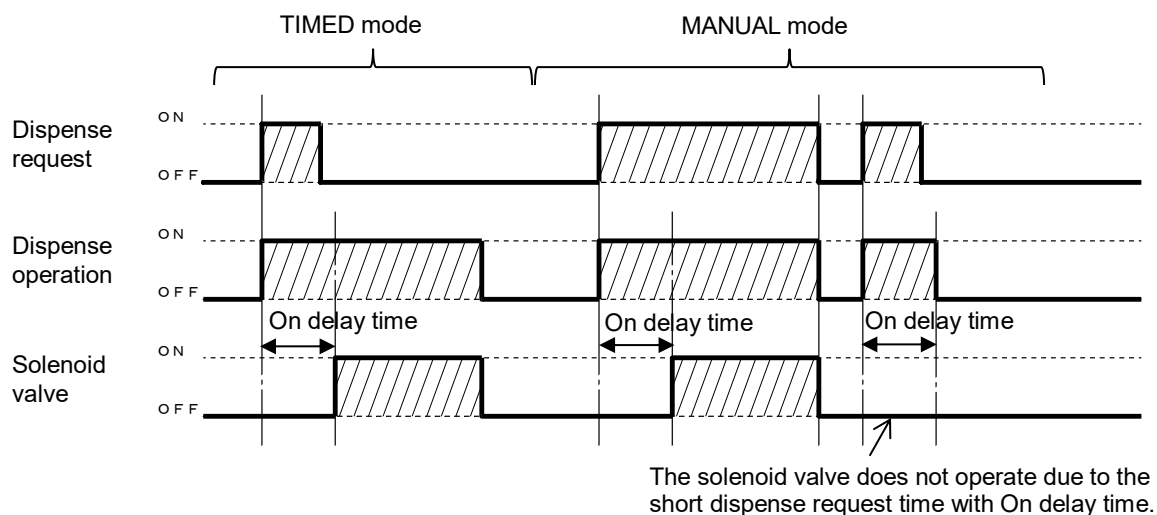
Adjusting the dispense start and end points (delay timer)

The delay timer is a function that delays the dispense operation after receiving a dispense request. This is useful when adjusting the dispense start and end points.

On delay time

On delay time sets the delay time from dispense request to dispense start. Use this when making minor adjustments to the dispense start position.

* Set this to 0 (zero) when the On delay time is not in use.

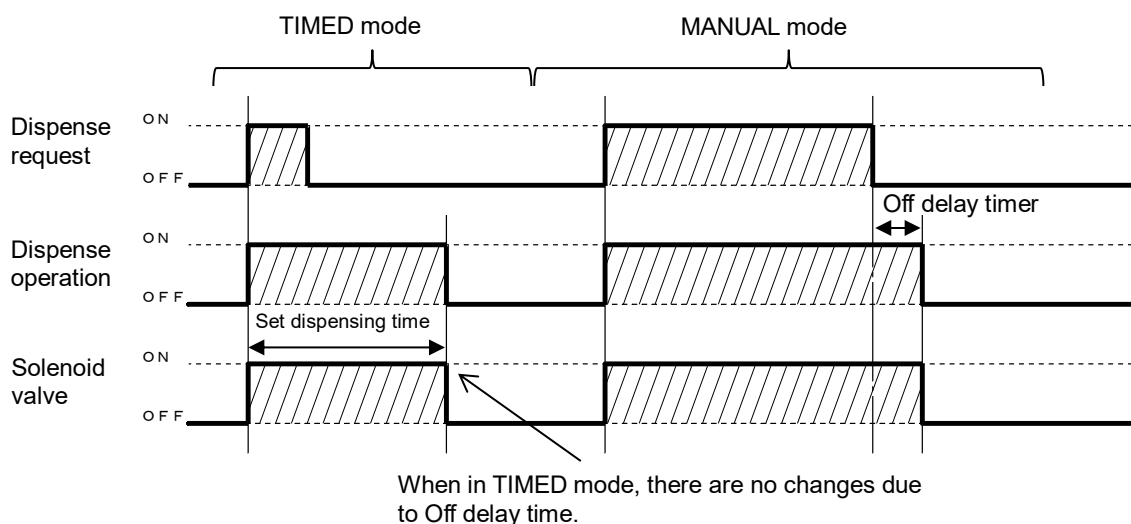


Off delay timer

When in MANUAL mode, Off delay timer sets the delay time from dispense end request to dispense end. Use this when making minor adjustments to the dispense end position.

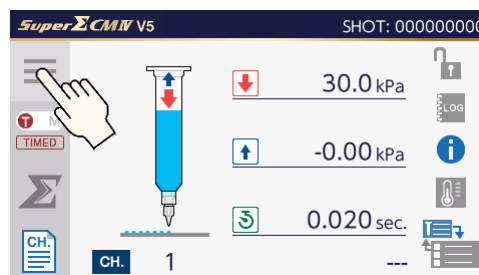
* Set this to 0 (zero) when the OFF delay timer is not in use.

* This setting is only for MANUAL mode.

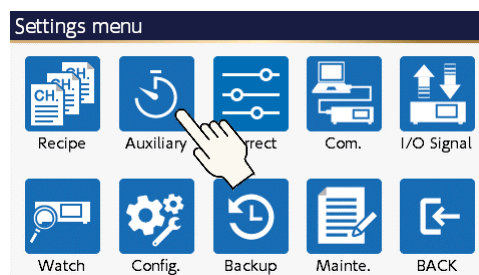


Delay Timer Setting

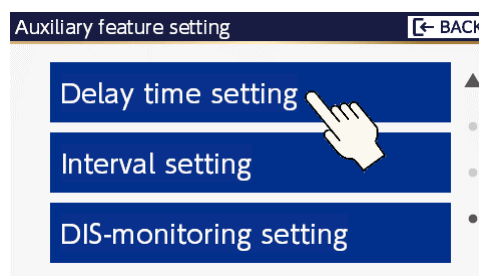
- 1 Tap the [≡] icon in Dispense screen A or B.



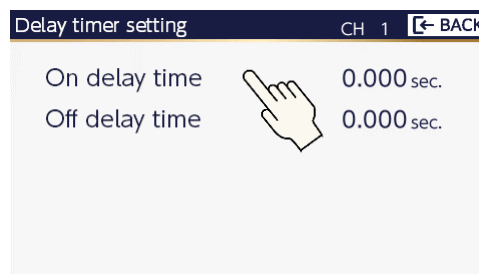
- 2 The "Settings menu" screen is displayed. Tap the [⚙️] icon.



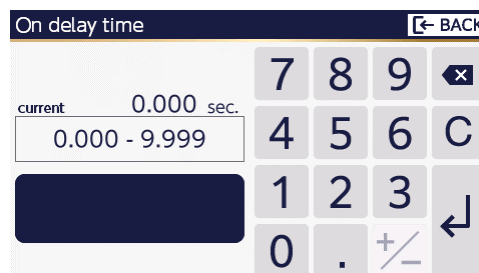
- 3 The "Auxiliary feature setting" screen is displayed. Tap "Delay timer setting."



- 4 The "Delay timer setting" screen is displayed. Tap the parameters to change settings for.



- 5 The tapped parameter input screen is displayed. Set the parameters. When settings are completed, the screen returns to the previous screen.



Repeating Dispensing at Fixed Intervals (Interval)

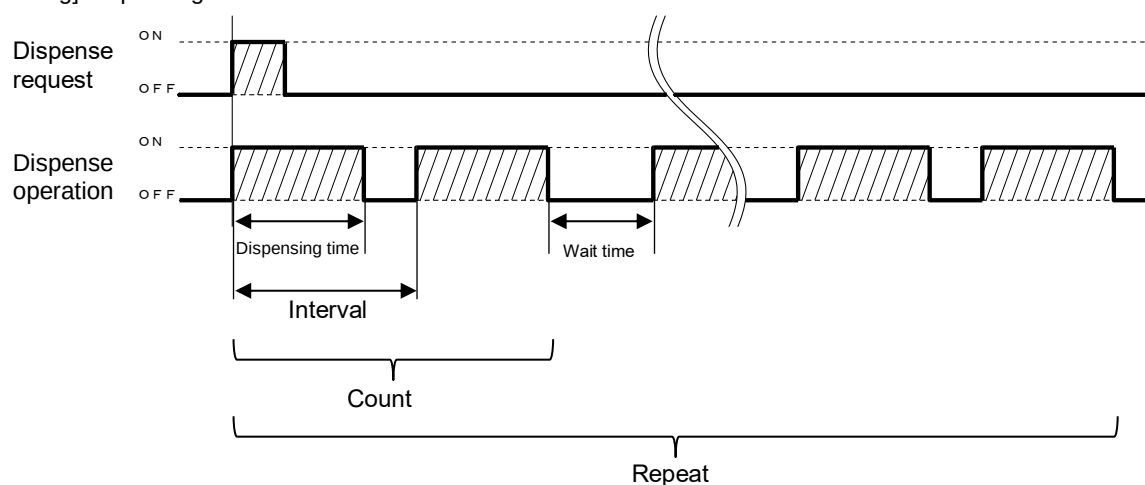
This mode repeatedly dispenses and stops dispensing at the set interval. This is useful for preventing liquid material adhering to the nozzle tip.

Operation of the Interval feature is set as follows.

Item	Description
Interval feature	Sets the Interval feature ON/OFF.
"Channel link"	Sets the current channel or a Link CH. is used as the dispensing conditions for the Interval feature. OFF : Uses dispensing conditions of the current channel. ON : Uses dispensing conditions of a Link CH. channel.
"Link CH."	Sets the channel to use as the dispensing conditions for the Interval feature.
"Count"	Sets the dispense operation count.
"Interval"	Sets the dispense operation interval.
"Repeat"	Sets how many times the set count operation is repeated.
"Wait time"	This is the Wait time from the set count end to the next dispense start.
"Interval mode"	SET : Dispenses at the set interval for the set number of Repeats. MAN : Dispenses at the set interval while the dispense request is ON.

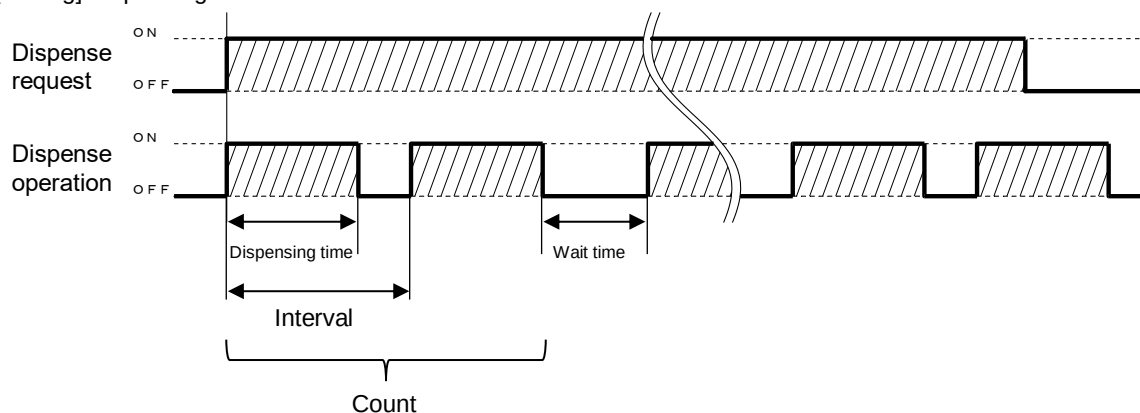
SET Mode

[Setting] Dispensing count: 2 shot



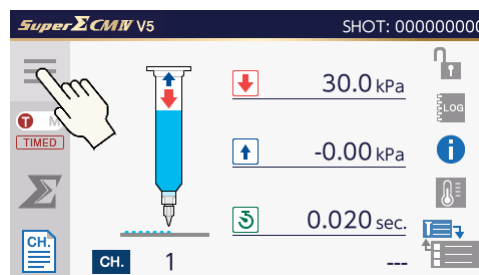
MAN Mode

[Setting] Dispensing count: 2 shot

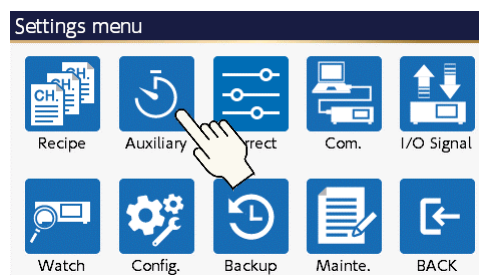


Interval Feature Setting

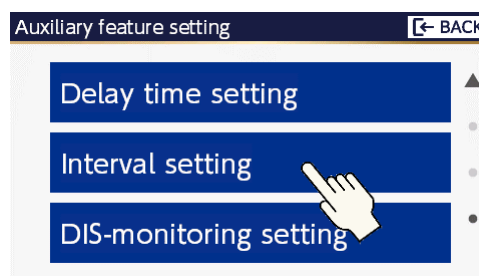
- 1 Tap the [] icon in Dispense screen A or B.



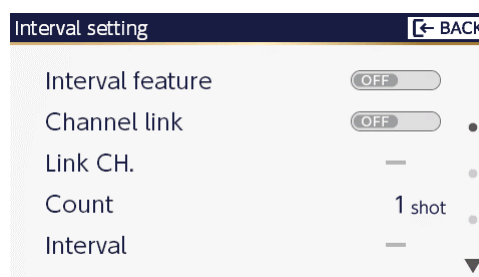
- 2 Tap the [] icon in the "Settings menu" screen.



- 3 The "Auxiliary feature setting" screen is displayed. Tap "Interval setting."



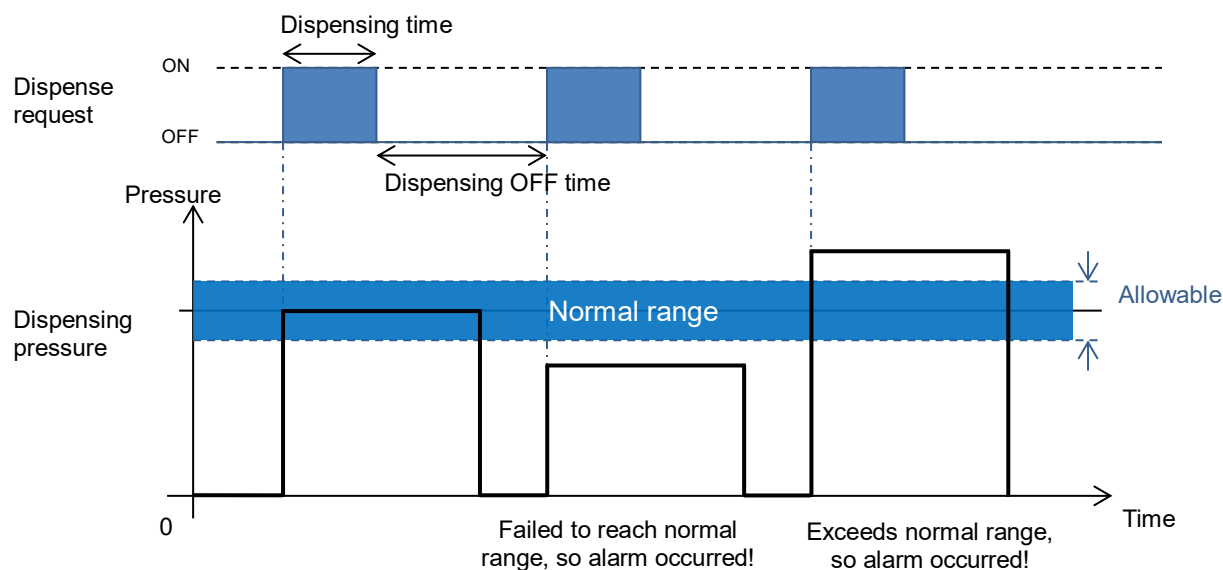
- 4 The "Interval setting" screen is displayed. Configure each setting.



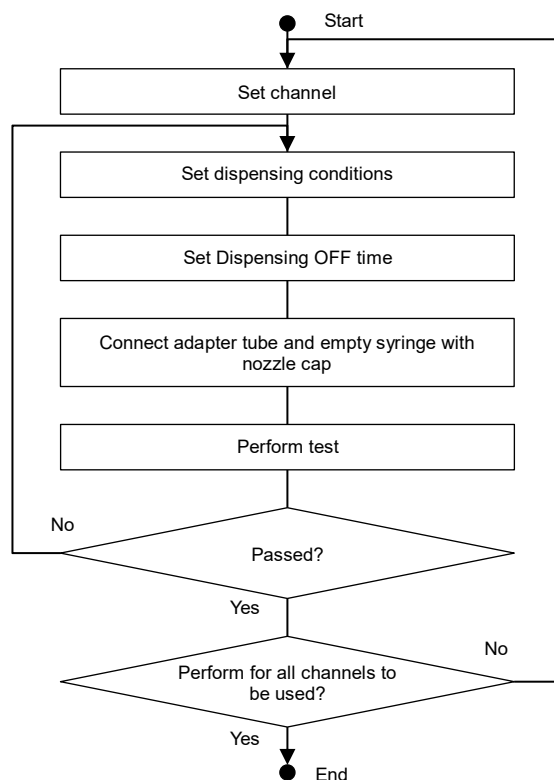
Monitoring Missed Dispensing (DIS-monitoring)

During dispense operations, the dispensing pressure can be monitored to check whether it has reached the set pressure. If an alarm occurs, relevant information is also recorded in the log.

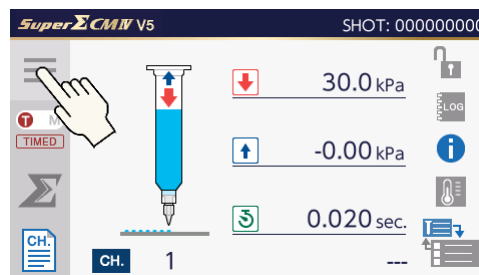
- * This setting is only for TIMED mode.
- * This might be able to be used depending on dispensing conditions. Perform a preliminary test operation, and only use this if the test passes.
- * If dispensing conditions have been changed, a test operation needs to be performed again.
- * Please perform test operations for all channels that will be used.
- * If the actual dispensing in the dispensing interval is 10 seconds or more, set "Dispensing OFF time" to the maximum value of 9.999 seconds.



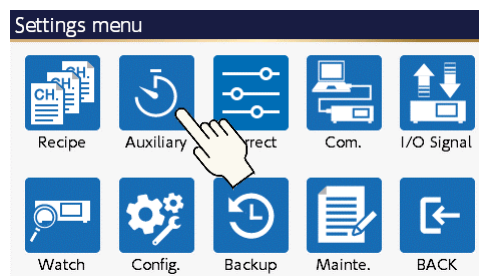
Operating flow



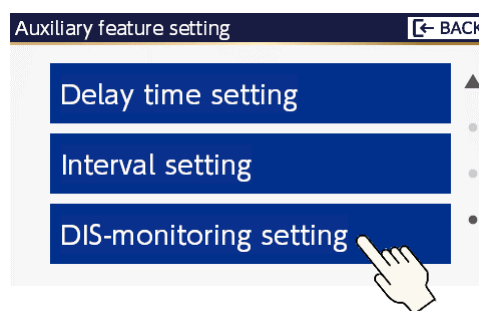
- 1 Tap the [≡] icon in Dispense screen A or B.



- 2 Tap the [⌚] icon in the "Settings menu" screen.



- 3 The "Auxiliary feature setting" screen is displayed. Tap "DIS-monitoring setting."

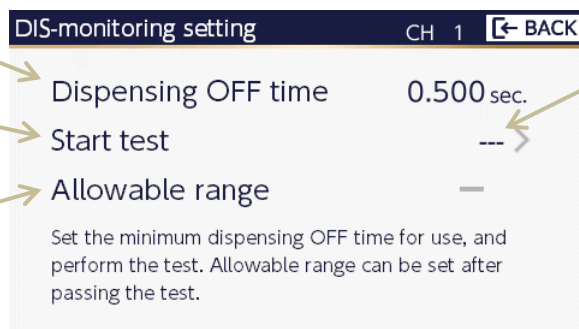


Description of Screens

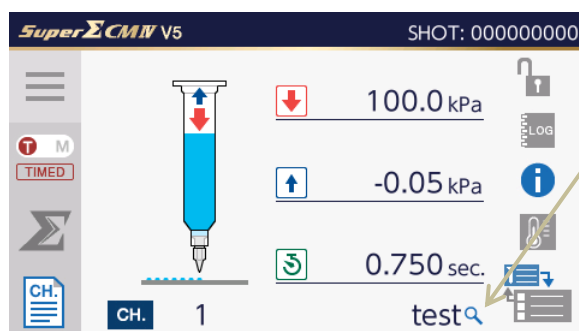
Sets "OFF time."

Tapping this starts the test.

Sets "Allowable." This can only be set if the test passes.



Test results are displayed.



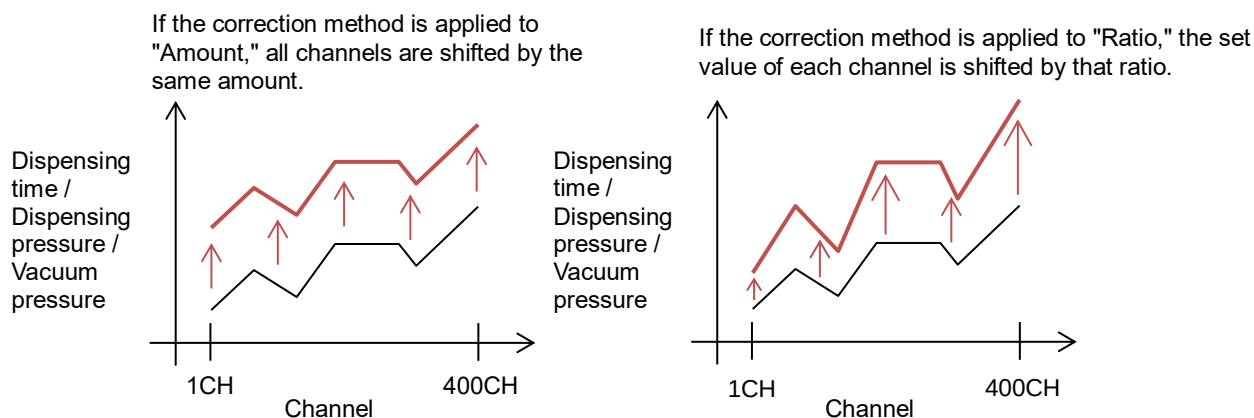
The [🔍] icon is displayed for channels with DIS-monitoring enabled.

Making Minor Adjustments to Dispensing Conditions as a Batch (DIS-correction)

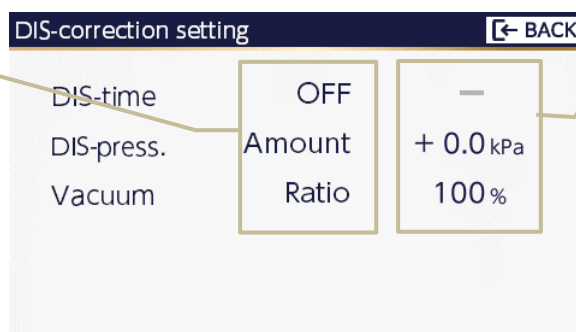
DIS-correction is useful for shifting all set dispensing conditions as a batch due to differences in liquid material lots. This can temporarily shift the dispensing conditions without changing the original settings.

Individual settings can be made to each Dispensing time, Dispensing pressure and Vacuum pressure.

*This function is not reflected in the Σ mode CH.



Each tap changes the correction method in the following order.
"OFF" → "Amount" →
"Ratio" → "OFF" → ...



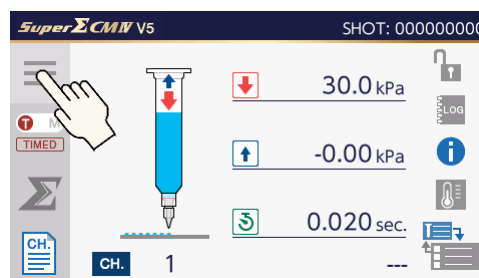
Moves to the Number Input Screen corresponding to the correction method.

Setting DIS-correction

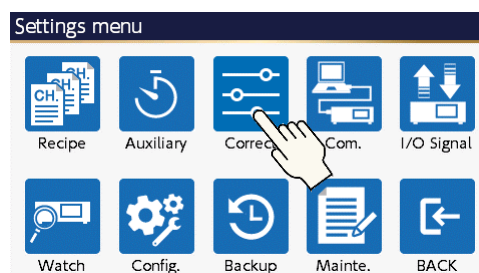
The following two methods can be used to move to the DIS-correction setting screen.

- Tap the  icon in the "Settings menu" screen

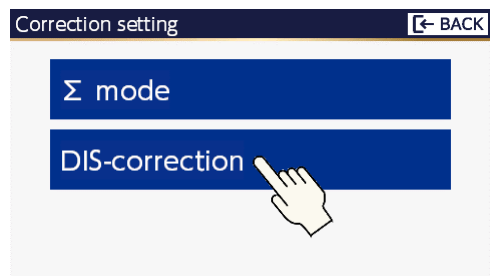
- 1 Tap the  icon in Dispense screen A or B.



- 2 The "Settings menu" screen is displayed. Tap the  icon.

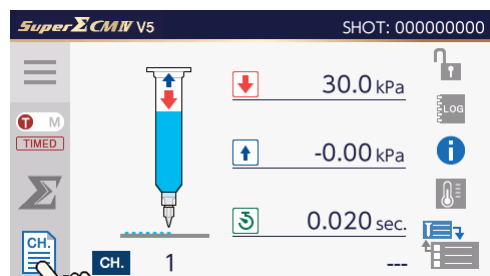


- 3 "The correction setting screen is displayed. Tap "Dispensing conditions."

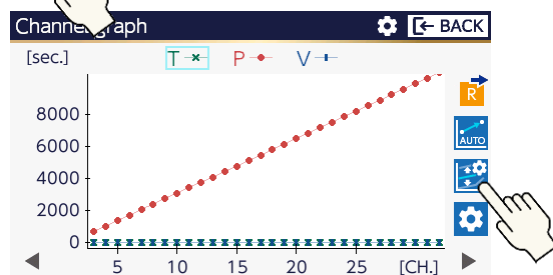


• Tap the icon in the Channel graph screen.

- 1 Tap the  icon in Dispense screen A or B.
* If the  icon is not displayed, see Icon allocation (P.78) and change the allocation.



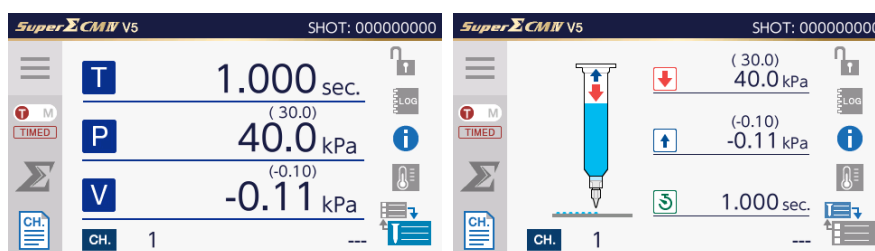
- 2 The Channel list screen or the Channel graph screen is displayed. Tap the  icon in the Channel graph screen.
* If the Channel list screen is displayed, see CH. display setting (P.48) and change the setting.



Displaying Dispense Screen A or B by DIS-correction

If DIS-correction is being used, Dispense screen A or B is displayed as follows.

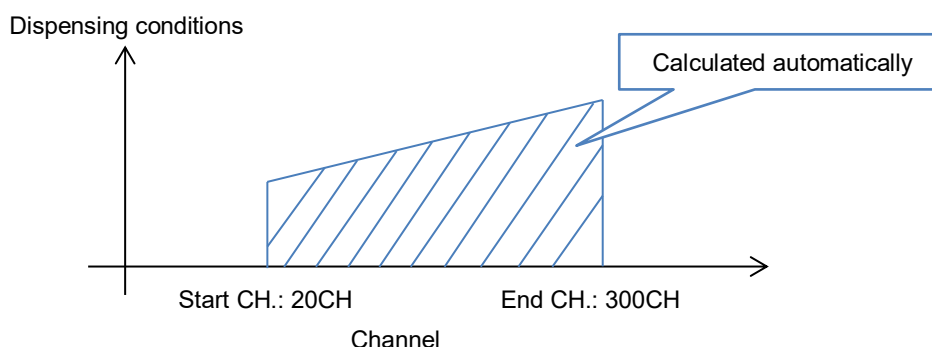
Upper: Set value before correction. (values in round parentheses)
Lower: Set value after correction.



Setting Dispensing Conditions for a Specified Range as a Batch (Auto Slope function)

The Dispensing time, Dispensing pressure, Vacuum pressure, ON delay timer, OFF delay timer, Dispensing OFF time, and Allowable range are calculated automatically and set for any specified channel section or any recipe Start CH. to End CH. section.

It is also possible to use uniform settings for Dispensing mode, Syringe size, Tube length, Plunger, and ECO mode in the specified channel section.



Set Auto Slope using the following.

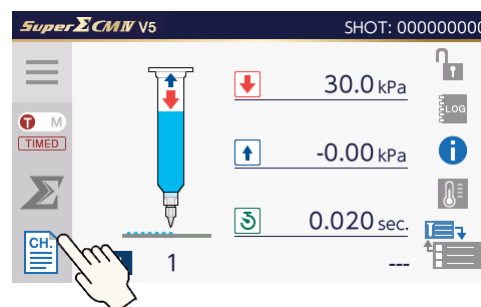
Item	Description
"Setting"	"CH" : Specifies the channel and performs Auto Slope. "Recipe" : Performs Auto Slope at Start and End CH. of the specified recipe. Each tap changes the setting in the following order. "CH" → "Recipe" → "CH" → ...
"Recipe number"	Sets the recipe number for performing Auto Slope. Enabled when the "Setting" is set to "Recipe."
Start CH.	Sets the channel to start Auto Slope.
End CH.	Sets the channel to end Auto Slope.
"DIS-condition setting"	Moves to the dispensing conditions detailed setting screen for automatically calculating Auto Slope. Four items can be set: Dispensing mode, Dispensing time, Dispensing pressure, and Vacuum pressure.
"DIS-auxiliary setting"	Moves to the dispensing conditions detailed setting screen for automatically calculating Auto Slope. Four items can be set: Syringe size, Tube length, Plunger, and ECO mode.
"Timer setting"	Moves to the dispensing conditions detailed setting screen for automatically calculating Auto Slope. Two items can be set: ON delay timer, and OFF delay timer.
"DIS-monitoring setting"	Moves to the dispensing conditions detailed setting screen for automatically calculating Auto Slope. Two items can be set: Dispensing OFF time, and Allowable range.

Move to the Auto slope setting screen.

The following four methods can be used to move to the "Auto slope setting" screen.

- Tap the label part of the list in the Channel list screen.

- 1 Tap the  icon in Dispense screen A or B.



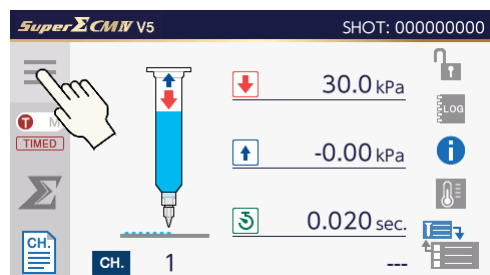
- The "Channel list" screen is displayed.
Tap inside the box in the figure on the right.

Channel list

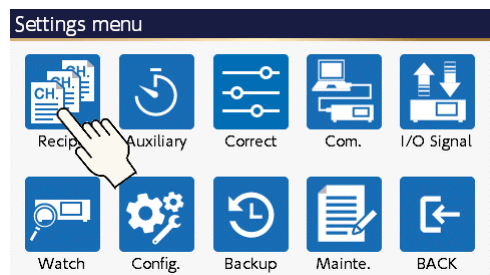
CH.	T[sec.]	P[kPa]	V[kPa]	Name
1	1.000	30.0	0.00	Part 1
2	2.000	31.0	0.00	Part 2
3	3.000	32.0	-0.00	Part 3
4	4.000	33.0	-0.00	---
5	5.000	34.0	-0.00	---
6	1.000	50.0	-0.05	test

•Press and hold an existing recipe in the "Recipe list" screen, and tap "Auto Slope".

- Tap the [≡] icon in Dispense screen A or B.



- The "Settings menu" screen is displayed. Tap the [≡] icon.



- The "Recipe list" screen is displayed. Press and hold the existing recipe.

Recipe list

Recipe	CH.	Mode	Name
2	undefined		
3	undefined		
4	undefined		
5	90 - 100	OFF	Work #001
6	undefined		
7	undefined		

- "Recipe Info." is displayed. Tap "Auto Slope."

Recipe list

Recipe	CH.	Mode	Name
2	undefined		
3	undefined		
4	undefined		
5	90 - 100	OFF	Work #001
6	undefined		
7	undefined		

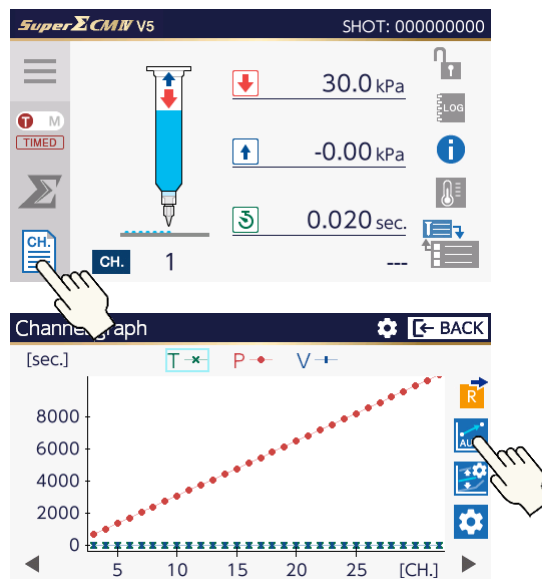
Recipe Info. R 5

Start	90 ch	Edit
End	100 ch	Delete
Switch	—	Auto Slope

• Tap the icon in the Channel graph screen.

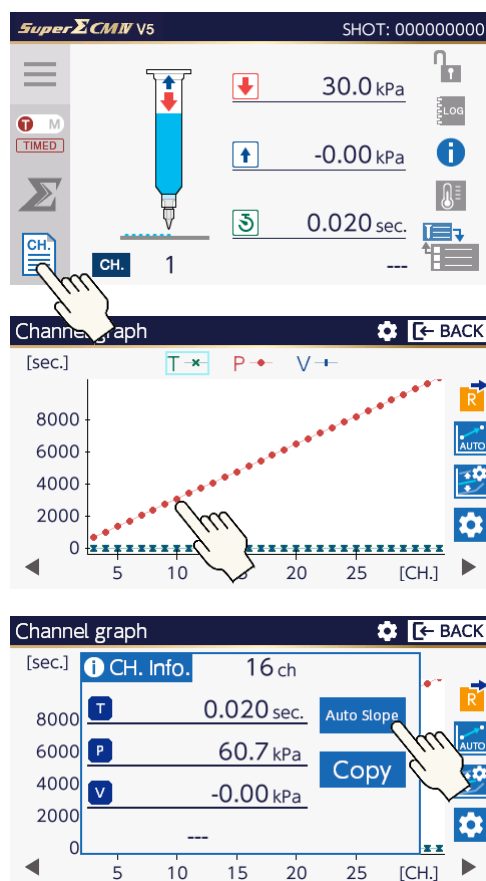
- 1 Tap the  icon in Dispense screen A or B.
* If the  icon is not displayed, see Icon allocation (P.78) and change the allocation.

- 2 The Channel list screen or the Channel graph screen is displayed. Tap the  icon.
* If the Channel list screen is displayed, see CH. display setting (P.48) and change the setting.



• Tap inside the graph in the Channel graph screen, and tap "Auto Slope" in "CH. Info.".

- 1 Tap the  icon in Dispense screen A or B.
* If the  icon is not displayed, see Icon allocation (P.78) and change the allocation.
- 2 The Channel list screen or the Channel graph screen is displayed. Tap inside the graph in the Channel graph screen.
* If the Channel list screen is displayed, see CH. display setting (P.48) and change the setting.
- 3 "CH. Info." for the associated channel in the tapped position is displayed. Tap "Auto Slope."

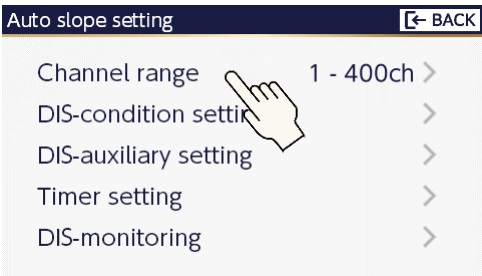


Determining the Range for performing Auto Slope.

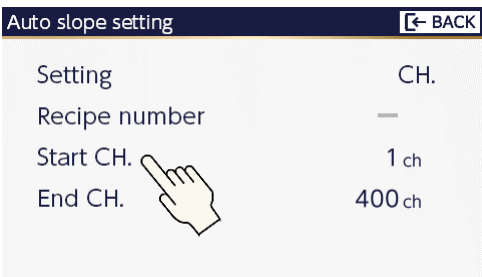
The channel range for performing Auto Slope can be determined by the following two methods.

•Directly specifying channel

- 1 Move to the “Auto slope setting” screen, and tap “Channel range”.

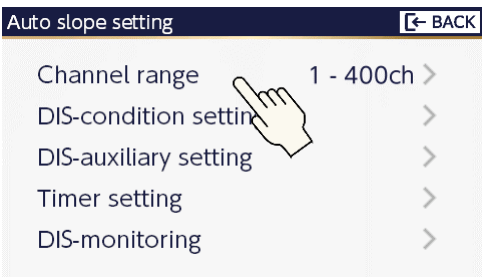


- 2 Tap “Setting”, switch to “CH.”, and set the Start CH. and End CH.

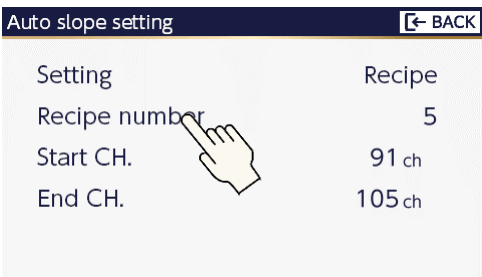


• Specifying from the recipe

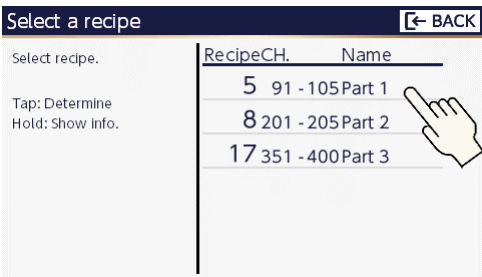
- 1 Move to the “Auto slope setting” screen, and tap “Channel range”.



- 2 Tap “Setting”, switch to “Recipe”, and tap “Recipe number”.

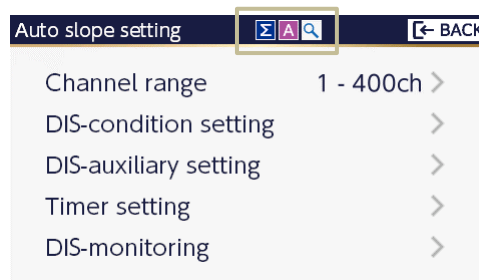


- 3 The “Recipe selection” screen is displayed. Select the recipe number for performing Auto Slope.



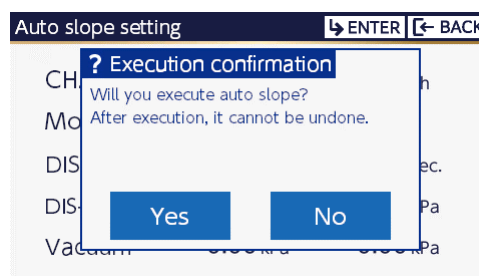
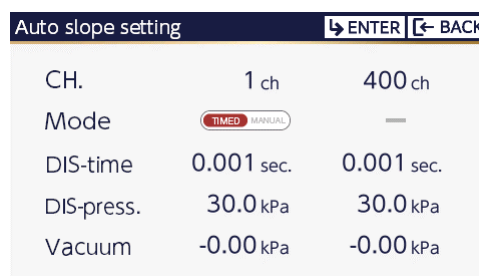
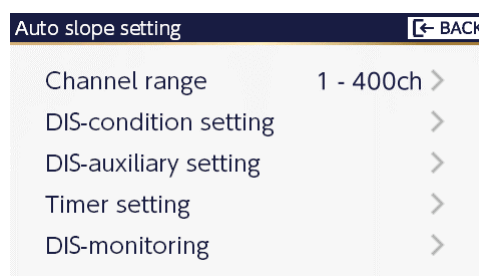
Icons are displayed at the top of the screen if Σ mode CH, recipes with auto-increment function ON, or pass CH of DIS-monitoring function are included in the Auto Slope range.

- [Σ] :
 Σ mode CH is included.
 Some Auto Slope cannot be performed.
- [A] :
 Recipes with auto-increment function ON are included.
- [Q] :
 Pass CH of DIS-monitoring function are included.
 If dispensing conditions have been changed due to running Auto Slope, test results are cleared.



Performs Auto slope

- 1 Select either "DIS-condition setting", "DIS-auxiliary setting", "Timer setting", or "DIS-monitoring setting" in the Auto slope setting screen.
- 2 The figure on the right is the case where "DIS-condition setting" has been selected. Input set values for Start CH. and End CH., and tap the [ENTER] icon.
- 3 A confirmation screen appears. Tapping "Yes" runs Auto Slope.



Express Shot Function

Express Shot function recalls the preset pressure in each channels quickly to support your dispensing operation.

*Available in Manual mode only.

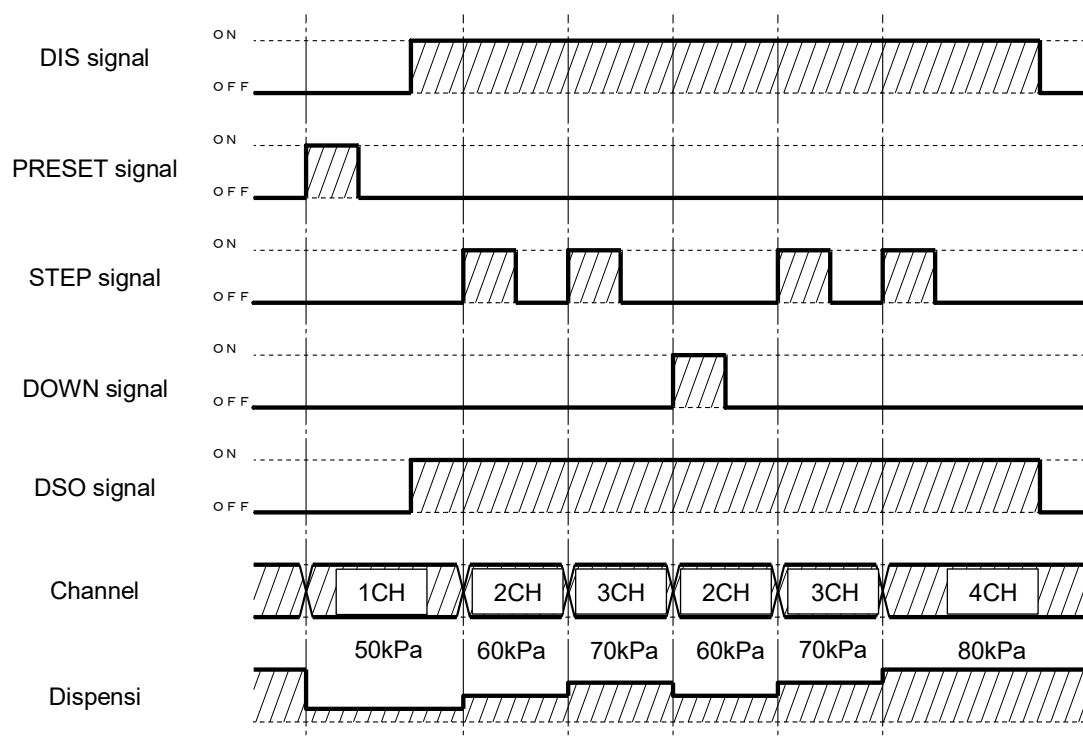
This function is used in the following cases.

- When dispensing lines of different thicknesses
- When dispensing dots of different volumes

* The STEP/PRESET/DOWN signals are recommended for switching channels when using this function.

* RS-232C communications are unable to change channels while In-dispensing, and cannot be used for this function.

* Switching channels while performing the Auto-increment function resets the switching values.



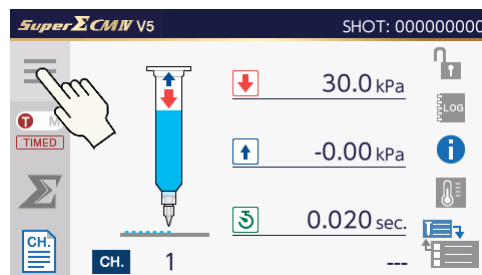
Changing the RS-232C Communication Setting (Communication Setting)

RS-232C communications between this device and host devices can be used to configure communication settings.

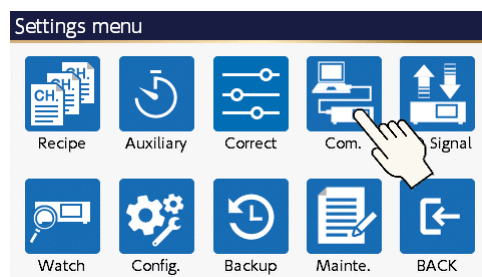
For details on each item, refer to the communication (RS-232C) section of the separate instruction manual.

Item	Description
"Communication speed"	Sets the communication speed for RS-232C communications. Each tap changes the setting in the following order. "38400bps" → "57600bps" → "9600bps" → "19200bps" → "38400bps" → ...
"Checksum"	Sets whether a checksum is used in the communication command format.
"Com. protocol"	Sets the Com. protocol. Each tap changes the setting in the following order. "Type A" → "Type C" → "Type A" → ...
"Com. check"	The communication transmission data history can be checked.

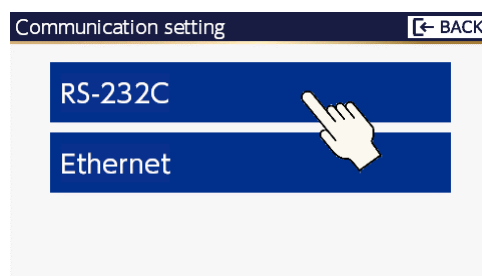
- 1 Tap the [≡] icon in Dispense screen A or B.



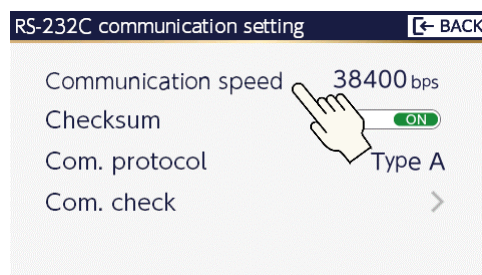
- 2 The "Settings menu" screen is displayed.
Tap the [Settings] icon.



- 3 The "Communication setting" screen is displayed. Tap "RS-232C".



- 4 The "RS-232C communication setting" screen is displayed. Configure each setting.



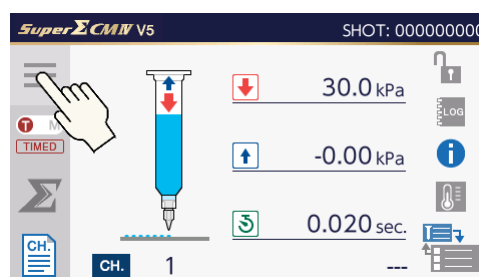
Changing the Ethernet Communication Setting (Communication Setting)

Ethernet communications between this device and host devices can be used to configure communication settings.

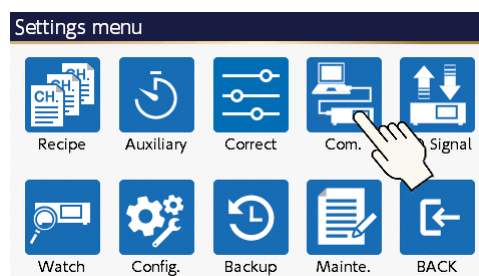
For details on each item, refer to the communication section (Ethernet) of the separate instruction manual.

Item	Description
"Controller setting"	Edits and displays various settings for Ethernet communication such as the IP address.
"Administrator setting"	Sets a password to login as an administrator. Displays the date and time of the last login.
"Read only"	When this is ON, restricts operations to only getting controller settings.
"Web page refresh interval"	Sets the interval for automatically updating the web page screen when using the web server.
"Com. check"	Allows you to check the connection status of Ethernet communication.

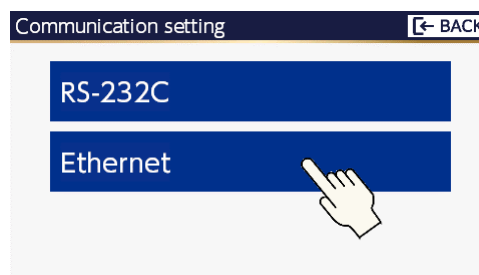
- 1 Tap the [≡] icon in Dispense screen A or B.



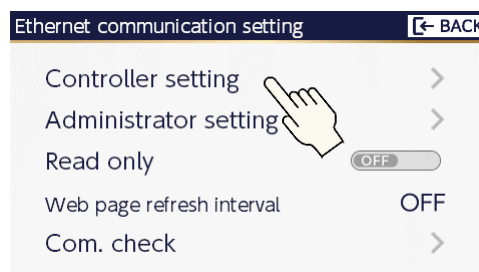
- 2 The "Settings menu" screen is displayed. Tap the [Com.] icon.



- 3 The "Communication setting" screen is displayed. Tap "Ethernet".



- 4 The "Ethernet communication setting" screen is displayed. Configure each setting.

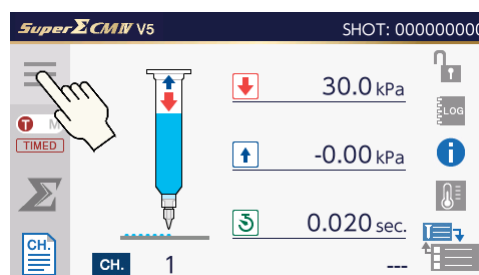


Checking the Controller Status (Watch)

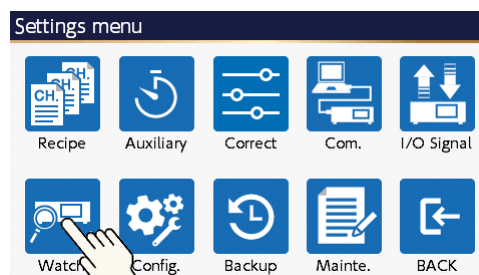
Operation history allows the dispense count number and log information to be checked.

Item	Description
"User counter"	This is the value recorded for Count. It can be cleared with the following User counter clear.
"Total counter"	This is the value recorded for the solenoid valve operation count. * This value cannot be cleared.
"Air supply pressure"	Displays the value of the air supply pressure supplied from the air input port.
"User counter clear"	Clears the User counter. * Cannot be restored to the original condition after deleting. Please take care with this function.
"Sol. valve alarm OFF"	Turns OFF solenoid valve alarm that is occurring. *After execution, solenoid valve alarm does not occur until restart.
"Move to log"	Moves to the "Log" screen. See P.71 "Checking the Operation History (Log)" for more details on log.

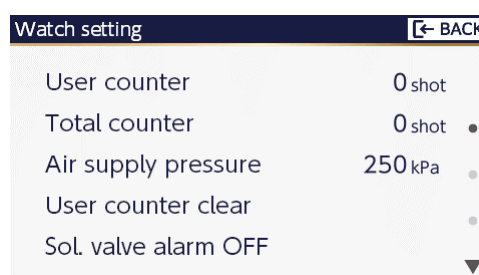
- 1 Tap the [] icon in Dispense screen A or B.



- 2 The "Settings menu" screen is displayed.
Tap the [] icon.

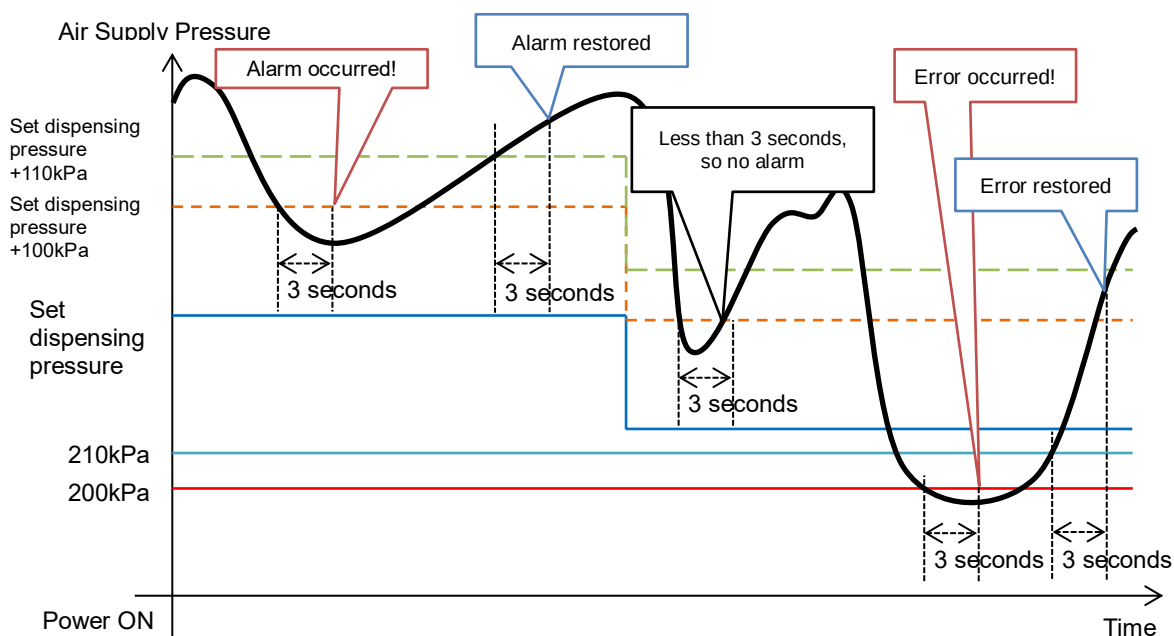


- 3 The "Watch setting" screen is displayed.



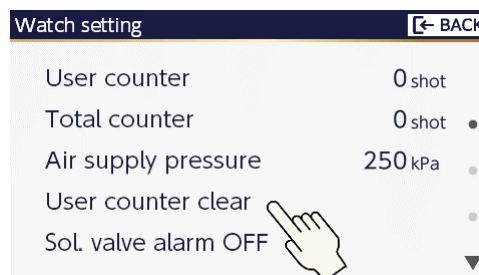
Air Supply Pressure

Ensure that the air supply pressure is "Set dispensing pressure +100 kPa or more." If the air supply pressure is below the "Set dispensing pressure +100 kPa" for 3 seconds continuously, an air supply pressure alarm occurs. The air supply pressure alarm is restored when the air supply pressure exceeds "Set dispensing pressure +110 kPa" for 3 seconds continuously. If the air supply pressure is below "200 kPa" for 3 seconds continuously, an air supply pressure error occurs. The air supply pressure error is restored when the air supply pressure exceeds "210 kPa" for 3 seconds continuously.



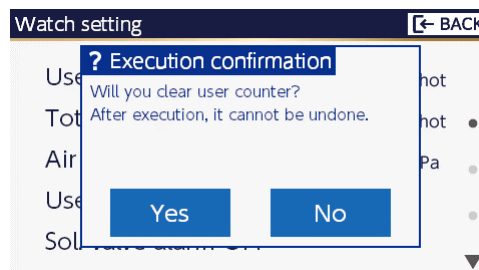
User Counter Clear

- 1 Tap "User counter clear" in the "Watch setting" screen.



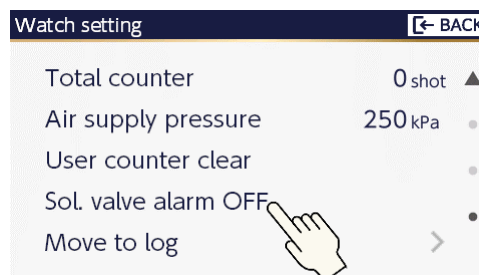
- 2 "Execution confirmation" is displayed. Tap "Yes" or "No" to confirm.

* Cannot be restored to the original condition after deleting. Please take care with this function.



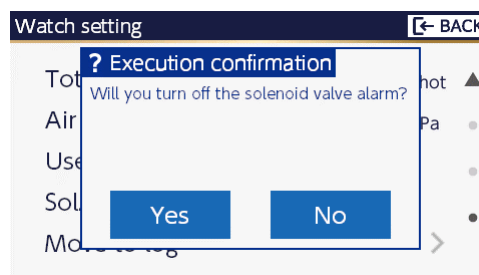
Solenoid valve alarm OFF

- 1 Tap "Sol. valve alarm OFF" in the "Watch setting" screen.



- 2 "Execution confirmation" is displayed. Tap "Yes" or "No" to confirm.

*When executed, solenoid valve alarm does not occur until restart. Please take care with this function.

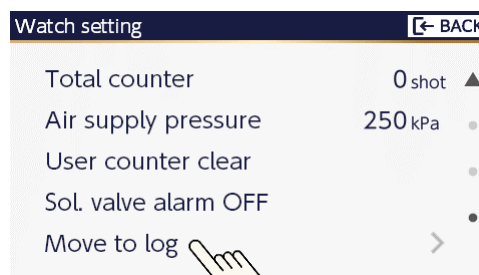


Checking the Operation History (Log)

The operation history of this device and the date and time can be checked. See P.131 "Troubleshooting" for more details on the log content.

* The newest 1000 entries are recorded.

Tap "Move to log" in the "Watch setting" screen. The screen moves to the log list screen.



Tapping each log entry moves to the "Log details" screen.

Log			2023/10/15 16:00:00	BACK
Date	Code	Message		
10/14 13:20	006	Air supply pressure error		
10/14 13:20	100	Air supply pressure warning		
10/14 13:05	202	Temp. unit error clear		
10/14 13:00	302	User counter clear		
10/14 13:00	012	Temp. unit error: Timeout		

The detailed log content is displayed.

The date of occurrence and the total counter at the time of the log are displayed.

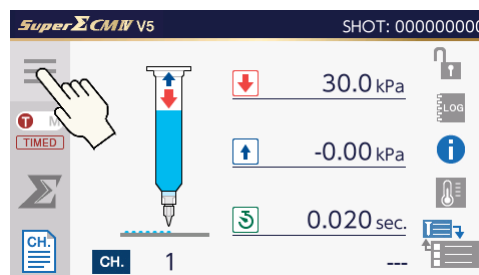
Log details 100		2023/10/15 16:00:26	← BACK
Air supply pressure is low.			
Date of occurrence	Supply	0 kPa	
2023/10/14	DIS-mode	Sigma Timed	
13:20:23	DIS-time	1.000 sec.	
Total counter	DIS-press.	30.0 kPa	
63 shot	Vacuum	-0.00 kPa	
	DIS-state	Idle	

Data and parameters related to the recorded log are displayed. Use this information when contacting our sales staff.

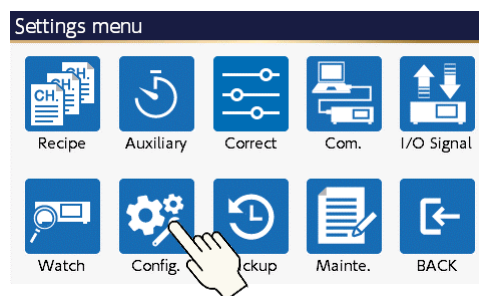
Changing Controller Configure Setting (Config.)

Use the following procedure to move to the "Configure setting" screen if changes are made to the controller environment.

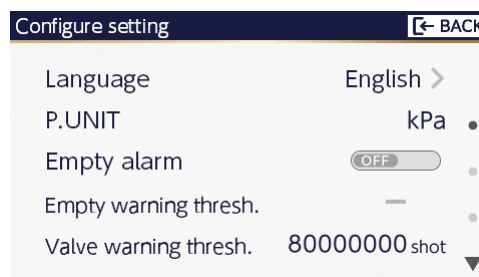
- 1 Tap the [≡] icon in Dispense screen A or B.



- 2 The "Settings menu" screen is displayed.
Tap the [⚙️] icon.



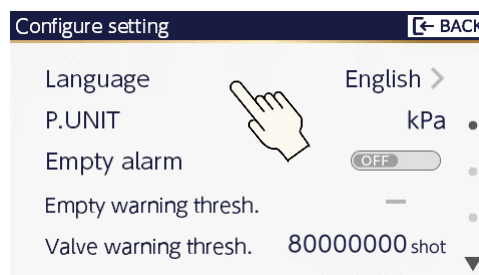
- 3 The "Configure setting" screen is displayed.



Changing the Display Language

Three languages can be configured to be displayed on the LCD touch panel: English, Japanese and Chinese.

- 1 Tap "Language" in the "Configure setting" screen.



- 2 The "Select Language" screen is displayed. Select the language to change.



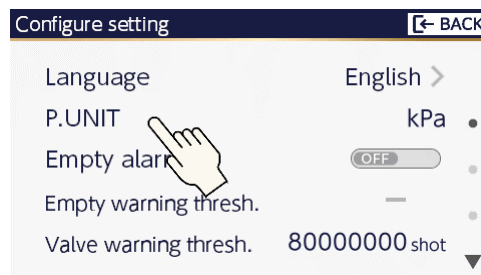
- 3 After several seconds the setting is completed, and the screen returns to the "Configure setting" screen.

Changing P.UNIT

Two units of pressure can be displayed for all pressure values: "kPa" and "bar."

Tap "P.UNIT" in the "Configure setting" screen.
Each tap changes the setting in the following order.

"kPa" → "bar" → "kPa" → ...

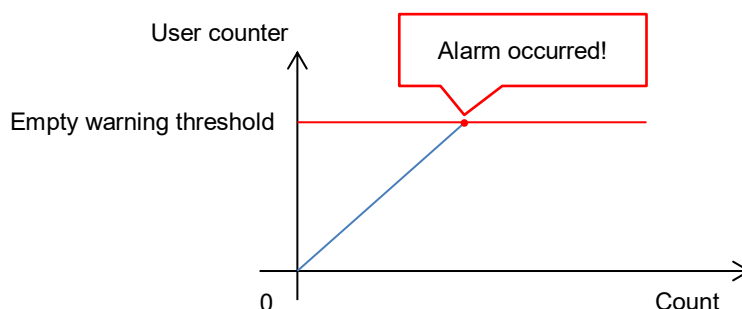


Detecting the Remaining Volume of Liquid Material in the Syringe (Empty Alarm)

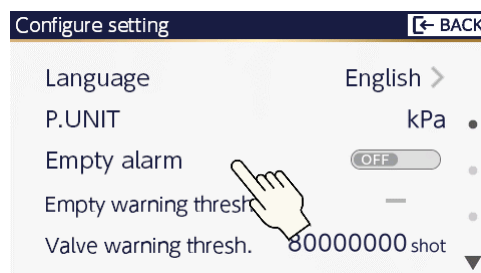
If the User counter value reaches the set Empty warning threshold, the device is in a Remaining volume warning status. Setting a warning threshold to suit operations in advance means it can be used as a simple empty alarm.

The conditions for clearing the remaining volume warning status are as follows.

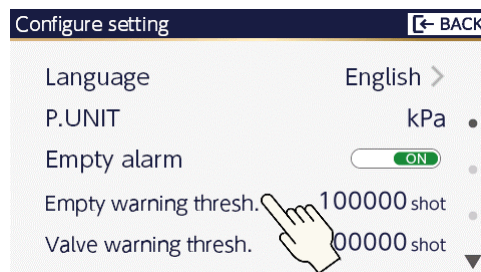
- If Empty warning threshold > User counter by using User counter clear
- If Empty warning threshold > User counter by changing the setting of Empty warning threshold
- If Empty alarm is set to OFF



- 1 Tap "Empty alarm" in the "Configure setting" screen. This allows switching between ON/OFF.



- 2 Next, tap "Empty warning thresh."

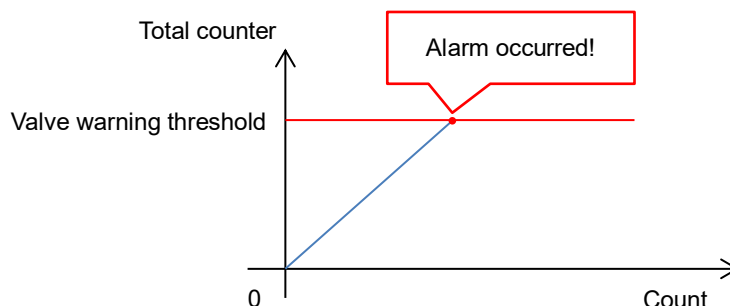


- 3 The "Empty warning thresh." input screen is displayed. Set the "Empty warning thresh." When settings are completed, the screen returns to the previous screen.

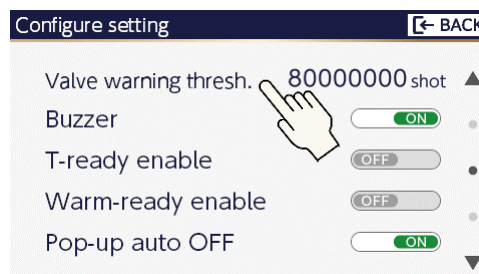
Changing the Solenoid Valve Replacement Reference (Valve Warning Threshold)

If the Total counter value reaches the set Valve warning threshold, an alarm occurs.

The alarm status is cleared if the Valve warning threshold > Total counter by changing the setting of Valve warning threshold.



- 1 Tap "Valve warning thresh." in the "Configure setting" screen.

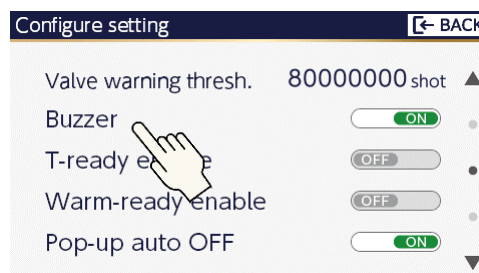


- 2 The "Valve warning thresh." input screen is displayed. Set the "Valve warning threshold". When settings are completed, the screen returns to the previous screen.

Changing the Buzzer Setting (Buzzer)

The operating sound when tapping the LCD touch panel and the warning sound when an alarm or error occurs can be changed.

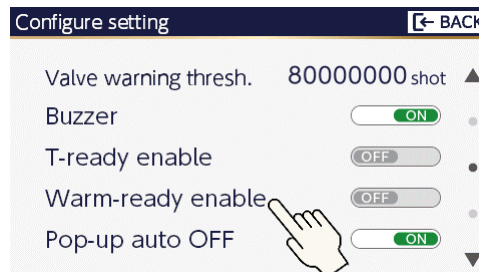
Tap "Buzzer" in the "Configure setting" screen.
This allows switching between ON/OFF.



Restricting dispensing (T-ready enable, Warm-ready enable)

Settings of conditions for restricting dispensing operation can be changed.

Tap "T-ready enable" or "Warm-ready enable" in the "Configure setting" screen.
This allows switching between ON/OFF.

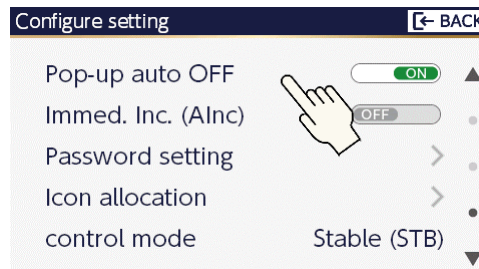


Item	Description
"T-ready enable"	Dispensing operation is prohibited if the current CH is in a recipe for which temperature unit function is set to ON, or if a Temp. unit error or Temp. unit alarm has occurred.
"Warm-ready enable"	Dispensing operation is prohibited while warming up (during display of the warm-up remaining timer at the top right of the Dispense screen).

Automatically Hiding Pop-ups (Pop-up Auto OFF)

Setting	Operation
OFF	Pop-ups when an alarm or error occurs remain displayed until tapping the LCD touch panel.
ON	Pop-ups when an alarm or error occurs are hidden automatically when the alarm or error is cleared.

Tap "Pop-up auto OFF" in the "Configure setting" screen.
This allows switching between ON/OFF.

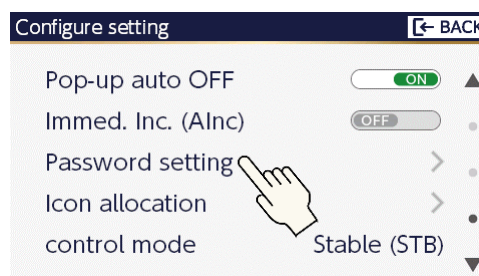


Restricting Setting Changes (Password Setting)

The device can be locked with a password to restrict settings from being changed by anyone other than the administrator.
All settings other than changing channels are restricted.

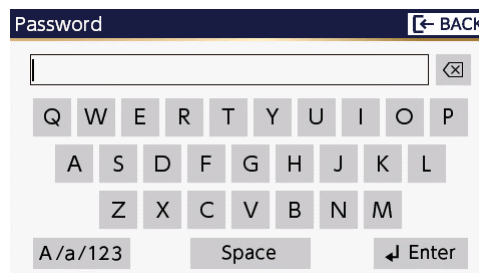
Steps for setting password lock

- 1 Tap "Password setting" in the "Configure setting" screen.



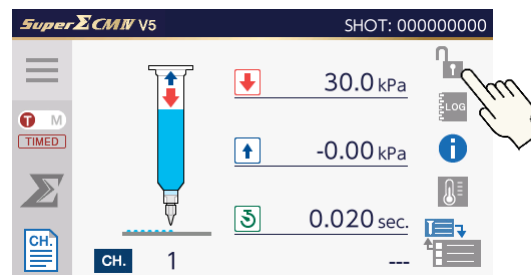
- 2 The current password needs to be entered. Enter the current password.

The default password is 0000 (four zeros).

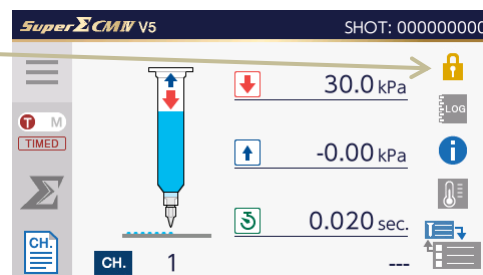


- 3 If the entered password is correct, a new password needs to be set. Enter a new password.
- 4 Tap the [BACK] icon or [] icon to move to Dispense screen A or B.

- 5 Tap the [] icon in Dispense screen A or B.

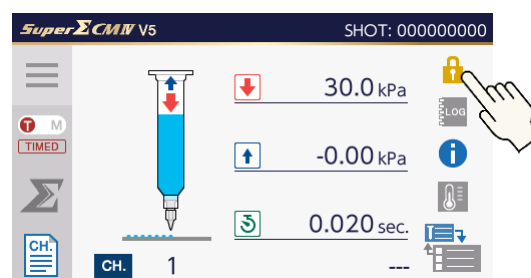


- 6 The password input screen is displayed. Enter a set password.
- 7 If the password is correct, the device is password locked, and it returns to Dispense screen A or B. The [] icon is also displayed to indicate the lock status.



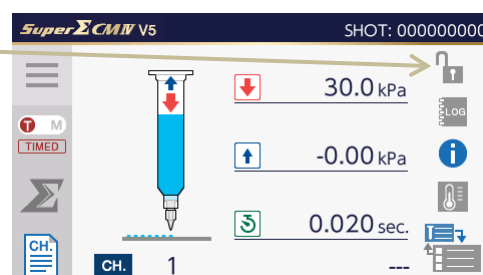
Steps for canceling password lock

- 1 Tap the [] icon in Dispense screen A or B.



- 2 The password input screen is displayed. Enter a set password.

If the password is correct, the device is password locked, and it returns to Dispense screen A or B. The [] icon is also displayed to indicate the lock canceled status.



Forgotten Password

Entering "AllMemoryClear" in the password input screen performs "All Memory Clear", and the password is reset to default.

Changing Icon Allocation (Icon Allocation)

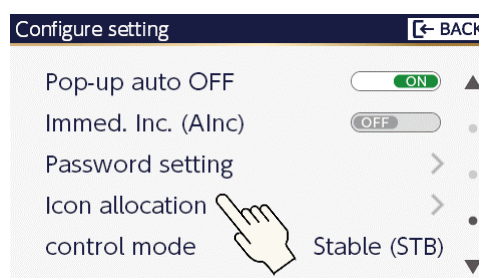
Allocation of the main icons on the Dispense screen can be changed.

Icon List

Main Icon	Function	Edit Allocation	JSON set value
	Moves to the "Settings menu."	×	0
 	Switches the Dispensing mode.  : Displayed if in TIMED mode.  : Displayed if in MANUAL mode. Tapping  in Σ mode switches to dummy shot mode  .	○ (default value)	1
 	The Σ mode state is indicated, and the screen moves to the Σ related screen.  : Displayed if in Normal mode. Tap to move to the " Σ basic setting" screen.  : Displayed if in Σ mode. Tap to move to the " Σ basic setting" screen. Press and hold to move to the " Σ mode setting" screen.	○ (default value)	11
	Moves to the "Channel list" screen or the "Channel graph" screen.	○ (default value)	2
	Moves to the "DIS-correction setting" screen.	○	3
	Moves up one channel. * Cannot move up beyond 401CH.	○	4
	If channel belongs to a recipe: Channel changes to Start CH. of the associated recipe. If channel does not belong to a recipe: Channel changes to 1CH.	○	5
	Moves down one channel. * Cannot move down below 0CH.	○	6
	If a channel belongs to a recipe and the Auto-increment function is ON, this changes the Start CH. and starts Auto-increment operation.	○	7
	Clears the User counter.	○	8
 	Switches the Interval feature between ON/OFF.  : Displayed if Interval feature is OFF.  : Displayed if Interval feature is ON.	○	9
	Moves to the "Select Start CH." screen. Selecting the recipe moves to the Start CH.	○	10
	No function available.	○	99

Icon Allocation

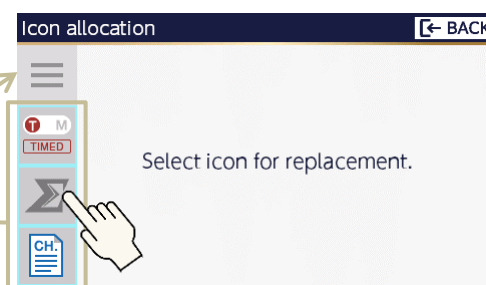
- 1 Tap "Icon allocation" in the "Configure setting" screen.



- 2 The "Icon allocation" screen is displayed. Tap the icon to change the allocation for.

The [] icon cannot be changed

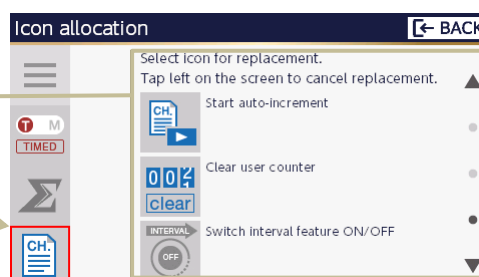
Select either



- 3 A list of icons that can be allocated is displayed. Tap the icon to change allocation. The icon is allocated.

List of icons that can be allocated

Selected icon

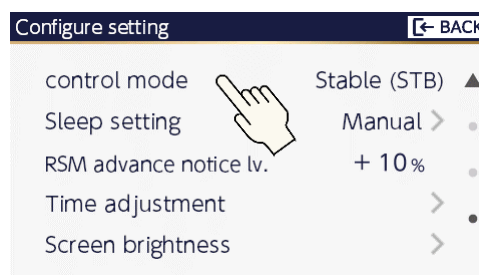


Changing solenoid valve control mode (Control mode)

Allows changing the dispensing control mode.

- 1 Each tap of "Control mode" in the "Configure setting" screen changes the setting in the following order.

"Stable (STB)" → "Speedy (SPD)" → "Stable (STB)" → ...



Select the desired control mode after referencing the following table.

Control mode	Details
"STB"	This mode reduces variance in the dispensing volume.
"SPD"	This mode is used for high-speed Takt operation.

Changing the sleep function setting (Sleep setting)

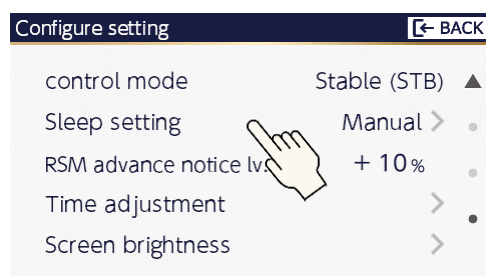
This device can be transitioned to a Sleep mode with low power and air consumption.

In the Sleep mode, the screen light goes OFF. Also, all input operation is disabled, except for LCD Touch Panel tap operation and SLEEP signal input. The READY signal goes OFF (Dispense not possible), but other output signals are output like normal.

To return from Sleep mode, it is necessary to tap the LCD Touch Panel of the device, or switch the SLEEP signal from OFF to ON.

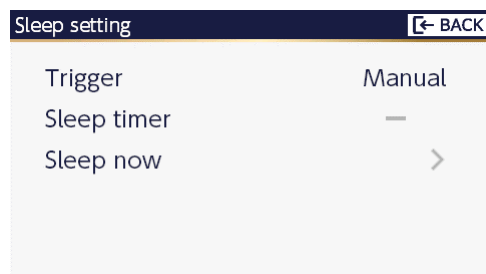
Item	Description
"Trigger setting"	Sets trigger for transitioning to Sleep mode. "Manual" : Transition immediately to Sleep mode by tapping "Sleep now" or switching the SLEEP signal from OFF to ON. "Auto" : Transition to Sleep mode when the set sleep timer elapses, after operation of the device ceases.
"Sleep timer"	Changing is possible if "Trigger setting" has been set to "Auto." Transition to Sleep mode when the set sleep timer elapses, after operation of the device ceases.
"Sleep now"	Operation is enabled if the "Trigger setting" has been set to "Manual." Tapping and selecting "Yes" at the confirmation screen transitions immediately to Sleep mode.

- 1 Tap "Sleep setting" in the "Configure setting" screen.



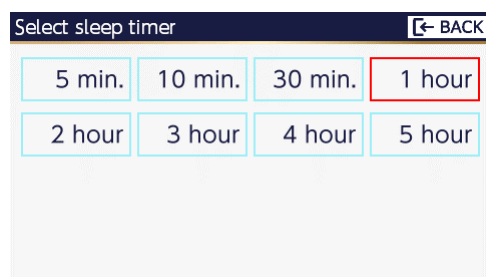
- 2 The "Sleep setting" screen is displayed. Each tap of "Trigger setting" changes the setting in the following order.

"Manual" → "Auto." → "Manual" → ...

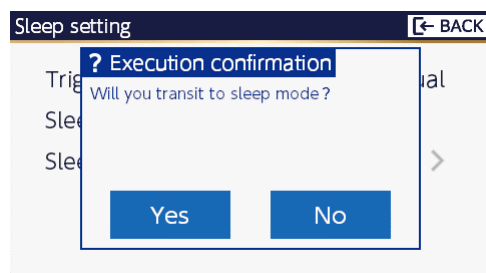


- 3 If the "Trigger setting" has been set to "Auto.", then tapping "Sleep timer" moves to the "Sleep timer selection" screen.

* The image on the right shows the case when "1 hour" is selected.



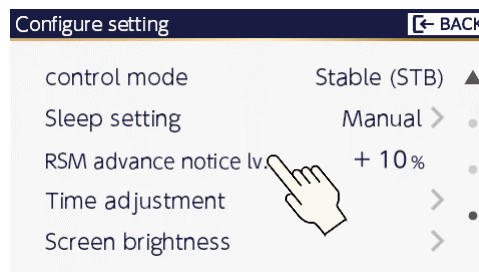
- 4 If the "Trigger setting" has been set to "Manual", then tapping "Sleep timer" moves to the confirmation screen. Tap "Yes" and it immediately enters Sleep mode.



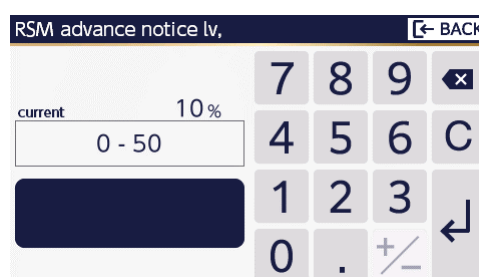
Discovering reduction in remaining volume quickly from the remaining volume warning position (Remaining volume detection notice level)

The Automatic remaining volume alert function allows detection that the remaining volume of liquid material in the syringe has decreased prior to occurrence of Σ mode remaining volume alert. This can be used as a reference for the timing of starting preparation for syringe replacement.

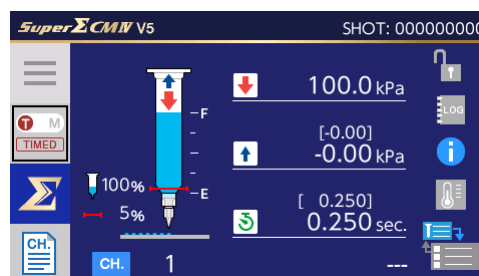
- 1 Tap "RSM advance notice lv." in the "Configure setting" screen.



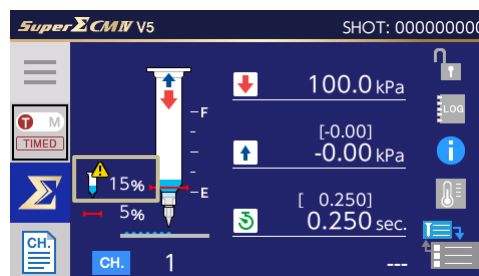
- 2 The "RSM advance notice lv." input screen is displayed. Set the Remaining volume detection notice level.



- 3 If the Automatic remaining volume alert function is ON and the RSM level has been set to 5%, then when the liquid volume is 100%, the Dispense screen appears like the image on the right.



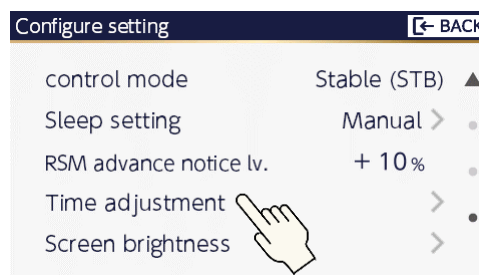
- 4 If the Remaining volume detection notice level is set to 10%, then the alert mark is displayed when the remaining volume of liquid reaches 15% or less. Also, the RSM_P signal goes ON at this time. (P.106)



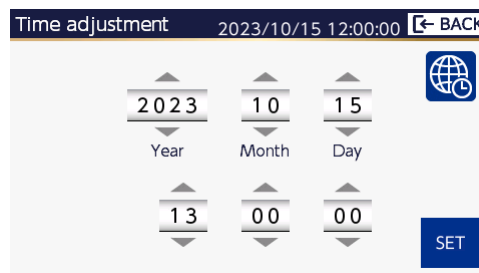
Changing the Time (Time Adjustment)

The time displayed for this device can be changed.

- 1 Tap "Time adjustment" in the "Configure setting" screen.



- 2 The "Time adjustment" screen is displayed. Tap the up/down arrows for the year, month, day, hour, minute, and second, and after adjusting the time, tap "SET" to save.



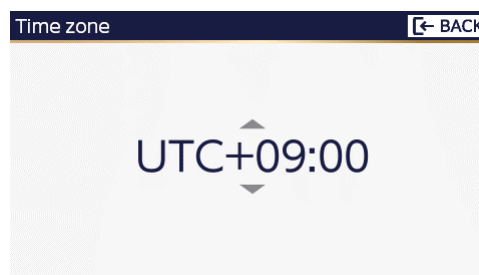
This unit displays the current local time.

To change or obtain the current time of this unit using Ethernet communication, it is necessary to set the time zone.

- 1 Tap the  icon on the "Time adjustment" screen.



- 2 "Time zone" setting screen will appear. Select the time zone by tapping the top and bottom of the difference time from the standard time.
*The change will be reflected immediately.
*The figure on the right shows the settings when selecting Japan time.



The time zones that can be set are as follows. The initial value is UTC+09:00.

ID assigned to each time zone is used for RS-232C communication and Ethernet communication.

ID	Time zone
0	UTC -12:00
1	UTC -11:00
2	UTC -10:00
3	UTC -09:30
4	UTC -09:00
5	UTC -08:00
6	UTC -07:00
7	UTC -06:00
8	UTC -05:00
9	UTC -04:00

ID	Time zone
10	UTC -03:30
11	UTC -03:00
12	UTC -02:00
13	UTC -01:00
14	UTC +00:00
15	UTC +01:00
16	UTC +02:00
17	UTC +03:00
18	UTC +03:30
19	UTC +04:00

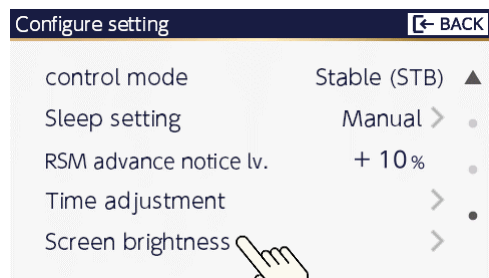
ID	Time zone
20	UTC +04:30
21	UTC +05:00
22	UTC +05:30
23	UTC +05:45
24	UTC +06:00
25	UTC +06:30
26	UTC +07:00
27	UTC +08:00
28	UTC +08:45
29	UTC +09:00

ID	Time zone
30	UTC +09:30
31	UTC +10:00
32	UTC +10:30
33	UTC +11:00
34	UTC +12:00
35	UTC +12:45
36	UTC +13:00
37	UTC +14:00

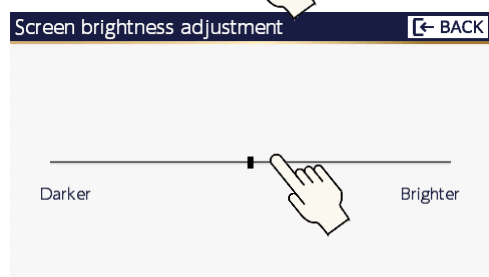
Changing the LCD Screen Brightness (Screen Brightness)

The brightness of the LCD touch panel can be changed.

- 1 Tap "Screen brightness" in the "Configure setting" screen.



- 2 The "Screen brightness adjustment" screen is displayed. Tap on the slider to adjust the brightness.
* Care must be taken as life of the backlight may be shortened if a bright setting is used.



Backing Up the Controller Settings (Backup)

Connecting USB memory with the USB-2 connector allows the device settings to be backed up.

Available Operation (Import)	Description
"the specified recipe"	Imports the specified recipe data from USB memory. * If the specified recipe is already registered on the device, it will be overwritten. * If channels included in the specified recipe are included in other recipes, the other recipes will be deleted.
"the specified CH"	Imports the specified channel data from USB memory. * Channel data will be overwritten.
"all data"	Imports all data saved on the device from USB memory. * All data will be overwritten.

Available Operation (Export)	Description
"all data"	Exports all data saved on the device to USB memory.
"log data"	Exports log data saved on the device to USB memory.

* The USB memory to be used must be formatted with FAT32.

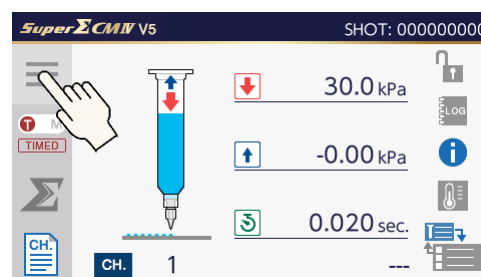
* Any USB memory with encryption function cannot be used.

* Some types of USB memory are not compatible, and operation of all USB devices cannot be guaranteed.

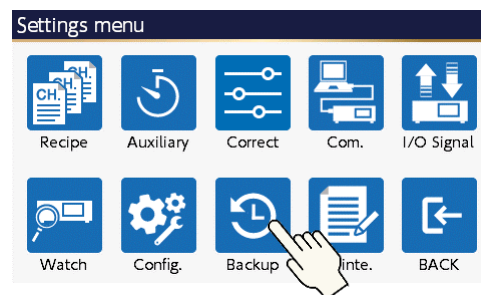
* Available files to be read is up to 20.

Use the following procedure to move to the "Backup" screen if a backup is to be performed.

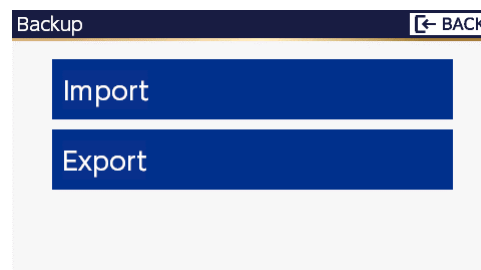
- 1 Tap the [≡] icon in Dispense screen A or B.



- 2 The "Settings menu" screen is displayed.
Tap the [Backup] icon.



- 3 The "Backup" screen is displayed.



Import

The following three types of data can be selected for import.

- "Import specified recipe"
- "Import specified CH"
- "Import all data"

"Import specified recipe"

Copies the specified recipe data from USB memory.

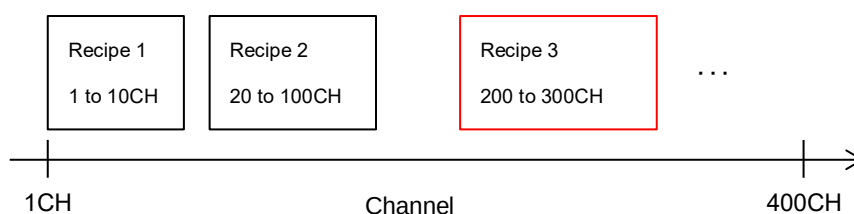
* If the specified recipe is already registered on the device, it will be overwritten.

* If channels included in the specified recipe are included in other recipes, the other recipes will be deleted.

* The DIS-monitoring results of all copied channels will be reset to default. To use DIS-monitoring, perform tests for all channels again.

[If the specified recipe is already registered on the device]

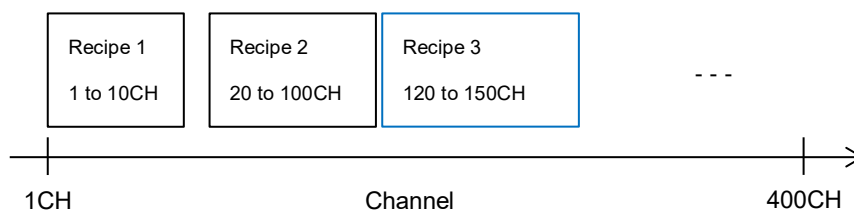
Before import



Import Recipe 3 (120 to 150CH)

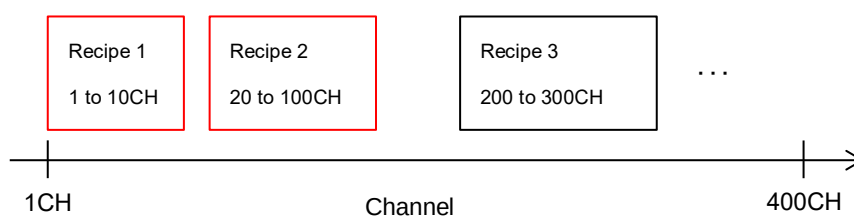


After import



[If channels included in the specified recipe are included in other recipes]

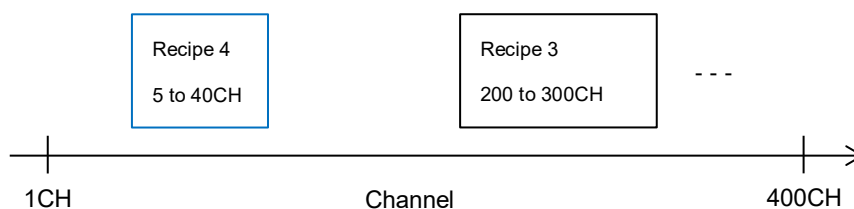
Before import



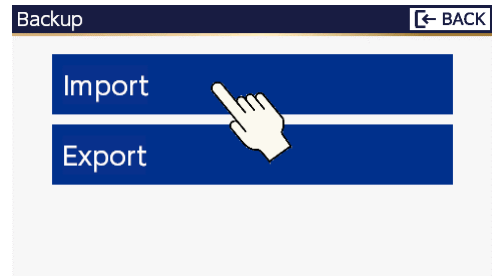
Import Recipe 4 (5 to 40CH)



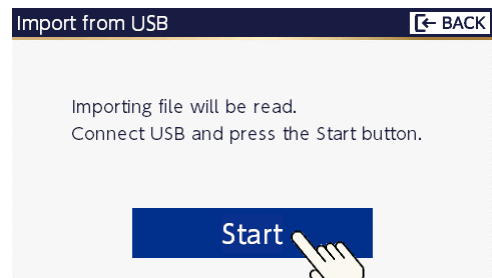
After import



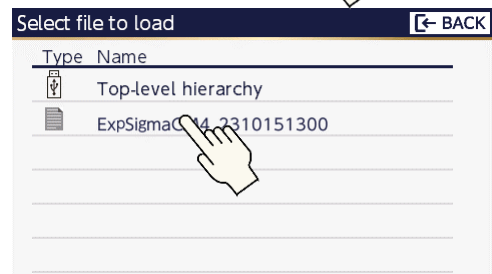
- 1 Tap "Import" in the "Backup" screen.



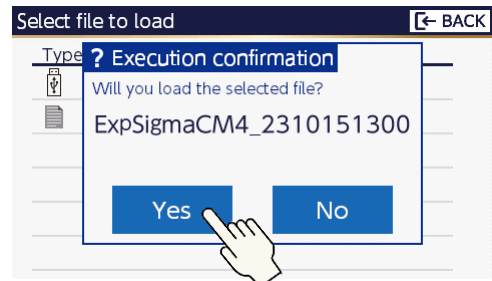
- 2 The "Import from USB" screen is displayed. Connect USB memory that has the designated data file saved with the USB-2 connector, and tap "Start."



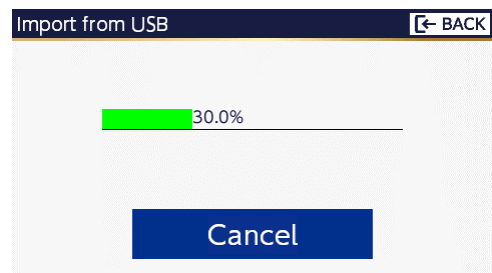
- 3 The "Select file to load" screen is displayed. Tap the file to load.



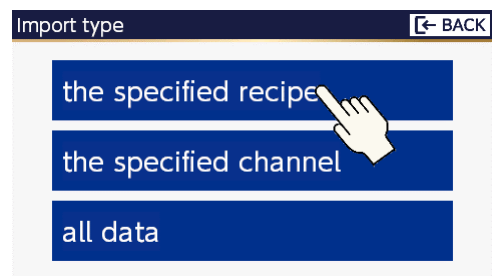
- 4 The "Execution confirmation" screen is displayed. Tap "Yes."



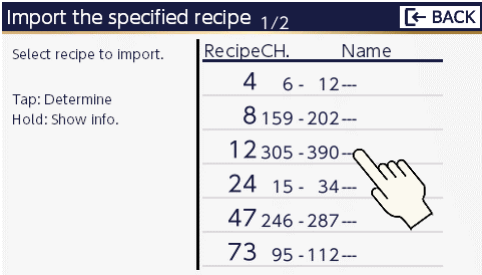
- 5 This starts loading the file.
* The data in the controller has not yet been changed at this point.



- 6 When the file has been loaded, the "Import type" screen is displayed. Tap "the specified recipe"



- 7 The screen for selecting the recipe is displayed. Tap the recipe to import in the recipe list to select it.



- 8 Tap "Start" to start the import.



- 9 The import is complete when the import complete notification is displayed.

"Import specified CH"

Copies the specified channel data from USB memory.

* The specified channel will be overwritten.

* The DIS-monitoring results of the copied channel will be reset to default. To use DIS-monitoring, perform the test again.

Before import

Channel	Channel	Channel			Channel	Channel
1 CH	2 CH	3 CH	...		399 CH	400 CH

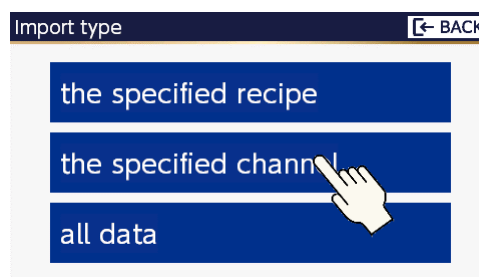
Import
3CH



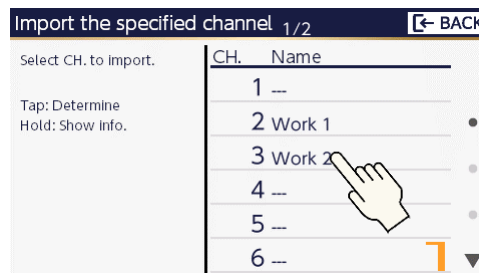
After import

Channel	Channel	Channel			Channel	Channel
1 CH	2 CH	3 CH	...		399 CH	400 CH

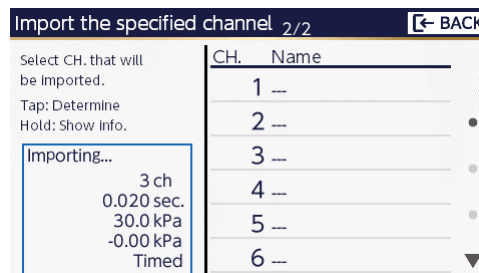
- 1 Load the file from the USB memory using the same procedures as P.85 "Import specified recipe".
- 2 When the file has been loaded, the "Import type" screen is displayed. Tap "the specified channel"



- 3 The screen for selecting the channel is displayed. Tap the channel to import in the channel list to select it.



- 4 Tap the channel to import to.



- 5 The import is complete when the import complete notification is displayed.

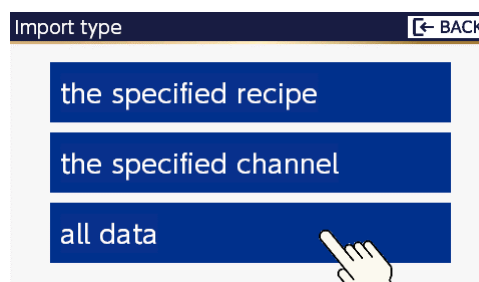
"Import all data"

Copies all data saved on the device from USB memory.

* All data except User counter and Total counter will be overwritten.

* The DIS-monitoring results of all channels will be reset to default. To use DIS-monitoring, perform tests for all channels being used again.

- 1 Load the file from the USB memory using the same procedures as P.85 "Import specified **recipe**".
- 2 When the file has been loaded, the "Import type" screen is displayed. Tap "all data."



- 3 The import is complete when the import complete notification is displayed.

Export

The following two types of data can be exported.

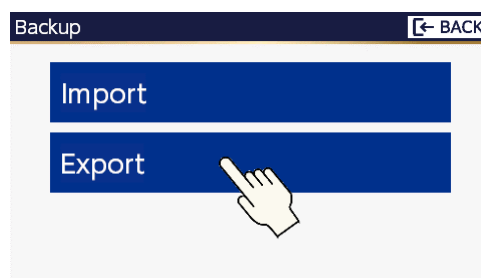
- All saved settings data
- Log data

Export all saved settings data

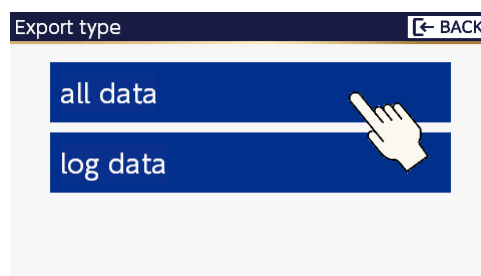
Copies all settings data saved on the device to USB memory.

* If the all settings data file is already on the USB memory, it will be overwritten.

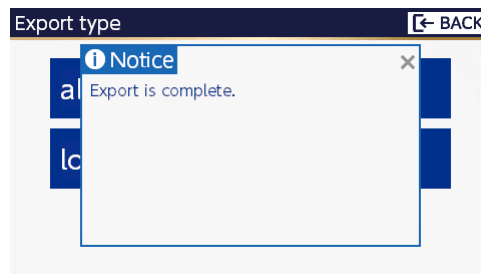
- 1 Tap "Export" in the "Backup" screen.



- 2 The "Export type" screen is displayed. Tap "all data."



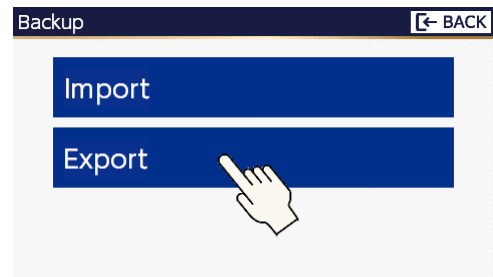
- 3 The export is complete when the export complete notification is displayed.



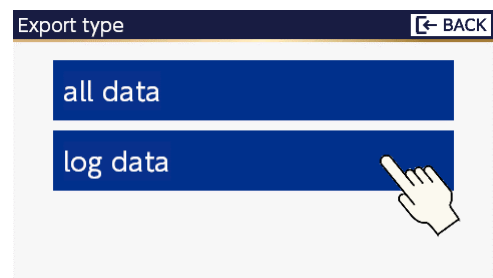
Export log data

Copies all log data saved on the device to USB memory.

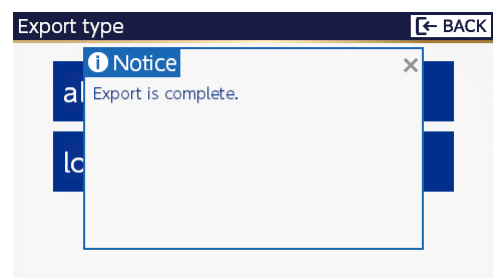
- 1 Tap "Export" in the "Backup" screen.



- 2 The "Export type" screen is displayed. Tap "log data."



- 3 The export is complete when the export complete notification is displayed.



5 Connections using External Input/Output Terminals

The external input/output terminals can be used to switch Dispensing mode or channels.

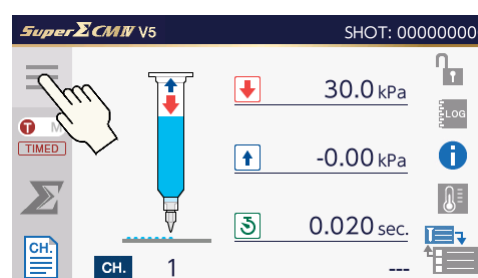
Connector Specifications

Controller side	D-Sub 25-pin socket (fastener: M2.6)	
User side (recommended)	Name: Connector set	Model: MC—DSUB25P

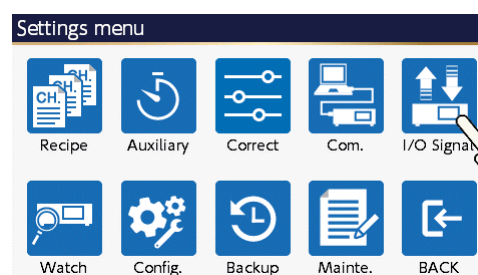
Changing External Input/Output Terminal Settings ("I/O Signal")

Use the following procedure to move to the "I/O signal setting" screen to change settings of the external input/output terminals.

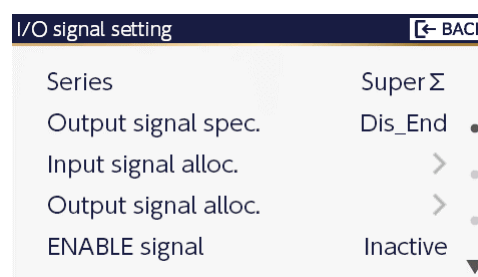
- 1 Tap the [] icon in Dispense screen A or B.



- 2 The "Settings menu" screen is displayed.
Tap the [] icon.



- 3 The "I/O signal setting" screen is displayed.



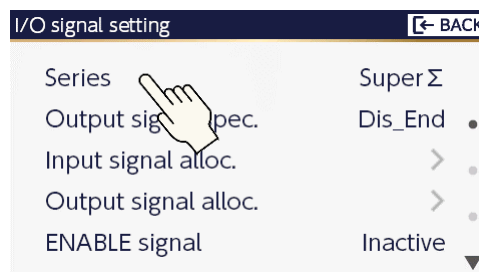
Pin Allocation

Changing Pin Allocation

The pin allocation of the I/O connector can be changed.
Changing the pin allocation will reset the input/output signal allocation to default.

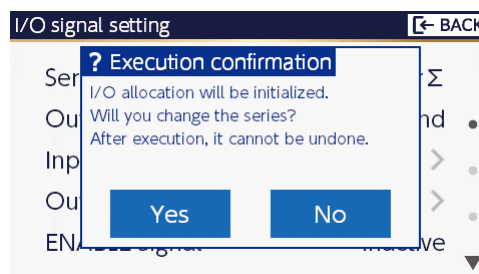
Default value: SuperΣ

- 1 Tap "Series" in the "I/O signal setting" screen.



- 2 The "Execution confirmation" screen is displayed. Tap "Yes" or "No" to confirm.

The following order changes each time this is executed.
"SuperΣ" → "ML-808GX" → "SuperΣ" → ...



Series: SuperΣ

PIN	I/O	Signal Name	Function		Edit Allocation
1 *1	-	-	-	-	-
2	I	DIS	Dispense	Inputs dispense ON/OFF	×
3	I	TMS	Dispensing mode switching	Sets the dispensing mode	○
4	I	CHS	Channel strobe	Sets the channel	○
5	I	CD02	Specify channel	Specifies the last digit of the channel as binary number 2	○
6	I	CD08	Specify channel	Specifies the last digit of the channel as binary number 8	○
7	I	CD20	Specify channel	Specifies the first digit of the channel as binary number 2	○
8	I	ENABLE	Dispensing enabled	Enables the dispense operation	○
9	O	NDSO *2	DSO inverse	Outputs a DSO inverse signal	×
		END *2	Dispense complete	Output when a dispense signal is input after dispense end	
10	O	EXE1	Solenoid valve alarm	Output when total counter reaches the valve warning threshold.	○
11	O	PON	Power on	Output when POWER is supplied to the device	×
12	O	READY *2*3	Ready	Output when all the following are satisfied (dispense possible state) - When not In-dispensing cycle - When there is no air supply pressure alarm - Dispense screen A or B is being displayed - When "T-ready enable" is On and there are no temperature unit errors - When "T-ready enable" is On and there are no temperature unit alarms	×
		NDVO *2	DVO inverse	Outputs a DVO inverse signal	
13	FG	GND	-	-	-
14 *1	-	-	-	-	-
15	I	CD80	Specify channel	Specifies the first digit of the channel as binary number 8	○
16	I	TMD	Specify mode	Specifies the dispensing mode	○
17	I	CD01	Specify channel	Specifies the last digit of the channel as binary number 1	○
18	I	CD04	Specify channel	Specifies the last digit of the channel as binary number 4	○
19	I	CD10	Specify channel	Specifies the first digit of the channel as binary number 1	○
20	I	CD40	Specify channel	Specifies the first digit of the channel as binary number 4	○
21	O	DSO	In-dispensing	Output during dispense operations	×
22	O	RSM	Remaining volume warning	Output under either of the following statuses. - When a Σ mode remaining volume alert occurs - When User counter reaches the Empty warning threshold	○
23	O	PSE1	Error	Output under either of the following statuses - When total counter reaches the designated number - When an exhaust error occurs - When a vacuum pressure error occurs - When an air supply pressure error occurs	○
24	O	DVO	In-dispensing cycle	Output during in-dispensing cycle (from start of dispensing until disposal after dispensing).	×
25	FG	GND	-	-	-

* Do not connect DC24V to the signal line. Doing so may cause a malfunction.

*1 Connections are prohibited on pin numbers 1 and 14. DC24V is output on these pins, so do not connect any device.

*2 See P.109 "Switching Output Signal Function (Output Signal Specifications)" for information on switching signal specifications.

*3 See P.70 "Air Supply Pressure" for information on the air supply pressure alarm.

Series: ML-808GX

PIN	I/O	Signal Name	Function		Edit Allocation
1 *1	-	-	-	-	-
2	I	DIS	Dispense	Inputs dispense ON/OFF	×
3	I	PRESET	Channel preset	Changes the channel to 1CH	○
4	I	CHS	Channel strobe	Sets the channel	○
5	I	CD02	Specify channel	Specifies the last digit of the channel as binary number 2	○
6	I	CD08	Specify channel	Specifies the last digit of the channel as binary number 8	○
7	I	CD20	Specify channel	Specifies the first digit of the channel as binary number 2	○
8	I	ENABLE	Dispensing enabled	Enables the dispense operation	○
9	O	S_END	Switching complete	Output when a channel switching signal or dispensing mode switching signal is input	○
10	O	ERROR	Error	Output under either of the following statuses - When total counter reaches the designated number - When a vacuum pressure error occurs - When an air supply pressure error occurs - When a temperature unit error occurs	○
11	O	PON	Power on	Output when POWER is supplied to the device	×
12	O	AI END	Auto-increment end	Output when Auto-increment has completed	○
13	FG	GND	-	-	-
14 *1	-	-	-	-	-
15	I	CD80	Specify channel	Specifies the first digit of the channel as binary number 8	○
16	I	STEP	Channel step	Moves the channel up 1CH	○
17	I	CD01	Specify channel	Specifies the last digit of the channel as binary number 1	○
18	I	CD04	Specify channel	Specifies the last digit of the channel as binary number 4	○
19	I	CD10	Specify channel	Specifies the first digit of the channel as binary number 1	○
20	I	CD40	Specify channel	Specifies the first digit of the channel as binary number 4	○
21	O	ALARM	Alarm	Output under either of the following statuses - When total counter reaches the valve warning threshold - When the air supply pressure < Set dispensing pressure +100 kPa	○
22	O	PCS	Pressure is being switched	Output for 1 second after changing the dispensing pressure	○
23	O	READY*3	Ready	Output when all the following are satisfied - When not In-dispensing cycle - When there is no air supply pressure alarm - Dispense screen A or B is being displayed - When "T-ready enable" is On and there are no temperature unit errors - When "T-ready enable" is On and there are no temperature unit alarms	○
24	O	DSO*2	In-dispensing	Output during dispense operations	×
		END*2	Dispense complete	Output when a dispense signal is input after dispense end	
25	FG	GND	-	-	-

* Do not connect DC24V to the signal line. Doing so may cause a malfunction.

*1 Connections are prohibited on pin numbers 1 and 14. DC24V is output on these pins, so do not connect any device.

*2 See P.109 "Switching Output Signal Function (Output Signal Specifications)" for information on switching signal specifications.

*3 See P.70 "Air Supply Pressure" for information on the air supply pressure alarm.

Allocating Input/Output Signals

The input/output signal allocation can be changed.

Allocating Input Signals

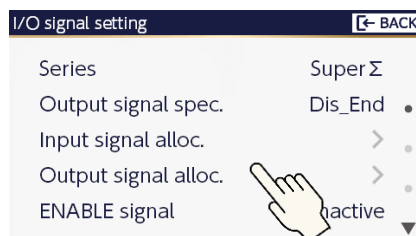
Out of all the following input signals, signals other than those in the bold frame can be selected and allocated to input pins.

The default values are the signals outlined in the P.93"Series: SuperΣ" chart or the P.94"Series: ML-808GX" chart.

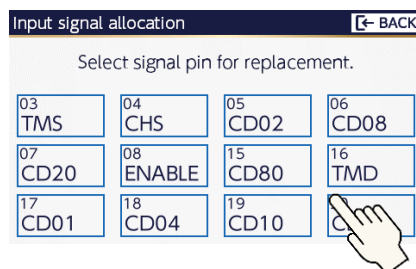
Signal Name	Function		Reference
DIS	Dispense	Inputs dispense ON/OFF	P.98
ENABLE	Dispensing enabled	Enables the dispense operation	P.98
STEP	Channel step	Moves the channel up 1CH * Cannot move up beyond 401CH	P.99
PRESET	Channel preset	If channel belongs to a recipe: Changes to Start CH. of the recipe that the channel belongs to If channel does not belong to a recipe: Changes the channel to 1CH	
DOWN	Channel down	Moves the channel down 1CH * Cannot move down below 0CH	
TMD	Specify mode	Specifies the dispensing mode	P.99
TMS	Dispensing mode switching	Sets the dispensing mode	
C_CNT	Counter clear	Clears the User counter.	P.100
EDIT	Edit	Switches between I/O EDIT mode enabled/disabled	P.112
RESET	Reset	Turns ALARM2 signal output OFF	P.100
CHS	Channel switching	Sets the channel	P.102
CD01	Specify channel	Specifies the last digit of the channel as binary number 1	
CD02	Specify channel	Specifies the last digit of the channel as binary number 2	
CD04	Specify channel	Specifies the last digit of the channel as binary number 4	
CD08	Specify channel	Specifies the last digit of the channel as binary number 8	
CD10	Specify channel	Specifies the first digit of the channel as binary number 1	
CD20	Specify channel	Specifies the first digit of the channel as binary number 2	
CD40	Specify channel	Specifies the first digit of the channel as binary number 4	
CD80	Specify channel	Specifies the first digit of the channel as binary number 8	
SLEEP	Sleep	In operation: Transits to sleep. In sleep: Returns from Sleep mode.	P.101

Allocation Method

- 1 Tap "Input signal alloc." in the "I/O signal setting" screen.



- 2 The "Input signal allocation" screen is displayed. Tap the signal pin for replacement (initial replacement signal pin).

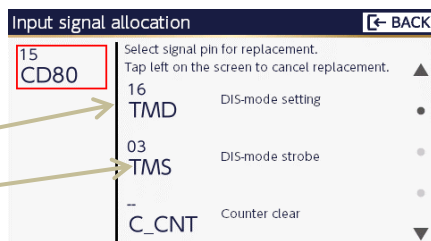


- 3 Tap the target replacement signal pin. The signal allocation can be replaced.

Initial replacement signal pin and signal name

Currently allocated input pin number

Signal Name



Function overview

Allocating Output Signals

Out of all the following output signals, signals other than those in the bold frame can be selected and allocated to output pins.

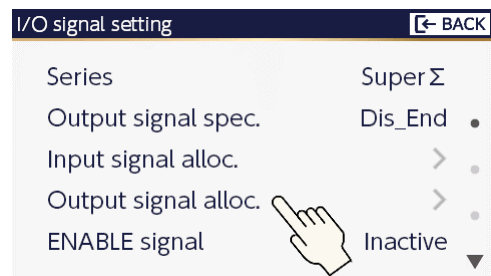
The default values are the signals outlined in the P.93"Series: SuperΣ" chart or the P.94"Series: ML-808GX" chart.

Signal Name	Function		Reference
PON	Power on	Output when POWER is supplied to the device	P.106
DSO	In-dispensing	Output during dispense operations	P.105
END	Dispense complete	Output when a dispense signal is input after dispense end	
NDSO	DSO inverse	Outputs a DSO inverse signal	
DVO	In-dispensing cycle	Output during in-dispensing cycle	
NDVO	DVO inverse	Outputs a DVO inverse signal	
ERROR	Error	Output under either of the following statuses - When total counter reaches the designated number - When an air supply pressure error occurs - When an exhaust error occurs - When a vacuum pressure error occurs - When an internal communication error occurs - When a temperature unit error occurs	P.105
ALARM	Alarm	Output under either of the following statuses - When total counter reaches the valve warning threshold - When an air supply pressure Alarm occurs	P.105,106
EXE1	Solenoid valve alarm	When total counter reaches the valve warning threshold	P.105
EXE2	Error	Output under either of the following statuses - When total counter reaches the designated number - When an exhaust error occurs	P.105
PSE1	Error	Output under either of the following statuses - When total counter reaches the designated number - When an air supply pressure error occurs - When an exhaust error occurs - When a vacuum pressure error occurs	P.106
PSE2	Error	Output under either of the following statuses - When an air supply pressure error occurs - When a vacuum pressure error occurs	P.106
ALARM2	Alarm2	Output when dispensing pressure outside of the allowable range are detected with the DIS-monitoring function	P.107
RSM	Remaining volume warning	Output under either of the following statuses. - When a Σ mode remaining volume alert occurs - When User counter reaches the Empty warning threshold	P.105
READY	Ready	Output when all the following are satisfied - When not In-dispensing cycle - When there is no air supply pressure alarm - Dispense screen A or B is being displayed - When "T-ready enable" is On and there are no temperature unit errors - When "T-ready enable" is On and there are no temperature unit alarms	P.105
PCS	Pressure is being switched	Output for 1 second after changing the dispensing pressure	P.108

Signal Name	Function		Reference
AI_END	Auto-increment complete	Output when Auto-increment has completed	P.107
S_END	Switching complete	Output when the following input signals are functioning properly, and while signals are input - STEP signals - PRESET signal - DOWN signals - TMS signals - C_CNT signal - CHS signals - SLEEP signal	P.99,100,102
ESTS	During I/O EDIT mode	Output when I/O EDIT mode is ON	P.112
EWAIT	Waiting for I/O EDIT data	Output while waiting for data signal input in I/O EDIT mode	
T_RDY	Temperature Ready	Output when there are no errors or alarms occur with temperature controllers for which "Control" is ON.	P.108
RSM_P	Remaining volume detection notice	Output when liquid remaining volume is below the set detection level + remaining volume detection notice level, when the automatic remaining volume alert function is ON in Σ mode.	P.106

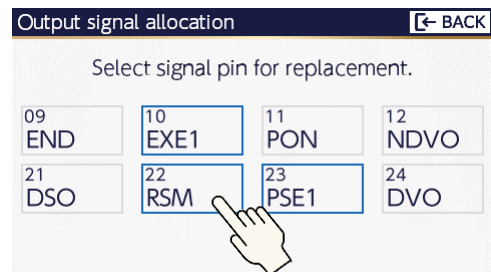
Allocation Method

- 1 Tap "Output signal alloc." in the "I/O signal setting" screen.



- 2 The "Output signal allocation" screen is displayed. Tap the signal pin for replacement (initial replacement signal pin).

* The signal pins surrounded by the gray box cannot be replaced.

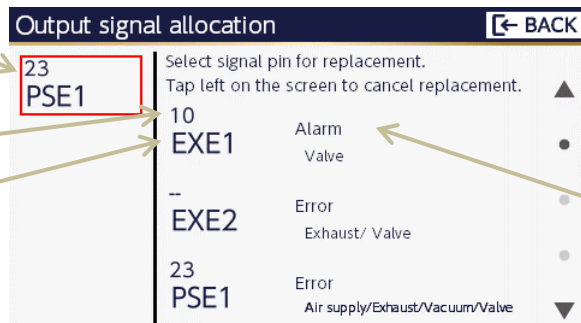


- 3 Tap the target replacement signal pin. The signal allocation can be replaced.

Initial replacement signal pin and signal name

Currently allocated input pin number

Signal Name



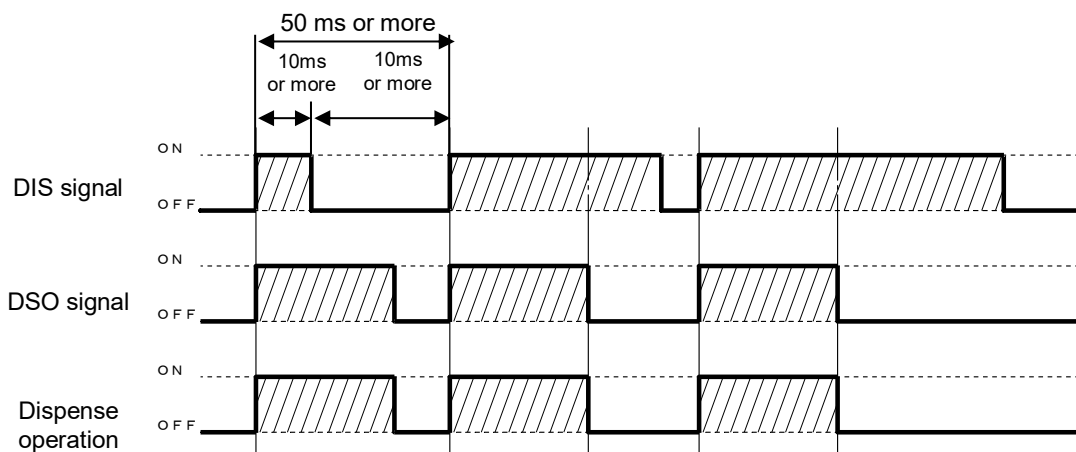
Function overview

Input

DIS signal

PIN	Signal Name	Allocation	Function
2	DIS	Fixed	Inputs dispense ON/OFF

Timing Chart



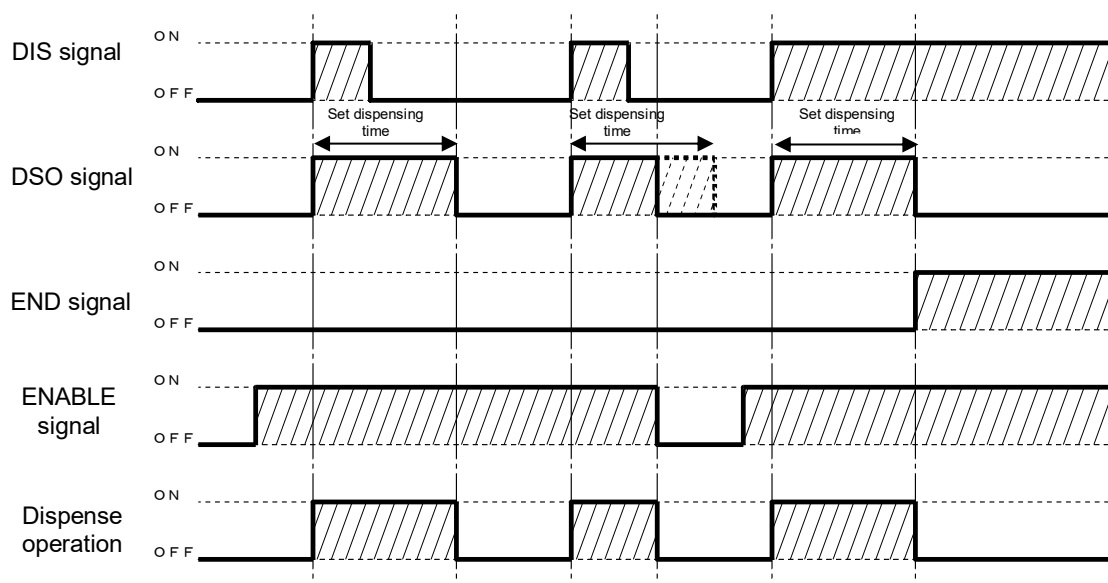
ENABLE signal

PIN	Signal Name	Allocation	Function
8 (default value)	ENABLE	Replaceable	Enables the dispense operation

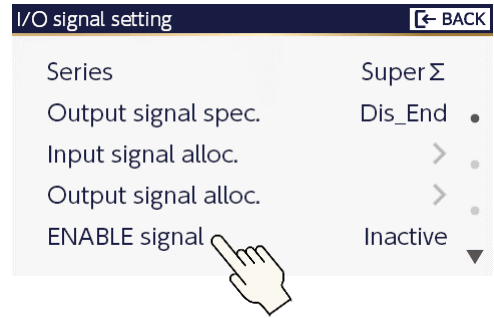
The ENABLE signal controls whether dispense operations are enabled/disabled. If the ENABLE signal function is Active, dispense operations are only possible while the ENABLE signal is input. Turning the ENABLE signal input OFF during In-dispensing stops dispense operations. If the ENABLE signal function is Inactive, the ENABLE signal input does not affect others.

Timing Chart

If the ENABLE signal function is ON



Tap "ENABLE signal" in the "I/O signal setting" screen.
This allows switching between Active/Inactive.



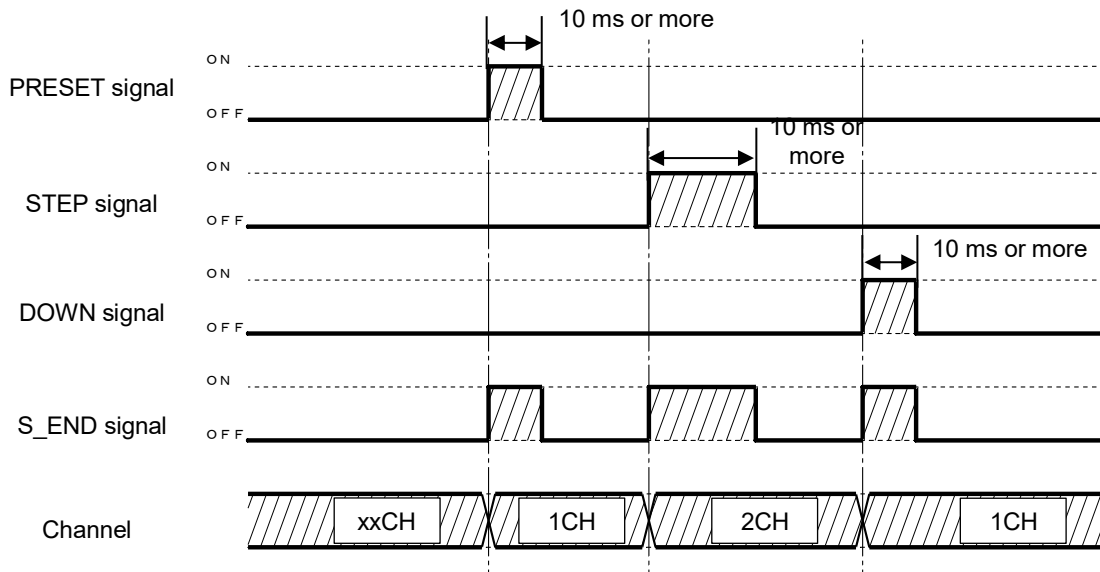
STEP/PRESET/DOWN Signals

"Series"	PIN	Signal Name	Allocation	Function
SuperΣ	*	PRESET	Replaceable	If channel belongs to a recipe: Changes to Start CH. of the recipe that the channel belongs to If channel does not belong to a recipe: Changes the channel to 1CH
ML-808GX	3 (default value)			
SuperΣ	*	STEP	Replaceable	Moves the channel up 1CH * Cannot move up beyond 401CH
ML-808GX	16 (default value)			
SuperΣ	*	DOWN	Replaceable	Moves the channel down 1CH * Cannot move down below 0CH
ML-808GX	*			

* Becomes available to be used for input signal allocation.

The STEP/PRESET/DOWN signals function with the ON edge.

Timing Chart



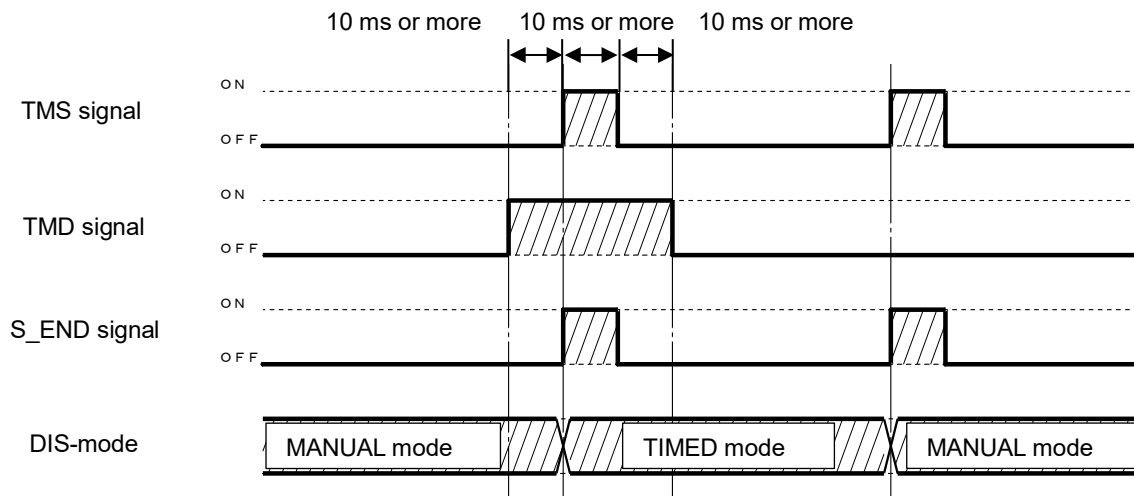
TMS/TMD signals

"Series"	PIN	Signal Name	Allocation	Function
SuperΣ	3 (default value)	TMS	Replaceable	Sets the Dispensing mode
ML-808GX	*			
SuperΣ	16 (default value)	TMD	Replaceable	Dispensing mode data signal (specifies the dispensing mode) OFF: MANUAL mode ON: TIMED mode
ML-808GX	*			

* Becomes available to be used for input signal allocation

The Dispensing mode switches to TMD signal status with the TMS signal ON edge. If the TMS signal status is normal, the Dispensing mode does not change even when changing the TMD signal.

Timing Chart



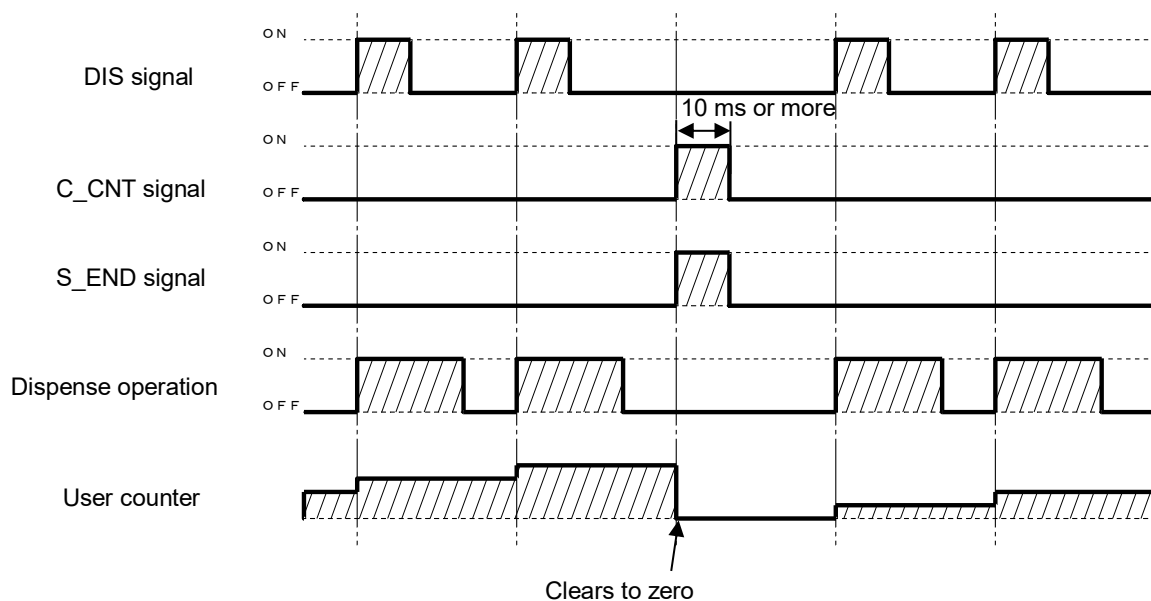
C CNT signal

PIN	Signal Name	Allocation	Function
*	C_CNT	Replaceable	Clears the User counter.

* Becomes available to be used for input signal allocation

Functions with the ON edge.

Timing Chart



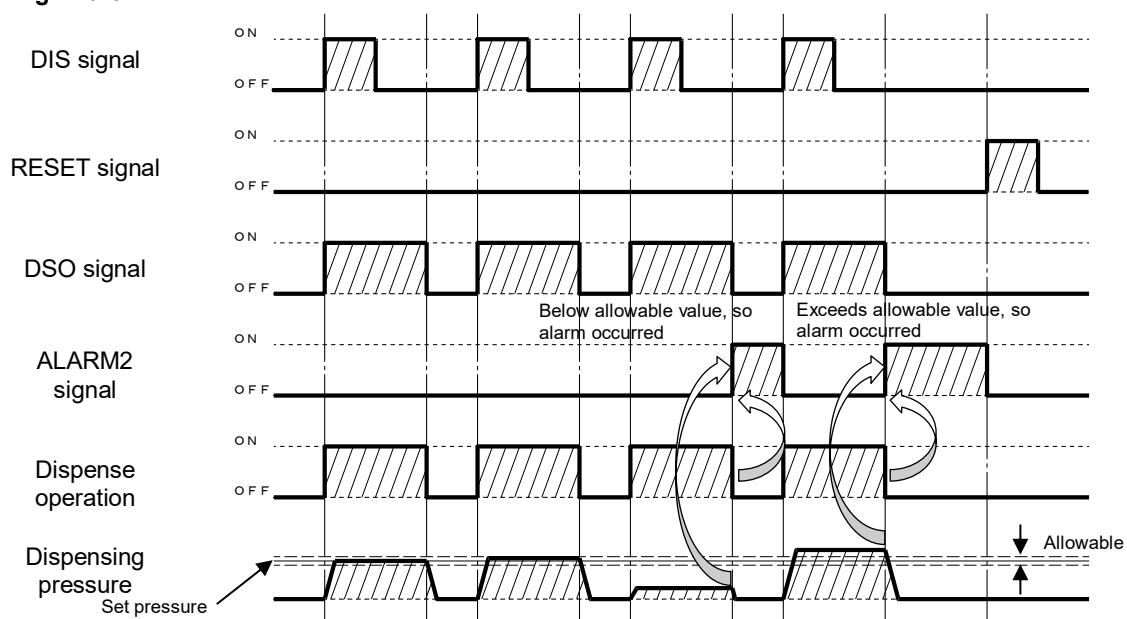
RESET signal

PIN	Signal Name	Allocation	Function
*	RESET	Replaceable	Turns ALARM2 signal output OFF

* Becomes available to be used for input signal allocation

Functions with the ON edge.

Timing Chart



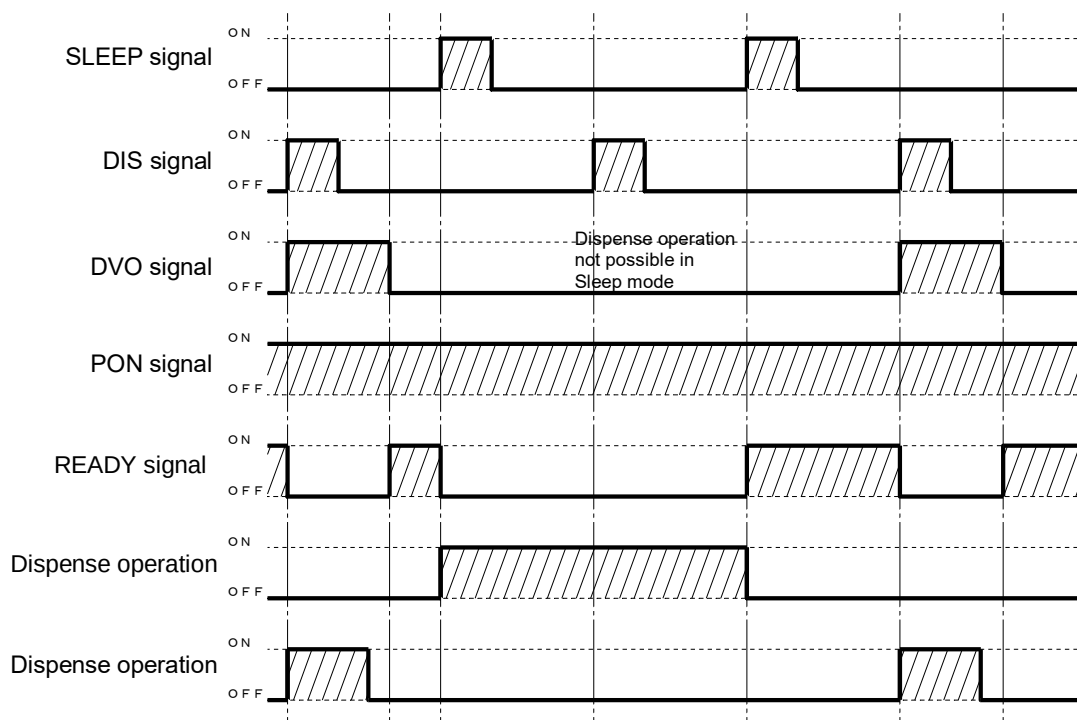
SLEEP signal

PIN	Signal Name	Allocation	Function
*	SLEEP	Replaceable	When normal: Enters Sleep mode when set to ON while the READY signal is ON. In Sleep mode: Recovers from Sleep mode.

* Becomes available to be used for input signal allocation

Functions with the ON edge.

Timing Chart



Channel Switching Signals

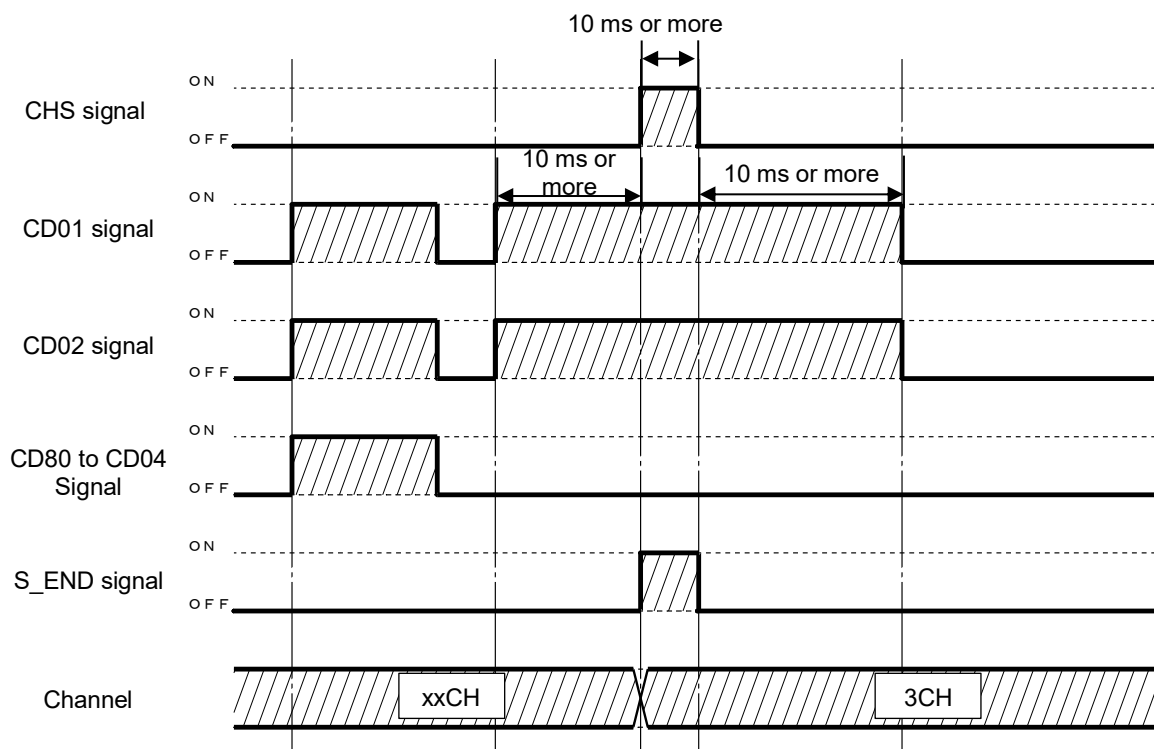
PIN	Signal Name	Allocation	Function
4 (default value)	CHS	Replaceable	Channel switching set signal
17 (default value)	CD01	Replaceable	Specifies the last digit of the channel as binary number 1
5 (default value)	CD02	Replaceable	Specifies the last digit of the channel as binary number 2
18 (default value)	CD04	Replaceable	Specifies the last digit of the channel as binary number 4
6 (default value)	CD08	Replaceable	Specifies the last digit of the channel as binary number 8
19 (default value)	CD10	Replaceable	Specifies the first digit of the channel as binary number 1
7 (default value)	CD20	Replaceable	Specifies the first digit of the channel as binary number 2
20 (default value)	CD40	Replaceable	Specifies the first digit of the channel as binary number 4
15 (default value)	CD80	Replaceable	Specifies the first digit of the channel as binary number 8

◇ Channel switching method

The channel switches to CD01 to CD80 signal status with the CHS signal ON edge.

If the CHS signal status is normal, the channel does not change even when changing the CD01 to CD80 signal. CD01 to CD80 signal has a BCD input. Specify the last digit of CD01 to CD08, and the first digit of CD10 to CD80 (see the next page for corresponding channels).

Timing Chart



Channel Switching Signal Combination Chart

No.	I/O Signal Name							
	CD80	CD40	CD20	CD10	CD08	CD04	CD02	CD01
01 CH								•
02 CH							•	
03 CH							•	•
04 CH						•		
05 CH						•		•
06 CH						•	•	
07 CH						•	•	•
08 CH					•			
09 CH					•			•
10 CH				•				
11 CH				•				•
12 CH				•			•	
13 CH				•			•	•
14 CH				•		•		
15 CH				•		•		•
16 CH				•		•	•	
17 CH				•		•	•	•
18 CH				•	•			
19 CH				•	•			•
20 CH			•					
21 CH			•					•
22 CH			•				•	
23 CH			•				•	•
24 CH			•			•		
25 CH			•			•		•
26 CH			•			•	•	
27 CH			•			•	•	•
28 CH			•		•			
29 CH			•		•			•
30 CH			•	•				
31 CH			•	•				•
32 CH			•	•			•	
33 CH			•	•			•	•
34 CH			•	•		•		
35 CH			•	•		•		•
36 CH			•	•		•	•	
37 CH			•	•		•	•	•
38 CH			•	•	•			
39 CH			•	•	•			•
40 CH		•						
41 CH		•						•
42 CH		•					•	
43 CH		•					•	•
44 CH		•				•		
45 CH		•				•		•
46 CH		•				•	•	
47 CH		•				•	•	•
48 CH		•			•			
49 CH		•			•			•
50 CH		•		•				

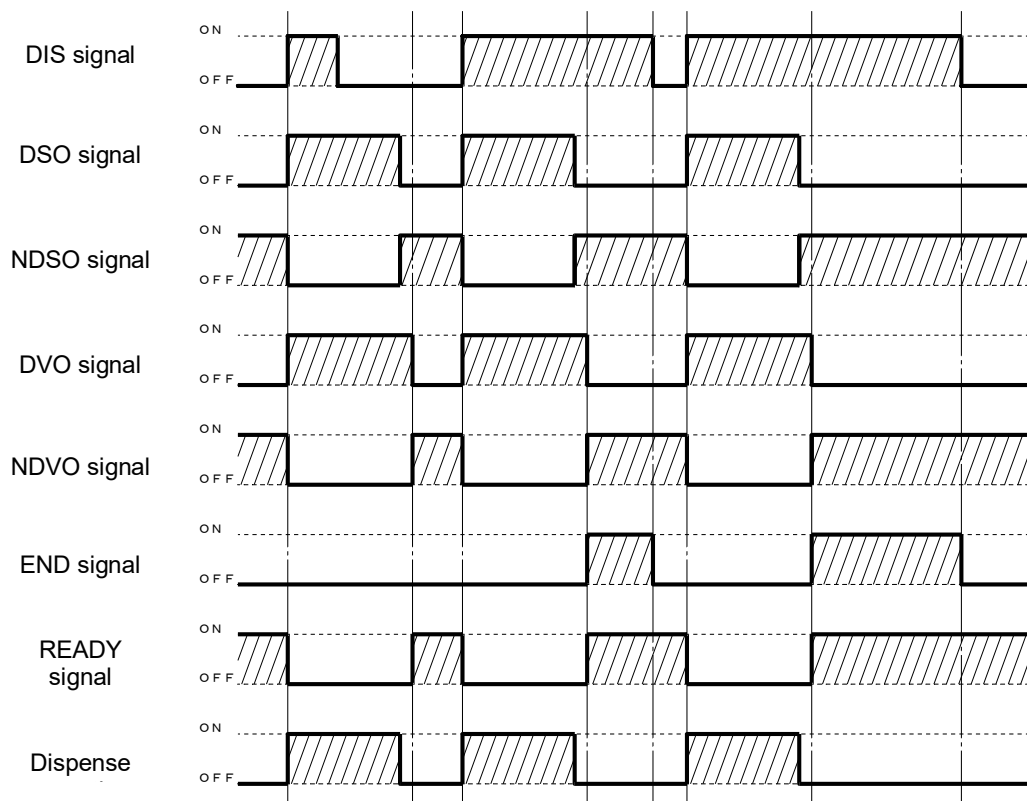
No.	I/O Signal Name							
	CD80	CD40	CD20	CD10	CD08	CD04	CD02	CD01
51 CH		•		•				•
52 CH		•		•			•	
53 CH		•		•			•	•
54 CH		•		•		•		
55 CH		•		•		•		•
56 CH		•		•		•	•	
57 CH		•		•		•	•	•
58 CH		•		•	•			
59 CH		•		•	•			•
60 CH		•	•					
61 CH		•	•					•
62 CH		•	•				•	
63 CH		•	•				•	•
64 CH		•	•			•		
65 CH		•	•			•		•
66 CH		•	•			•	•	
67 CH		•	•			•	•	•
68 CH		•	•		•			
69 CH		•	•		•			•
70 CH		•	•	•				
71 CH		•	•	•				•
72 CH		•	•	•			•	
73 CH		•	•	•			•	•
74 CH		•	•	•		•		
75 CH		•	•	•		•		•
76 CH		•	•	•		•	•	
77 CH		•	•	•		•	•	•
78 CH		•	•	•	•			
79 CH		•	•	•	•			•
80 CH	•							
81 CH	•							•
82 CH	•						•	
83 CH	•						•	•
84 CH	•					•		
85 CH	•					•		•
86 CH	•					•	•	
87 CH	•					•	•	•
88 CH	•				•			
89 CH	•				•			•
90 CH	•			•				
91 CH	•			•				•
92 CH	•			•			•	
93 CH	•			•			•	•
94 CH	•			•		•		
95 CH	•			•		•		•
96 CH	•			•		•	•	
97 CH	•			•		•	•	•
98 CH	•			•	•			
99 CH	•			•	•			•
100 CH								

※ Cells with "•" symbol in the chart indicates input is ON.

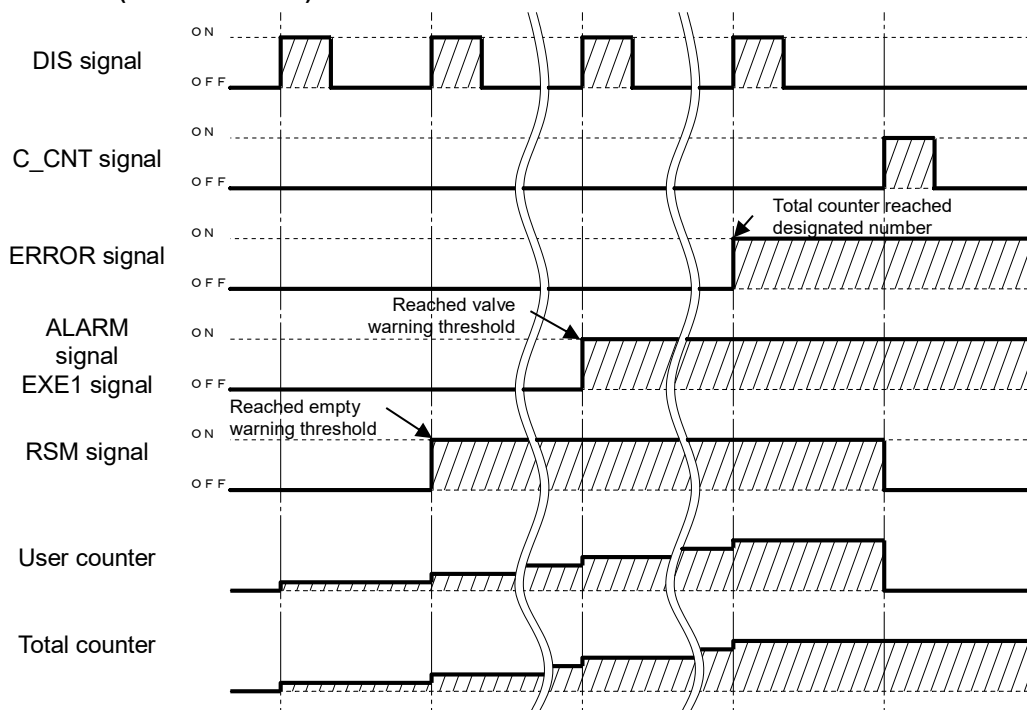
Output

Timing Chart

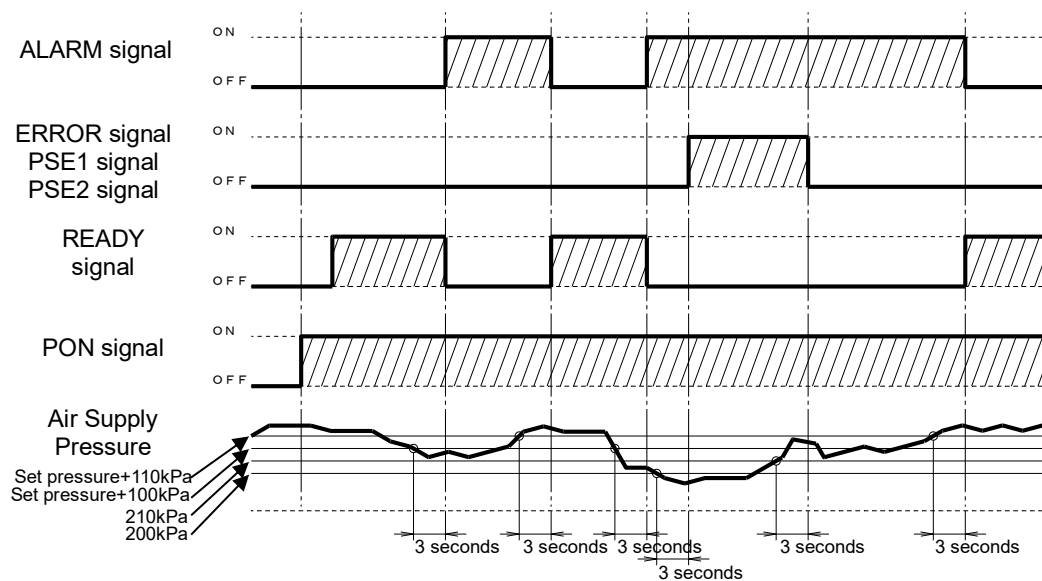
Dispense-related



Error/Alarm (Counter-related)



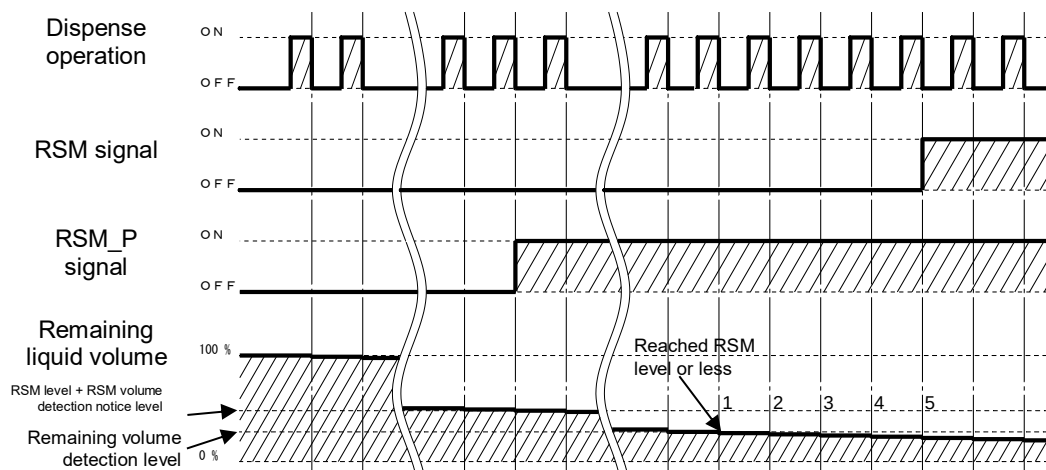
Alarm (Air Supply Pressure-related)



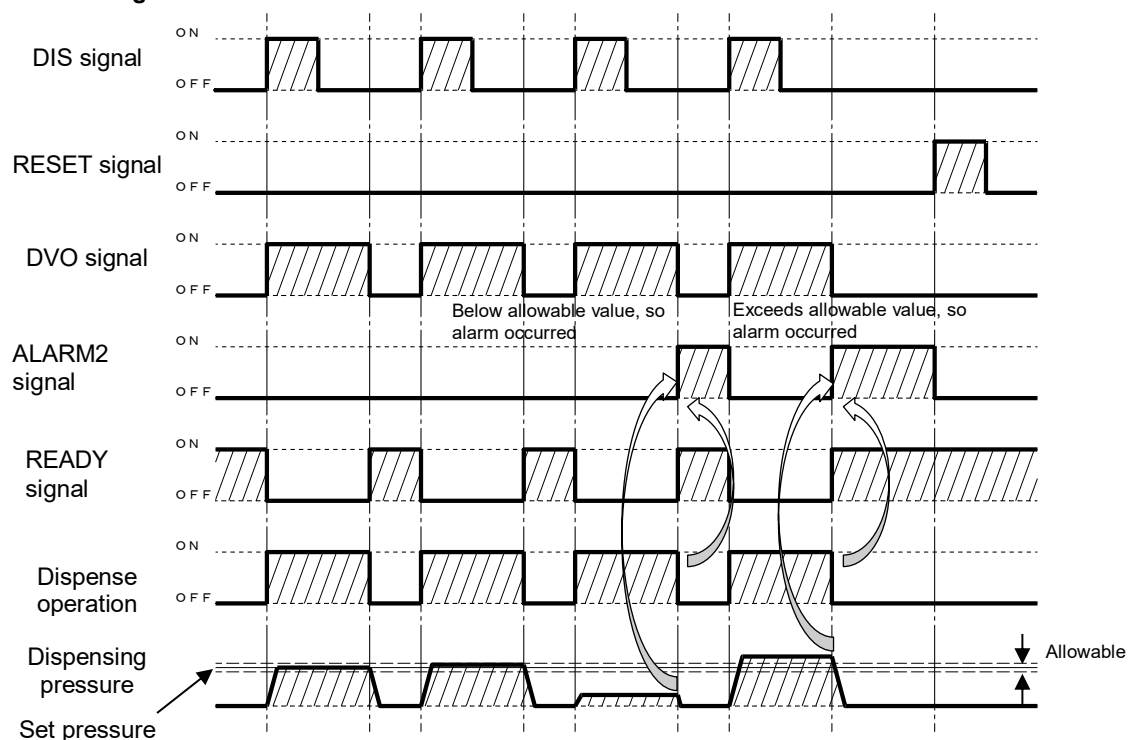
Σ mode remaining amount alert

[Setting]

"RSM count": 5 times



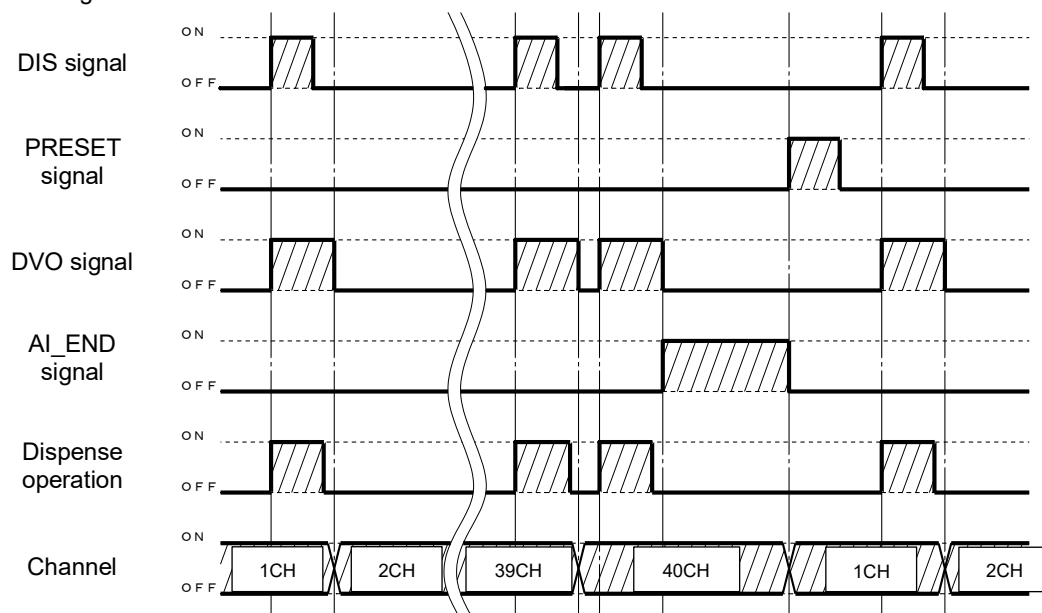
DIS-monitoring-related



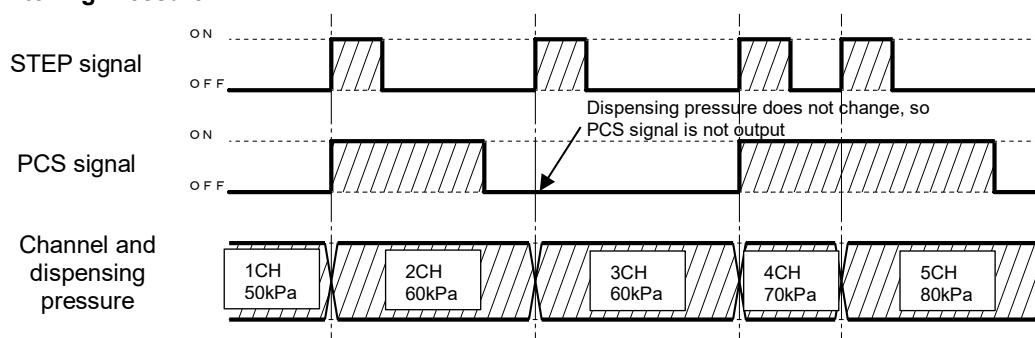
Auto-increment-related

[Setting]

Auto-increment function: ON, Start CH.: 1CH, End CH.: 40CH, Switching condition: Dispense count, Switching value: 1 shot



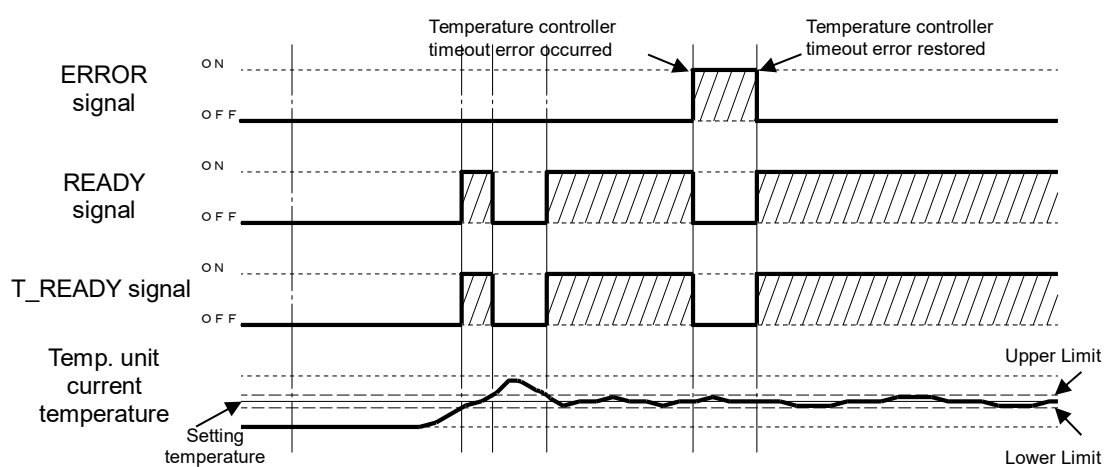
Switching Pressure



Temperature unit-related

[Setting]

T-ready enable: ON



Switching Output Signal Function (Output Signal Specifications)

The following output signal functions can be switched.

Default value: Dispense complete setting

Series: For SuperΣ

PIN	I/O	"Output signal spec.": In-dispensing setting			"Output signal spec.": Dispense complete setting		
		Signal Name	Function		Signal Name	Function	
9	O	NDSO	DSO inverse	Outputs a DSO inverse signal	END	Dispense complete	Output when a dispense signal is input after dispense end
12	O	READY	Ready	Output when all the following are satisfied - Not In-dispensing cycle - An air supply pressure error or alarm has not occurred. - Dispense screen A or B is being displayed - When "T-ready enable" is On and there are no temperature unit errors - When "T-ready enable" is On and there are no temperature unit alarms - When "Warm-ready enable" is On and not warming up	NDVO	DVO inverse	Outputs a DVO inverse signal

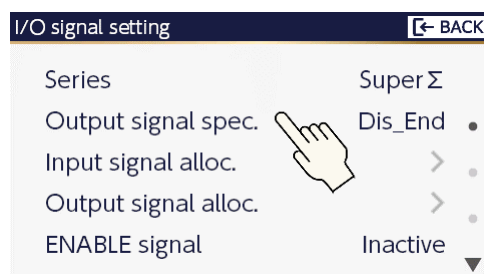
Series: For ML-808GX

PIN	I/O	"Output signal spec.": In-dispensing setting			"Output signal spec.": Dispense complete setting		
		Signal Name	Function		Signal Name	Function	
24	O	DSO	In-dispensing	Output during dispense operations	END	Dispense complete	Output when a dispense signal is input after dispense end

Setting Method

Tap "Output signal spec." in the "I/O signal setting" screen.
Each tap changes the setting in the following order.

"Dis_Now" → "Dis_End" → "Dis_Now" → ...

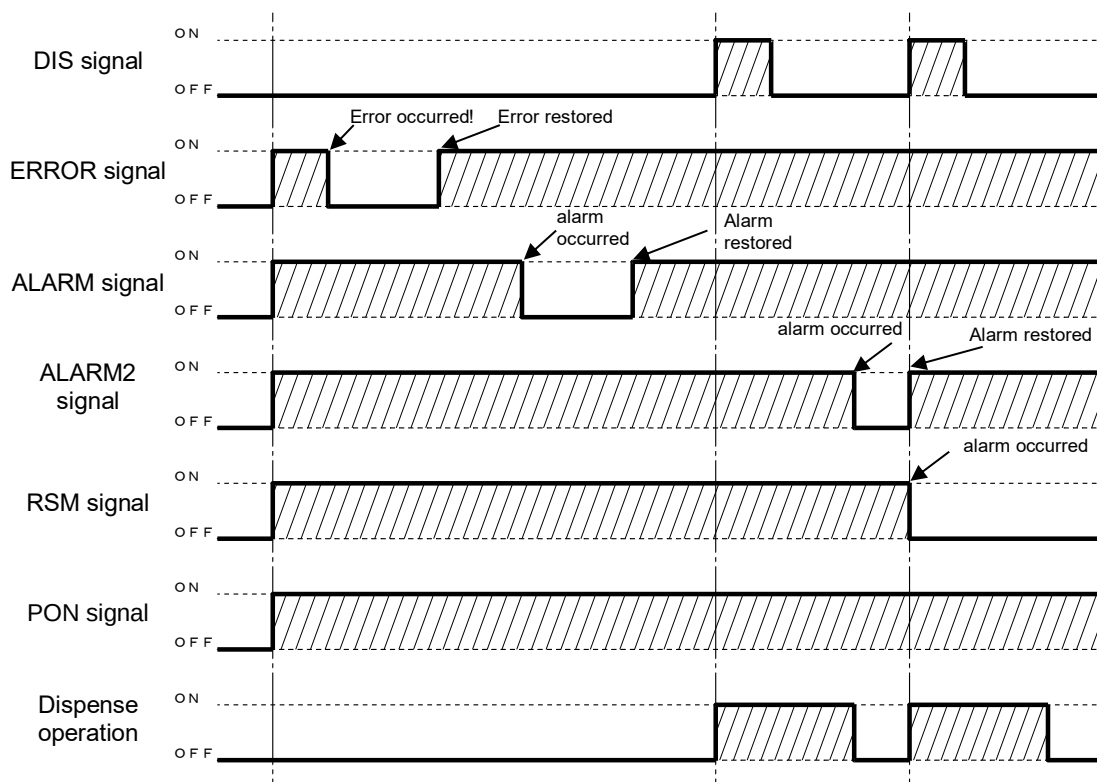


Switching Error and Alarm-related Signal Output Status

The ERROR, ALARM, EXE1, EXE2, PSE1, PSE2, ALARM2, RSM, and RSM P signal output statuses can be inverted.

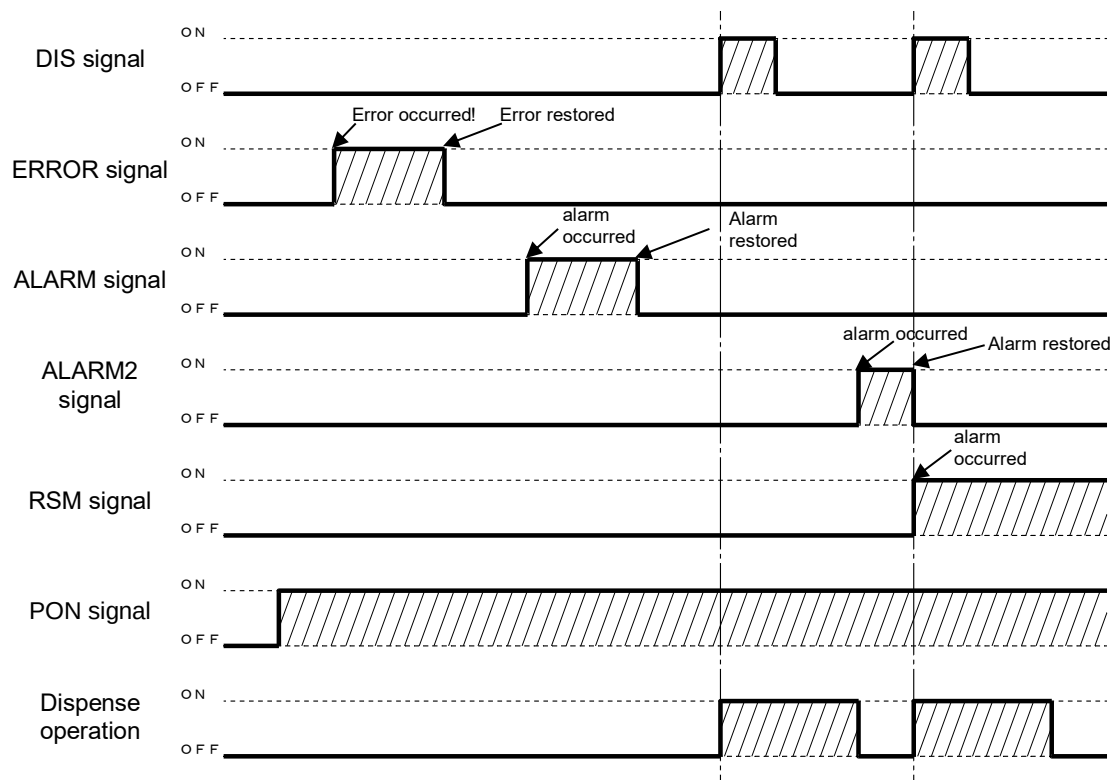
<<On Error: OFF (default value)>>

Alarm or error status	ERROR	ALARM	EXE1	EXE2	PSE1	PSE2	ALARM2	RSM	RSM_P
No occurrence	ON	ON	ON	ON	ON	ON	ON	ON	ON
Occurring	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF



<<On Error: "ON">>

Alarm or error status	ERROR	ALARM	EXE1	EXE2	PSE1	PSE2	ALARM2	RSM	RSM_P
No occurrence	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
Occurring	ON	ON	ON	ON	ON	ON	ON	ON	ON



This function uses the I/O connector to change controller parameters externally.

```
graph TD; A[EDIT signal ON] --> B[CD80 to CD01 signal ON corresponding to command]; B --> C[CHS signal ON]; C --> D[CD80 to CD01 signal ON corresponding to binary number bit 8 to 1 of value to specify]; D --> E[CHS signal ON]; E --> F[CD80 to CD01 signal ON corresponding to binary number bit 16 to 9 of value to specify]; F --> G[CHS signal ON]; G --> H[CD80 to CD01 signal ON corresponding to binary number bit 24 to 17 of value to specify]; H --> I[CHS signal ON]; I --> J[EDIT signal OFF];
```

The flowchart illustrates the sequence of operations for setting parameters. It begins with the EDIT signal ON, followed by a series of steps for setting the command, CHS, and CD80 to CD01 signals. The sequence is repeated for parameters to specify, and ends with the EDIT signal OFF. A bracket on the left indicates that the steps from CHS signal ON to CD80 to CD01 signal ON corresponding to binary number bit 24 to 17 of value to specify can be omitted if not available.

Command

0: CDxx signal OFF 1: CDxx signal ON

Signal	CD80	CD40	CD20	CD10	CD08	CD04	CD02	CD01
Data1	bit 8	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1
Data2	bit 16	bit 15	bit 14	bit 13	bit 12	bit 11	bit 10	bit 9
Data3	bit 24	bit 23	bit 22	bit 21	bit 20	bit 19	bit 18	bit 17

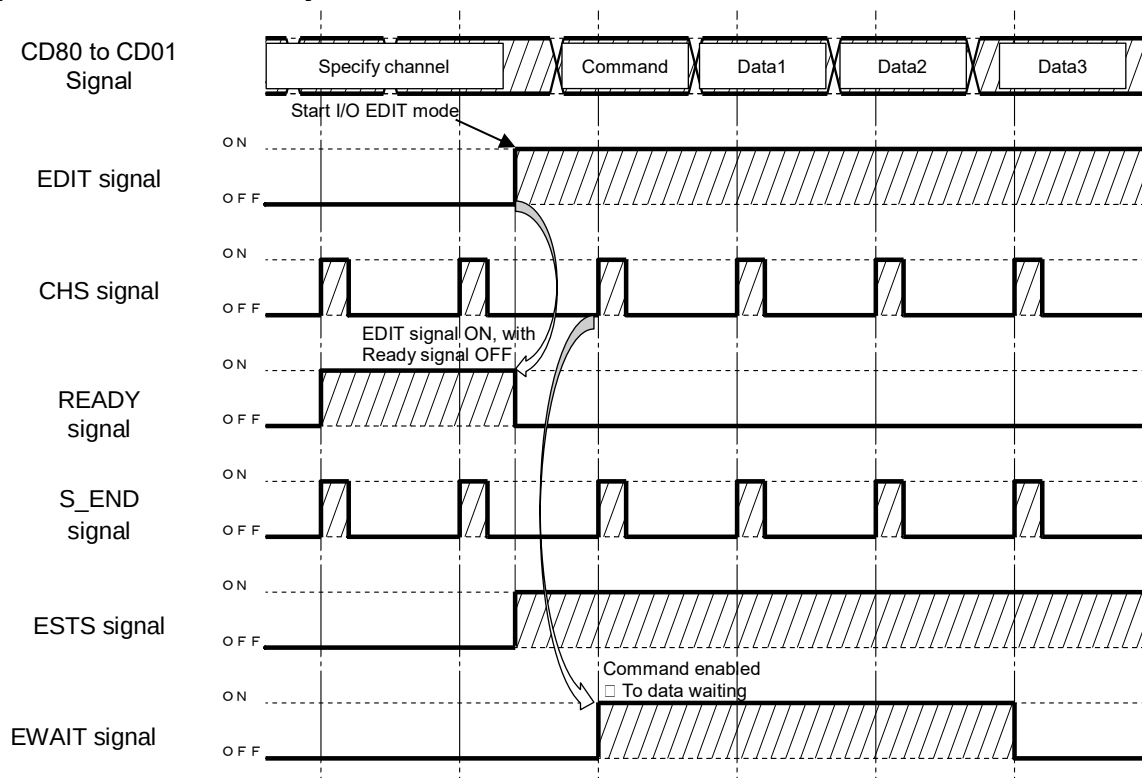
List

The following parameters can be changed.

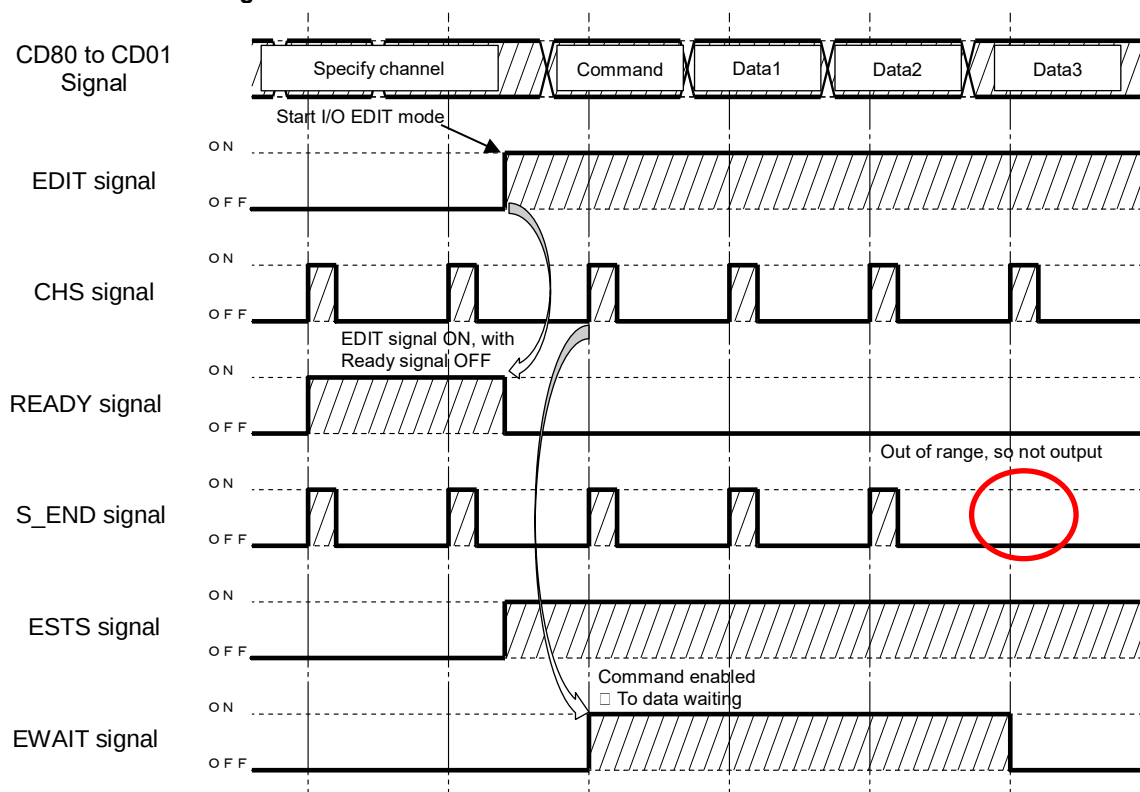
Item	Command	Data1	Data2	Data3	Range
Channel	0000 0001	bit 1 to 8	bit 9 to 16	-	1 to 400 [CH]
Dispensing time	0000 0010	bit 1 to 8	bit 9 to 16	bit 17 to 24	10 to 9999999 [0.001 sec.]
Dispensing pressure	V5 V2	0000 0011	bit 1 to 8	bit 9 to 16	300 to 5000 [0.1 kPa]
			bit 1 to 8	bit 9 to 16	50 to 2000 [0.1 kPa]
Vacuum pressure	0000 0100	bit 1 to 8	bit 9 to 16	-	0 to 500 [-0.01 kPa]
Dispensing mode	0000 0101	bit 1 to 8	-	-	TIMED mode (0)/MANUAL mode (1)

Timing Chart

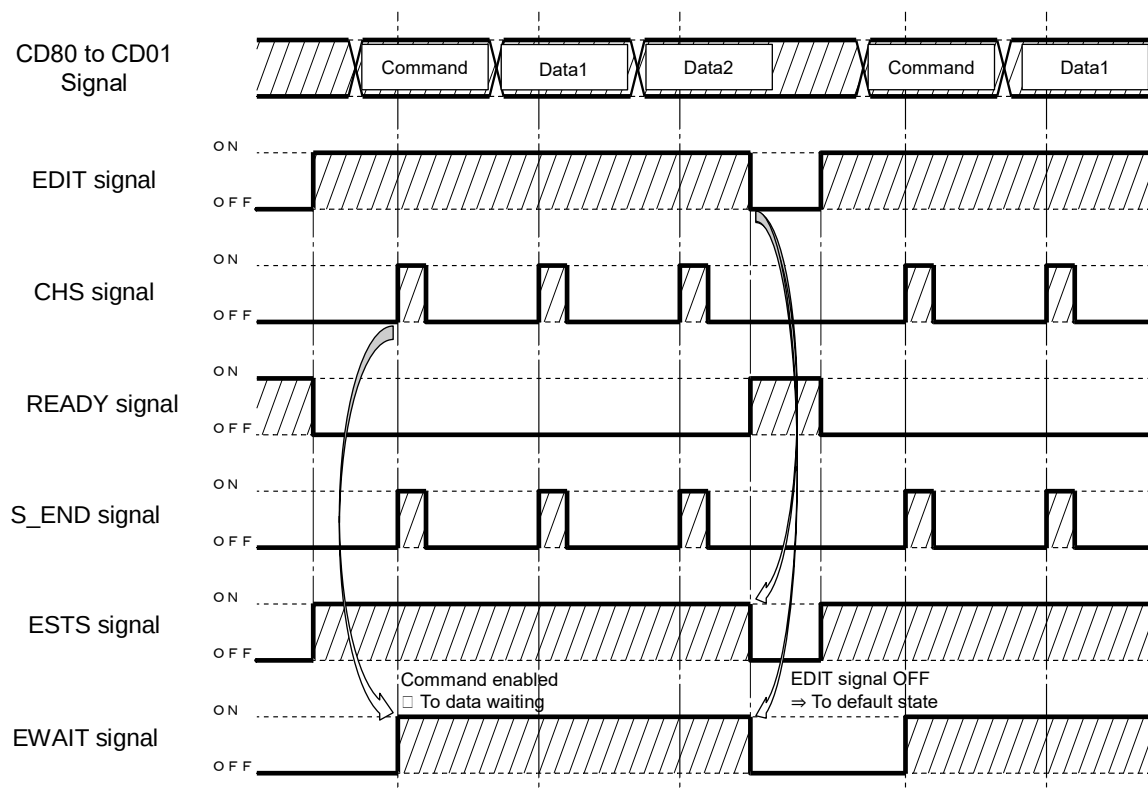
If parameters are set normally



If parameters out of range are set



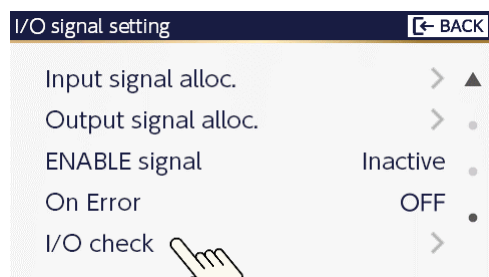
To Cancel



I/O check

This device has a function that checks the connection of external input/output terminals.

Tap "I/O check" in the "I/O signal setting" screen.



Displays the current status of I/O Connector input pins.

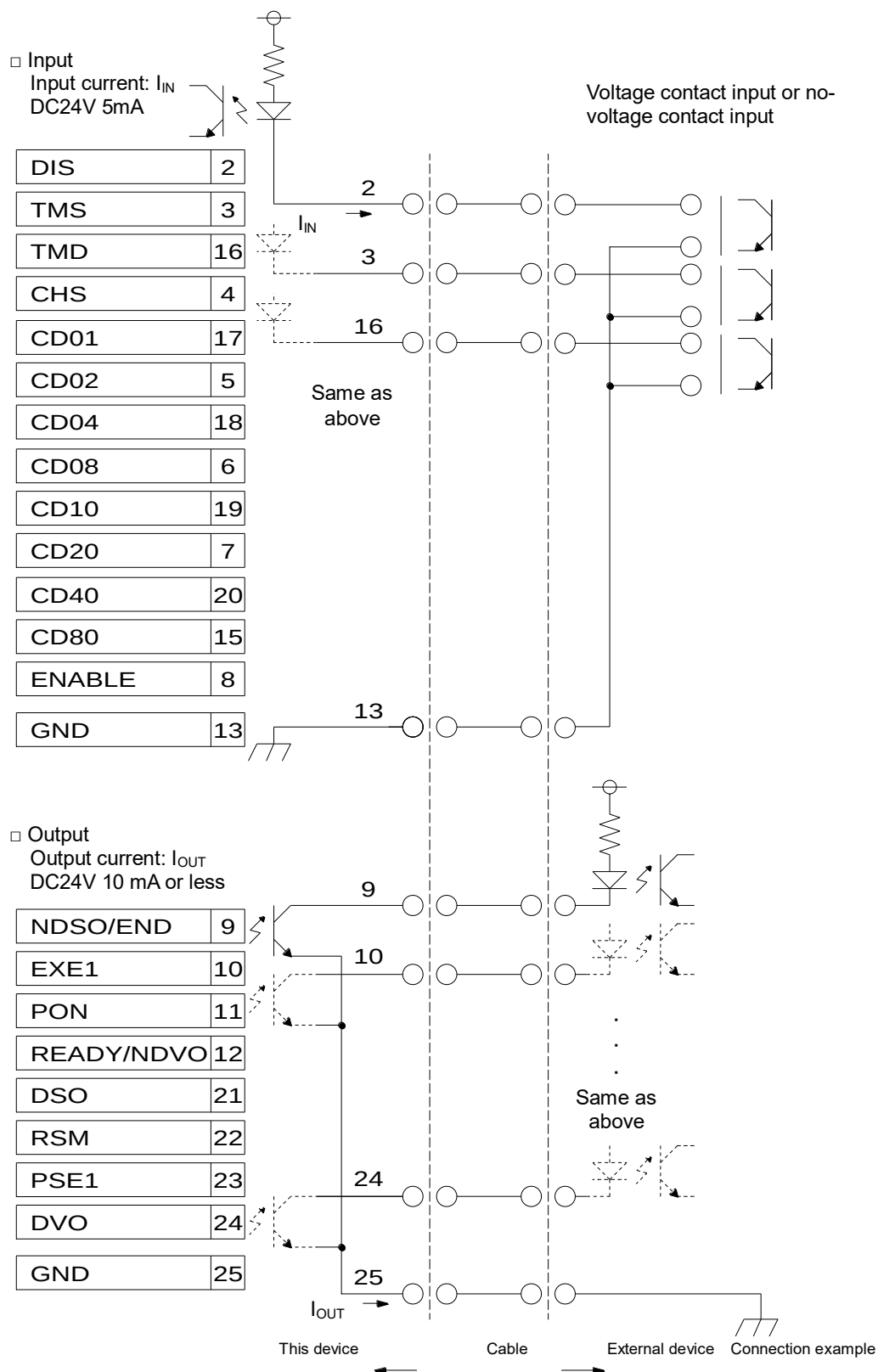
[]:ON
[]:OFF



Displays the current status of I/O Connector output pins. Tapping inverts the output status of each pin.

[]:ON
[]:OFF

Example of Connection with External Device {S-SIGMA-CM4-(V5/V2)-N}



Connections are prohibited on pin numbers 1 and 14. DC24V is output on these pins, but do not use this as a power supply for the external device.

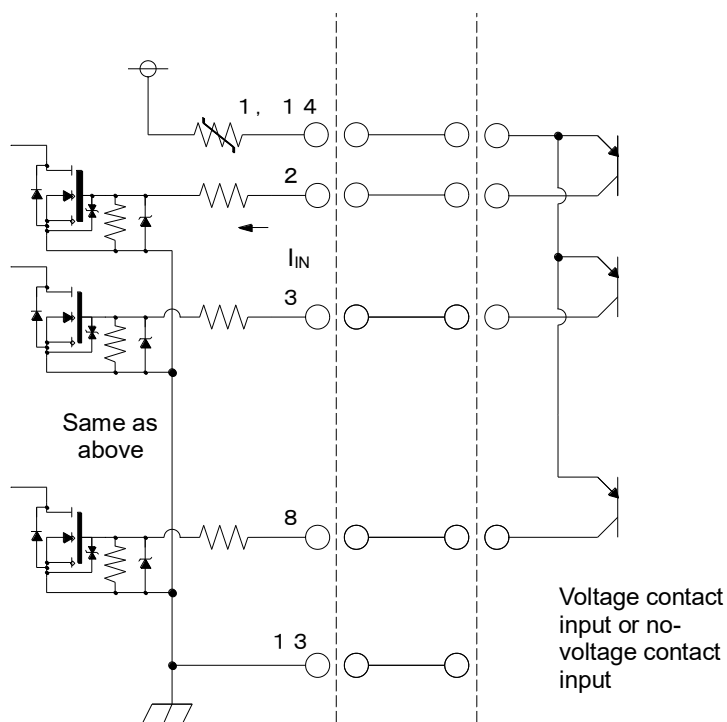
Pin numbers 13 and 25 (GND) are connected to the internal circuit GND, DSUB9 GND (pin number 5), and chassis.

Example of Connection with External Device {S-SIGMA-CM4-(V5/V2)- P}

□ Input

Input current: I_{IN}
Less than DC24V 5 mA

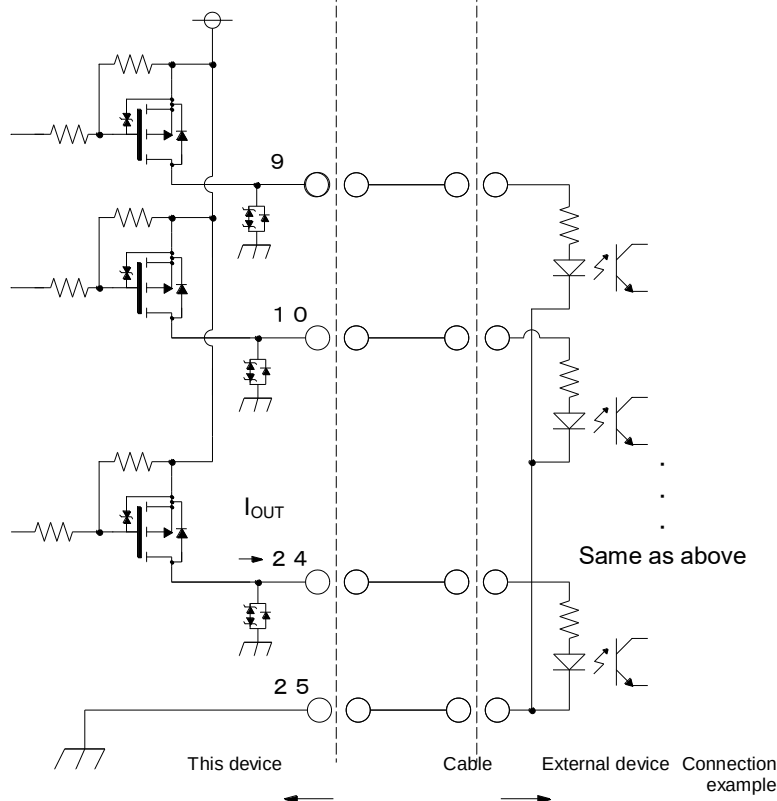
DIS	2
TMS	3
TMD	16
CHS	4
CD01	17
CD02	5
CD04	18
CD08	6
CD10	19
CD20	7
CD40	20
CD80	15
ENABLE	8
GND	13



□ Output

Output current: I_{OUT}
DC24V 10 mA or less

NDSO/END	9
EXE1	10
PON	11
READY/NDVO	12
DSO	21
RSM	22
PSE1	23
DVO	24
GND	25



Use pin numbers 1 and 14 within their rated range only for powering IO.
Pin numbers 13 and 25 (GND) are connected to the internal circuit GND, DSUB9 GND (pin number 5), and chassis.

6 Setting List

Settings for Each Channel

Name		Allowable Setting Range	Default Value	All Memory Clear Applies	Parameter Clear Applies	Reference Page
Dispensing mode		TIMED/MANUAL	TIMED	○	○	P.20
Dispensing time		0.001 to 9999.999 [sec.]	0.010 [sec.]	○	○	P.21
Dispensing pressure	V5	30.0 to 500.0 [kPa]	30.0 [kPa]	○	○	P.19
	V2	5.0 to 200.0 [kPa]	30.0 [kPa]	○	○	
Vacuum pressure		-5.00 to -0.00 [kPa]	-0.00 [kPa]	○	○	P.20
Channel name		0 to 14 characters	No settings	○	○	P.41
Syringe size		3mL/5mL/10mL/20mL/30mL/ 50mL/70mL/W5mL/W10mL/ W30mL/W55mL	10 mL	○	○	P.50
Adapter tube length		0.5/1.0/1.5/2.0 m	1.0 m	○	○	P.50
Plunger usage		Used / None	None	○	○	P.50
ECO mode		OFF/ON	ON	○	○	P.51
Σ (sigma) mode		Normal/Σ	Normal	○	○	P.22
Σ dummy shot mode		OFF/ON	ON	○	○	P.16
"Vol. reduction corr."		OFF/ON	ON	○	○	P.24
Alpha correction value		-200 to 999	0	○	○	P.24
Delta correction		5 to 200 [%]	100 [%]	○	○	P.25
Fluid-drip prevention		OFF/ON	ON	○	○	P.25
Vacuum correction value		-5.00 to +5.00 [kPa] *	+ 0.00 [kPa]	○	○	P.25
Remaining vol. alert		OFF/ON	ON	○	○	P.26
Remaining volume detection level		0 to 99 [%]	10 [%]	○	○	P.26
RSM count		1 to 10 [times]	5 [time]	○	○	P.26
Remaining vol. corr.		OFF/ON	ON	○	○	P.27
Correction value		-99 to +99 [%]	0 [%]	○	○	P.27
On delay time		0.000 to 9.999 [sec.]	0.000 [sec.]	○	○	P.52
Off delay timer		0.000 to 9.999 [sec.]	0.000 [sec.]	○	○	P.52
Dispensing OFF time		0.001 to 9.999 [sec.]	0.500 [sec.]	○	○	P.56
Allowable		10.0 to 100.0 [kPa]	10.0 [kPa]	○	○	
DIS-monitoring function test results		Cannot be set (Not tested/Fail/Pass)	Not tested	○	○	

○: Applicable x: Not Applicable

* The maximum and minimum value of the VAC correction value are determined by the vacuum pressure set when Σ mode was enabled.

When Σ mode is enabled while the vacuum pressure is set to -3.0 kPa, the configurable range is between -3.0 kPa and +3.0 kPa.

Settings for Each Recipe

Name	Allowable Setting Range	Default Value	All Memory Clear Applies	Parameter Clear Applies	Reference Page
Registered Recipes	1 to 100 [Recipes]	Not registered	○	○	P.35
Start CH.	1 to 400 [CH]	1 [CH]	○	○	
End CH.	1 to 400 [CH]	400 [CH]	○	○	
Recipe name	0 to 14 characters	No settings	○	○	P.35
Liquid material	0 to 14 characters	No settings	○	○	
Nozzle	0 to 14 characters	No settings	○	○	
Auto-increment Function	OFF/ON	ON	○	○	P.36
Auto-increment dispensing mode	TIMED/MANUAL	TIMED	○	○	
Auto-increment ECO mode	OFF/ON	ON	○	○	
Auto-increment switching condition	Total time /Total count /Pot life	Total time	○	○	
Auto-increment Switching Value	Total time: 1 to 9999 [sec.] Total count: 1 to 9999 [shot] Pot life: 1 to 9999 [sec.]	Total time: 9999 [sec.] Total count: 9999 [shot] Pot life: 9999 [sec.]	○	○	
Temperature controller control	OFF/ON	ON	○	○	P.38
Temperature controller setting temperature	0.0 to 100.0 [°C]	25.0 [°C]	○	○	
Temperature controller upper limit	0.0 to 100.0 [°C]	1.0 [°C]	○	○	
Temperature controller lower limit	0.0 to 100.0 [°C]	1.0 [°C]	○	○	

○: Applicable x: Not Applicable

Settings Common to All Channels

Name	Allowable Setting Range	Default Value	All Memory Clear Applies	Parameter Clear Applies	Reference Page
Channel	1 to 400 [CH]	1 [CH]	○	×	P.41
User counter	0 to 999,999,999 [shot]	0 [shot]	○	×	P.69
Total counter *1	0 to 999,999,999 [shot]	0 [shot]	×	×	P.69
Interval feature	OFF/ON	ON	○	×	P.54
Interval; Link CH.	0 to 400 [CH]	0 [CH]	○	×	
Interval; Count	1 to 9999 [shot]	1 [shot]	○	×	
Interval; Interval	0.010 to 9999.999 [sec.]	0.010 [sec.]	○	×	
Interval; Repeat	1 to 9999 [times]	1 [time]	○	×	
Interval; Wait time	0.001 to 9999.999 [sec.]	0.010 [sec.]	○	×	
Interval mode	MAN/SET	SET	○	×	P.59
DIS-correction Dispensing time method	OFF/Amount/Ratio	OFF	○	×	
DIS-correction Dispensing time Amount	-9.999 to 9.999 [sec.]	+0.000 [sec.]	○	×	
DIS-correction Dispensing time Ratio	1 to 999 [%]	100 [%]	○	×	
DIS-correction Dispensing pressure Method	OFF/Amount/Ratio	OFF	○	×	
DIS-correction Dispensing pressure Amount	V5	-500.0 to 500.0 [kPa]	○	×	
	V2	-200.0 to 200.0 [kPa]	○	×	
DIS-correction Dispensing pressure Ratio	1 to 999 [%]	100 [%]	○	×	
DIS-correction Vacuum pressure Method	OFF/Amount/Ratio	OFF	○	×	
DIS-correction Vacuum pressure Amount	-5.00 to 5.00 [kPa]	+0.00 [kPa]	○	×	
DIS-correction Vacuum pressure Ratio	1 to 999 [%]	100 [%]	○	×	P.67
Communication speed	9600/19200/38400/57600 [bps]	38400 [bps]	○ *3	×	
Checksum	OFF/ON	ON	○ *3	×	
Com. protocol	Type A/Type C	Type A	○ *3	×	
IP address	0.0.0.0 to 255.255.255.255	192.168.0.10	○ *4	×	P.68
Subnet mask	0.0.0.0 to 255.255.255.255	255.255.255.0	○ *4	×	
Default gateway	0.0.0.0 to 255.255.255.255	192.168.0.1	○ *4	×	
Administrator password	0 to 14 characters	0000	○ *4	×	
Read only	OFF/ON	ON	○ *4	×	
Web page refresh interval	0 to 60 [min.]	0 [min.]	○ *4	×	
Series	ML-808GX/SuperΣ	SuperΣ	○	×	P.92
Output signal spec.	Dispense complete setting / In-dispensing setting	Dispense complete setting	○	×	P.109
Input signal allocation	See reference page		○	×	P.95
Output signal allocation	See reference page		○	×	P.96
ENABLE signal	Inactive/Active	Inactive	○	×	P.98
On Error	OFF/ON	ON	○	×	P.110

○: Applicable x: Not Applicable

Name	Allowable Setting Range	Default Value	All Memory Clear Applies	Parameter Clear Applies	Reference Page
Language	English/Japanese/Chinese	*2	×	×	P.73
P.UNIT	kPa/bar	kPa	○	×	P.74
Empty alarm	OFF/ON	ON	○	×	P.74
Empty warning threshold	1 to 999999999 [shot]	100000 [shot]	○	×	
Valve warning threshold	1 to 999999999 [shot]	80000000 [shot]	○	×	P.75
Buzzer	OFF/ON	ON	○	×	P.75
T-ready Enable	OFF/ON	OFF	○	×	P.40
Warm-ready enable	OFF/ON	OFF	○	×	P.40
Pop-up auto OFF	OFF/ON	ON	○	×	P.76
Immed. Inc. (AInc)	OFF/ON	OFF	○	×	P.37
Password lock	OFF/ON	OFF	○	×	P.76
Password	0 to 14 characters	0000	○	×	
Icon allocation	See reference page		○	×	P.78
Control mode	STB/SPD	STB	○	×	P.79
"Sleep timer"	OFF (Manual) / 5 min. / 10 min. / 30 min./1 hour/2 hour/ 3 hour/4 hour/5 hour	OFF	○	×	P.80
RSM advance notice lv.	0 to 50 [%]	10 [%]	○	×	P.81
Time	2000/01/01 00:00:00 to 2099/12/31 23:59:59	*2	×	×	P.82
Time zone	See reference page		×	×	P.82
Screen brightness	5 to 100 [%]	50 [%]	○	×	P.83
Controller name	0 to 10 characters	No settings	○	×	P.124

○: Applicable x: Not Applicable

*1 Total counter cannot be cleared.

*2 Set when turning POWER SUPPLY ON for the first time

*3 It is not cleared by all memory clear via RS-232C communication.

*4 It is not cleared by all memory clear via Ethernet communication.

7 Maintenance & Inspection

Replacing Consumable Parts

The solenoid valve units of this device require regular replacement due to wear. This device will be collected for replacement. Contact our sales department.

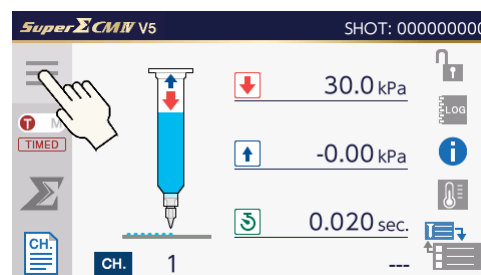
Potential Symptoms of Wear	Consumable Parts	Replacement Reference*
Variations in dispensing amount. Variations in start and end points when dispensing lines.	Solenoid valve	ERROR signal is output. (output when the Count reaches a specific number)
The date and time settings will be cleared.	Built-in battery	About 7 years
The screen is dark.	Backlight	About 5 years

* This is for reference purposes only, and the timing of symptoms caused by wear may differ depending on operating conditions. Note that normal operations may also be possible even after the replacement reference timing.

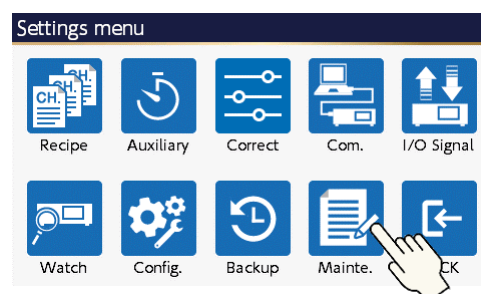
Performing Controller Maintenance (Maintenance)

Use the following procedure to move to the "Maintenance" screen if maintenance is to be performed.

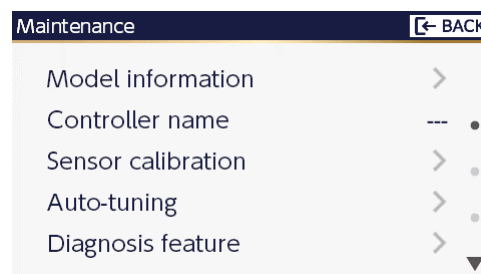
- 1 Tap the [≡] icon in Dispense screen A or B.



- 2 The "Settings menu" screen is displayed.
Tap the [Maintenance] icon.



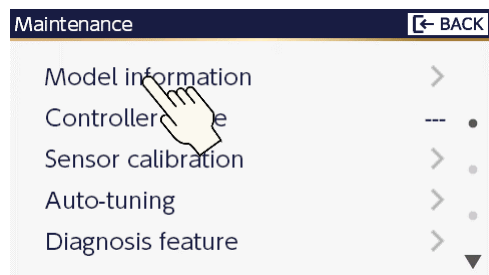
- 3 The "Maintenance" screen is displayed.



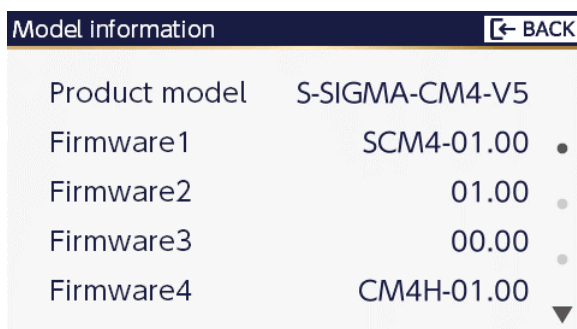
Checking the Version (Model Information)

Provide the following information when contacting our sales staff.

- 1 Tap "Model information" in the "Maintenance" screen.



- 2 The "Model Information" screen is displayed.

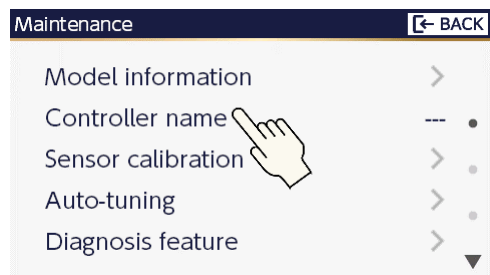


Item	Description
"Product model"	The model of this device.
"Firmware1"	The firmware 1 version of this device.
"Firmware2"	The firmware 2 version of this device.
"Firmware3"	The firmware 3 version of this device.
"Firmware4"	The firmware 4 version of this device.
"I/O specifications"	These are the I/O specifications of the external input/output terminals of this unit.

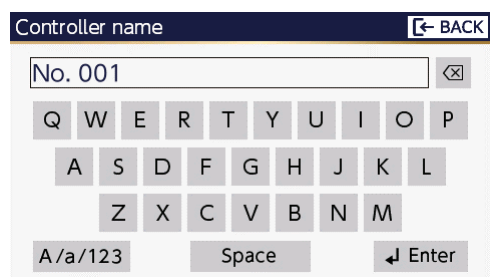
Sets Controller name (Controller name)

You can set a unique name for this unit. A name of up to 10 characters can be used.

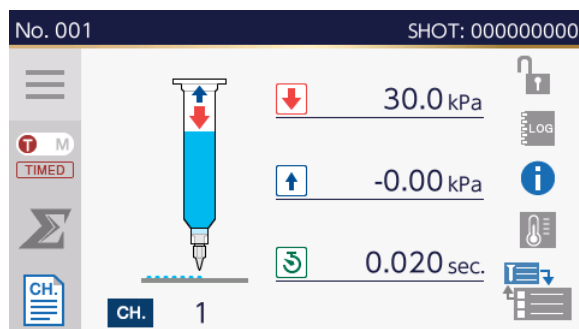
- 1 Tap "Controller name" in the "Maintenance" screen.



- 2 The controller name input screen appears. Set a controller name.



The controller name that has been set is displayed in the upper left corner of the Dispense screen.

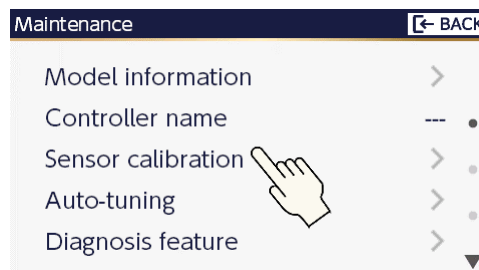


Calibrating the Pressure Sensor (Sensor Calibration)

Perform pressure sensor calibration in either of the following cases. Doing so may improve the symptoms. If there are no improvements, there may be a fault with the internal parts. Contact our sales department.

- Variations in all operations regardless of the set vacuum pressure
- Using the Diagnosis feature resulted in failure, and performing sensor calibration was notified

- 1 Tap "Sensor calibration" in the "Maintenance" screen.

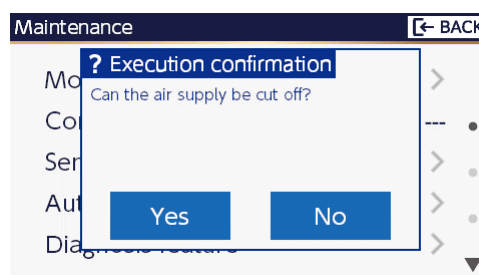


- 2 "Execution confirmation" is displayed for cutting off the air supply pressure. Tap "Yes" or "No" depending on the conditions.

* Cutting off the air supply pressure and performing the calibration is recommended.

When selecting "Yes" : Calibrates the sensors for dispensing pressure, vacuum pressure, and air supply pressure.

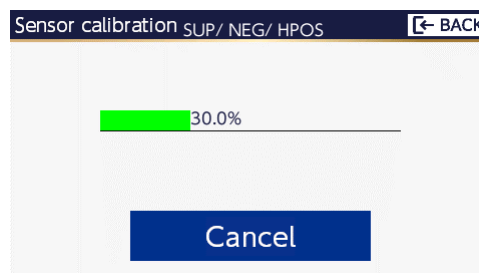
When selecting "No" : Calibrates the sensors for dispensing pressure and vacuum pressure.



- 3 The "Sensor calibration" screen corresponding to the selection is displayed. Follow the instructions displayed and tap "Start."



- 4 Sensor calibration starts. (Sensor calibration takes around 10 seconds)



- 5 When sensor calibration is completed, the "Notice" screen is displayed. Tapping the LCD Touch Panel displays the "Maintenance" screen.

Diagnosing the Controller Status (Diagnosis Feature)

This diagnoses any faults with the dispensing pressure, vacuum pressure and solenoid valve. If a failure is diagnosed using this function, check the details displayed on the screen and perform diagnosis again.

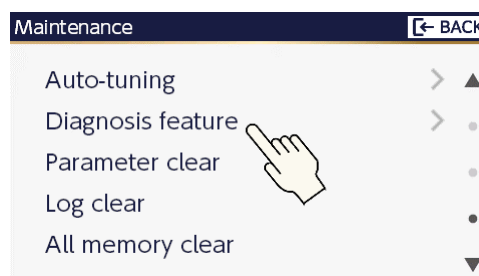
Contact our sales department if there is no improvement to symptoms.

* Normal operations are possible even if there is a failure diagnosed using this function.

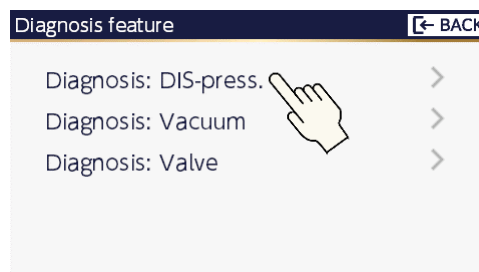
* The device may be diagnosed as normal, even if there is a decrease in dispensing precision.

Dispensing Pressure Diagnosis

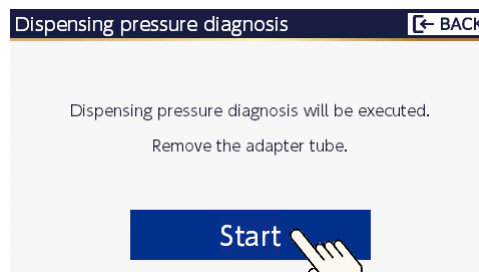
- 1 Tap "Diagnosis feature" in the "Maintenance" screen.



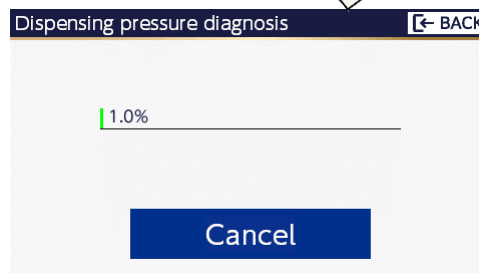
- 2 The "Diagnosis feature" screen is displayed. Tap "Diagnosis: DIS-pres."



- 3 The "Dispensing pressure diagnosis" screen is displayed. Remove the adapter tube, then tap "Start."



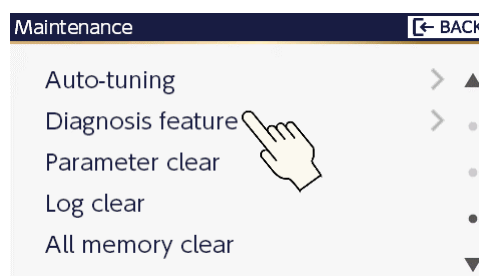
- 4 Dispensing pressure diagnosis starts. (Dispensing pressure diagnosis takes up to 10 minutes)



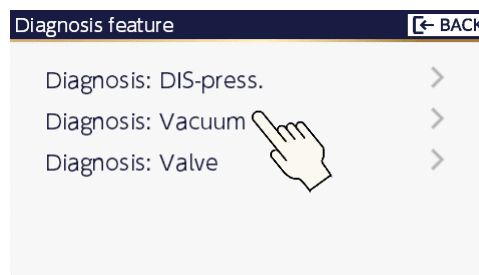
- 5 When diagnosis is completed, "Result" is displayed. Tapping the LCD Touch Panel displays the "Maintenance" screen.

Vacuum Pressure Diagnosis

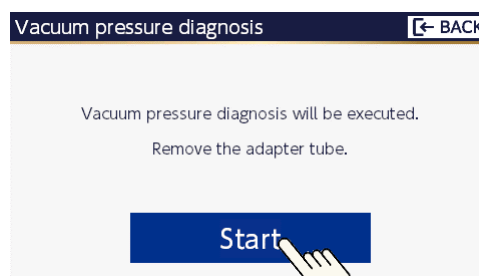
- 1 Tap "Diagnosis feature" in the "Maintenance" screen.



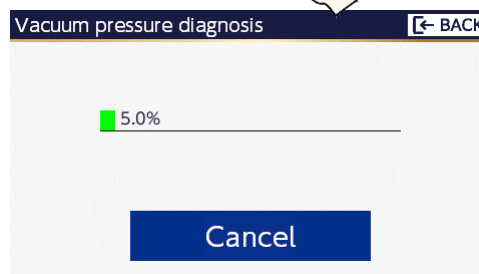
- 2 The "Diagnosis feature" screen is displayed. Tap "Diagnosis: Vacuum."



- 3 The "Vacuum pressure diagnosis" screen is displayed. Remove the adapter tube, then tap "Start."



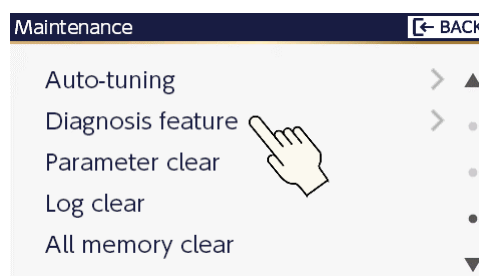
- 4 Vacuum pressure diagnosis starts. (Vacuum pressure diagnosis takes around 1 minute)



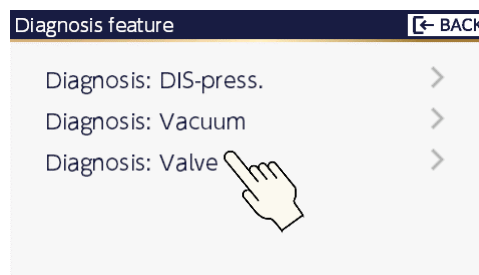
- 5 When diagnosis is completed, "Result" is displayed. Tapping the LCD Touch Panel displays the "Maintenance" screen.

Solenoid Valve Diagnosis

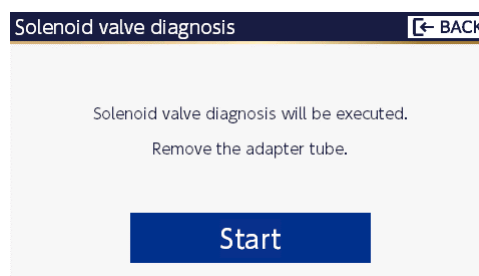
- 1 Tap "Diagnosis feature" in the "Maintenance" screen.



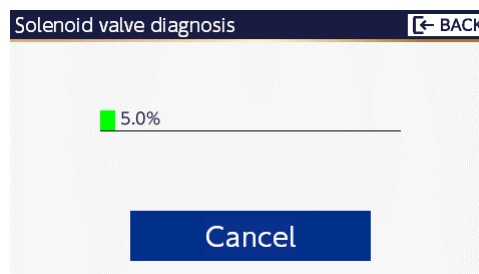
- 2 The "Diagnosis feature" screen is displayed. Tap "Solenoid valve diagnosis."



- 3 The "Solenoid valve diagnosis" screen is displayed. Remove the adapter tube, then tap "Start."



- 4 Solenoid valve diagnosis starts.
(Solenoid valve diagnosis takes around 30 seconds)



- 5 When diagnosis is completed, "Result" is displayed. Tapping the LCD Touch Panel displays the "Maintenance" screen.

* This function is used to detect solenoid valve failures. Use the ERROR signal output as a reference for the replacement of solenoid valves due to wear.

Resetting to Default

The following four types of default resets are available.

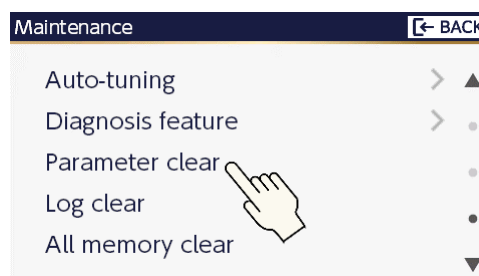
- User counter clear
- Parameter clear
- Log clear
- All memory clear

See P.70 "User Counter Clear" for details on User counter clear.

Parameter clear

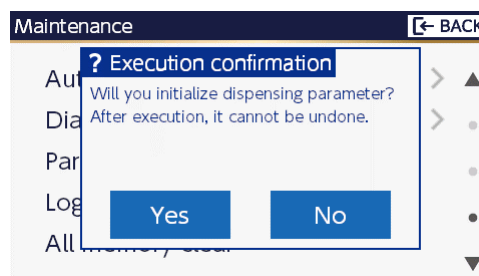
Settings saved for each channel are reset to default. See P.118 "Settings for Each Channel" for more details of settings that are reset to default.

- 1 Tap "Parameter clear" in the "Maintenance" screen.



- 2 "Execution confirmation" is displayed. Tap "Yes" or "No" to confirm.

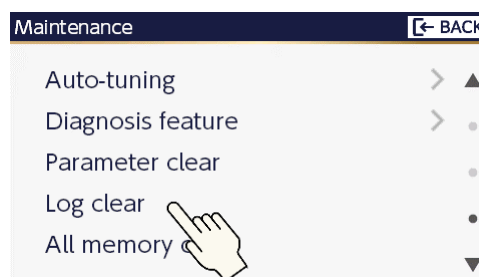
* Cannot be restored to the original condition after deleting.
Please take care with this function.



Log clear

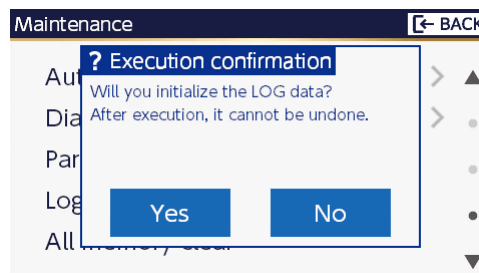
Clear the operation history (log).

- 1 Tap "Log clear" in the "Maintenance" screen.



- 2 "Execution confirmation" is displayed. Tap "Yes" or "No" to confirm.

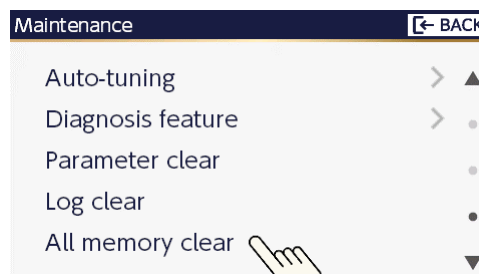
* Cannot be restored to the original condition after deleting.
Please take care with this function.



All memory clear

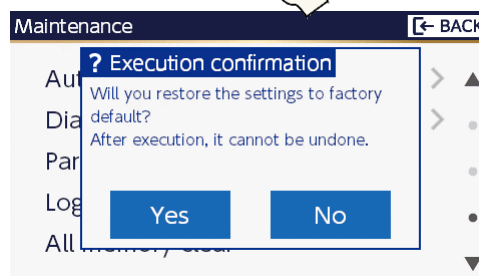
All settings are reset to default. See P.118 "Settings for Each Channel" and P.120 "Settings Common to All Channels" for more details of settings that are reset to default.

- 1 Tap "All memory clear" in the "Maintenance" screen.



- 2 "Execution confirmation" is displayed. Tap "Yes" or "No" to confirm.

* Cannot be restored to the original condition after deleting.
Please take care with this function.



Upgrading the Firmware

Use the following procedure if the firmware needs to be upgraded.

- 1 Connect USB memory that has the firmware saved with the USB-2 connector.
- 2 Tap "Firmware update" in the "Maintenance" screen.

Troubleshooting

If an error or alarm has occurred, check the log and take appropriate action. If there are problems that are not recorded in the log, read the following carefully and take appropriate action.
Contact our sales department if the problem is not resolved with the log and actions outlined below.

Log List

Code	Message	Details / Action
001	"Setting data load error"	○: "Setting data cannot be loaded." ✓: The device is being affected by noise. Move the controller away from the noise source. If this error occurs frequently, contact our sales staff. * Data is reset to default after an error occurs. Configure the settings again.
002	"Startup error"	○: "Failed to start up the program." ✓: "If this error occurs, contact our sales staff."
003	"Image data error"	○: "Image data cannot be loaded." ✓: "If this error occurs, contact our sales staff."
004	"RTC error"	○: "Time information cannot be loaded." ✓: "Set the time again. If this error occurs repeatedly, contact our sales staff."
005	"Solenoid valve error"	○: "This error occurs when the solenoid valve unit has worn and is at increased risk of failure." ✓: "Replace the solenoid valve."
006	"Air supply pressure error"	○: "Air supply pressure error has occurred." ✓: Check the air supply pressure and input air supply pressure that meets the specifications on P.134.
007	"Exhaust error"	○: "Exhaust error has occurred." ✓: "Check the exhaust port for clogging."
008	"Vacuum pressure error"	○: "Vacuum pressure error has occurred." ✓: "Check the exhaust port for clogging." ✓: Check the air supply pressure and input air supply pressure that meets the specifications on P.134.
009	"Temp. unit error: HS warning"	○: "Internal parts error." ✓: "Contact our sales staff."
010	"Temp. unit error: Input error"	○: "Temp. sensor connection error." ✓: "Check the wiring."
011	"Temp. unit error: HB warning"	○: "Heater connection error." There is an error with the heater connection. ✓: "Check the wiring."
012	"Temp. unit error: Timeout"	○: "Com. cable connection error." ✓: "Check the wiring." ○: "Temp. unit is not turned on." ✓: "Turn on the power."
013	"Temp. unit error: Setting error"	○: Settings cannot be written to the temp. unit controller. ✓: "Operate temp. unit so that, the settings can be changed."
014	"Internal error 1"	○: "Internal error: Connection" ✓: If this error occurs frequently contact our sales staff.
015	"Internal error 2"	○: "Internal error: Connection" ✓: Contact our sales staff.
016	"Internal error 3"	○: "Internal error: Connection" ✓: Contact our sales staff.
017	"Internal error 4"	○: "Version Error has been detected." ✓: Perform Firmware update of the dispensing unit by following the screen display.
018	Reserved	-
019	Reserved	-
020	"Internal error 7"	○: "Internal error: Communication" ✓: If this error occurs frequently contact our sales staff.

Code	Message	○ Details / ✓ Action
100	"Air supply pressure warning"	○: "Air supply pressure is low." ✓: Check the air supply pressure and input air supply pressure that meets the specifications on P.134.
101	"Solenoid valve alarm"	○: "Solenoid valve replacement timing." ✓: See P.75 "Changing the Solenoid Valve Replacement Reference (Valve Warning Threshold)" and change the replacement timing. ✓: Contact our sales staff and replace the solenoid valve.
102	"Empty alarm"	○: "Empty alarm has occurred." ✓: See P.70 "User Counter Clear" and clear the User counter.
103	"Temp. unit alarm: Warning 1"	○: "Temp. unit alarm (Warning 1) has occurred." ✓: See the Temperature Controller [Type 05 MINI Peltier with Communications Function] instruction manual.
104	"Temp. unit alarm: Warning 2"	○: "Temp. unit alarm (Warning 2) has occurred." ✓: See the Temperature Controller [Type 05 MINI Peltier with Communications Function] instruction manual.
105	"DIS-monitoring alarm"	○: "Dispensing pressure is out of allowable range." ✓: Perform dispensing pressure diagnosis and solenoid valve diagnosis to check if there are any errors. Contact our sales department if there are errors, and request an overhaul.
106	"Σ mode remaining amount alert"	○: "Σ mode remaining volume alert has occurred." ✓: Change the syringe.
107	Reserved	-
200	"Air supply resume: Error clear"	○: "Air supply has resumed. Error has been cleared."
201	"Air supply resume: Alarm clear"	○: "Air supply has resumed. Alarm has been cleared."
202	"Temp. unit error clear"	○: "Recovered from temperature unit error."
203	"Vacuum is normal. Error clear"	○: "Vacuum pressure returned to normal. Error has been cleared."
204	"Empty alarm clear"	○: "Empty alarm has been cleared."
205	"Exhaust error clear"	○: "Exhaust error has been cleared."
206	"Internal error clear"	○: "Internal error has been cleared."
207	Reserved	-
208	Reserved	-
209	"Warm-up operation complete"	○: "Warm-up operation has been completed."
210	"Transition to Sleep mode"	○: "The system has now entered Sleep mode."
211	"Recovery from Sleep mode"	○: "The system has now recovered from Sleep mode."
300	"All memory clear"	○: "All memory clear has been executed."
301	"Parameter clear"	○: "Parameter clear has been executed."
302	"User counter clear"	○: "User counter clear has been executed."
303	"ENABLE signal OFF"	○: "ENABLE signal has been turned OFF."
304	"Dispensing pressure diagnosis"	○: "Dispensing pressure diagnosis has been executed."
305	"Vacuum pressure diagnosis"	○: "Vacuum pressure diagnosis has been executed."
306	"Solenoid valve diagnosis"	○: "Solenoid valve diagnosis has been executed."
307	"Replace sol. valve"	○: "Solenoid valve replace has been executed."
308	"Dispensing unit firmware update"	○: "The firmware of the dispensing unit has been updated."

Other

No.	Details	Cause	Action
1	Press the controller	POWER CORD connection error.	● Check and insert the POWER CORD to the AC ADAPTER.
2	POWER to turn the POWER ON.	AC ADAPTER connection error.	● Check and insert the AC ADAPTER plug to the device.
3	Device does not operate	Controller error.	● Check the dispense request signal. Repair is required if there is an error. Contact our sales staff.
4		External input device error such as foot switch.	● Check the external input device. Repair is required if there is an error. Contact our sales staff.
5	Does not dispense	Dispensing pressure is insufficient, or is lower than the set value.	● Increase the dispensing pressure. ● Check if the required air supply pressure is being supplied.
6		The dispensing time set value is too short.	● Increase the dispensing time.
7		Liquid material is blocked in the nozzle or other parts, or inside the container.	● Wash the affected parts and accessories, or replace them.
8		There is no more liquid material.	● Replace the liquid material with fresh material.
9		Dispense screen A or B is not displayed.	● Tap the [← BACK] or [←] icon to display Dispense screen A or B.
10		The adapter tube is not inserted.	● Check if the adapter tube quick joint is connected properly to AIR OUT.
11		The ENABLE signal input status is OFF while the ENABLE signal function is Active.	● Turn the ENABLE signal function Inactive, or turn the ENABLE signal input status ON.
12	The external input terminal does not receive an input signal.	The connector (plug) connected to the I/O terminal is loose or disconnected.	● Insert the connector and tighten the securing ring (or screw).
13		The connection signal polarity is in reverse.	● Check the connection signal polarity. ● Check the connected pin number.
14		The signal input time is too short.	● See P.91"5 Connections using External Input/Output Terminals" and input the signal longer than the specified time.
15	Material is dispensed, but is not stable.	Either 5 to 8 above.	● See 5 to 8 above.
16		The setting parameter has been changed.	● Check the parameter, then change it to the correct value.
17		There are air bubbles present in the liquid material.	● Replace with the liquid material that has been sufficiently defoamed.
18		The liquid material viscosity has increased, or is starting to harden up.	● Wash the parts sufficiently, and use fresh liquid material.
19		The solenoid valve unit has become worn.	● Repair or replace the unit.
20	The date and time settings are cleared	The battery is dead.	● It will be a battery replacement work at Musashi.
21	Screen was dark from the time of purchase	Backlight is dead.	● Adjust brightness with P.83"Changing the LCD Screen Brightness (Screen Brightness)". ● If it cannot be effectively adjusted, it will be a panel replacement work at Musashi.

8 Other

Product Specifications

Model		S-SIGMA-CM4-CTR-V5-N	S-SIGMA-CM4-CTR-V5-P	S-SIGMA-CM4-CTR-V2-N	S-SIGMA-CM4-CTR-V2-P
Dispensing system		Air-pulse system			
Control system		Micro-computer controlled electronic-pneumatic system			
Display		4.3-inch touch screen LCD (Japanese / English / Chinese)			
Dispensing mode		TIMED mode, MANUAL mode			
Dispensing pressure setting range	Σ function (*1)ON	30.0 to 500.0 [kPa]		20.0 to 200.0 [kPa]	
	Σ function OFF	30.0 to 500.0 [kPa]		5.0 to 200.0 [kPa]	
Dispensing duration setting range	Σ function ON	TIMED mode 0.020 to 9999.999 [s] (depends on tube length)			
	Σ function OFF	MANUAL mode: 0.035 to 9999.999 [s]			
Vacuum pressure setting range		0.001 to 9999.999 [s]			
Main functions		Automatic correction of volume reduction, Automatic fluid-drip prevention, Automatic remaining volume alert, Alpha correction, Delta correction, Vacuum pressure correction, Remaining volume display correction, Auto-increment, Auto slope, Express shot, Interval dispensing, Date/time (*2), Operation history, Dispensing shot counter, Stopwatch, Dispensing parameter correction, Self-diagnosis, Pressure monitoring, ECO mode, Sleep mode, I/O assignment change, Password lock, Display unit switching (kPa/bar), Solenoid valve alarm, Solenoid valve error, Exhaust error, Vacuum error, Supply pressure error, Channel copy, Parameter backup			
Channel number		400 [CH]			
Expandability	USB host	Type-A x1 port (To connect to mass storage device)			
	USB device	Type-C x1 port (To connect to PC) (*3)			
	LAN	RJ45 x1 port (10BASE-T/100BASE-TX)			
	RS-232C	D-Sub9 pin (plug) x1 port			
	Device cooperation	e-CON (4-pin socket) x1 port (Temperature controller)			
External input/output signal D-Sub	Connector	D-Sub25 pin (socket) x1 port			
	Rated value	24 [V], 10 [mA] or less			
	Input (Photo-coupler)	24 [V] common Input resistance 5 [kΩ]	GND common Input resistance 13 [kΩ]	24 [V] common Input resistance 5 [kΩ]	GND common Input resistance 13 [kΩ]
		Dispense, Channel switching, Channel select, Dispensing mode select, Dispense enable (assignable: Channel preset, channel step, Sleep, Counter clear,etc.)			
	Output (Open collector)	GND common	24 [V] common	GND common	24 [V] common
Source pressure range		200 to 700 [kPa]		200 to 400 [kPa]	
Supply power (*4)	Connector	Dedicated connector for ACAT-4, e-CON (3-pin socket) (*5)			
	Rated voltage	DC 24 [V] ±10 [%]			
	Power consumption	26 [W]			
Operating environment	Temperature	In operation: 15 to 35 [°C], Not in operation: 1 to 60 [°C]			
	Humidity	In operation: 30 to 75 [%], Not in operation: 30 to 80 [%] (no condensation in any condition)			
	Altitude	2000 [m] or less			
	Ingress protection rating	IP20			
	Pollution degree	Pollution degree 2			
	Other condition	Indoor use only			
External dimensions (Excluding protrusions)		Width (W) 250 [mm] x Depth (D) 250 [mm] x Height (H) 108 [mm]			
Weight		4.0 [kg]			
Conformity standards		EU RoHS2, CE-marking (EMC directive)			
Options		AC ADAPTER (ACAT-4), Hand switch (LS-2T-DSUB25-1.5M, LS-3T-DSUB25-1.5M), Foot switch (FS/W-2-DSUB25-2.0M), Various connection cables, etc.			

*1 Σ function refers to automatic correction volume reduction, automatic fluid-drip prevention and automatic remaining volume alert.

*2 Date/time data is saved by built-in lithium coin battery (primary battery).

*3, *5 Please use 3m or less length USB and power cable (out of CE conformity when exceeding 3m length).

*4 AC adapter is not supplied with the product. Supply power in one of the following methods:

- By using our ACAT-4 (input rating of AC100 to 240 [V] 50/60 [Hz]).

- By using the SELV power supply that satisfies the LVD harmonized standards for CE marking (applicable plug: e-CON 3-pin).

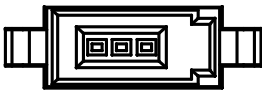
DC IN/OUT specifications

Connector Specifications

Compatible plug	e-CON (3-pin plug)
Recommended cable diameter	AWG#20~22

- * If power is supplied from the DC IN/OUT port, use a 3 m or less cable length.
Cables longer than 3 m are out of range of this device's CE marking and declaration of conformity.
- * An inrush current flows when the power is turned ON. Care must be taken as depending on the type of power supply the device might not start up normally even if the power supply meets the power supply ratings.
- * e-CON is an industry standard connector that has been standardized by numerous FA device manufacturers. It is widely used for sensors, PLC and other purposes.

Plug insertion socket shape and polarity position



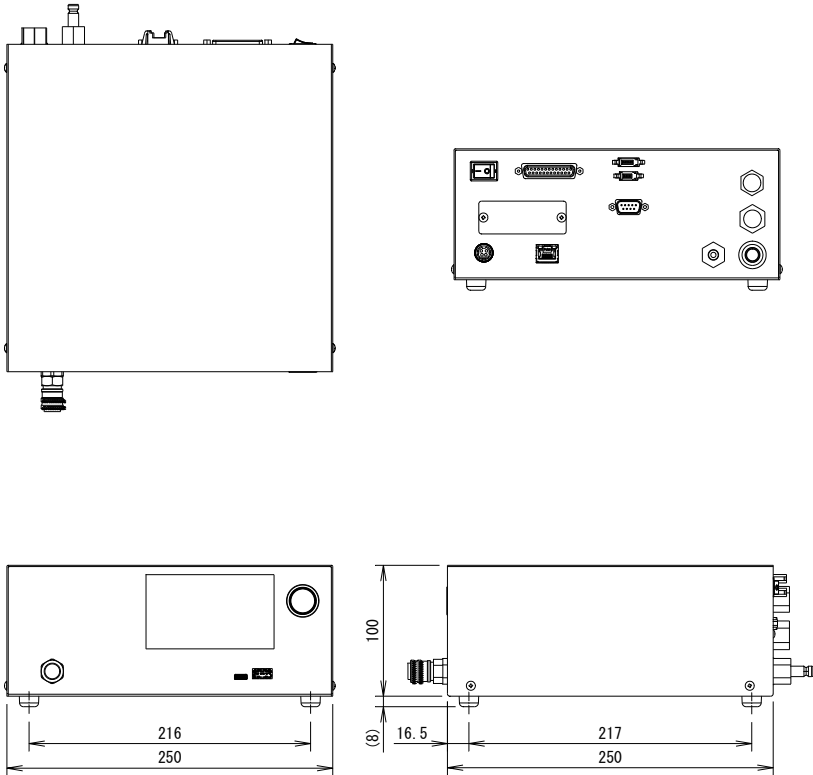
[3] [2] [1]

Pin Number	Polarity
1	+DC24V IN
2	GND(COM)
3	N.C.

External dimensions

S-SIGMA-CM4-CTR-(V5/V2)-(N/P)

Units: mm



Communications RS-232C

Read this after checking Basic Operations.

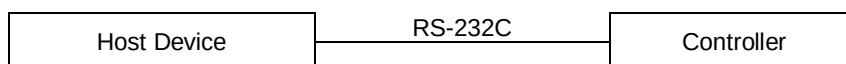
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Connections using Serial Communication

1 Overview

This device can be controlled by host devices using RS-232C communications. Host devices can set and acquire dispense start and end, and dispensing conditions with the use of commands.



RS-232C Communication Specifications

The communication specifications used by this device are as follows.

Communication mode		Half duplex
Interface		RS-232C compliant
Synchronization method		Start stop synchronous type
Flow control		None
Character code		ASCII code
Communication speed		9600 [bps]
		19200 [bps]
		38400 [bps] (default value)
		57600 [bps]
Com. protocol		Type A (default value)
		Type C
Communication format	Start bit	1
	Data length	8
	Parity bit	None
	Stop bit	1

RS-232C Communication Connection Specifications

Connector Specifications

Controller side	USB Type C receptacle (front) D-Sub 9-pin plug (inch screw fastener: No.4) (rear)
-----------------	--

Cable Used for RS-232C Communications

■ Front

Product Name	Model
CONNECT CABLE [USB C-C TYPE 2.0M]	CB-USB-C-C-2.0M

Commercially available USB Type C - USB Type C cable can be used.

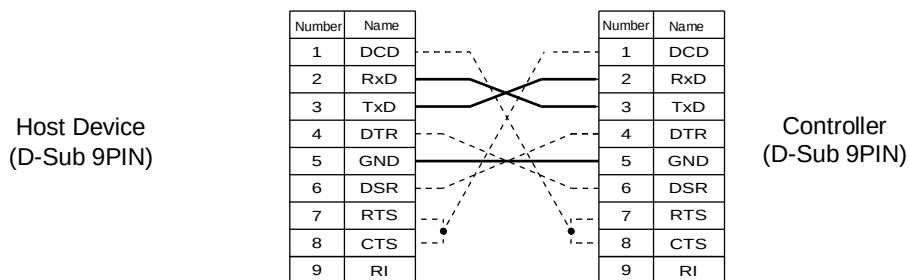
■ Rear

Product Name	Model
CONNECTION CABLE [DSUB 9PIN INTERLINK 1.8M]	MUCAD-C7-SMM-1.8M
CONNECTION CABLE [DSUB 9PIN INTERLINK 3M]	MUCAD-C8-SMM

Outlined in the connection cable image below.

A commercially available RS-232C cable (D-Sub 9 PIN female - 9 PIN female crossover cable) can be used.

* This device does not perform flow control, so only the solid line connections are used.

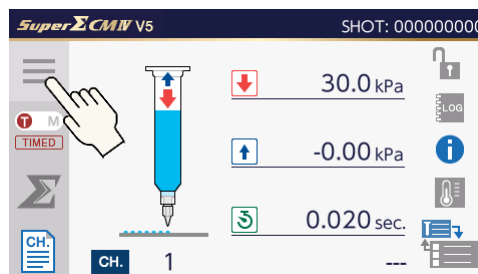


Changing RS-232C Communication Settings

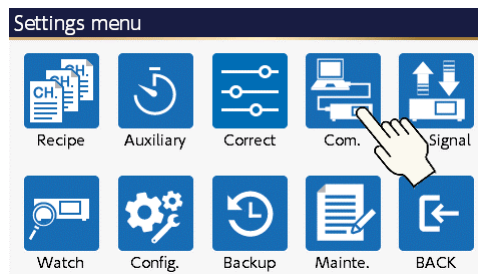
Item	Description
"Communication speed"	Sets the communication speed for RS-232C communications. Each tap changes the setting in the following order. "38400bps" → "57600bps" → "9600bps" → "19200bps" → "38400bps" → ...
"Checksum"	Sets whether a checksum is used in the communication command format.
"Com. protocol"	Sets the Com. protocol. Each tap changes the setting in the following order. "Type A" → "Type C" → "Type A" → ...

Use the following procedure to move to the "RS-232C communication setting" screen to change RS-232C communication settings.

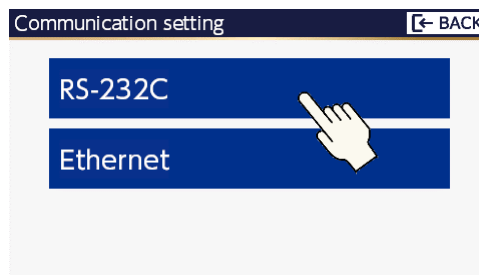
- 1 Tap the [≡] icon in Dispense Screen A or B.



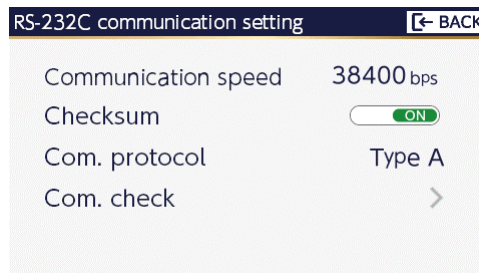
- 2 The "Settings menu" screen is displayed. Tap the [Settings icon] icon.



- 3 The "Communication setting" screen is displayed. Tap "RS-232C communication setting".



- 4 The "RS-232C communication setting" screen is displayed. Configure each setting.



Communication speed

Tap "Communication speed" in the "RS-232C communication setting" screen. Each tap changes the setting in the following order.

"38400bps" → "57600bps" → "9600bps" → "19200bps" → "38400bps" → ...

Checksum

The Checksum in "Packet format" on P.8 and P.61 can be omitted.

Tap "Checksum" in the "RS-232C communication setting" screen. This allows switching between OFF/ON.

ON	:	Checksum is required.
(default value)	:	
OFF	:	Checksum is not required.

Com. protocol

The protocol used with this device and RS-232C communication can be switched. See P.6"2 Com. protocol: Type A" or P.59"3 Com. Protocol: Type C" for information on each protocol.

Tap "Com. protocol" in the "RS-232C communication setting" screen. Each tap changes the setting in the following order.

"Type A" → "Type C" → "Type A" ...

2 Com. protocol: Type A

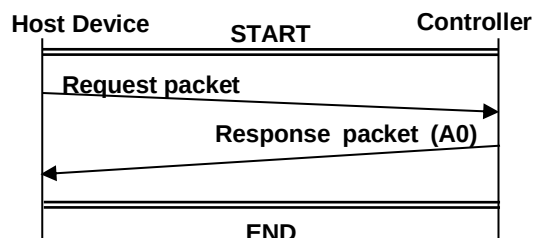
Communication Procedure

Communications between this device and the host device is performed as follows.

Communication Sequence

1. Starts when request packet is sent from the host device.
2. The controller receiving the request packet sends a response packet to the host device to complete the communication.

If there are no problems with the packet content or controller status, and the request packet is processed correctly, the response packet error type is "0," indicating a normal response.



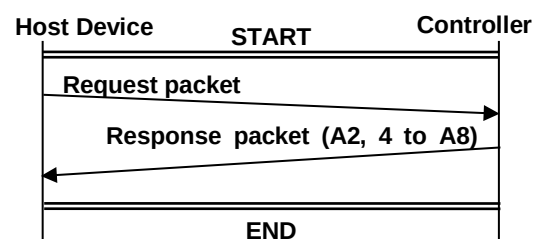
Cases Causing Errors

- Incorrect packet received or cannot be received

If the request packet cannot be received for any reason, the controller sends a response packet to the host device to complete the communication.

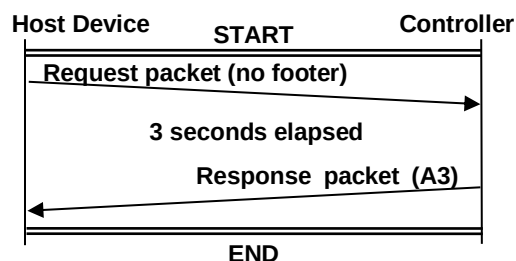
The response packet error type is "2, 4 to 8," indicating the cause of the error.

See "Error Types" below to take the appropriate action.



- Timeout

After the controller receives a request packet preceded by "M" and the footer cannot be received within 3 seconds, the response packet error type is "3," indicating a timeout.



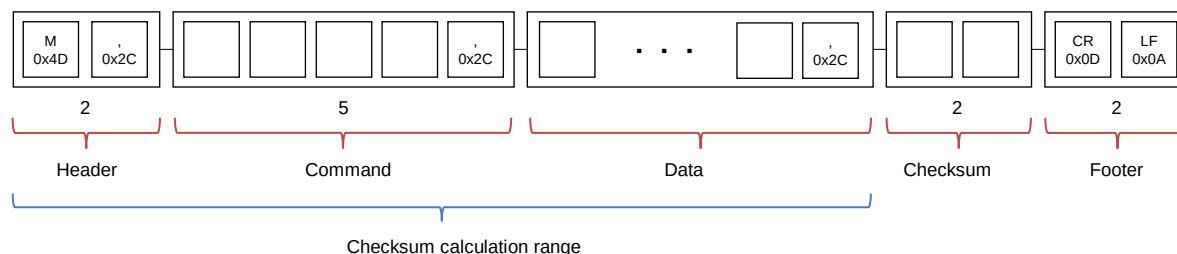
Error Types

Error Types	Details	Cause	Action
0	Normal response	-	-
1	Reserved	-	-
2	Data error	Either of the following. ● Differs to the data quantity defined by each command ● Differs to the defined format ● Does not meet the receiving conditions for each command ● Omitted data that cannot be omitted	See each command (P.20) and correct the values.
3	Timeout error	After receiving a request packet, the footer was not received within 3 seconds	Check the connection status
4	Checksum error	The request packet checksum does not match, or cannot be read (there is data other than hexadecimal numbers)	See P.11"Checksum Calculation" and correct the checksum value.
5	Reserved	-	-
6	Command undefined error	There is an error in the request packet command	See commands that this device can handle (P.12).
7	Out of setting range error	The request packet data value is out of the range defined for each command, or cannot be read	See commands that this device can handle (P.12) and correct the value.
8	Inexecutable error	Not in receive status	● See the receive statuses outlined in P.8"Command Types" and perform the following ➤ Display Dispense screen A or B ➤ If in In-dispensing, wait until dispense end (P.50).

Packet Format

• Request packet

Packet configuration



Block	Data length	Role
Header	2-character fixed length	Indicated by ASCII code "M(0x4D)" + ", (0x2C)" at the start of the packet.
Command	5-character fixed length	Indicates a command. The following five types of commands are available. • Various operations • Set parameter • Acquire parameter • Acquire model information • Acquire other information
Data	Variable length	Indicates data defined by each command. Enter ", (0x2C)" to the end of each data.
Checksum*	2-character fixed length	Data for detecting incorrect transmission data. The borrow of data from the header block to the final data of the data block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	2-character fixed length	Indicated by ASCII code "CR(0x0D)" + "LF(0x0A)" in the at the end of the packet.

* Can be omitted with settings. The checksum is set as required in the factory default.

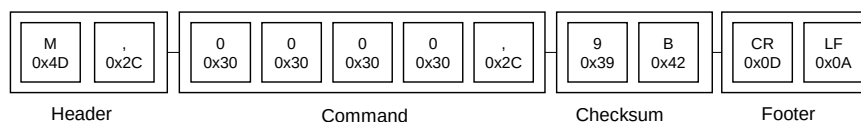
Command Types

Type	Details	Receive Statuses
Various operations	Command for operating the device.	Can be received with Dispense screen A/B, or Communication check screen, and dispense operation is not being performed.
Set parameter	Command for setting the device parameters.	Can be received with Dispense screen A/B, or Communication check screen, and dispense operation is not being performed.
Acquire parameter	Command for acquiring the current parameters set with the device.	Can be received with Dispense screen A/B, or Communication check screen.
Acquire model information	Command for acquiring information other than parameters that can be set such as dispense status or alarm status.	Can be received with Dispense screen A/B, or Communication check screen.
Acquire other information	Command for acquiring the maximum number of commands and version information.	Can be received with Dispense screen A/B, or Communication check screen.

When data does not exist

Depending on the command type, data may not be included in the packet (E.g.: Command 0000: Dispense request, etc.) In these cases, follow up with a checksum immediately after the command.

E.g.: Command 0000 (Dispense request)

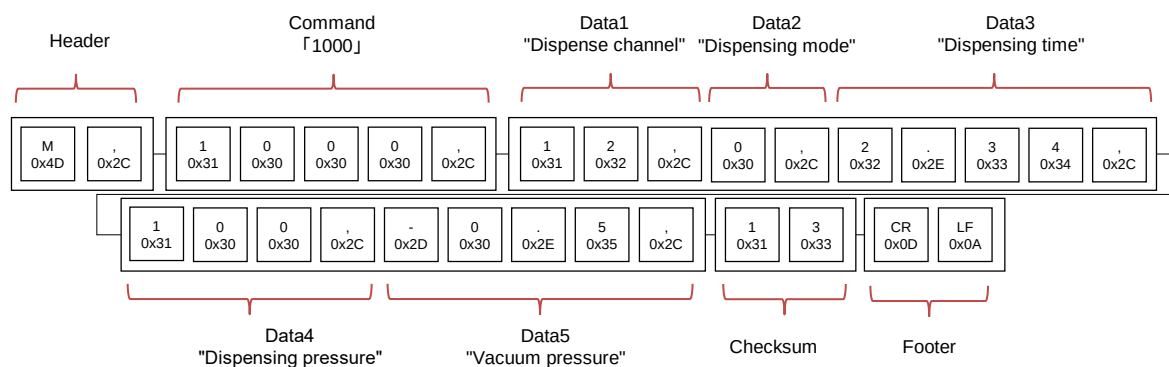


Omitting values in data

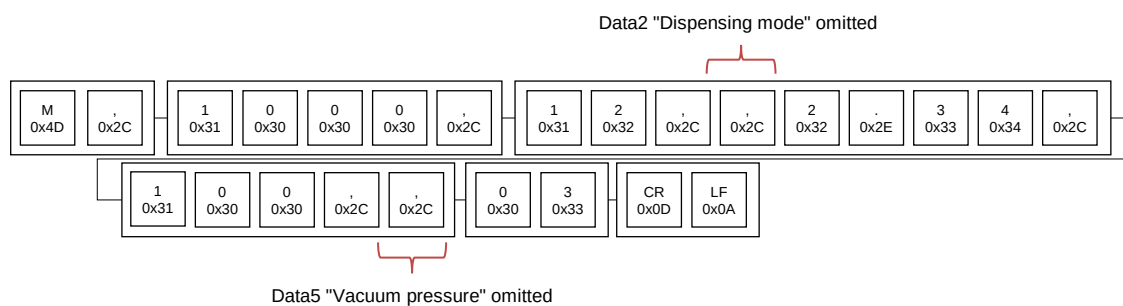
In the request packet, the values of data can be omitted. If the values are omitted, the values for the relevant parameter are not changed, and the values set at the time are retained.

E.g.: Command 1000 (Set parameter)

Specifying all values

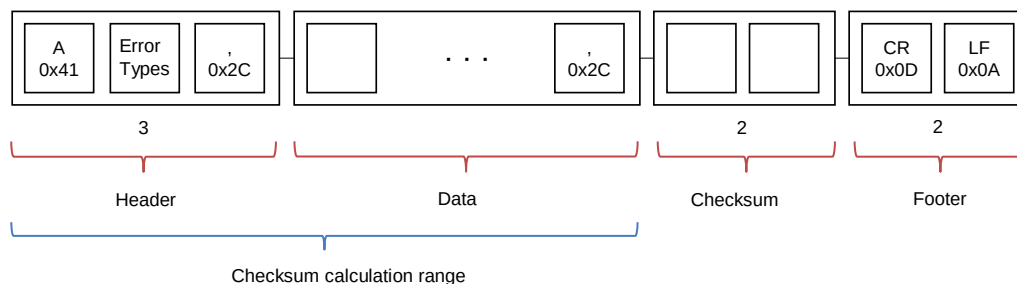


If some parameter values are omitted (here, Dispensing mode and vacuum pressure are omitted)



• Response packet

Packet configuration



Block	Data length	Role
Header	3-character fixed length	Indicated by ASCII code "A(0x41)" + Error Type + " , (0x2C)" at the start of the packet.
Data	Variable length	Indicates data corresponding to each command or error response. " , (0x2C)" is added to the end of each data.
Checksum*	2-character fixed length	Data for detecting incorrect transmission data. The borrow of data from the header block to the final data of the data block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	2-character fixed length	Indicated by ASCII code "CR(0x0D)" + "LF(0x0A)" in the at the end of the packet.

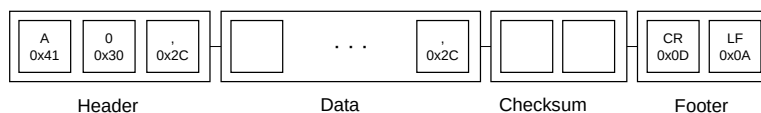
* Can be omitted with settings. See Basic Operations. The checksum is set as required in the factory default.

Response packet data

The following three patterns of response packets are available.

A. Normal response

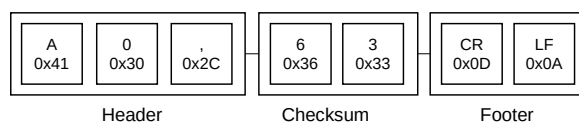
Data corresponding to each command is stored. See P.16 "Description of Commands" for more details on data.



When data does not exist

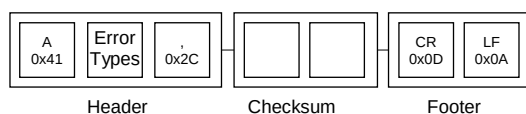
Depending on the command type, data may not be included in the packet (E.g.: Normal response to Command 0000: Dispense request). In these cases, a checksum is followed up immediately after the header for the reply.

E.g.: Normal response to Command 0000 (Dispense request)



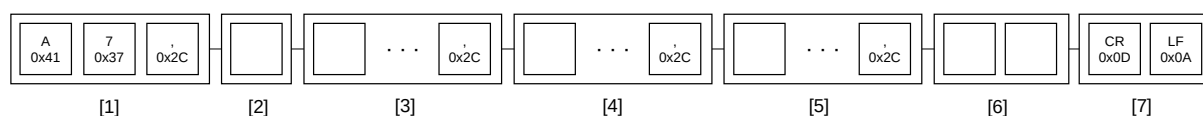
B. Error response other than Error Type 7

No data is included in the packet, and the "Error Type" indicating the cause of the error is stored in the header. See P.7"Error Types" for more details.



C. Error response with Error Type 7

This is sent as the reply if the request packet data value is out of the range defined for each command. Details of the cause of the error are stored in the response packet data. See P.11"C. Error response with Error Type 7".



No.	Data length	Details
[1]	3-character fixed length	Header
[2]	1-character fixed length	Item number of out of range data
[3]	Variable length	Set value
[4]	Variable length	Minimum value of setting range
[5]	Variable length	Maximum value of setting range
[6]	2-character fixed length	Checksum
[7]	2-character fixed length	Footer

Checksum Calculation

The following outlines the dispense request command packet sample and the checksum calculation method.

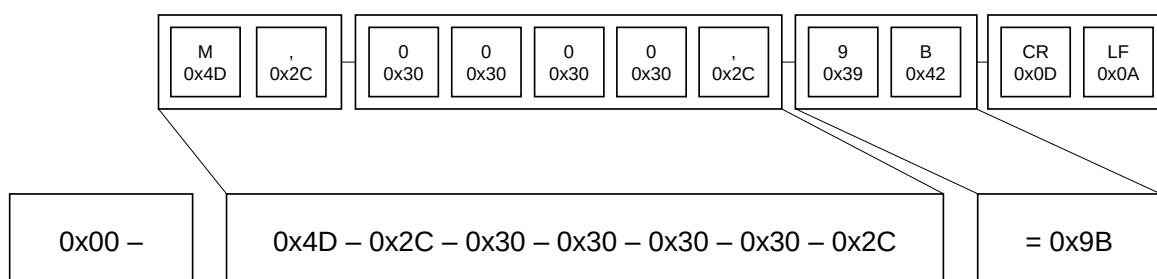
Dispense request command packet sample

The dispense request command is 0000 and has no data, so the packet is as follows.



Checksum calculation

The borrow of header data of the header block to the final data of the data block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal. The result converted to an ASCII code is the checksum



Request Packet List

Type	Command	Details	Data											
			1	2	3	4	5	6	7	8	9	10	11	12
Various operations	0000	Dispense request	-	-	-	-	-	-	-	-	-	-	-	-
	0001	STEP	-	-	-	-	-	-	-	-	-	-	-	-
	0002	DOWN	-	-	-	-	-	-	-	-	-	-	-	-
	0003	PRESET	-	-	-	-	-	-	-	-	-	-	-	-
	0004	TIMED mode switching	-	-	-	-	-	-	-	-	-	-	-	-
	0005	MANUAL mode switching	-	-	-	-	-	-	-	-	-	-	-	-
	0006	DIS-monitoring test	-	-	-	-	-	-	-	-	-	-	-	-
	0007	Channel load	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	0008	Channel copy	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	0009	Recipe load	Recipe number	-	-	-	-	-	-	-	-	-	-	-
	0010	Delete recipe	Recipe number	-	-	-	-	-	-	-	-	-	-	-
	0011	Diagnosis feature	Applicable	Start/Cancel	-	-	-	-	-	-	-	-	-	-
	0012	Temperature unit auto-tuning	Temperature unit number	Start/Cancel	-	-	-	-	-	-	-	-	-	-
	0013	User counter clear	-	-	-	-	-	-	-	-	-	-	-	-
	0014	Error log clear	-	-	-	-	-	-	-	-	-	-	-	-
	0015	Parameter clear	-	-	-	-	-	-	-	-	-	-	-	-
	0016	All memory clear	-	-	-	-	-	-	-	-	-	-	-	-
	0017	Full sampling	-	-	-	-	-	-	-	-	-	-	-	-
	0018	Empty sampling	-	-	-	-	-	-	-	-	-	-	-	-
	0019	Detailed sampling	Remaining volume in syringe	-	-	-	-	-	-	-	-	-	-	-
	0020	Cancel Σ mode	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	0021	ECO mode	OFF/ON	-	-	-	-	-	-	-	-	-	-	-
	0022	Solenoid valve alarm Signal OFF	-	-	-	-	-	-	-	-	-	-	-	-
Set parameter	1000	Dispensing conditions	Dispense channel	DIS-mode	Dispensing time	Dispensing pressure	Vacuum pressure	-	-	-	-	-	-	-
	1001	Delay timer	Dispense channel	On delay time	Off delay timer	-	-	-	-	-	-	-	-	-
	1002	DIS-monitoring function	Dispense channel	Allowable	Dispensing OFF time	-	-	-	-	-	-	-	-	-
	1003	Channel name	Dispense channel	Channel name	-	-	-	-	-	-	-	-	-	-
	1004	DIS-auxiliary	Syringe size	Adapter tube length	Plunger enabled/disabled	ECO mode	-	-	-	-	-	-	-	-
	1005	Σ function: Vol. reduction corr.	Dispense channel	OFF/ON	Alpha correction value	Delta correction	-	-	-	-	-	-	-	-
	1006	Σ function: Fluid-drip prevention	Dispense channel	OFF/ON	Vacuum correction value	-	-	-	-	-	-	-	-	-
	1007	Σ function: Remaining vol. alert	Dispense channel	OFF/ON	Detection level	Detection count	-	-	-	-	-	-	-	-
	1008	Σ function: Remaining vol. corr.	Dispense channel	OFF/ON	Correction value	-	-	-	-	-	-	-	-	-
	1009	Recipe	Recipe number	Start CH.	End CH.	-	-	-	-	-	-	-	-	-
	1010	Auto-increment Function	Recipe number	Function OFF/ON	DIS-mode	ECO mode	Switching Condition	Switching value	-	-	-	-	-	-
	1011	Recipe details	Recipe number	Recipe name	Nozzle	Liquid material	-	-	-	-	-	-	-	-
	1012	Temperature unit conditions	Recipe number	Temp. unit controller number	Control OFF/ON	Setting temperature	Upper Limit	Lower Limit	-	-	-	-	-	-
	1013	DIS-correction	Applicable	Correction method	Correction amount	Correction ratio	-	-	-	-	-	-	-	-
	1014	Interval feature	Function OFF/ON	Link CH.	Count	Interval	Repeat	Wait time	Interval mode	-	-	-	-	-
	1015	I/O signal	Series	Output signal spec.	ENABLE signal	On Error	-	-	-	-	-	-	-	-
	1016	Environment	Language	P.UNIT	Buzzer OFF/ON	T-ready Enable	Warm-ready enable	Pop-up auto OFF	Immed. Inc. (Alnc)	Control mode	Sleep timer	-	-	-
	1017	Warning function	Empty alarm OFF/ON	Empty warning threshold	Valve warning threshold	RSM advance notice lv.	-	-	-	-	-	-	-	-
	1018	Password lock	Function OFF/ON	Password	-	-	-	-	-	-	-	-	-	-
	1019	Input signal allocation	D-Sub 3PIN	D-Sub 4PIN	D-Sub 5PIN	D-Sub 6PIN	D-Sub 7PIN	D-Sub 8PIN	D-Sub 15PIN	D-Sub 16PIN	D-Sub 17PIN	D-Sub 18PIN	D-Sub 19PIN	D-Sub 20PIN
	1020	Output signal allocation	808	D-Sub 9PIN	D-Sub 10PIN	D-Sub 12PIN	D-Sub 21PIN	D-Sub 22PIN	D-Sub 23PIN	-	-	-	-	-
			Σ	D-Sub 10PIN	D-Sub 22PIN	D-Sub 23PIN	-	-	-	-	-	-	-	-
	1021	Date and time	Year	Month	Day	Hour	Minute	Second	-	-	-	-	-	-
	1022	Time zone	Time zone	-	-	-	-	-	-	-	-	-	-	-
	1023	Controller name	Controller name	-	-	-	-	-	-	-	-	-	-	-
	1024	IP address	1st byte from the top	2nd byte from the top	3rd byte from the top	4th byte from the top	-	-	-	-	-	-	-	-
	1025	Subnet mask	1st byte from the top	2nd byte from the top	3rd byte from the top	4th byte from the top	-	-	-	-	-	-	-	-
	1026	Default gateway	1st byte from the top	2nd byte from the top	3rd byte from the top	4th byte from the top	-	-	-	-	-	-	-	-
	1027	Administrator password	Administrator password	-	-	-	-	-	-	-	-	-	-	-
	1028	Web page refresh interval	Web page refresh interval	-	-	-	-	-	-	-	-	-	-	-
	1029	Read only	Read only	-	-	-	-	-	-	-	-	-	-	-

Type	Command	Details	Data											
			1	2	3	4	5	6	7	8	9	10	11	12
Acquire parameter	2000	Dispensing conditions	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2001	Delay timer	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2002	DIS-monitoring function	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2003	Channel name	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2004	DIS-auxiliary	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2005	Σ function: Vol. reduction corr.	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2006	Σ function: Fluid-drip prevention	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2007	Σ function: Remaining vol. alert	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2008	Σ function: Remaining vol. corr.	Dispense channel	-	-	-	-	-	-	-	-	-	-	-
	2009	Recipe	Recipe number	-	-	-	-	-	-	-	-	-	-	-
	2010	Auto-increment Function	Recipe number	-	-	-	-	-	-	-	-	-	-	-
	2011	Recipe details	Recipe number	-	-	-	-	-	-	-	-	-	-	-
	2012	Temperature unit conditions	Recipe number	-	-	-	-	-	-	-	-	-	-	-
	2013	DIS-correction	Applicable	-	-	-	-	-	-	-	-	-	-	-
	2014	Interval feature	-	-	-	-	-	-	-	-	-	-	-	-
	2015	I/O signal	-	-	-	-	-	-	-	-	-	-	-	-
	2016	Environment	-	-	-	-	-	-	-	-	-	-	-	-
	2017	Warning function	-	-	-	-	-	-	-	-	-	-	-	-
	2018	Password lock	-	-	-	-	-	-	-	-	-	-	-	-
	2019	Input signal allocation	-	-	-	-	-	-	-	-	-	-	-	-
	2020	Output signal allocation	-	-	-	-	-	-	-	-	-	-	-	-
		808 Σ	-	-	-	-	-	-	-	-	-	-	-	-
	2021	Date and time	-	-	-	-	-	-	-	-	-	-	-	-
	2022	Time zone	-	-	-	-	-	-	-	-	-	-	-	-
	2023	Controller name	-	-	-	-	-	-	-	-	-	-	-	-
	2024	IP address	-	-	-	-	-	-	-	-	-	-	-	-
	2025	Subnet mask	-	-	-	-	-	-	-	-	-	-	-	-
	2026	Default gateway	-	-	-	-	-	-	-	-	-	-	-	-
	2027	Administrator password	-	-	-	-	-	-	-	-	-	-	-	-
	2028	Web page refresh interval	-	-	-	-	-	-	-	-	-	-	-	-
	2029	Read only	-	-	-	-	-	-	-	-	-	-	-	-
Acquire model information	3000	Dispense status	-	-	-	-	-	-	-	-	-	-	-	-
	3001	Air Supply Pressure	-	-	-	-	-	-	-	-	-	-	-	-
	3002	Stopwatch	-	-	-	-	-	-	-	-	-	-	-	-
	3003	Current Channel	-	-	-	-	-	-	-	-	-	-	-	-
	3004	Current recipe	-	-	-	-	-	-	-	-	-	-	-	-
	3005	Auto-increment status	-	-	-	-	-	-	-	-	-	-	-	-
	3006	DIS-monitoring test status	-	-	-	-	-	-	-	-	-	-	-	-
	3007	Remaining volume detection status	-	-	-	-	-	-	-	-	-	-	-	-
	3008	Temperature unit status	-	-	-	-	-	-	-	-	-	-	-	-
	3009	Value after DIS-correction	-	-	-	-	-	-	-	-	-	-	-	-
	3010	Interval status	-	-	-	-	-	-	-	-	-	-	-	-
	3011	Counter	-	-	-	-	-	-	-	-	-	-	-	-
	3012	Diagnosis status	Applicable	-	-	-	-	-	-	-	-	-	-	-
	3013	Temperature unit auto-tuning	Applicable	-	-	-	-	-	-	-	-	-	-	-
	3014	Total error log number	-	-	-	-	-	-	-	-	-	-	-	-
	3015	Error log (main)	Error log number	-	-	-	-	-	-	-	-	-	-	-
	3016	Error log (sub)	Error log number	-	-	-	-	-	-	-	-	-	-	-
	3017	D-sub25 output status	-	-	-	-	-	-	-	-	-	-	-	-
Acquire other information	9996	MAC address	-	-	-	-	-	-	-	-	-	-	-	-
	9997	I/O specifications	-	-	-	-	-	-	-	-	-	-	-	-
	9998	Command information	-	-	-	-	-	-	-	-	-	-	-	-
	9999	Version information	-	-	-	-	-	-	-	-	-	-	-	-

Response Packet List

Type	Command	Details	Data												Pattern
			1	2	3	4	5	6	7	8	9	10	11	12	
Various operations	0000	Dispense request	-	-	-	-	-	-	-	-	-	-	-	-	A
	0001	STEP	-	-	-	-	-	-	-	-	-	-	-	-	A
	0002	DOWN	-	-	-	-	-	-	-	-	-	-	-	-	A
	0003	PRESET	-	-	-	-	-	-	-	-	-	-	-	-	A
	0004	TIMED mode switching	-	-	-	-	-	-	-	-	-	-	-	-	A
	0005	MANUAL mode switching	-	-	-	-	-	-	-	-	-	-	-	-	A
	0006	DIS-monitoring test	-	-	-	-	-	-	-	-	-	-	-	-	A
	0007	Channel load	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0008	Channel copy	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0009	Recipe load	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0010	Delete recipe	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0011	Diagnosis feature	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0012	Temperature unit auto-tuning	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0013	User counter clear	-	-	-	-	-	-	-	-	-	-	-	-	A
	0014	Error log clear	-	-	-	-	-	-	-	-	-	-	-	-	A
	0015	Parameter clear	-	-	-	-	-	-	-	-	-	-	-	-	A
	0016	All memory clear	-	-	-	-	-	-	-	-	-	-	-	-	A
	0017	Full sampling	-	-	-	-	-	-	-	-	-	-	-	-	A
	0018	Empty sampling	-	-	-	-	-	-	-	-	-	-	-	-	A
	0019	Detailed sampling	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0020	Cancel Σ	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0021	ECO mode OFF/ON	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	0022	Solenoid valve alarm Signal OFF	-	-	-	-	-	-	-	-	-	-	-	-	A
Set parameter	1000	Dispensing conditions	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1001	Delay timer	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1002	DIS-monitoring function	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1003	Channel name	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1004	DIS-auxiliary	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1005	Σ function: Vol. reduction corr.	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1006	Σ function: Fluid-drip prevention	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1007	Σ function: Remaining vol. alert	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1008	Σ function: Remaining vol. corr.	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1009	Recipe	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1010	Auto-increment Function	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1011	Recipe details	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1012	Temperature unit conditions	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1013	DIS-correction	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1014	Interval feature	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1015	I/O signal	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1016	Environment	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1017	Warning function	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1018	Password lock	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1019	Input signal allocation	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1020	Output signal allocation	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1021	Date and time	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1022	Time zone	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1023	Controller name	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1024	IP address	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1025	Subnet mask	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1026	Default gateway	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1027	Administrator password	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1028	Web page refresh interval	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A
	1029	Read only	Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
			-	-	-	-	-	-	-	-	-	-	-	-	A

Type	Command	Details	Data												Pattern
			1	2	3	4	5	6	7	8	9	10	11	12	
Acquire parameter	2000	Dispensing conditions	Dispense channel	DIS-mode	Dispensing time	Dispensing pressure	Vacuum pressure	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2001	Delay timer	Dispense channel	On delay timer	Off delay timer	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2002	DIS-monitoring function	Dispense channel	Allowable	Dispensing OFF time	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2003	Channel name	Dispense channel	Channel name	-	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2004	DIS-auxiliary	Dispense channel	Syringe size	Adapter tube length	Plunger enabled/disabled	ECO mode	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2005	Σ function: Vol. reduction corr.	Dispense channel	OFF/ON	Alpha correction value	Delta correction	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2006	Σ function: Fluid-drip prevention	Dispense channel	OFF/ON	Vacuum correction value	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2007	Σ function: Remaining vol. alert	Dispense channel	OFF/ON	Detection level	Detection count	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2008	Σ function: Remaining vol. corr.	Dispense channel	OFF/ON	Correction value	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2009	Recipe	Recipe number	Start CH.	End CH.	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2010	Auto-increment Function	Recipe number	Function OFF/ON	DIS-mode	ECO mode	Switching Condition	Switching value	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2011	Recipe details	Recipe number	Recipe name	Nozzle	Liquid material	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2012	Temperature unit conditions	Recipe number	Temperature unit Number	Control OFF/ON	Setting temperature	Upper Limit	Lower Limit	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2013	DIS-correction	Applicable	Correction method	Correction amount	Correction ratio	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	2014	Interval feature	Function OFF/ON	Link CH.	Count	Interval	Repeat	Wait time	Interval mode	-	-	-	-	-	A
	2015	I/O signal	Series	Output signal spec.	ENABLE signal	On Error	-	-	-	-	-	-	-	-	A
	2016	Environment	Language	P.UNIT	Buzzer OFF/ON	T-ready Enable	Warm-ready enable	Pop-up auto OFF	Immed. Inc. (AInc)	Control mode	Sleep timer	-	-	-	A
	2017	Warning function	Empty alarm OFF/ON	Empty warning threshold	Valve warning threshold	RSM advance notice lv.	-	-	-	-	-	-	-	-	A
	2018	Password lock	Function OFF/ON	Password	-	-	-	-	-	-	-	-	-	-	A
	2019	Input signal allocation	D-Sub 3PIN	D-Sub 4PIN	D-Sub 5PIN	D-Sub 6PIN	D-Sub 7PIN	D-Sub 8PIN	D-Sub 15PIN	D-Sub 16PIN	D-Sub 17PIN	D-Sub 18PIN	D-Sub 19PIN	D-Sub 20PIN	A
	2020	Output signal allocation	D-Sub 9PIN	D-Sub 10PIN	D-Sub 12PIN	D-Sub 21PIN	D-Sub 22PIN	D-Sub 23PIN	-	-	-	-	-	-	A
			D-Sub 10PIN	D-Sub 22PIN	D-Sub 23PIN	-	-	-	-	-	-	-	-	-	A
	2021	Date and time	Year	Month	Day	Hour	Minute	Second	-	-	-	-	-	-	A
	2022	Time zone	Time zone	-	-	-	-	-	-	-	-	-	-	-	A
	2023	Controller name	Controller name	-	-	-	-	-	-	-	-	-	-	-	A
	2024	IP address	1st byte from the top	2nd byte from the top	3rd byte from the top	4th byte from the top	-	-	-	-	-	-	-	-	A
	2025	Subnet mask	1st byte from the top	2nd byte from the top	3rd byte from the top	4th byte from the top	-	-	-	-	-	-	-	-	A
	2026	Default gateway	1st byte from the top	2nd byte from the top	3rd byte from the top	4th byte from the top	-	-	-	-	-	-	-	-	A
	2027	Administrator password	Administrator password	-	-	-	-	-	-	-	-	-	-	-	A
	2028	Web page refresh interval	Web page refresh interval	-	-	-	-	-	-	-	-	-	-	-	A
	2029	Read only	Read only	-	-	-	-	-	-	-	-	-	-	-	A
Acquire model information	3000	Dispense status	Dispense status	Dispense availability	Error status	Alarm status	-	-	-	-	-	-	-	-	A
	3001	Air Supply Pressure	Air Supply Pressure	-	-	-	-	-	-	-	-	-	-	-	A
	3002	Stopwatch	Stopwatch	-	-	-	-	-	-	-	-	-	-	-	A
	3003	Current Channel	Dispense channel	-	-	-	-	-	-	-	-	-	-	-	A
	3004	Current recipe	Recipe number	-	-	-	-	-	-	-	-	-	-	-	A
	3005	Auto-increment status	Status	Remaining volume	-	-	-	-	-	-	-	-	-	-	A
	3006	DIS-monitoring function status	Status	Result	-	-	-	-	-	-	-	-	-	-	A
	3007	Empty alarm state	Current remaining volume	Current detection count	-	-	-	-	-	-	-	-	-	-	A
	3008	Temperature unit status	Current temperature of temperature controller 1	Current temperature of temperature controller 2	Current temperature of temperature controller 3	Current temperature of temperature controller 4	Current temperature of temperature controller 5	-	-	-	-	-	-	-	A
	3009	Value after DIS-correction	Dispensing time	Dispensing pressure	Vacuum pressure	-	-	-	-	-	-	-	-	-	A
	3010	Interval status	Count	Repeat	-	-	-	-	-	-	-	-	-	-	A
	3011	Counter	User counter	Total counter	-	-	-	-	-	-	-	-	-	-	A
	3012	Diagnosis status	Applicable	Status	Result	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	3013	Temperature unit auto-tuning	Temperature unit number	Status	-	-	-	-	-	-	-	-	-	-	A
	3014	Total error log number	Total error log number	-	-	-	-	-	-	-	-	-	-	-	A
	3015	Error log (main)	Date of occurrence	Error code	Total counter	-	-	-	-	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
	3016	Error log (sub)	Data pattern	Related data 1	Related data 2	Related data 3	Related data 4	Related data 5	Related data 6	-	-	-	-	-	A
			Out of range data number	Set value	Minimum value of setting range	Maximum value of setting range	-	-	-	-	-	-	-	-	C
Acquire other information	9996	MAC address	MAC address	-	-	-	-	-	-	-	-	-	-	-	A
	9997	I/O specifications	I/O specifications	-	-	-	-	-	-	-	-	-	-	-	A
	9998	Command information	Various operations Command Maximum number	Parameter Setting command Maximum number	Parameter Acquire command Maximum number	Acquire model information Command Maximum number	-	-	-	-	-	-	-	-	A
	9999	Model information	Product model	Firmware 1	Firmware 2	Firmware 3	-	-	-	-	-	-	-	-	A

Description of Commands

Example of details

◇ Command

[Function] Function overview of the command.

[Data] Description of the data block. If multiple data blocks exist, always specify in order from the top. The chart outlines the following details.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit

No. : Item number of data.

Details : Details of data.

Can be Omitted Indicates if data can be omitted.

○: Yes

X: No

Minimum value : Minimum value that can be specified.

Maximum value : Maximum value that can be specified.

Unit : Unit of data.

Models with specification differences are named according to the following chart.

Model	Name
S-SIGMA-CM4-CTR-V5-(N/P)	V5
S-SIGMA-CM4-CTR-V2-(N/P)	V2

【Notice】

Please do not send unauthorized command to avoid a malfunction.

Example of creating a request packet

◇ Command 1000

[Function] Sets the dispensing conditions of the specified dispense channel as a batch.

* Error A8 will be come out when all of below conditions are applied.

•Indicated channel is belonging to the recipe

•Status of Auto-Increment function is ON

•Dispensing mode is changed

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	DIS-mode 0 = TIMED mode 1 = MANUAL mode	○	0	1	[-]
3	Dispensing time	○	0.010	9999.999	[s]
4	Dispensing pressure	V5	30.0	500.0	[kPa]
		V5	0.300	5.000	[bar]
		V2	5.0	200.0	[kPa]
		V2	0.050	2.000	[bar]
5	Vacuum pressure	○	-5.00	-0.00	[kPa]
		○	-0.0500	-0.0000	[bar]

Response packet data during normal response: None

◆ To change the dispensing conditions of dispense channel 13CH to the following, create a request packet using the flow below.

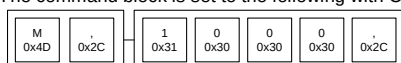
Dispensing mode: Retain current Dispensing time: 1.234 s Dispensing pressure: 111.1 kPa Vacuum pressure: -0.1 kPa

* If P.UNIT is set to kPa

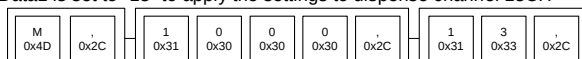
- The request packet begins with the following header indicating the start



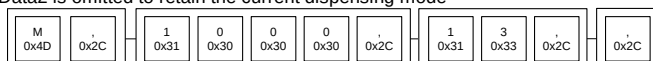
- The command block is set to the following with Command 1000 that changes dispensing conditions



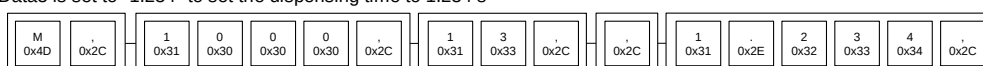
- Data1 is set to "13" to apply the settings to dispense channel 13CH



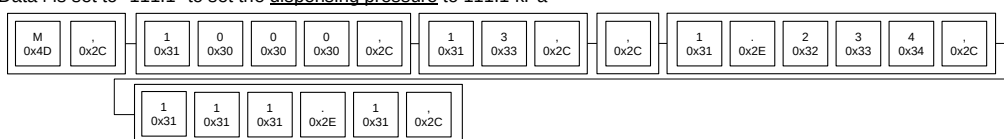
- Data2 is omitted to retain the current dispensing mode



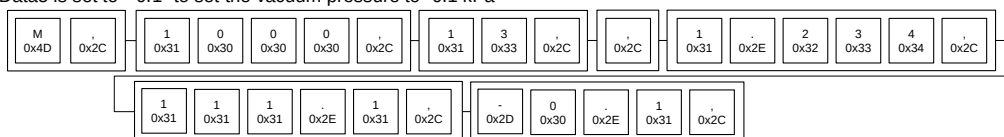
- Data3 is set to "1.234" to set the dispensing time to 1.234 s



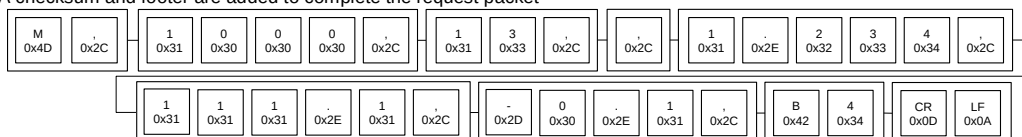
- Data4 is set to "111.1" to set the dispensing pressure to 111.1 kPa



- Data5 is set to "-0.1" to set the vacuum pressure to -0.1 kPa



- A checksum and footer are added to complete the request packet



Example of reading a request packet

Normal response

◇ Command 2000

[Function] Acquires the dispensing conditions of the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	DIS-mode 0 = TIMED mode 1 = MANUAL mode	0	1	[-]
3	Dispensing time	0.010	9999.999	[s]
4	Dispensing pressure	V5	30.0	[kPa]
			0.300	[bar]
		V2	5.0	[kPa]
			0.050	[bar]
5	Vacuum pressure	-5.00	-0.00	[kPa]
		-0.0500	-0.0000	[bar]

◆If acquire parameter Command 2000 is received under the following status

Device status

Dispense channel : 13CH

Dispensing mode : TIMED mode

Dispensing time : 6.34 s

Dispensing pressure : 75 kPa

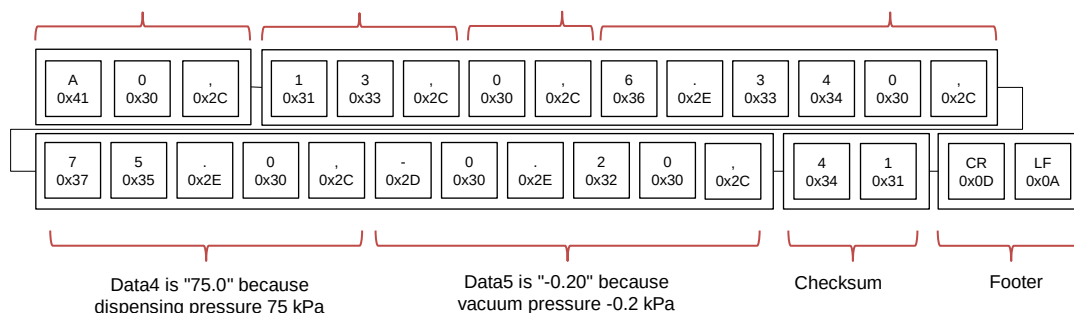
Vacuum pressure : -0.2 kPa

Header error
type is "0" for
normal response

Data1 is "13" because
dispense channel 13CH

Data2 is "0" because
Dispensing mode TIMED
mode

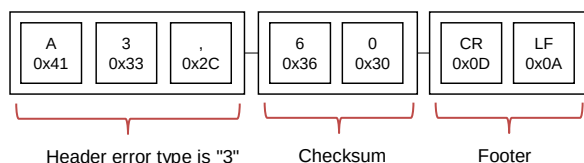
Data3 is "6.340" because
dispensing time 6.34 s



Error response other than Error Type 7

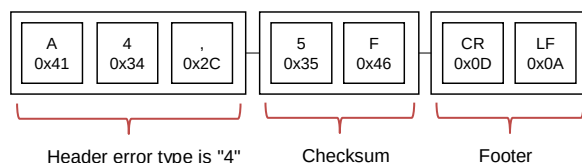
Error type 3:

After receiving a request packet header, the footer was not received within 3 seconds



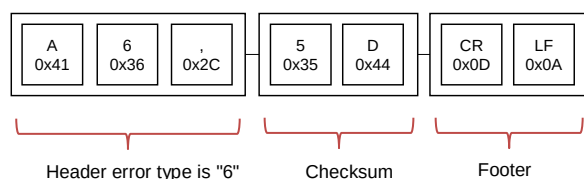
Error type 4:

The request packet checksum does not match, or cannot be read (there is data other than hexadecimal numbers)



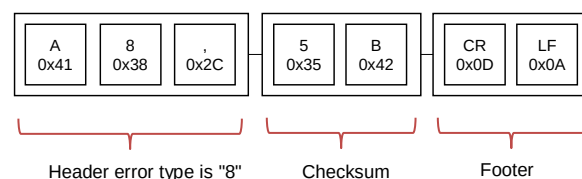
Error type 6:

Request packet command is undefined



Error type 8:

Received a request packet in status other than wait status



Error response with Error Type 7

◇ Command 1014

[Function] Sets the Interval feature.

[Data]

Request packet data

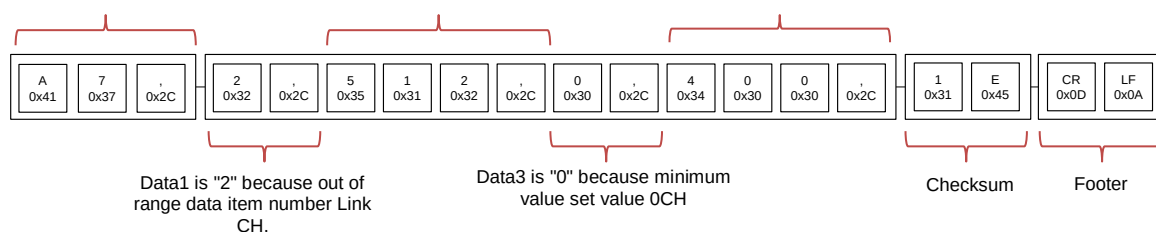
No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Interval feature OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
2	Link CH.	○	0	400	[CH]
3	Count	○	1	9999	[times]
4	Interval	○	0.010	9999.999	[s]
5	Repeat	○	1	9999	[times]
6	Wait time	○	0.010	9999.999	[s]
7	Interval mode 0 = SET 1 = MAN	○	0	1	[-]

Response packet data during normal response: None

◆ If a set parameter command 1014 was received, but the data value corresponding to Link CH. was 512, which was out of range

Header error type is out of setting range error "7"

Data2 is "512" because set value 512 Data4 is "400" because maximum set value 400CH



Command

◇ Command 0000

[Function] This is a dispense request.
In TIMED mode, it is sent once to perform dispense for the set time.
In MANUAL mode, the first transmission is to start dispense, and the second is to stop dispense.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0001

[Function] Moves the dispense channel up 1CH. Same function as STEP signal.
* Error response A8 if dispense channel is 400CH.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0002

[Function] Moves the dispense channel down 1CH. Same function as DOWN signal.
* Error response A8 if dispense channel is 1CH.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0003

[Function] Changes the dispense channel to 1CH. If the current channel belongs to a recipe, it changes to Start CH.
Same function as PRESET signal.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0004

[Function] Sets the dispensing mode of the current channel to TIMED mode.
* Error response A8 if the current channel belongs to a recipe and the Auto-increment function is ON.
*Error response A8 if in MANUAL mode of Σ (sigma) mode.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0005

[Function] Sets the dispensing mode of the current channel to MANUAL mode.
 * Error response A8 if the current channel belongs to a recipe and the Auto-increment function is ON.
 * Enters dummy shot mode if in TIMED mode of Σ (sigma) mode.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0006

[Function] Performs a DIS-monitoring test.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0007

[Function] Changes to the specified dispense channel.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	x	1	400	[CH]

Response packet data during normal response: None

◇ Command 0008

[Function] Copies settings of the current dispense channel to the specified dispense channel.
 * Take care, as if this command is performed, the status cannot be restored to the previous status.
 * Error response A8 if the specified channel belongs to a recipe and the Auto-increment function is ON.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	x	1	400	[CH]

Response packet data during normal response: None

◇ Command 0009

[Function] Changes to Start CH. of the specified recipe.
 * Error response A2 if an unregistered recipe number is specified.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	x	1	100	[-]

Response packet data during normal response: None

◇ Command 0010

[Function] Deletes the specified recipe.

* Take care, as if this command is performed, the status cannot be restored to the previous status.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]

Response packet data during normal response: None

◇ Command 0011

[Function] This diagnoses any faults with the dispensing pressure, vacuum pressure and solenoid valve.

* Perform this after removing the adapter tube.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing pressure 1 = Vacuum pressure 2 = Solenoid valve	×	0	2	[-]
2	Start/Cancel diagnosis 0 = Cancel 1 = Start	×	0	1	[-]

Response packet data during normal response: None

◇ Command 0012

[Function] Performs auto-tuning of the connected temperature controller.

* Perform this after removing the syringe from the temperature unit block.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Temperature unit number	×	1	5	[-]
2	Start/cancel Auto-tuning 0 = Cancel 1 = Start	×	0	1	[-]

Response packet data during normal response: None

◇ Command 0013

[Function] Resets the User counter to default (zero).

* Take care, as if this command is performed, the status cannot be restored to the previous status.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ **Command 0014**

[Function] Clears the error log.
* Take care, as if this command is performed, the status cannot be restored to the previous status.
* Please do not power off in 20 seconds after the command execution.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ **Command 0015**

[Function] Resets the settings saved for each channel to their default values.
* Take care, as if this command is performed, the status cannot be restored to the previous status.
* Please do not power off in 20 seconds after the command execution.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ **Command 0016**

[Function] Restores all settings except language and communication settings to default.
* Take care, as if this command is performed, the status cannot be restored to the previous status.
* Please do not power off in 20 seconds after the command execution.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ **Command 0017**

[Function] [Function] Performs the sampling process necessary to enable Σ (sigma) mode for the current dispense channel.
The syringe connected to the device must be full for the sampling process.
* Error response A8 if in Σ (sigma) mode.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ **Command 0018**

[Function] Performs the sampling process necessary to enable automatic setting of remaining vol. corr. for the current dispense channel.
The syringe connected to the device must be empty for the sampling process.
* Error response A8 if in Normal mode.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 0019

[Function] Performs the sampling process necessary to enable more detailed correction for remaining vol. corr. for the current dispense channel.

The syringe connected to the device must have the remaining volume specified in the request packet data for the sampling process.

* Error response A8 if in Normal mode.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Remaining volume in syringe	×	1	99	[%]

Response packet data during normal response: None

◇ Command 0020

[Function] Cancels the Σ (sigma) mode of the specified dispense channel.

* Take care, as if this command is performed, the status cannot be restored to the previous status.

[Data]

* If zero is specified for the dispense channel, this is applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response: None

◇ Command 0021

[Function] Sets the ECO (sigma) mode of the Current dispense channel.

* Error response A8 if the current channel belongs to a recipe and the Auto-increment function is ON.

* Error response A8 if in Σ (sigma) mode.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	ECO mode 0 = OFF 1 = ON	×	0	1	[-]

Response packet data during normal response: None

◇ Command 0022

[Function] Turns off the solenoid valve alarm signal.

* This setting resets to the default setting of enabled when the power supply is turned ON again.

[Data]

Request packet data: None

Response packet data during normal response: None

◇ Command 1000

[Function] Sets the dispensing conditions of the specified dispense channel as a batch.

* Error A8 will be come out when all of below conditions are applied.

- Indicated channel is belonging to the recipe
- Status of Auto-Increment function is ON
- Dispensing mode is changed

* Error response A8 if in Σ (sigma) mode.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

Request packet data						
No.	Details		Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel		○	0	400	[CH]
2	DIS-mode 0 = TIMED mode 1 = MANUAL mode		○	0	1	[-]
3	Dispensing time		○	0.010	9999.999	[s]
4	Dispensing pressure	V5	○	30.0	500.0	[kPa]
			○	0.300	5.000	[bar]
		V2	○	5.0	200.0	[kPa]
			○	0.050	2.000	[bar]
5	Vacuum pressure		○	-5.00	-0.00	[kPa]
			○	-0.0500	-0.0000	[bar]

Response packet data during normal response: None

◇ Command 1001

[Function] Sets the delay timer for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	On delay timer	○	0.000	9.999	[s]
3	Off delay timer	○	0.000	9.999	[s]

Response packet data during normal response: None

◇ Command 1002

[Function] Sets DIS-monitoring for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Allowable	○	10.0	100.0	[kPa]
		○	0.100	1.000	[bar]
3	Dispensing OFF time	○	0.050	9.999	[s]

Response packet data during normal response: None

◇ Command 1003

[Function] Sets channel Name for the specified dispense channel.

[Data]

* Place the character string within double quotation marks.

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Channel name*	○	0	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

* For "double quotation marks (0x22)," set "\" (0x5C) + double quotation marks (0x22)."

* For "\" (0x5C)," set "\" (0x5C) + \" (0x5C)."

Response packet data during normal response: None

◇ Command 1004

[Function] Sets the DIS-auxiliary setting of the specified dispense channel.

* Error response A8 if in Σ (sigma) mode.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL 7 = W 5mL 8 = W 10mL 9 = W 30mL 10 = W 55mL	○	0	10	[-]
3	Adapter tube length 1 = 0.5 m 2 = 1.0 m 3 = 1.5 m 4 = 2.0 m	○	1	4	[-]
4	Plunger enabled/disabled 0 = None 1 = Use	○	0	1	[-]
5	ECO mode 0 = OFF 1 = ON	○	0	1	[-]

Response packet data during normal response: None

◇ Command 1005

[Function] Sets Automatic correction of volume reduction for Σ function for the specified dispense channel.
* Error response A8 if in Normal mode.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Vol. reduction corr. OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
3	Alpha correction value	○	-200	999	[-]
4	Delta correction	○	50	200	[%]

Response packet data during normal response: None

◇ Command 1006

[Function] Sets fluid-drip prevention for Σ function for the specified dispense channel.
* Error response A8 if in Normal mode.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Fluid-drip prevention OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
3	Vacuum correction value	○	-5.00 ^{*1}	5.00 ^{*2}	[kPa]
			-0.0500 ^{*1}	0.0500 ^{*2}	[bar]

*1 The maximum value changes to “(-1) x vacuum pressure setting” when the Σ (sigma) mode is started.

*2 The minimum value changes to “vacuum pressure setting” when the Σ (sigma) mode is started.

Response packet data during normal response: None

◇ Command 1007

[Function] Sets automatic remaining volume alert function for Σ function for the specified dispense channel.
* Error response A8 if in Normal mode.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Remaining vol. alert OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
3	Remaining volume detection level	○	0	99	[%]
4	RSM count	○	1	10	[times]

Response packet data during normal response: None

◇ Command 1008

[Function] Sets remaining vol. corr. for Σ function for the specified dispense channel.
* Error response A8 if in Normal mode.

[Data]

* If zero is specified for the dispense channel, the changes are applied to the selected dispense channel when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]
2	Remaining vol. corr. OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
3	Correction value	○	-99	99	[%]

Response packet data during normal response: None

◇ Command 1009

[Function] Sets the recipe to the specified recipe number.
* Error response A2 if a channel already registered to a recipe is included between the specified Start CH. and End CH.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]
2	Start CH.	×	1	400	[CH]
3	End CH.	×	Start CH.	400	[CH]

Response packet data during normal response: None

◇ Command 1010

[Function] Sets the Auto-increment function of the specified recipe number.
* Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]
2	Auto-increment Function OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
3	Auto-increment Function dispensing mode 0 = TIMED mode 1 = MANUAL mode	○	0	1	[-]
4	ECO mode 0 = OFF 1 = ON	○	0	1	[-]
5	Switching Condition 0 = Total time 1 = Total count 2 = Pot life	○	0	2	[-]
6	Switching value	Total time	○	1	9999 [s]
		Total count	○	1	9999 [shot]
		Pot life	○	1	9999 [s]

Response packet data during normal response: None

◇ Command 1011

[Function] Sets the details of the specified recipe number.
 * Error response A2 if an non-registered recipe number is specified.

[Data]
 * Place the character string within double quotation marks.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]
2	Recipe name*	○	0	14	[Character]
3	Nozzle*	○	0	14	[Character]
4	Liquid material*	○	0	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

* For "double quotation marks (0x22)," set "\" (0x5C) + double quotation marks (0x22)."

* For "\" (0x5C)," set "\" (0x5C) + \" (0x5C)."

Response packet data during normal response: None

◇ Command 1012

[Function] Sets the temperature controller conditions of the specified recipe number.
 * Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]
2	Temp. unit controller number	×	1	5	[-]
3	Control OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
4	Setting temperature	○	0.0	100.0	[°C]
5	Upper Limit	○	0.0	100.0	[°C]
6	Lower Limit	○	-100.0	-0.0	[°C]

Response packet data during normal response: None

◇ **Command 1013**

[Function] Sets DIS-correction.

[Data]

Request packet data

Request packet data							
No.	Details			Can be Omitted	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing time 1 = Dispensing pressure 2 = Vacuum pressure			×	0	2	[-]
2	Correction method 0 = OFF 1 = Correction amount 2 = Correction ratio			○	0	2	[-]
3	Correction amount	For the applicable dispensing time		○	-9.999	9.999	[s]
		For the applicable dispensing pressure	V5	○	-500.0	500.0	[kPa]
				○	-5.000	5.000	[bar]
			V2	○	-200.0	200.0	[kPa]
				○	-2.000	2.000	[bar]
		For the applicable vacuum pressure		○	-5.00	5.00	[kPa]
				○	-0.0500	0.0500	[bar]
4	Correction ratio			○	1	999	[%]

Response packet data during normal response: None

◇ **Command 1014**

[Function] Sets the Interval feature.

[Data]

Request packet data

No.	Details		Can be Omitted	Minimum value	Maximum value	Unit
1	Interval feature OFF/ON 0 = OFF 1 = ON		○	0	1	[-]
2	Link CH. 0 = Link CH. Unused Non-zero = Designated channel		○	0	400	[CH]
3	Count		○	1	9999	[times]
4	Interval		○	0.010	9999.999	[s]
5	Repeat		○	1	9999	[times]
6	Wait time		○	0.010	9999.999	[s]
7	Interval mode 0 = MAN 1 = SET		○	0	1	[-]

Response packet data during normal response: None

◇ Command 1015

[Function] Changes the I/O signal settings.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Series 0 = ML-808GX 1 = SuperΣ	○	0	1	[-]
2	Output signal spec. 0 = In-dispensing 1 = Dispense complete	○	0	1	[-]
3	ENABLE function 0 = Inactive 1 = Active	○	0	1	[-]
4	On Error 0 = OFF 1 = ON	○	0	1	[-]

Response packet data during normal response: None

◇ Command 1016

[Function] Changes Configure setting.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Language 0 = English 1 = Japanese 2 = Chinese	○	0	2	[-]
2	P.UNIT 0 = kPa 1 = bar	○	0	1	[-]
3	Buzzer OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
4	T-ready Enable 0 = OFF 1 = ON	○	0	1	[-]
5	Warm-ready enable 0 = OFF 1 = ON	○	0	1	[-]
6	Pop-up auto OFF 0 = OFF 1 = ON	○	0	1	[-]
7	Immed. Inc. (Alnc) 0 = OFF 1 = ON	○	0	1	[-]
8	Control mode 0 = SPD 1 = STB	○	0	1	[-]
9	Sleep timer 0 = OFF 5 = 2 hours (Manual) 1 = 5 minutes 6 = 3 hours 2 = 10 minutes 7 = 4 hours 3 = 30 minutes 8 = 5 hours 4 = 1 hours	○	0	8	[-]

Response packet data during normal response: None

◇ Command 1017

[Function] Changes alert function setting

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Empty alarm 0 = OFF (disabled) 1 = ON (enabled)	○	0	1	[-]
2	Empty warning threshold	○	1	999999999	[shot]
3	Valve warning threshold	○	1	999999999	[shot]
4	RSM advance notice lv.	○	0	50	[%]

Response packet data during normal response: None

◇ Command 1018

[Function] Sets the password lock.

[Data]

* Place the character string within double quotation marks.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Password lock OFF/ON 0 = OFF 1 = ON	○	0	1	[-]
2	Password*	○	0	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

* For "double quotation marks (0x22)," set "\" (0x5C) + double quotation marks (0x22)."

* For "\" (0x5C)," set "\" (0x5C) + \" (0x5C)."

Response packet data during normal response: None

◇ **Command 1019**

[Function] Sets the input signal function allocation.

* Allocated functions cannot be duplicated. Error response A2 if duplicated.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	D-Sub3 PIN	○	0	18	[-]
2	D-Sub4 PIN	○	0	18	[-]
3	D-Sub5 PIN	○	0	18	[-]
4	D-Sub6 PIN	○	0	18	[-]
5	D-Sub7 PIN	○	0	18	[-]
6	D-Sub8 PIN	○	0	18	[-]
7	D-Sub15 PIN	○	0	18	[-]
8	D-Sub16 PIN	○	0	18	[-]
9	D-Sub17 PIN	○	0	18	[-]
10	D-Sub18 PIN	○	0	18	[-]
11	D-Sub19 PIN	○	0	18	[-]
12	D-Sub20 PIN	○	0	18	[-]

Functions that can be allocated to each data are as follows.

0 =	ENABLE signal	9 =	CHS signal
1 =	STEP signal	10 =	CD01 signal
2 =	PRESET signal	11 =	CD02 signal
3 =	DOWN signal	12 =	CD04 signal
4 =	TMD signal	13 =	CD08 signal
5 =	TMS signal	14 =	CD10 signal
6 =	C_CNT signal	15 =	CD20 signal
7 =	EDIT signal	16 =	CD40 signal
8 =	RESET signal	17 =	CD80 signal
		18 =	SLEEP signal

Response packet data during normal response: None

◇ Command 1020

[Function] Sets the output signal allocation.

* Allocated functions cannot be duplicated. Error response A2 if duplicated.

[Data]

Request packet data

• Series settings for ML-808GX

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	D-Sub9 PIN	○	0	15	[-]
2	D-Sub10 PIN	○	0	15	[-]
3	D-Sub12 PIN	○	0	15	[-]
4	D-Sub21 PIN	○	0	15	[-]
5	D-Sub22 PIN	○	0	15	[-]
6	D-Sub23 PIN	○	0	15	[-]

• Series settings for SuperΣ

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	D-Sub10 PIN	○	0	15	[-]
2	D-Sub22 PIN	○	0	15	[-]
3	D-Sub23 PIN	○	0	15	[-]

Functions that can be allocated to each data are as follows.

0	=	ERROR signal	8	=	ESTS signal
1	=	ALARM signal	9	=	EWAIT signal
2	=	ALARM2 signal	10	=	T_READY signal
3	=	RSM signal	11	=	EXE1 signal
4	=	READY signal	12	=	EXE2 signal
5	=	PCS signal	13	=	PSE1 signal
6	=	AI_END signal	14	=	PSE2 signal
7	=	S_END signal	15	=	RSM_P signal

Response packet data during normal response: None

◇ Command 1021

[Function] Sets the date and time.

* Error response A2 if date and time that does not exist is set.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Year	○	2000	2099	[Year]
2	Month	○	1	12	[Month]
3	Day	○	1	31	[Day]
4	Hour	○	0	23	[Hour]
5	Minute	○	0	59	[Minute]
6	Second	○	0	59	[Second]

Response packet data during normal response: None

◇ Command 1022

[Function] Set your time zone.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Time zone	○	0	37	[-]

The selectable time zones are as follows.

0 = UTC -12:00	10 = UTC -03:30	20 = UTC +04:30	30 = UTC +09:30
1 = UTC -11:00	11 = UTC -03:00	21 = UTC +05:00	31 = UTC +10:00
2 = UTC -10:00	12 = UTC -02:00	22 = UTC +05:30	32 = UTC +10:30
3 = UTC -09:30	13 = UTC -01:00	23 = UTC +05:45	33 = UTC +11:00
4 = UTC -09:00	14 = UTC +00:00	24 = UTC +06:00	34 = UTC +12:00
5 = UTC -08:00	15 = UTC +01:00	25 = UTC +06:30	35 = UTC +12:45
6 = UTC -07:00	16 = UTC +02:00	26 = UTC +07:00	36 = UTC +13:00
7 = UTC -06:00	17 = UTC +03:00	27 = UTC +08:00	37 = UTC +14:00
8 = UTC -05:00	18 = UTC +03:30	28 = UTC +08:45	
9 = UTC -04:00	19 = UTC +04:00	29 = UTC +09:00	

Response packet data during normal response: None

◇ Command 1023

[Function] Set a controller name.

[Data]

* Place the character string within double quotation marks.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Controller name	×	0	10	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

* For "double quotation marks (0x22)," set "\" (0x5C) + double quotation marks (0x22)."

* For "\" (0x5C)," set "\" (0x5C) + \" (0x5C)."

Response packet data during normal response: None

◇ Command 1024

[Function] Set IP address.

* The setting changes will take effect after reboot.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	1st byte from the top	○	0	255	[-]
2	2nd byte from the top	○	0	255	[-]
3	3rd byte from the top	○	0	255	[-]
4	4th byte from the top	○	0	255	[-]

Response packet data during normal response: None

◇ Command 1025

[Function] Set a subnet mask.
* The setting changes will take effect after reboot.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	1st byte from the top	○	0	255	[-]
2	2nd byte from the top	○	0	255	[-]
3	3rd byte from the top	○	0	255	[-]
4	4th byte from the top	○	0	255	[-]

Response packet data during normal response: None

◇ Command 1026

[Function] Set a default gateway.
* The setting changes will take effect after reboot.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	1st byte from the top	○	0	255	[-]
2	2nd byte from the top	○	0	255	[-]
3	3rd byte from the top	○	0	255	[-]
4	4th byte from the top	○	0	255	[-]

Response packet data during normal response: None

◇ Command 1027

[Function] Set the administrator password.
[Data]
* Place the character string within double quotation marks.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Administrator password	○	0	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

* For "double quotation marks (0x22)," set "\" (0x5C) + double quotation marks (0x22)."

* For "\" (0x5C)," set "\" (0x5C) + \" (0x5C)."

Response packet data during normal response: None

◇ Command 1028

[Function] Set the web page refresh interval.
[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Web page refresh interval	○	0	60	[Minute]

Response packet data during normal response: None

◇ Command 1029

[Function] Set OFF/ ON of Read only.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	OFF/ON of Read only 0 = OFF 1 = ON	○	0	1	[-]

Response packet data during normal response: None

◇ Command 2000

[Function] Acquires the dispensing conditions of the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

Response packet data during normal response

No.	Details		Minimum value	Maximum value	Unit
1	Dispense channel		0	400	[CH]
2	DIS-mode 0 = Normal / TIMED mode 1 = NORMAL / MANUAL mode: 2 Σ / TIMED mode 3 Σ / MANUAL mode 4 Σ / Dummy shot mode		0	4	[-]
3	Dispensing time		0.001	9999.999	[s]
4	Dispensing pressure	V5	30.0	500.0	[kPa]
			0.300	5.000	[bar]
		V2	5.0	200.0	[kPa]
			0.050	2.000	[bar]
5	Vacuum pressure		-5.00	-0.00	[kPa]
			-0.0500	-0.0000	[bar]

◇ Command 2001

[Function] Acquires settings of the delay timer for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	On delay timer	0.000	9.999	[s]
3	Off delay timer	0.000	9.999	[s]

◇ Command 2002

[Function] Acquires DIS-monitoring function settings of the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

Response packet data during normal response					
No.	Details		Minimum value	Maximum value	Unit
1	Dispense channel		0	400	[CH]
2	Allowable	V5	10.0	100.0	[kPa]
			0.100	1.000	[bar]
		V2	10.0	100.0	[kPa]
			0.100	1.000	[bar]
3	Dispensing OFF time		0.010	9.999	[s]

◇ Command 2003

[Function] Acquires channel name of the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details		Minimum value	Maximum value	Unit
1	Dispense channel		0	400	[CH]
2	Channel name * The response is placed within double quotation marks.		0	14	[Character]

◇ Command 2004

[Function] Acquires the DIS-auxiliary setting of the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL 7 = W 5mL 8 = W 10mL 9 = W 30mL 10 = W 55mL	0	10	[-]
3	Adapter tube length 1 = 0.5 m 2 = 1.0 m 3 = 1.5 m 4 = 2.0 m	1	4	[-]
4	Plunger enabled/disabled 0 = None 1 = Use	0	1	[-]
5	ECO mode 0 = OFF 1 = ON	0	1	[-]

◇ Command 2005

[Function] Acquires settings of automatic correction of volume reduction function for Σ function for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	Vol. reduction corr. OFF/ON 0 = OFF 1 = ON	0	1	[-]
3	Alpha correction value	-200	999	[-]
4	Delta correction	50	200	[%]

◇ Command 2006

[Function] Acquires setting of fluid-drip prevention for Σ function for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	Fluid-drip prevention OFF/ON 0 = OFF 1 = ON	0	1	[-]
3	Vacuum correction value	-5.00 -0.0500	5.00 0.0500	[kPa] [bar]

◇ Command 2007

[Function] Acquires settings of automatic remaining volume alert function for Σ function for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	Remaining vol. alert OFF/ON 0 = OFF 1 = ON	0	1	[-]
3	Remaining volume detection level	0	99	[%]
4	RSM count	1	10	[times]

◇ Command 2008

[Function] Acquires settings of remaining vol. corr. for Σ function for the specified dispense channel.

[Data]

* If zero is specified for the dispense channel, the settings from the selected dispense channel are acquired when the packet is received.

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Dispense channel	○	0	400	[CH]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense channel	0	400	[CH]
2	Remaining vol. corr. OFF/ON 0 = OFF 1 = ON	0	1	[-]
3	Correction value	-99	99	[%]

◇ Command 2009

[Function] Acquires the settings of the specified recipe number.
If the specified recipe number is undefined, data is omitted and the response will only be " , " (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Recipe number	1	100	[-]
2	Start CH.	1	400	[CH]
3	End CH.	1	400	[CH]

◇ Command 2010

[Function] Acquires the Auto-increment function settings of the specified recipe number.
If the specified recipe number is undefined, data is omitted and the response will only be " , " (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]

Response packet data during normal response

Response packet data during normal response					
No.	Details		Minimum value	Maximum value	Unit
1	Recipe number		1	100	[-]
2	Auto-increment Function OFF/ON 0 = OFF 1 = ON		0	1	[-]
3	Auto-increment Function dispensing mode 0 = TIMED mode 1 = MANUAL mode		0	1	[-]
4	Auto-increment Function ECO mode 0 = OFF 1 = ON		0	1	[-]
5	Switching Condition 0 = Total time 1 = Total count 2 = Pot life		0	2	[-]
6	Switching value	Total time	1	9999	[s]
		Total count	1	9999	[shot]
		Pot life	1	9999	[s]

◇ **Command 2011**

[Function] Acquires detailed information of the specified recipe number.
If the specified recipe number is undefined, data is omitted and the response will only be " , " (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Recipe number	1	100	[-]
2	Recipe name * The response is placed within double quotation marks.	0	14	[Character]
3	Nozzle * The response is placed within double quotation marks.	0	14	[Character]
4	Liquid material * The response is placed within double quotation marks.	0	14	[Character]

◇ Command 2012

[Function] Acquires the specified temperature controller conditions within the specified recipe.
If the specified recipe number is undefined, data is omitted and the response will only be " , " (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Recipe number	×	1	100	[-]
2	Temp. unit controller number	×	1	5	[-]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Recipe number	1	100	[-]
2	Temp. unit controller number	1	5	[-]
3	Control OFF/ON 0 = OFF 1 = ON	0	1	[-]
4	Setting temperature	0.0	100.0	[°C]
5	Upper Limit	0.0	100.0	[°C]
6	Lower Limit	0.0	100.0	[°C]

◇ Command 2013

[Function] Acquires the DIS-correction settings.

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing time 1 = Dispensing pressure 2 = Vacuum pressure	×	0	2	[-]

Response packet data during normal response

Response packet data during normal response						
No.	Details			Minimum value	Maximum value	Unit
1	Applicable 0 = Dispensing time 1 = Dispensing pressure 2 = Vacuum pressure			0	2	[-]
2	Correction method 0 = OFF 1 = Correction amount 2 = Correction ratio			0	2	[-]
3	Correction amount	For the applicable dispensing time		-99.999	99.999	[s]
		For the applicable dispensing pressure	V5	-500.0	500.0	[kPa]
				-5.000	5.000	[bar]
			V2	-200.0	200.0	[kPa]
				-2.000	2.000	[bar]
		For the applicable vacuum pressure		-5.00	5.00	[kPa]
		-0.0500	0.0500	[bar]		
4	Correction ratio			1	999	[%]

◇ Command 2014

[Function] Acquires the Interval feature settings.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Interval feature OFF/ON 0 = OFF 1 = ON	0	1	[-]
2	Link CH.	0	400	[CH]
3	Count	1	9999	[times]
4	Interval	0.010	9999.999	[s]
5	Repeat	1	9999	[times]
6	Wait time	0.010	9999.999	[s]
7	Interval mode 0 = MAN 1 = SET	0	1	[-]

◇ Command 2015

[Function] Acquires the I/O signal settings.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Series 0 = ML-808GX 1 = SuperΣ	0	1	[-]
2	Output signal spec. 0 = In-dispensing 1 = Dispense complete	0	1	[-]
3	ENABLE function 0 = Inactive 1 = Active	0	1	[-]
4	On Error 0 = OFF 1 = ON	0	1	[-]

◇ **Command 2016**

[Function] Acquires the Configure setting.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Language 0 = English 1 = Japanese 2 = Chinese	0	2	[-]
2	P.UNIT 0 = kPa 1 = bar	0	1	[-]
3	Buzzer OFF/ON 0 = OFF 1 = ON	0	1	[-]
4	T-ready Enable 0 = OFF 1 = ON	0	1	[-]
5	Warm-ready enable 0 = OFF 1 = ON	0	1	[-]
6	Pop-up auto OFF 0 = OFF 1 = ON	0	1	[-]
7	Immed. Inc. (Alnc) 0 = OFF 1 = ON	0	1	[-]
8	Control mode 0 = STB 1 = SPD	0	1	[-]
9	Sleep timer 0 = OFF (Manual) 1 = 5 minutes 2 = 10 minutes 3 = 30 minutes 4 = 1 hours 5 = 2 hours 6 = 3 hours 7 = 4 hours 8 = 5 hours	0	8	[-]

◇ **Command 2017**

[Function] Acquires alert function setting.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Empty alarm 0 = OFF (disabled) 1 = ON (enabled)	0	1	[-]
2	Empty warning threshold	1	999999999	[shot]
3	Valve warning threshold	1	999999999	[shot]
4	RSM advance notice lv.	0	50	[%]

◇ **Command 2018**

[Function] Acquires the password lock settings.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Password lock OFF/ON 0 = OFF 1 = ON	0	1	[-]
2	Password * The response is placed within double quotation marks.	0	14	[Character]

◇ Command 2019

[Function] Acquires the input signal function allocation settings.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	D-Sub3 PIN	0	18	[-]
2	D-Sub4 PIN	0	18	[-]
3	D-Sub5 PIN	0	18	[-]
4	D-Sub6 PIN	0	18	[-]
5	D-Sub7 PIN	0	18	[-]
6	D-Sub8 PIN	0	18	[-]
7	D-Sub15 PIN	0	18	[-]
8	D-Sub16 PIN	0	18	[-]
9	D-Sub17 PIN	0	18	[-]
10	D-Sub18 PIN	0	18	[-]
11	D-Sub19 PIN	0	18	[-]
12	D-Sub20 PIN	0	18	[-]

◇ Command 2020

[Function] Acquires the output signal function allocation settings.

[Data]

Request packet data: None

Response packet data during normal response

• Series settings for ML-808GX

No.	Details	Minimum value	Maximum value	Unit
1	D-Sub9 PIN	0	15	[-]
2	D-Sub10 PIN	0	15	[-]
3	D-Sub12 PIN	0	15	[-]
4	D-Sub21 PIN	0	15	[-]
5	D-Sub22 PIN	0	15	[-]
6	D-Sub23 PIN	0	15	[-]

• Series settings for SuperΣ

No.	Details	Minimum value	Maximum value	Unit
1	D-Sub10 PIN	0	15	[-]
2	D-Sub22 PIN	0	15	[-]
3	D-Sub23 PIN	0	15	[-]

◇ Command 2021

[Function] Acquires the date and time.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Year	2000	2099	[Year]
2	Month	1	12	[Month]
3	Day	1	31	[Day]
4	Hour	0	23	[Hour]
5	Minute	0	59	[Minute]
6	Second	0	59	[Second]

◇ **Command 2022**

[Function] Import the time zone.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Time zone	0	37	[-]

◇ **Command 2023**

[Function] Import the controller name.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Controller name * The response is placed within double quotation marks.	0	10	[Character]

◇ **Command 2024**

[Function] Acquires the IP address.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	1st byte from the top	0	255	[-]
2	2nd byte from the top	0	255	[-]
3	3rd byte from the top	0	255	[-]
4	4th byte from the top	0	255	[-]

◇ **Command 2025**

[Function] Import the subnet mask.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	1st byte from the top	0	255	[-]
2	2nd byte from the top	0	255	[-]
3	3rd byte from the top	0	255	[-]
4	4th byte from the top	0	255	[-]

◇ **Command 2026**

[Function] Import the default gateway.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	1st byte from the top	0	255	[-]
2	2nd byte from the top	0	255	[-]
3	3rd byte from the top	0	255	[-]
4	4th byte from the top	0	255	[-]

◇ **Command 2027**

[Function] Import the administrator password.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Administrator password * The response is placed within double quotation marks.	0	14	[Character]

◇ **Command 2028**

[Function] Import the web page refresh interval.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Web page refresh interval	0	60	[Minute]

◇ **Command 2029**

[Function] Import the OFF/ON of Read only.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	OFF/ON of Read only 0 = OFF 1 = ON	0	1	[-]

◇ Command 3000

[Function] Acquires the controller status.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispense status 0 = Dispense waiting 1 = In-dispensing	0	1	[-]
2	Dispense availability 0 = Dispense possible 1 = Dispense not possible	0	1	[-]
3	Error status	0	2147483647	[-]
4	Alarm status	0	2147483647	[-]

bit	Error status	bit	Error status
0	"Setting data load error"	10	"Temp. unit error: HB warning"
1	"Startup error"	11	"Temp. unit error: Timeout"
2	"Image data error"	12	"Temp. unit error: Setting error"
3	"RTC error"	13	"Internal error 1"
4	"Solenoid valve error"	14	"Internal error 2"
5	"Air supply pressure error"	15	"Internal error 3"
6	"Exhaust error"	16	"Internal error 4"
7	"Vacuum pressure error"	17	Reserved
8	"Temp. unit error: HS warning"	18	Reserved
9	"Temp. unit error: Input error"	19	"Internal error 7"

bit	Alarm status	bit	Alarm status
0	"Air supply pressure warning"	3	"Temp. unit alarm: Warning 1"
1	"Solenoid valve alarm"	4	"Temp. unit alarm: Warning 2"
2	"Empty alarm"	5	"DIS-monitoring alarm"
		6	"Σ (sigma) mode remaining amount alert"

◇ Command 3001

[Function] Acquires the current air supply pressure.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Current air supply pressure	0	999	[kPa]
		0	9.99	[bar]

◇ Command 3002

[Function] Acquires the stopwatch (previous dispensing time) value.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Stopwatch	0.001	9999.999	[s]

◇ Command 3003

[Function] Acquires the current dispense channel.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Current dispense channel	1	400	[CH]

◇ Command 3004

[Function] Acquires the current channel recipe number.

If it does not belong to a recipe, data is omitted and the response will only be " , (0x2C)" (comma).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Current recipe number	1	100	[-]

◇ Command 3005

[Function] Acquires the Auto-increment function progress.

If the function is OFF, data is omitted and the response will only be " , (0x2C)" (comma).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details		Minimum value	Maximum value	Unit
1	Progress status 0 = Waiting 1 = Progressing 2 = Complete		0	2	[-]
2	Remaining volume	Total time	0	9999	[s]
		Total count	0	9999	[shot]
		Pot life	0	9999	[s]

◇ Command 3006

[Function] Acquires the DIS-monitoring test status.

During testing, result data is omitted and the response will only be " , " (comma).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details		Minimum value	Maximum value	Unit
1	DIS-monitoring test status 0 = Waiting 1 = Testing		0	1	[-]
2	DIS-monitoring test result 0 = Not performed 1 = NG 2 = OK		0	2	[-]

◇ Command 3007

[Function] Acquires empty alarm state.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Current remaining volume	0	100	[%]
2	Current detection count*	0	10	[times]

* If the automatic remaining volume alert function is OFF, data is omitted and the response will only be " , " (comma).

◇ Command 3008

[Function] Acquires the current temperature of the temperature unit.

Error A2 will be come out when current channel is not belonging to any recipe.

When current channel is belonging to the recipe and also temperature control function is OFF, feedback data will not be displayed but show " , (0x2C)" (comma).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Current temperature of temperature controller 1	0.0	999.9	[°C]
2	Current temperature of temperature controller 2	0.0	999.9	[°C]
3	Current temperature of temperature controller 3	0.0	999.9	[°C]
4	Current temperature of temperature controller 4	0.0	999.9	[°C]
5	Current temperature of temperature controller 5	0.0	999.9	[°C]

◇ Command 3009

[Function] Acquires the value after DIS-correction.

If the DIS-correction correction method is OFF, data is omitted and the response will only be " , (0x2C)" (comma).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Dispensing time after DIS-correction	0.010	9999.999	[s]
2	Dispensing pressure after DIS-correction	V5	30.0	[kPa]
			0.300	[bar]
		V2	5.0	[kPa]
			0.050	[bar]
3	Vacuum pressure after DIS-correction	-5.00	-0.00	[kPa]
		-0.0500	-0.0000	[bar]

◇ Command 3010

[Function] Acquires the Interval feature progress status.

If the function is OFF, data is omitted and the response will only be " , " (comma).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Current count	0	9999	[times]
2	Current repeat	0	9999	[times]

◇ Command 3011

[Function] Acquires the counter (User counter, Total counter).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	User counter	0	999999999	[shot]
2	Total counter	0	999999999	[shot]

◇ Command 3012

[Function] Acquires the Diagnosis feature status.

During diagnosis, diagnosis result data is omitted and the response will only be " , " (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing pressure 1 = Vacuum pressure 2 = Solenoid valve	×	0	2	[-]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing pressure 1 = Vacuum pressure 2 = Solenoid valve	0	2	[-]
2	Diagnosis status 0 = Waiting 1 = Diagnosing	0	1	[-]
3	Diagnosis result 0 = Errors 1 = No errors	0	1	[-]

◇ Command 3013

[Function] Acquires the temperature unit auto-tuning status.

[Data]

Request packet data:

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Temperature unit number	×	1	5	[-]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Temperature unit number	1	5	[-]
2	Auto-tuning status 0 = NG 1 = OK 2 = Not performed 3 = Performing	0	3	[-]

◇ Command 3014

[Function] Acquires the total error log number.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Total error log number	0	1000	[logs]

◇ Command 3015

[Function] Acquires the specified error log (main).

If there is no specified error log, data is omitted and the response will only be " , (0x2C)" (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Error log number	×	1	1000	[logs]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Date of occurrence * The response is placed within double quotation marks.	2000/1/1 0:0:0	2099/12/31 23:59:59	[Character]
2	Error code	1	999	[-]
3	Total counter	0	999999999	[shot]

◇ **Command 3016**

[Function] Acquires the specified error log (sub).

If there is no specified error log, data is omitted and the response will only be " , (0x2C)" (comma).

If there is no related data, data is omitted and the response will be " , (0x2C)" (comma).

[Data]

Request packet data

No.	Details	Can be Omitted	Minimum value	Maximum value	Unit
1	Error log number	×	1	1000	[logs]

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Error pattern	0	9	[-]
2	Related data 1	See below.		
3	Related data 2			
4	Related data 3			
5	Related data 4			
6	Related data 5			
7	Related data 6			

If error pattern is "0"

Related data	Details	Minimum value	Maximum value	Unit
1 to 6	None	-	-	-

If error pattern is "1"

Related data	Details	Minimum value	Maximum value	Unit
1	Air Supply Pressure	0	800	[kPa]
		0.00	8.00	[bar]
2	DIS-mode 0 = Normal / TIMED mode 1 = NORMAL / MANUAL mode: 2 = Σ / TIMED mode 3 = Σ / MANUAL mode 4 = Σ / Dummy shot mode	0	4	[-]
3	Dispensing time	0.010	9999.999	[s]
4	Dispensing pressure	30.0	500.0	[kPa]
		0.300	5.000	[bar]
		5.0	200.0	[kPa]
		0.050	2.000	[bar]
5	Vacuum pressure	-5.00	-0.00	[kPa]
		-0.0500	-0.0000	[bar]
6	Dispense status when error occurred	0	1	[-]

If error pattern is "2"

Related data	Details	Minimum value	Maximum value	Unit
1	Default mode	1	2	[-]
	1 = Screen operation 2 = RS-232C communications			
2 to 6	None	-	-	-

If error pattern is "3"

Related data	Details	Minimum value	Maximum value	Unit
1	Temperature unit number	1	5	[-]
2	Current temperature	0.0	999.9	[°C]
3	Setting temperature	0.0	100.0	[°C]
4	Upper Limit	0.0	100.0	[°C]
5	Lower Limit	0.0	100.0	[°C]
6	None	-	-	-

If error pattern is "4"

Related data		Details		Minimum value	Maximum value	Unit
1	Air Supply Pressure			0	800	[kPa]
				0.00	8.00	[bar]
2	Dispensing time			0.010	9999.999	[s]
3	Dispensing pressure	V5		30.0	500.0	[kPa]
				0.300	5.000	[bar]
		V2		5.0	200.0	[kPa]
				0.050	2.000	[bar]
4	Vacuum pressure			-5.00	-0.00	[kPa]
				-0.0500	-0.0000	[bar]
5	User counter			0	999999999	[-]
6	Pressure reached			0	999.9	[kPa]
				0.000	9.999	[bar]

If error pattern is "5"

Related data	Details	Minimum value	Maximum value	Unit
1	Air Supply Pressure	0	800	[kPa]
		0.00	8.00	[bar]
2	Diagnosis result 0 = Failure 1 = Pass	0	1	[-]
3 to 6	None	-	-	-

If error pattern is "6"

Related data	Details	Minimum value	Maximum value	Unit
1	Error response	0	7	[-]
2 to 6	None	-	-	-

◇ Command 9996

[Function] Acquires the MAC address.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	MAC address * The response is placed within double quotation marks.	00-00-00-00-00-00	FF-FF-FF-FF-FF-FF	[-]

◇ Command 9997

[Function] Import the I/O specifications.

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	I/O specifications * The response is placed within double quotation marks.	NPN	PNP	[-]

◇ Command 9998

[Function] Acquires command information (maximum number of various operation commands, maximum number of set parameter commands, maximum number of acquire parameter commands, maximum number of acquire model information commands).

[Data]

Request packet data: None

Response packet data during normal response

No.	Details	Minimum value	Maximum value	Unit
1	Maximum number of various operation commands	0	999	[-]
2	Maximum number of set parameter commands	0	999	[-]
3	Maximum number of acquire parameter commands	0	999	[-]
4	Maximum number of acquire model information commands	0	999	[-]

◇ Command 9999

[Function] Acquires model information (product model, Firmware 1, Firmware 2, Firmware 3).

[Data] Placed within double quotation marks and stored.

Request packet data: None

Response packet data during normal response

No.	Details		Minimum value	Maximum value	Unit
1	Product model	V5	S-SIGMA-CM4-V5	S-SIGMA-CM4-V5	[-]
		V2	S-SIGMA-CM4-V2	S-SIGMA-CM4-V2	
2	Firmware 1		SCM4-01.00	SCM4-99.99	[-]
3	Firmware 2		1.0	FFFF.FFFF	[-]
4	Firmware 3		0.0	FFFF.FFFF	[-]

Troubleshooting

If there are any problems, first read the following carefully and take appropriate action.
Contact our sales department if the problem is not resolved with the actions outlined below.
See the rear cover of this manual for contact details.

Symptoms	No.	Cause	Action
No response packet is received.	1	A cable is disconnected.	● Check the cable connection conditions.
	2	There is an error with the connection specifications.	● See P.3"RS-232C Communication Connection Specifications".
	3	There is an error with the communication speed.	● Adjust the communication speed to match the host device.
	4	There is an error with the communication specifications and connection signal.	● See P.3"RS-232C Communication Specifications".
	5	There is an error with the communication port.	● Check the communication port of the host device.
The response packet error type is "3," indicating a timeout error.	6	Unable to receive footer ("CR(0x0D)"+"LF(0x0A)."	● Check the connection status.
The response packet error type is "4," indicating a checksum error.	7	There is an error with the checksum.	● See P.11"Checksum Calculation" and check if there are any errors with the checksum calculation method.
	8	A checksum has not been appended, even though the checksum setting is "ON."	● See P.11"Checksum Calculation" and append a checksum. ● See P.5"Checksum" and turn the checksum setting to OFF.
The response packet error type is "6," indicating a command undefined error.	9	The header does not have " , (0x2C)."	● The header is comprised of ASCII code "M(0x4D)" + " , (0x2C)." Check the request packet.
	10	There is an error with the command.	● See P.12"Request Packet List" and correct the command.
The response packet error type is "7," indicating an out of setting range error.	11	The data is outside the allowable setting range.	● See the relevant command in P.20"Command" and correct the data.
	12	There is no " , (0x2C)" at the end of each data.	● " , (0x2C)" is required at the end of each data. See P.17"Example of creating a request packet" and create the request packet.
The response packet error type is "8," indicating an inexecutable error.	13	Not in receive status	● See the receive statuses outlined in P.8"Command Types" and perform the following operation as necessary. ➤ Tap the [←BACK] or [←] icon to display the Dispense screen. ➤ If in In-dispensing, wait until dispense end. (P.50)

3 Com. Protocol: Type C

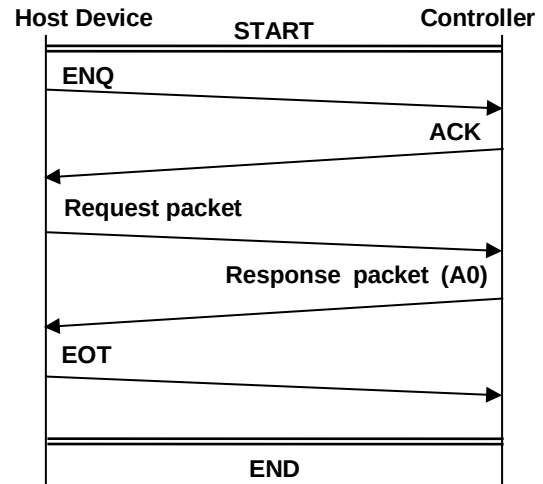
Communication Procedure

Communications between this device and the host device is performed as follows.

Communication Sequence

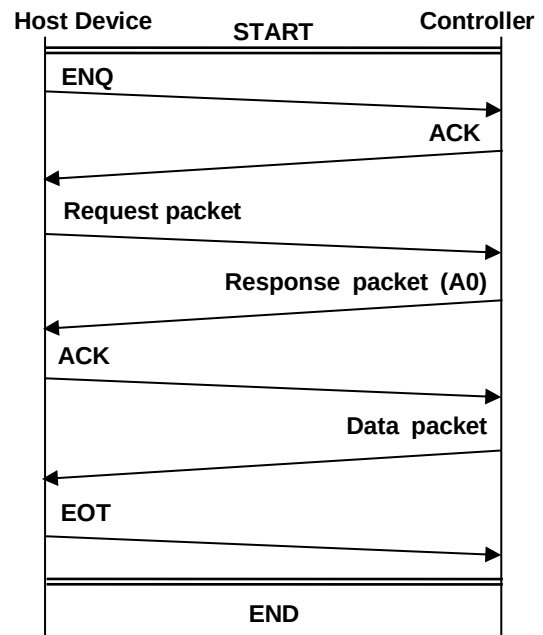
Download sequence

1. Starts when the control code "ENQ" is sent from the host device.
2. The controller receiving the control code "ENQ" sends the control code "ACK" to the host device, and is in packet waiting status.
3. Send a request packet from the host device.
4. The controller receiving the request packet sends a response packet to the host device.
If there are no problems with the packet content or controller status, and the request packet is processed correctly, the response packet error type is "0," indicating a normal response.
5. Send the control code "EOT" from the host device to end the communication.



Upload sequence

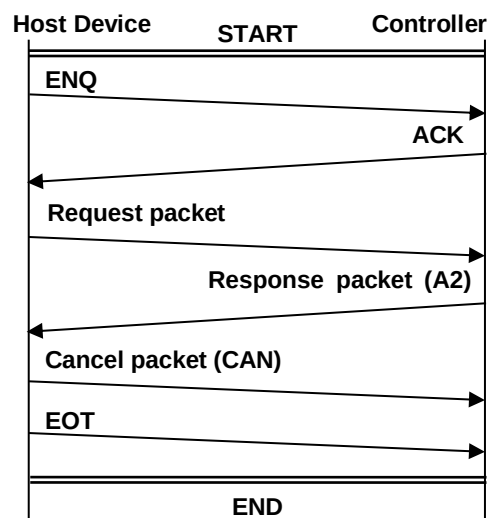
1. Starts when the control code "ENQ" is sent from the host device.
2. The controller receiving the control code "ENQ" sends the control code "ACK" to the host device, and is in packet waiting status.
3. Send a request packet from the host device.
4. The controller receiving the request packet sends a response packet to the host device.
If there are no problems with the packet content or controller status, and the request packet is processed correctly, the response packet error type is "0," indicating a normal response.
5. Send the control code "ACK" from the host device.
6. The controller receiving the control code "ACK" sends a data packet.
7. Send the control code "EOT" from the host device to end the communication.



Cases Causing Errors

- Incorrect packet received or cannot be received

If the request packet cannot be received for any reason, the controller sends a response packet to the host device. The response packet error type is "2" indicating an error response. Send the cancel packet and control code "EOT" from the host device to end the communication.



- Timeout

If the controller is unable to receive the request packet or control code during communication, the communication will be forced to end after 3 seconds.

Control Code

Control Code	Data	Application
STX	0x02	Always use for the packet header.
ETX	0x03	Always use for the packet footer.
EOT	0x04	Used when all sequences end.
ENQ	0x05	Used for transmission start request from the host computer.
ACK	0x06	Used for the response to the transmission start request from the host computer.

Error Types

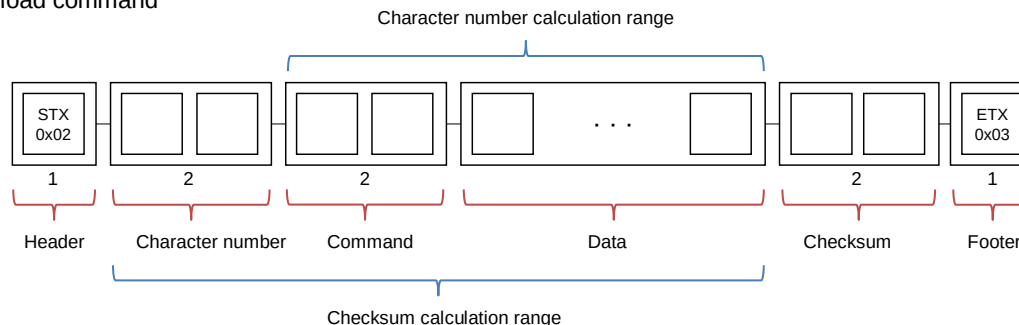
Error Types	Details	Cause	Action
0	Normal response	-	-
2	Error	There is an error with the communication procedure or packet format.	Correct the communication procedure or request packet.

Packet Format

• Request packet

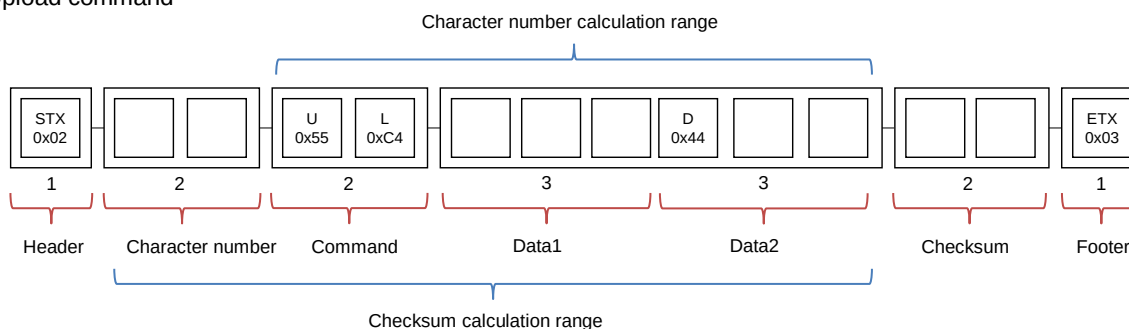
Packet configuration

Download command



Block	Data length	Role
Header	1-character fixed length	The packet header is indicated by ASCII code "STX(0x02)".
Character number	2-character fixed length	The sum of the command block and data block characters is indicated as a 2-digit hexadecimal. Each digit character is handled as ASCII code.
Command	2-character fixed length	Indicates a command.
Data	Variable length	Indicates data defined by each command. Separate with the specified delimiter or " , (0x2C)" (comma) between data if there are multiple blocks of data. Enter "SP(0x20)" + "SP(0x20)" if there is no data.
Checksum	2-character fixed length	Data for detecting incorrect transmission data. The borrow of header data from the character number to the final data of the data block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	1-character fixed length	The packet footer is indicated by ASCII code "ETX(0x03)".

Upload command

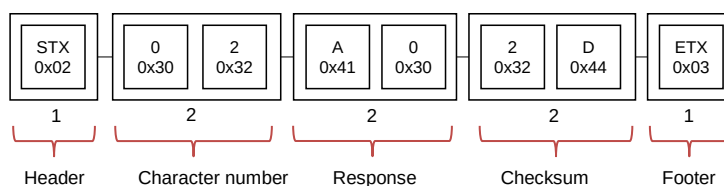


Block	Data length	Role
Header	1-character fixed length	The packet header is indicated by ASCII code "STX(0x02)".
Character number	2-character fixed length	The sum of the command block and data block characters is indicated as a 2-digit hexadecimal. Each digit character is handled as ASCII code.
Command	2-character fixed length	Indicates a command.
Data1	3-character fixed length	Indicates data defined by UL command.
Data2	3-character fixed length	Indicates data packet no. defined by UL command.
Checksum	2-character fixed length	Data for detecting incorrect transmission data. The borrow of header data from the character number to the final data of the data block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	1-character fixed length	The packet footer is indicated by ASCII code "ETX(0x03)".

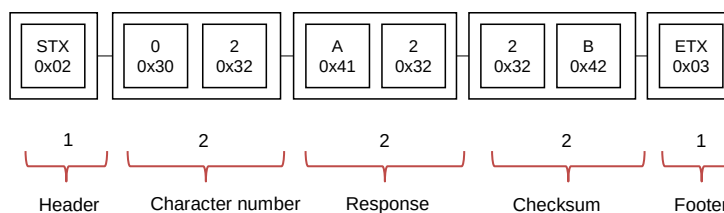
- Response packet

Packet configuration

Normal response



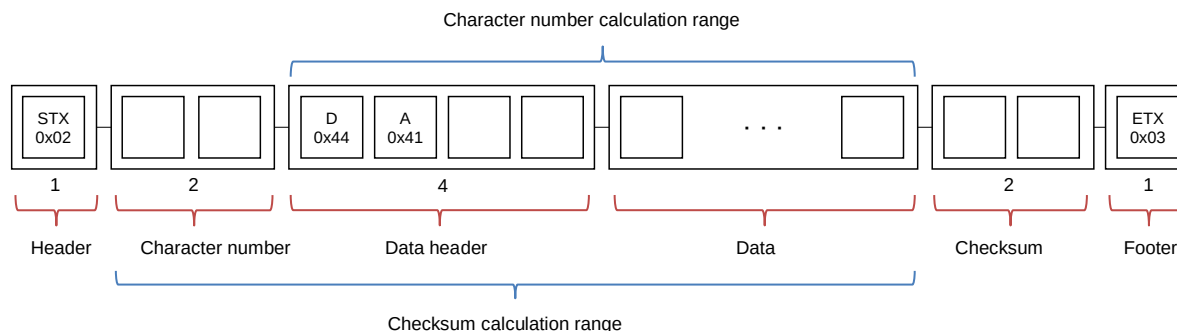
On Error response



Block	Data length	Role
Header	1-character fixed length	The packet header is indicated by ASCII code "STX(0x02)".
Character number	2-character fixed length	The sum of the response block characters is indicated as a 2-digit hexadecimal. Each digit character is handled as ASCII code.
Response	2-character fixed length	Normal response: ASCII code "A(0x41)" + "0(0x30)" is stored. Error response: ASCII code "A(0x41)" + "2(0x32)" is stored.
Checksum	2-character fixed length	Data for detecting incorrect transmission data. The borrow of header data from the character number to the final data of the response block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	1-character fixed length	The packet footer is indicated by ASCII code "ETX(0x03)".

- Data packet

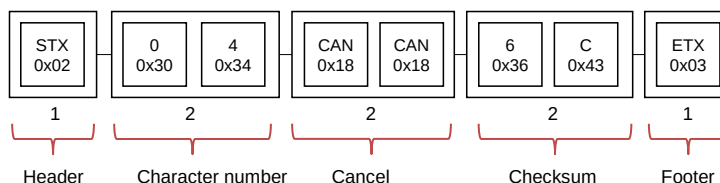
Packet configuration



Block	Data length	Role
Header	1-character fixed length	The packet header is indicated by ASCII code "STX(0x02)".
Character number	2-character fixed length	The sum of the data header block and data block characters is indicated as a 2-digit hexadecimal. Each digit character is handled as ASCII code.
Data header	4-character fixed length	Indicates start of the data block. The 2 characters of the first half indicate the ASCII code "D(0x44)" + "A(0x41)", and the 2 characters of the second half indicate the data packet no. specified in ASCII code.
Data	Variable length	Indicates data defined by each command. A prefix is appended to data that requires a prefix. ", (0x2C)" (comma) is appended to data that does not require a prefix.
Checksum	2-character fixed length	Data for detecting incorrect transmission data. The borrow of header data from the character number to the final data of the response block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	1-character fixed length	The packet footer is indicated by ASCII code "ETX(0x03)".

- Cancel packet

Packet configuration



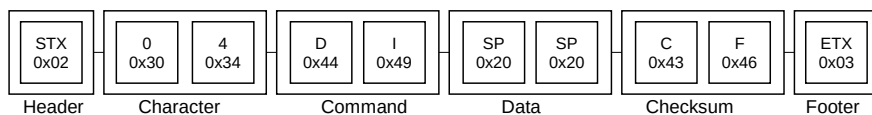
Block	Data length	Role
Header	1-character fixed length	The packet header is indicated by ASCII code "STX(0x02)".
Character number	2-character fixed length	The sum of the cancel block characters is indicated as a 2-digit hexadecimal. Each digit character is handled as ASCII code.
Cancel	2-character fixed length	Enter ASCII code "CAN(0x18)" + "CAN(0x18)."
Checksum	2-character fixed length	Data for detecting incorrect transmission data. The borrow of header data from the character number to the final data of the cancel block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal ASCII code.
Footer	1-character fixed length	The packet footer is indicated by ASCII code "ETX(0x03)".

Checksum Calculation

The following outlines the dispense request command packet sample and the checksum calculation method.

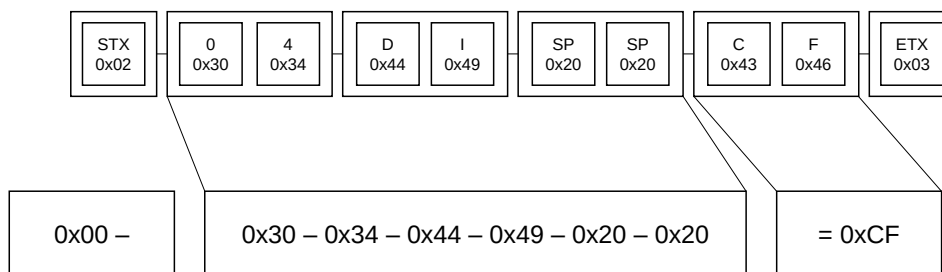
Dispense request command packet sample

For the dispense request DI command, the packet is as follows.



Checksum calculation

The borrow of header data of the header block to the final data of the data block is ignored from zero for 8bit subtraction, with the result indicated as a 2-digit hexadecimal. The result converted to an ASCII code is the checksum



Request Packet List

Download command

No.	Details	Character number	Command	Data	Checksum
1	Cancel	04	*P.63	-	6E
2	Dispense request	04	DI	□□	CF
3	Start Σ sampling	04	SM	□□□	BC
4	Dispensing mode switching	04	TM	□□	Variable
5	User counter clear	04	C8	□□	E1
6	Channel load (2 digits)	04	LH	□□	Variable
7	Channel load (3 digits)	05	LH	□□□	Variable
8	DIS-condition setting	14	PR	□□P□□□□T□□□□V-□□□□	Variable
9	DIS-condition setting	22	DL	□□□P□□□□T□□□□□V□□□□M□N△△△△△△△△△△	Variable
10	Syringe size setting	07	SY	□□□S□	Variable
11	Σ parameter setting	14	PM	□□□A+□□□D□□□V+□□□□	Variable
12	Remaining volume detection parameter setting	0B	RS	□□□LocC□□	Variable
13	Cancel Σ mode	05	CL	□□□	Variable
14	Solenoid valve alarm OFF	02	WO	-	F8
15	DIS-condition setting	24	AL	□□□P□□□□T□□□□□□□V□□□□M□N△△△△△△△△△△	Variable
16	Plunger setting	03	PL	□	Variable
17	Control mode setting	03	VC	□	Variable
18	Syringe size setting	08	SS	□□□S□□	Variable
19	Step up	04	01	□□	FB
20	Step down	04	02	□□	FA
21	preset	04	03	□□	F9
22	DIS-monitoring test	04	06	□□	F6
23	Channel copy	07	08	□□□□□	Variable
24	Recipe load	07	09	□□□□□	Variable
25	Delete recipe	07	0A	□□□□□	Variable
26	Diagnosis feature	07	0B	□□□□□	Variable
27	Temperature unit auto-tuning	07	0C	□□□□□	Variable
28	Error log clear	04	0E	□□	E7
29	Parameter clear	04	0F	□□	E6
30	All memory clear	04	0G	□□	E5
31	Empty sampling	04	0I	□□	E2
32	Detailed sampling	06	0J	□□□□□	Variable
33	ECO mode OFF/ON	05	0L	□□□	Variable
34	Delay timer setting	11	11	□□□□□,□□□□□,□□□□□	E2
35	DIS-monitoring function setting	11	12	□□□□□,□□□□□,□□□□□	Variable
34	Channel name setting	16	13	□□□□□,△△△△△△△△△△△△△△△△	Variable
35	DIS-auxiliary setting	10	14	□□□□□,□□□□□,□□□□□	Variable
36	Vol. reduction corr. setting	12	15	□□□□□,□,±□□□□,□□□□	Variable
37	Fluid-drip prevention setting	0E	16	□□□□□,□,±□□□□	Variable
38	Remaining vol. alert setting	0F	17	□□□□□,□□□□□	Variable
39	Remaining vol. corr. setting	0D	18	□□□□□,□,±□□□	Variable
40	Recipe settings	0F	19	□□□□□,□□□□□,□□□□□	Variable
41	Auto-increment function settings	14	1A	□□□□□,□□□□□,□□□□□	Variable
42	Recipe name settings	16	1B	□□□□□,△△△△△△△△△△△△△△△△	Variable
43	Nozzle name setting	16	1C	□□□□□,△△△△△△△△△△△△△△△△	Variable
44	Liquid material name setting	16	1D	□□□□□,△△△△△△△△△△△△△△△△	Variable
45	Temperature unit conditions setting	1A	1E	□□□□□,□□□□□,□□□□□,□□□□□	Variable
46	Setting DIS-correction	11	1F	□□□□□,□,±□□□□□,□□□□	Variable
47	Interval Feature Setting	25	1G	□□□□□,□□□□□□□□□□,□□□□□□□□□□,□□□□□□□□□□,□□□□□	Variable
48	Signal setting	0B	1H	□□□□□,□□□□□	Variable
49	Configure setting	14	1I	□□□□□,□□□□□,□□□□□,□□□□□	Variable
50	Remaining vol. alert function setting	1C	1J	□□□□□,□□□□□□□□□□,□□□□□□□□□□,□□	Variable
51	Password lock setting	14	1L	□□□□□,△△△△△△△△△△△△△△△△	Variable
52	Input signal allocation setting	27	1M	□□□□□,□□□□□,□□□□□,□□□□□,□□□□□,□□□□□,□□□□□,□□□□□	Variable
53	Output signal allocation setting	808	1N	□□□□□,□□□□□,□□□□□,□□□□□	Variable
		Σ	1N	□□□□□,□□□□□	Variable
54	Time adjust	17	1O	□□□□□□□□□□,□□□□□□□□□□	Variable
55	Controller name setting	0E	1Q	□□□□□,△△△△△△△△△△△△△△△△	Variable

* "□": ASCII code "SP(0x20)"

* "□": ASCII code "0(0x30) to 9(0x39)"

* "△": ASCII code "SP(0x20) to |(0x7E)"

Upload command

No.	Details	Character number	Command	Data1 and 2	Checksum
1	Acquires various settings and statuses	08	UI	000D00	Variable

Data Packet List

No.	Details	Corresponding data2	Data header and data	
1	Warnings and error occurrence status	D00	DA00□□□□□□□□	
2	Dispensing parameters (dispensing time is 5 digits)	D01	DA01P□□□□T□□□□□V□□□□M□N△△△△△△△△△△	
3	Syringe size and adapter tube length	D02	DA02S□A□	
4	Σ (sigma) mode correction values	D03	DA03A±□□□□D□□□V±□□□□	
5	Remaining volume detection parameters	D04	DA04L□□□□C□□□	
6	Currently detected remaining liquid material	D05	DA05R□□□	
7	Currently displayed channel number	D06	DA06CH□□□	
8	User counter value	D07	DA07CT□□□□□□□□	
9	Version information	D08	DA08VR□□□□□SP□	
10	Remaining volume detection correction information	D12	DA12F□C±□□□	
11	Warnings and error occurrence status	D13	DA13□□□□□□□□□□	
12	Plunger setting	D14	DA14P□	
13	Acquires multiple dispensing parameters at the same time	D15	DA15P□□□□T□□□□□□□□V□□□□M□N△△△△△△△△△△	
14	User counter value	D16	DA16UC□□□□□□□□□□	
15	Total counter value	D17	DA17TC□□□□□□□□□□	
16	Control mode setting	D18	DA18V□	
17	Syringe size and adapter tube length	D23	DA23S□□A□	
18	Acquires delay timer	D30	DA30,□□□□,□□□□	
19	Acquires DIS-monitoring function setting	D31	DA31,□□□□□□□□□□	
20	Acquisition of channel name	D32	DA32, △△△△△△△△△△△△△△	
21	Acquires ECO mode setting	D33	DA33,□	
22	Σ function: Acquires Vol. reduction corr. setting	D34	DA34,□,±□□□,□□□	
23	Σ function: Acquires fluid-drip prevention setting	D35	DA35,□,±□□□□	
24	Σ function: Acquires remaining vol. alert setting	D36	DA36,□,□□,□□	
25	Σ function: Acquires remaining vol. corr. setting	D37	DA37,□,±□□	
26	Acquires recipe	D38	DA38,□□□,□□□□	
27	Acquires auto-increment function setting	D39	DA39,□,□,□,□,□,□□□□,□□□□,□□□□	
28	Acquires recipe details	D40	DA40,△△△△△△△△△△,△△△△△△△△△△ ,△△△△△△△△△△	
29	Acquires temperature unit conditions	D41	DA41,□,□□□□,□□□□,□□□□,□,□□□□,□□□□,□□□□ ,□,□□□□,□□□□,□□□□,□□□□,□,□□□□,□□□□,□□□□ ,□,□□□□,□□□□,□□□□	
30	Acquires dispensing parameter correction function setting	D42	DA42,□,□,±□□□□□□□□,□□□	
31	Acquires Interval feature setting	D43	DA43,□,□□□,□□□□,□□□□□□□□,□□□□,□□□□□□□□,□	
32	Acquires I/O signal setting	D44	DA44,□,□,□,□	
33	Acquires configure setting	D45	DA45,□,□,□,□,□,□,□,□,□,□	
34	Acquires alert function setting	D46	DA46,□,□□□□□□□□□□,□□□□□□□□□□,□□	
35	Acquires password lock setting	D48	DA48,□,△△△△	
36	Acquires input signal allocation setting	D49	DA49,□□,□□,□□,□□,□□,□□,□□,□□,□□,□□,□□,□□,□□,□□	
37	Acquires output signal allocation setting	808 Σ	D50	DA50,□□,□□,□□,□□,□□,□□ DA50,□□,□□,□□
38	Acquires time (year, month, day, hour, minute, second)	D51	DA51,□□□□,□□,□□,□□,□□,□□,□□	
39	Acquires controller name	D52	DA52.□□□□□□□□□□	

* "□": ASCII code "0(0x30) to 9(0x39)"

* "Δ": ASCII code "SP(0x20) to |(0x7E)"

* "+": ASCII code "+(0x2B)" or "-(0x2D)"

No.	Details			Corresponding data2	Data header and data
40	Acquires dispense status			D53	DA53,0,0,0000000000,0000000000
41	Acquires air supply pressure			D54	DA54,0000
42	Acquires stopwatch value			D55	DA55,00000000
43	Acquires current recipe			D56	DA56,0000
44	Acquires Auto-increment status			D57	DA57,000,0000
45	Acquires DIS-monitoring test status			D58	DA58,0,0
46	Acquires empty alarm state			D59	DA59,0000
47	Acquires temperature unit status			D60	DA60,0000,0000,0000,0000,0000
48	Acquires value after DIS-correction			D61	DA61,00000000,0000,0000
49	Acquires interval status			D62	DA62,0000,0000
50	Acquires diagnosis status			D63	DA63,0,0,0,0
51	Acquires temperature unit auto-tuning status			D64	DA64,0,0,0,0,0
52	Acquires total error log number			D65	DA65,0000
53	Acquires error log (main)			D66	DA66,0000/00/00_00:00:00,000,0000000000
54	Acquires error log (sub)	Error pattern	0	D67	DA67,0, , , , ,
			1		DA67,0,000,0,00000000,0000,000,0
			2		DA67,0,0, , , , ,
			3		DA67,0,0,0000,0000,0000,0000,
			4		DA67,0,000,00000000,0000,000,0000000000,0000
			5		DA67,0,000,0, , , ,
			6		DA67,0,0, , , , ,
55	Acquires model information			D68	DA68,0000000000000000,0000000000,0000000000,0000000000
56	Acquires Firmware 4			D69	DA69,0000000000

* "□": ASCII code "0(0x30) to 9(0x39)"

Response Packet List

No.	Details	Character number	Response	Checksum
1	Normal response	04	A0	2D
2	Error response	04	A2	2B

Description of Commands

Example of details

◇ Command

Download command

[Function] Function overview of the command.

[Data] Description of the data block. If multiple data blocks exist, always specify in order from the top. The chart outlines the following details.

Request packet data

No.	Details	Digit number	Minimum value	Maximum value	Unit

Request packet legend

STX	x	x	○	○	□	?	?	ETX
-----	---	---	---	---	---	---	---	-----

No. : Item number of data.

Details : Details of data.

Prefix : The prefix designated for each data. This is omitted if it is not required.

Digit number : The specified digit number for parameters. If the target number does not reach the specified digit number, always use a 0 to meet the specified digit number. Specify the number with ASCII code.

Minimum value : Minimum value that can be specified.

Maximum value : Maximum value that can be specified.

Unit : Unit of data.

Request packet legend : Example format of request packet sent from the host device.

【Notice】

Please do not send unauthorized command to avoid a malfunction.

Upload command

[Function] Function overview of the command.

[Data] Description of the data block. If multiple data blocks exist, always specify in order from the top. The chart outlines the following details.

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit

Request packet legend

STX	0	8	U	L	O	O	D	□	□	?	?	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit

Data packet legend

STX	x	x	D	A	□	□	Δ	Δ	?	?	ETX
-----	---	---	---	---	---	---	---	---	---	---	-----

No. : Item number of data.

Details : Details of data.

Prefix : The prefix designated for each data. This is omitted if it is not required.

Digit number : The specified digit number for parameters. If the target number does not reach the specified digit number, always use a 0 to meet the specified digit number. Specify the number with ASCII code.

Minimum value : Minimum value that can be specified.

Maximum value : Maximum value that can be specified.

Unit : Unit of data.

Request packet legend : Example format of request packet sent from the host device.

Data packet legend : Example format of data packet sent from this device.

Models with specification differences are named according to the following chart.

Model	Name
S-SIGMA-CM4-CTR-V5-(N/P)	V5
S-SIGMA-CM4-CTR-V2-(N/P)	V2

Example of creating a request packet

◇ DL Command

[Function] Sets the dispensing conditions of the specified dispense channel as a batch.

* Error A2 will be come out when all of below conditions are applied.

- Indicated channel is belonging to the recipe
- Status of Auto-Increment function is ON
- Dispensing mode is changed

* Error response A2 if in Σ (sigma) mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Dispensing pressure	P	4	300	5000	[0.1kPa]
			4	300	5000	[0.001bar]
			4	50	2000	[0.1kPa]
			4	50	2000	[0.001bar]
3	Dispensing time	T	5	1	9999	[0.001s]
4	Vacuum pressure	V	4	0	500	[-0.01kPa]
						[-0.0001bar]
5	DIS-mode	M	1	0	1	[-]
	0 = TIMED mode					
	1 = MANUAL mode					
6	Channel name*	N	10	10	10	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

STX	2	2	D	L	0	0	1	P	1	2	3	4	T	1	2	3	4	5	V	0	0	0	0
M	0	N	M	U	S	A	S	H	I	SP	SP	SP	B	3	ETX								

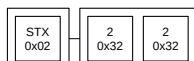
◆To change the dispensing conditions of dispense channel 13CH to the following, create a request packet using the flow below.

Dispensing pressure: 111.1 kPa Dispensing time: 1.234 s Vacuum pressure: -0.1 kPa
 Dispensing mode: TIMED mode Channel name: MUSASHI

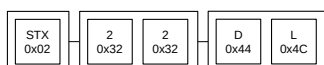
- The request packet begins with the following header indicating the start



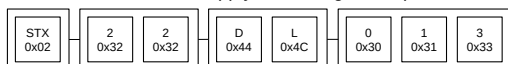
- The character number is set to "22" with DL Command that changes dispensing conditions



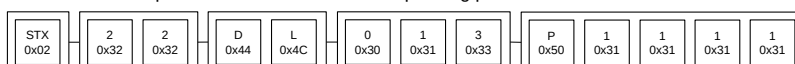
- Command is set to "DL"



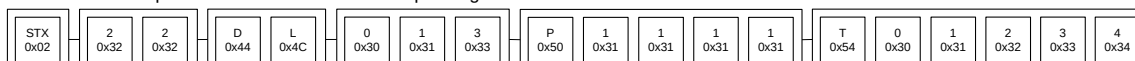
- Data1 is set to "013" to apply the settings to dispense channel 13CH



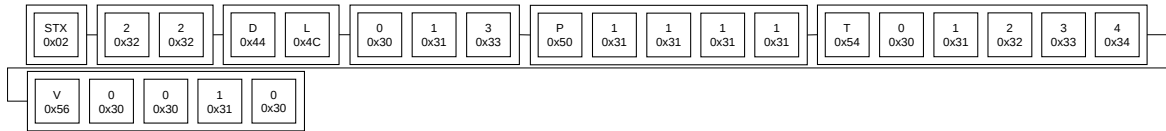
- Data2 is set with prefix to "P1111" to set the dispensing pressure to 111.1 kPa



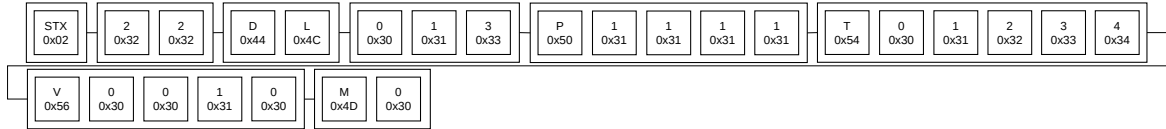
- Data3 is set with prefix to "T01234" to set the dispensing time to 1.234 s



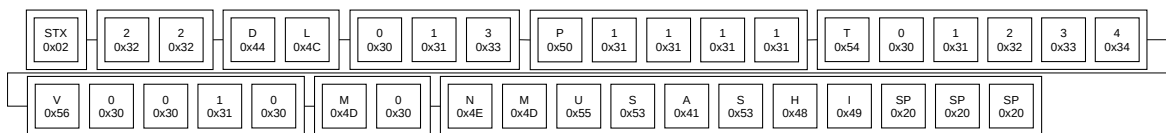
7. Data5 is set with prefix to "V0010" to set the vacuum pressure to -0.1 kPa



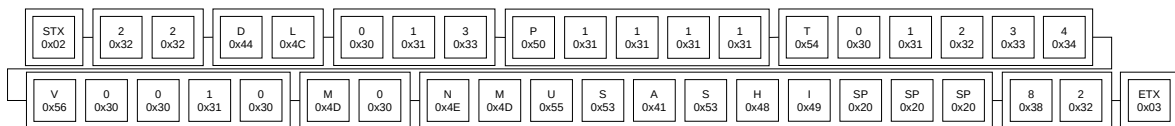
8. Data6 is set with prefix to "M0" to set the dispensing mode to TIMED mode.



9. Data7 is set with prefix to "NMUSASHI____" to set the channel name to "MUSASHI"
 * 10 characters are required for the channel name data, so when setting to 9 characters or less, fill the lacking part with spaces, etc.



10. A checksum and footer are added to complete the request packet



Data packet reading example

◇ UL Command

[Function] Acquires various settings and statuses

[Data]

[DA01 When acquiring dispensing parameters (dispensing time is 5 digits)]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	1	1	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	0	2	C	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

Data packet data							
No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispensing pressure	V5	P	4	300	5000	[0.1kPa]
						[0.001bar]	
	V2	50			2000	[0.1kPa]	
						[0.001bar]	
2	Dispensing time		T	5	1	99999	[0.001s]
3	Vacuum pressure		V	4	0	500	[-0.01kPa]
							[-0.0001kPa]
4	DIS-mode 0 = Normal / TIMED mode 1 = NORMAL / MANUAL mode: 2 = Σ / TIMED mode 3 = Σ / MANUAL mode 4 = Dummy shot mode		M	1	0	4	[-]
5	Channel name * Spaces are entered to fill blank space when there are less than 10 characters		N	10	10	10	[Character]

◆ If UL Command specifying dispense channel 13CH is received under the following status

Device status

Dispense channel: 13CH

Dispensing pressure : 75 kPa

Dispensing time : 1.234 s

Vacuum pressure: -0.10 kPa

Dispensing mode: Normal / TIMED mode

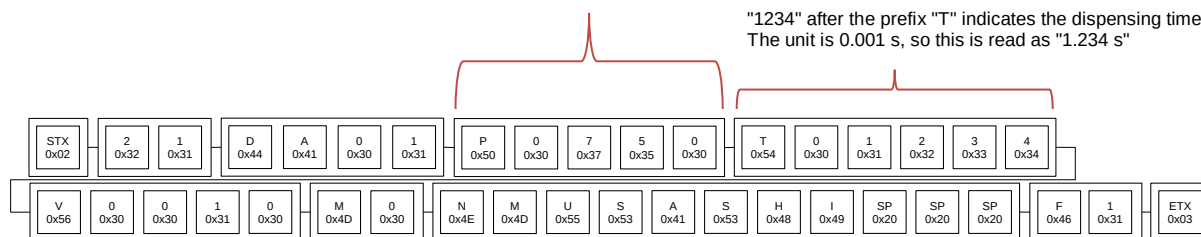
Channel name: MUSASHI

"0750" after the prefix "P" indicates the dispensing pressure

The unit is 0.1 kPa, so this is read as "75 kPa"

"1234" after the prefix "T" indicates the dispensing time

The unit is 0.001 s, so this is read as "1.234 s"



"0" after the prefix "M" indicates the dispensing mode
0 is read as "Normal / TIMED mode"

"MUSASHI" after the prefix "N" indicates the channel name

* 10 characters are required for the channel name data, so when setting to 9 characters or less, the lacking part is filled with spaces (0x20).

"0010" after the prefix "V" indicates the vacuum pressure

The unit is -0.01 kPa, so this is read as "-0.10 kPa"

Command

Command Type [1]

◇ DI Command

[Function] This is a dispense request.

In TIMED mode, it is sent once to perform dispense for the set time.

In MANUAL mode, the first transmission is to start dispense, and the second is to stop dispense.

[Data]

Request packet data: None

Request packet legend

STX	0	4	D	I	SP	SP	C	F	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ SM Command

[Function] [Function] Performs the sampling process necessary to enable Σ (sigma) mode for the current dispense channel.

The syringe connected to the device must be full for the sampling process.

* Error response A2 if in Σ (sigma) mode.

[Data]

Request packet data: None

Request packet legend

STX	0	4	S	M	SP	SP	B	C	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ TM Command

[Function] Changes the dispensing mode of the current channel.

* Error response A8 if the current channel belongs to a recipe and the Auto-increment function is ON.

[Data]

Request packet data: None

Request packet legend

STX	0	4	T	M	SP	SP	B	B	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ C8 Command

[Function] Resets the User counter to default (zero).

* Take care, as if this command is performed, the status cannot be restored to the previous status.

[Data]

Request packet data: None

Request packet legend

STX	0	4	C	8	SP	SP	2	3	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ **LH command (2 digits)**

[Function] Changes to the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	CH	2	1	99	[-]

Request packet legend

STX	0	4	L	H	1	2	A	5	ETX
-----	---	---	---	---	---	---	---	---	-----

◇ **LH command (3 digits)**

[Function] Changes to the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	CH	3	1	400	[CH]

Request packet legend

STX	0	5	L	H	1	2	3	7	1	ETX
-----	---	---	---	---	---	---	---	---	---	-----

◇ **PR Command**

[Function] Configures the dispensing pressure, dispensing time (4 digits), and vacuum pressure.

* Error response A2 if in Σ (sigma) mode.

* Use the AL command (P.77) to set dispensing time of 5 digits or more.

[Data]

Request packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispensing pressure	V5	P	4	300	5000	[0.1kPa]
				4	300	5000	[0.001bar]
		V2		4	50	2000	[0.1kPa]
				4	50	2000	[0.001bar]
2	Dispensing time		T	4	1	9999	[0.001s]
3	Vacuum pressure		V-	4	0	500	[0.01kPa] [0.0001bar]

Request packet legend

STX	1	4	P	R	SP	SP	P	1	2	3	4	T	1	2	3	4	V	-	0	0	0	0	3	E	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ DL Command

[Function] Sets the dispensing conditions of the specified dispense channel as a batch.

* Error A2 will be come out when all of below conditions are applied.

- Indicated channel is belonging to the recipe
- Status of Auto-Increment function is ON
- Dispensing mode is changed

* Error response A2 if in Σ (sigma) mode.

* Use the AL command (P.77) to set dispensing time of 6 digits or more.

[Data]

Request packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel		-	3	1	400	[CH]
2	Dispensing pressure	V5	P	4	300	5000	[0.1kPa]
				4	300	5000	[0.001bar]
		V2		4	50	2000	[0.1kPa]
				4	50	2000	[0.001bar]
3	Dispensing time		T	5	1	9999	[0.001s]
4	Vacuum pressure		V	4	0	500	[-0.01kPa] [-0.0001bar]
5	DIS-mode 0 = TIMED mode 1 = MANUAL mode		M	1	0	1	[-]
6	Channel name*		N	10	10	10	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	2	2	D	L	0	0	1	P	1	2	3	4	T	1	2	3	4	5	V	0	0	0	0
M	0	N	M	U	S	A	S	H	I	SP	SP	SP	B	3	ETX								

◇ SY Command

[Function] Configures the syringe size for the specified dispense channel.

* Use the SS command (P.78) to set the W-type syringe size.

[Data]

Request packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel		-	3	1	400	[CH]
2	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL		S	1	0	6	[-]

Request packet legend

STX	0	7	S	Y	0	0	1	S	0	D	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	-----

◇ PM Command

[Function] Configures parameters used in Σ (sigma) mode for the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Alpha correction value	A	4	-200	+999	[-]
3	Delta correction	D	3	50	200	[%]
4	Vacuum correction value	V	5	-500*1	+500*2	[0.01kPa] [0.0001bar]

*1 The maximum value changes to “(-1) x vacuum pressure setting” when the Σ (sigma) mode is started.

*2 The minimum value changes to “vacuum pressure setting” when the Σ (sigma) mode is started.

Request packet legend

STX	1	4	P	M	0	0	1	A	+	1	2	3	D	1	2	3	V	+	1	2	3	4	4	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ RS Command

[Function] Configures parameters used in remaining volume detection for the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Remaining volume detection level	L	2	0	99	[%]
3	RSM count	C	2	1	10	[times]

Request packet legend

STX	0	B	R	S	0	0	1	L	1	2	C	0	1	0	5	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ CL Command

[Function] Cancels the Σ (sigma) mode of the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]

Request packet legend

STX	0	5	C	L	0	0	1	7	B	ETX
-----	---	---	---	---	---	---	---	---	---	-----

◇ WO Command

[Function] Turns off the solenoid valve alarm signal.

[Data]

Request packet data: None

Request packet legend

STX	0	2	W	O	F	8	ETX
-----	---	---	---	---	---	---	-----

◇ AL Command

[Function] Sets the dispensing conditions of the specified dispense channel as a batch.

* Error A2 will be come out when all of below conditions are applied.

- Indicated channel is belonging to the recipe
- Status of Auto-Increment function is ON
- Dispensing mode is changed

* Error response A2 if in Σ (sigma) mode.

[Data]

Request packet data

Request packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel		-	3	1	400	[CH]
2	Dispensing pressure	V5	P	4	300	5000	[0.1kPa]
				4	300	5000	[0.001bar]
		V2		4	50	2000	[0.1kPa]
				4	50	2000	[0.001bar]
3	Dispensing time		T	7	1	9999999	[0.001s]
4	Vacuum pressure		V	4	0	500	[-0.01kPa] [-0.0001bar]
5	DIS-mode 0 = TIMED mode 1 = MANUAL mode		M	1	0	1	[-]
6	Channel name*		N	10	10	10	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	2	4	A	L	0	0	1	P	1	2	3	4	T	1	2	3	4	5	6	7
V	0	1	2	3	M	0	N	M	U	S	A	S	H	I	SP	SP	SP	4	1	ETX

◇ PL Command

[Function] Sets the plunger of the Current dispense channel.

[Data]

Request packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Plunger enabled/disabled 0 = None 1 = Use		-	1	0	1	[-]

Request packet legend

STX	0	3	P	L	0	D	1	ETX
-----	---	---	---	---	---	---	---	-----

◇ VC Command

[Function] Configures the control mode.

[Data]

Request packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Control mode 0 = SPD 1 = STB		-	1	0	1	[-]

Request packet legend

STX	0	3	V	C	0	D	4	ETX
-----	---	---	---	---	---	---	---	-----

◇ SS Command

[Function] Configures the syringe size for the specified dispense channel.
* Error response A2 if in Σ (sigma) mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL 7 = W 5mL 8 = W 10mL 9 = W 30mL 10 = W 55mL	S	2	0	10	[-]

Request packet legend

STX	0	8	S	S	0	0	1	S	0	0	A	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 01 Command

[Function] Moves the dispense channel up 1CH. Same function as STEP signal.
* Error response A2 if dispense channel is 400CH.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	1	SP	SP	F	B	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 02 Command

[Function] Moves the dispense channel down 1CH. Same function as DOWN signal.
* Error response A2 if dispense channel is 1CH.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	2	SP	SP	F	A	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 03 Command

[Function] Changes the dispense channel to 1CH. If the current channel belongs to a recipe, it changes to Start CH.
Same function as PRESET signal.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	3	SP	SP	F	9	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 06 Command

[Function] Performs a DIS-monitoring test.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	6	SP	SP	F	6	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 08 Command

[Function] Copies settings of the current dispense channel to the specified dispense channel.

* Take care, as if this command is performed, the status cannot be restored to the previous status.

* Error response A2 if the specified channel belongs to a recipe and the Auto-increment function is ON.

[Data]

Request packet data: None

Request packet legend

STX	0	7	0	8	SP	SP	1	2	3	5	B	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	-----

◇ 09 Command

[Function] Changes to Start CH. of the specified recipe.

* Error response A2 if an unregistered recipe number is specified.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]

Request packet legend

STX	0	7	0	9	SP	SP	0	0	1	5	F	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	-----

◇ 0A Command

[Function] Deletes the specified recipe.

* Take care, as if this command is performed, the status cannot be restored to the previous status.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]

Request packet legend

STX	0	7	0	A	SP	SP	0	0	1	5	7	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	-----

◇ 0B Command

[Function] This diagnoses any faults with the dispensing pressure, vacuum pressure and solenoid valve.
 * Perform this after removing the adapter tube.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing pressure 1 = Vacuum pressure 2 = Solenoid valve	-	1	0	2	[-]
2	Start/Cancel diagnosis 0 = Cancel 1 = Start	,	1	0	1	[-]

Request packet legend

STX	0	7	0	B	SP	SP	1	,	1	5	9	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	-----

◇ 0C Command

[Function] Performs auto-tuning of the connected temperature controller.
 * Perform this after removing the syringe from the temperature unit block.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Temperature unit number	-	1	1	5	[-]
2	Start/cancel Auto-tuning 0 = Cancel 1 = Start	,	1	0	1	[-]

Request packet legend

STX	0	7	0	C	□	□	1	,	1	5	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 0E Command

[Function] Clears the error log.
 * Take care, as if this command is performed, the status cannot be restored to the previous status.
 * Please do not power off in 20 seconds after the command execution.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	E	SP	SP	E	7	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 0F Command

[Function] Resets the settings saved for each channel to their default values.
 * Take care, as if this command is performed, the status cannot be restored to the previous status.
 * Please do not power off in 20 seconds after the command execution.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	F	SP	SP	E	6	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 0G Command

[Function] Restores all settings except language and communication settings to default.
 * Take care, as if this command is performed, the status cannot be restored to the previous status.
 * Please do not power off in 20 seconds after the command execution.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	G	SP	SP	E	5	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 0I (numeral zero, alphabetic capital "I") Command

[Function] Performs the sampling process necessary to enable automatic setting of remaining vol. corr. for the current dispense channel.
 The syringe connected to the device must be empty for the sampling process.
 * Error response A2 if in Normal mode.

[Data]

Request packet data: None

Request packet legend

STX	0	4	0	I	SP	SP	E	3	ETX
-----	---	---	---	---	----	----	---	---	-----

◇ 0J Command

[Function] Performs the sampling process necessary to enable more detailed correction for remaining vol. corr. for the current dispense channel.
 The syringe connected to the device must have the remaining volume specified in the request packet data for the sampling process.
 * Error response A2 if in Normal mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Remaining volume in syringe	-	2	1	99	[%]

Request packet legend

STX	0	6	0	J	SP	SP	4	0	7	C	ETX
-----	---	---	---	---	----	----	---	---	---	---	-----

◇ 0L Command

[Function] Sets the ECO (sigma) mode of the Current dispense channel.

* Error response A2 if the current channel belongs to a recipe and the Auto-increment function is ON.

* Error response A2 if in Σ (sigma) mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	ECO mode 0 = OFF 1 = ON	-	1	0	1	[-]

Request packet legend

STX	0	5	0	L	SP	SP	1	A	E	ETX
-----	---	---	---	---	----	----	---	---	---	-----

◇ 11 Command

[Function] Sets the delay timer for the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	On delay timer	,	4	0	9999	[0.001s]
3	Off delay timer	,	4	0	9999	[0.001s]

Request packet legend

STX	1	1	1	1	SP	SP	0	0	1	,	1	2	3	4	,	1	2	3	4	7	F	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 12 Command

[Function] Sets DIS-monitoring for the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Allowable	,	4	100	1000	[0.1kPa] [0.0001bar]
3	Dispensing OFF time	,	4	50	9999	[0.001s]

Request packet legend

STX	1	1	1	2	SP	SP	0	0	1	,	0	5	0	0	,	1	0	0	0	8	C	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 13 Command

[Function] Sets channel Name for the specified dispense channel.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Channel name*	,	14	14	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	1	6	1	3	SP	SP	0	0	1	,	D	I	S	P	E	N	S	E	R	1	2	3	4	5	8	C	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 14 Command

[Function] Sets the DIS-auxiliary setting of the specified dispense channel.
* Error response A2 if in Σ (sigma) mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL 7 = W 5mL 8 = W 10mL 9 = W 30mL 10 = W 55mL	,	10	0	10	[-]
3	Adapter tube length 1 = 0.5 m 2 = 1.0 m 3 = 1.5 m 4 = 2.0 m	,	1	1	4	[-]
4	Plunger enabled/disabled 0 = None 1 = Use	,	1	0	1	[-]
5	ECO mode 0 = OFF 1 = ON	,	1	0	1	[-]

Request packet legend

STX	0	F	1	4	SP	SP	0	0	1	,	1	,	2	,	0	,	0	E	1	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 15 Command

[Function] Sets Automatic correction of volume reduction for Σ function for the specified dispense channel.
* Error response A2 if in Normal mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Vol. reduction corr. OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
3	Alpha correction value	,	4	-200	+999	[-]
4	Delta correction	,	3	50	200	[%]

Request packet legend

STX	1	2	1	5	SP	SP	0	0	1	,	1	,	+	1	2	3	,	1	2	3	5	A	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 16 Command

[Function] Sets fluid-drip prevention for Σ function for the specified dispense channel.
* Error response A2 if in Normal mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Fluid-drip prevention OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
3	Vacuum correction value	,	4	-500 ^{*1}	+500 ^{*1}	[0.01kPa] [0.0001bar]

*1 The maximum/minimum value changes to vacuum pressure setting when the Σ (sigma) mode is started.

Request packet legend

STX	0	E	1	6	SP	SP	0	0	1	,	1	,	-	2	0	0	0	B	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 17 Command

[Function] Sets automatic remaining volume alert function for Σ function for the specified dispense channel.
* Error response A8 if in Normal mode.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Remaining vol. alert OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
3	Remaining volume detection level	,	2	0	99	[%]
4	RSM count	,	2	1	10	[times]

Request packet legend

STX	0	F	1	7	SP	SP	0	0	1	,	1	,	2	0	,	0	5	D	5	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 18 Command

[Function] Sets remaining vol. corr. for Σ function for the specified dispense channel.
* Error response A8 if in Normal mode.

[Data] Enter ", (0x2C)" between each data.

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Remaining vol. corr. OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
3	Correction value	,	3	-99	+99	[%]

Request packet legend

STX	0	D	1	8	SP	SP	0	0	1	,	1	,	+	0	4	3	A	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 19 Command

[Function] Sets the recipe to the specified recipe number.
 * If a channel already registered to a recipe is included between the specified Start CH. and End CH., error response A2 results.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Start CH.	,	3	1	400	[CH]
3	End CH.	,	3	Start CH.	400	[CH]

Request packet legend

STX	0	F	1	9	SP	SP	0	0	1	,	1	2	3	,	3	2	1	C	B	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1A Command

[Function] Sets the Auto-increment function of the specified recipe number.
 * Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Auto-increment function OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
3	Auto-increment function DIS-mode 0 = TIMED mode 1 = MANUAL mode	,	1	0	1	[-]
4	ECO mode 0 = OFF 1 = ON	,	1	0	1	[-]
5	Switching Condition 0 = Total time 1 = Total count 2 = Pot life	,	1	0	2	[-]
6	Switching value	,	4	1	9999	[s]
						[shot]
						[s]

Request packet legend

STX	1	4	1	A	SP	SP	0	0	1	,	1	,	0	,	0	,	1	,	1	2	3	4	F	0	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1B Command

[Function] Sets the recipe name of the specified recipe number.

* Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[.]
2	Recipe name	,	14	14	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	1	6	1	B	SP	SP	0	0	1	,	R	E	C	I	P	E	0	0	0	0	0	0	0	0	1	F	0	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1C Command

[Function] Sets the nozzle name of the specified recipe number.

* Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[.]
2	Nozzle name	,	14	14	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	1	6	1	C	SP	SP	0	0	1	,	N	O	Z	Z	L	E	0	0	0	0	0	0	0	0	1	C	6	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1D Command

[Function] Sets the Liquid material name of the specified recipe number.

* Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[.]
2	Liquid material name	,	14	14	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	1	6	1	D	SP	SP	0	0	1	,	L	I	Q	U	I	D	0	0	0	0	0	0	0	0	1	D	E	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1E Command

[Function] Sets the temperature controller conditions of the specified recipe number.
 * Error response A2 if an non-registered recipe number is specified.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Temp. unit controller number	,	1	1	5	[-]
3	Control OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
4	Setting temperature	,	4	0	1000	[0.1°C]
5	Upper Limit	,	4	0	1000	[0.1°C]
6	Lower Limit	,	4	0	1000	[-0.1°C]

Request packet legend

STX	1	A	1	E	SP	SP	0	0	1	,	1	,	1	,	0	5	0	0	,	0	0	2	0	,	0	0	2	0	C	0	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1F Command

[Function] Sets DIS-correction.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing time 1 = Dispensing pressure 2 = Vacuum pressure	-	1	0	2	[-]
2	Correction method 0 = OFF 1 = Correction amount 2 = Correction ratio	,	1	0	2	[-]
3	Correction amount	,	5	-9999	+9999	[0.001s]
				-5000	+5000	[0.1kPa]
				-2000	+2000	[0.001bar]
				-500	+500	[0.01kPa]
						[0.0001bar]
4	Correction ratio	,	3	1	999	[%]

Request packet legend

STX	1	1	1	F	SP	SP	1	,	1	,	+	0	1	0	0	,	1	0	0	8	4	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1G Command

[Function] Sets the Interval feature.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Interval feature OFF/ON 0 = OFF 1 = ON	-	1	0	1	[-]
2	Link CH.	,	3	0	400	[CH]
3	Count	,	4	1	9999	[times]
4	Interval	,	7	10	9999999	[0.001s]
5	Repeat	,	4	1	9999	[times]
6	Wait time	,	7	10	9999999	[0.001s]
7	Interval mode 0 = SET 1 = MAN	,	1	0	1	[-]

Request packet legend

STX	2	3	1	G	SP	SP	1	,	0	,	0	1	0	0	,	0	0	0	1	0	0	0
,	0	0	0	5	,	0	0	1	0	0	0	0	,	0	2	2	ETX					

◇ 1H Command

[Function] Changes the I/O signal settings.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Series 0 = ML-808GX 1 = SuperΣ	-	1	0	1	[-]
2	Output signal spec. 0 = In-dispensing 1 = Dispense complete	,	1	0	1	[-]
3	ENABLE function 0 = Inactive 1 = Active	,	1	0	1	[-]
4	On Error 0 = OFF 1 = ON	,	1	0	1	[-]

Request packet legend

STX	0	B	1	H	SP	SP	1	,	1	,	0	,	0	8	F	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	-----

◇ 1I (numeral one, alphabetic capital “I”) Command

[Function] Changes Configure setting.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Language 0 = English 1 = Japanese 2 = Chinese	-	1	0	2	[-]
2	P.UNIT 0 = kPa 1 = bar	,	1	0	1	[-]
3	Buzzer OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
4	T-ready Enable 0 = OFF 1 = ON	,	1	0	1	[-]
5	Warm-ready enable 0 = OFF 1 = ON	,	1	0	1	[-]
6	Pop-up auto OFF 0 = OFF 1 = ON	,	1	0	1	[-]
7	Immed. Inc. (Alnc) 0 = OFF 1 = ON	,	1	0	1	[-]
8	Control mode 0 = SPD 1 = STB	,	1	0	1	[-]
9	Sleep timer 0 = OFF (Manual) 1 = 5 minutes 2 = 10 minutes 3 = 30 minutes 4 = 1 hours 5 = 2 hours 6 = 3 hours 7 = 4 hours 8 = 5 hours	,	1	0	8	[-]

Request packet legend

STX	1	5	1	I	SP	SP	1	,	0	,	1	,	0	,	0	,	0	,	1	,	0	C	D	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1J Command

[Function] Changes alert function setting

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Empty alarm 0 = OFF (disabled) 1 = ON (enabled)	-	1	0	1	[-]
2	Empty warning threshold	,	9	1	999999999	[shot]
3	Valve warning threshold	,	9	1	999999999	[shot]
4	RSM advance notice lv.	,	2	0	50	[%]

Request packet legend

STX	1	C	1	J	SP	SP	1	,	0	0	0	0	0	1	0	0	0	,	0	5	0	0	0	0	0	0	0	,	1	0	5	5	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1L Command

[Function] Sets the password lock.

[Data]

* Place the character string within double quotation marks.

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Password lock OFF/ON 0 = OFF 1 = ON	-	1	0	1	[-]
2	Password*	,	Variable	0	14	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	0	A	1	L	SP	SP	1	,	1	2	3	4	A	B	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	-----

◇ 1M Command

[Function] Sets the input signal function allocation.

* Allocated functions cannot be duplicated. Error response A2 if duplicated.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	D-Sub3 PIN	-	2	0	18	[-]
2	D-Sub4 PIN	,	2	0	18	[-]
3	D-Sub5 PIN	,	2	0	18	[-]
4	D-Sub6 PIN	,	2	0	18	[-]
5	D-Sub7 PIN	,	2	0	18	[-]
6	D-Sub8 PIN	,	2	0	18	[-]
7	D-Sub15 PIN	,	2	0	18	[-]
8	D-Sub16 PIN	,	2	0	18	[-]
9	D-Sub17 PIN	,	2	0	18	[-]
10	D-Sub18 PIN	,	2	0	18	[-]
11	D-Sub19 PIN	,	2	0	18	[-]
12	D-Sub20 PIN	,	2	0	18	[-]

Functions that can be allocated to each data are as follows.

0	=	ENABLE signal	9	=	CHS signal
1	=	STEP signal	10	=	CD01 signal
2	=	PRESET signal	11	=	CD02 signal
3	=	DOWN signal	12	=	CD04 signal
4	=	TMD signal	13	=	CD08 signal
5	=	TMS signal	14	=	CD10 signal
6	=	C_CNT signal	15	=	CD20 signal
7	=	EDIT signal	16	=	CD40 signal
8	=	RESET signal	17	=	CD80 signal
			18	=	SLEEP signal

Request packet legend

STX	2	7	1	M	SP	SP	0	0	,	0	1	,	0	2	,	0	3	,	0	4	,	0	5
	,	0	6	,	0	7	,	0	8	,	0	9	,	1	0	,	1	1	4	5	ETX		

◇ 1N Command

[Function] Sets the output signal allocation.

* Allocated functions cannot be duplicated. Error response A2 if duplicated.

[Data]

Request packet data

• Series settings for ML-808GX

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	D-Sub9 PIN	-	2	0	15	[-]
2	D-Sub10 PIN	,	2	0	15	[-]
3	D-Sub12 PIN	,	2	0	15	[-]
4	D-Sub21 PIN	,	2	0	15	[-]
5	D-Sub22 PIN	,	2	0	15	[-]
6	D-Sub23 PIN	,	2	0	15	[-]

• Series settings for SuperΣ

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	D-Sub10 PIN	-	2	0	15	[-]
2	D-Sub22 PIN	,	2	0	15	[-]
3	D-Sub23 PIN	,	2	0	15	[-]

Functions that can be allocated to each data are as follows.

0	=	ERROR signal	8	=	ESTS signal
1	=	ALARM signal	9	=	EWAIT signal
2	=	ALARM2 signal	10	=	T_READY signal
3	=	RSM signal	11	=	EXE1 signal
4	=	READY signal	12	=	EXE2 signal
5	=	PCS signal	13	=	PSE1 signal
6	=	AI_END signal	14	=	PSE2 signal
7	=	S_END signal	15	=	RSM_P signal

Request packet legend

• Series settings for ML-808GX

STX	1	5	1	N	SP	SP	0	0	,	0	1	,	0	2	,	0	3	,	0	4	,	0	5	B	0	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

• Series settings for SuperΣ

STX	0	C	1	N	SP	SP	0	0	,	0	1	,	0	2	5	3	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	-----

◇ 1O (numeral one, alphabetic capital “O”) Command

[Function] Sets the date and time.

* Error response A2 if date and time that does not exist is set.

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Year	-	4	2000	2099	[Year]
2	Month	,	2	1	12	[Month]
3	Day	,	2	1	31	[Day]
4	Hour	,	2	0	23	[Hour]
5	Minute	,	2	0	59	[Minute]
6	Second	,	2	0	59	[Second]

Request packet legend

STX	1	7	1	0	SP	SP	2	0	2	3	,	0	1	,	0	1	,	1	2	,	0	0	,	0	0	5	0	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ 1Q Command

[Function] Controller name

[Data]

Request packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Controller name	-	10	10	10	[Character]

* Available set characters: "space (0x20)" to "~ (0x7E)"

Request packet legend

STX	0	E	1	Q	SP	SP	D	I	S	P	E	N	S	E	R	1	E	B	ETX
-----	---	---	---	---	----	----	---	---	---	---	---	---	---	---	---	---	---	---	-----

◇ UL Command

[Function] Acquires various settings and statuses

[Data]

[DA00 Acquisition of warnings and error occurrence status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	0	0	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	0	0	0	3	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Air supply pressure error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
2	Vacuum pressure error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
3	Exhaust error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
4	Σ (sigma) mode remaining amount alert 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
5	Solenoid valve error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
6	Solenoid valve alarm 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
7	Reserved ("0" (zero) is always acquired)	-	1	0	0	[-]

Data packet legend

Air supply pressure error = No occurrence
 Vacuum pressure error = No occurrence
 Exhaust error = No occurrence
 Σ (sigma) mode remaining amount alert = Occurring
 Solenoid valve error = No occurrence
 Solenoid valve alarm = Occurring

STX	0	B	D	A	0	0	0	0	0	1	0	1	0	5	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA01 Acquisition of dispensing parameters (dispensing time is 5 digits)]

* The value of the lower 5 digits is acquired when a dispensing time of 6 digits or more is set.

* Use[DA15 Acquisition of **multiple dispensing parameters at the same time**] (P.101) when a dispensing time of 6 digits or more is set.

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	1	1	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	0	2	C	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispensing pressure	V5	P	4	300	5000	[0.1kPa]
							[0.001bar]
	V2				50	2000	[0.1kPa]
							[0.001bar]
2	Dispensing time		T	5	1	99999	[0.001s]
3	Vacuum pressure		V	4	0	500	[-0.01kPa]
							[-0.0001kPa]
4	DIS-mode 0 = TIMED mode 1 = MANUAL mode		M	1	0	4	[-]
5	Channel name * A space is entered to fill blank space when there are less than 10 characters.		N	10	10	10	[Character]

Data packet legend

Dispensing pressure = 100.0kPa
 Dispensing time = 1.000s
 Vacuum pressure = -1.00kPa
 DIS-mode = Normal / TIMED mode
 Channel name = MUSASHI

STX	2	1	D	A	0	1	P	1	0	0	0	T	0	1	0	0	0	V	0	1	0	0
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

M	0	N	M	U	S	A	S	H	I	SP	SP	SP	0	5	ETX
---	---	---	---	---	---	---	---	---	---	----	----	----	---	---	-----

[DA02 Acquisition of syringe size and adapter tube length]

* Use[DA23 Acquisition of **syringe size and adapter tube length**](P.103) if W-type syringe size is set.

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	2	2	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	0	2	0	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL	S	1	0	6	[-]
2	Adapter tube length 1 = 0.5 m 2 = 1.0 m 3 = 1.5 m 4 = 2.0 m	A	1	1	4	[-]

Data packet legend

Syringe size = 5mL
Adapter tube length = 0.5m

STX	0	8	D	A	0	2	S	1	A	1	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA03 Acquisition of Σ (sigma) mode correction values]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	3	3	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	0	3	B	F	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Alpha correction value	A	4	-200	+999	[-]
2	Delta correction	D	3	50	200	[%]
3	Vacuum correction value	V	5	-500	+500	[0.01kPa] [0.0001bar]

Data packet legend

Alpha correction value = +123
Delta correction = 123
Vacuum correction value = +1.23kPa

STX	1	3	D	A	0	3	A	+	1	2	3	D	1	2	3	V	+	0	1	2	3	9	1	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA04 Acquisition of remaining volume detection parameters]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	4	4	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	0	4	C	3	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Remaining volume detection level	L	3	0	99	[%]
2	RSM count	C	3	1	10	[times]

Data packet legend

Remaining volume detection level = 12%
RSM count = 10 times

STX	0	C	D	A	0	4	L	0	1	2	C	0	1	0	F	1	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA05 Acquisition of currently detected remaining liquid material]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	5	5	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	0	5	B	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current remaining volume	R	3	0	100	[%]

Data packet legend

Current remaining volume = 12%

STX	0	8	D	A	0	5	R	0	1	2	C	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA06 Acquisition of currently displayed channel number]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	1	1	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	0	6	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current dispense channel	CH	3	1	400	[CH]

Data packet legend

Current dispense channel = 1CH

STX	0	9	D	A	0	6	C	H	0	0	1	9	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA07 Acquisition of user counter value]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	7	7	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	0	7	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	User counter	CT	8	0	99999999	[times]

Data packet legend

User counter = 12345678 times

STX	0	E	D	A	0	7	C	T	1	2	3	4	5	6	7	8	6	4	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA08 Acquisition of version information]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	8	8	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	0	8	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Firmware1	VR	4	100	9999	[-]
2	Pressure specification 2 = V2 5 = V5	SP	1	2	5	[-]

Data packet legend

Firmware1 = SCM4-01.00
Pressure type = V2

STX	0	D	D	A	0	8	V	R	0	1	0	0	S	P	2	6	1	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA12 Acquisition of remaining volume detection correction information]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	12	12	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	1	D	1	2	B	F	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Remaining vol. corr. OFF/ON 0 = OFF 1 = ON	F	1	0	1	[-]
2	Correction value	C	4	-99	+99	[%]

Data packet legend 1

Remaining vol. corr. = OFF
Correction value = 0%

STX	0	B	D	A	1	2	F	0	C	+	0	0	0	3	2	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

Remaining vol. corr. = ON
Correction value = 12%

STX	0	B	D	A	1	2	F	1	C	+	0	1	2	2	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA13 Acquisition of warnings and error occurrence status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	13	13	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	1	3	B	F	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Air supply pressure error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
2	Vacuum pressure error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
3	Exhaust error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
4	Σ (sigma) mode remaining amount alert 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
5	Solenoid valve error 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
6	Solenoid valve alarm 0 = No occurrence 1 = Occurring	-	1	0	1	[-]
7	Reserved ("0" (zero) is always acquired)	-	1	0	0	[-]
8	Reserved ("0" (zero) is always acquired)	-	1	0	0	[-]
9	Reserved ("0" (zero) is always acquired)	-	1	0	0	[-]

Data packet legend

Air supply pressure = Occurring
 error
 Vacuum pressure = No occurrence
 error
 Exhaust error = No occurrence
 Σ (sigma) mode = No occurrence
 remaining amount alert
 Solenoid valve error = No occurrence
 Solenoid valve alarm = Occurring

STX	0	D	D	A	1	3	1	0	0	0	0	1	0	0	0	F	1	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA14 Acquisition of plunger setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	14	14	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	1	D	1	4	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Plunger enabled/disabled 0 = None 1 = Use	P	1	0	1	[-]

Data packet legend

Plunger = None

STX	0	6	D	A	1	4	P	0	3	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	-----

[DA15 Acquisition of multiple dispensing parameters at the same time]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	15	15	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	1	D	1	5	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details		Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispensing pressure	V5	P	4	300	5000	[0.1kPa]
							[0.001bar]
	V2				50	2000	[0.1kPa]
							[0.001bar]
2	Dispensing time		T	7	1	9999999	[0.001s]
3	Vacuum pressure		V	4	0	500	[-0.01kPa]
							[-0.0001kPa]
4	DIS-mode 0 = TIMED mode 1 = MANUAL mode		M	1	0	4	[-]
5	Channel name * A space is entered to fill blank space when there are less than 10 characters.		N	10	10	10	[Character]

Data packet legend

Dispensing pressure = 123.4kPa
 Dispensing time = 1234.567s
 Vacuum pressure = -1.23kPa
 DIS-mode = Normal / TIMED mode
 Channel name = MUSASHI

STX	2	3	D	A	1	5	P	1	2	3	4	T	1	2	3	4	5	6	7	V	0	1	2	3
M	0	N	M	U	S	A	S	H	I	SP	SP	SP	7	5	ETX									

[DA16 Acquisition of user counter value]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	16	16	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	1	6	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	User counter	UC	9	0	999999999	[shot]

Data packet legend

User counter = 0shot

STX	0	F	D	A	1	6	U	C	0	0	0	0	0	0	0	0	0	5	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA17 Acquisition of total counter value]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	17	17	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	1	7	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Total counter	TC	9	0	999999999	[shot]

Data packet legend

Total counter = 0shot

STX	0	F	D	A	1	7	T	C	0	0	0	0	0	0	0	0	0	0	5	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA18 Acquisition of control mode setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	18	18	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	1	8	B	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Control mode 0 = SPD 1 = STB	V	1	0	1	[-]

Data packet legend

Control mode = SPD

STX	0	6	D	A	1	8	V	0	2	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	-----

[DA23 Acquisition of syringe size and adapter tube length]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	23	23	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	1	D	2	3	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Syringe size 0 = 3mL 1 = 5mL 2 = 10mL 3 = 20mL 4 = 30mL 5 = 50mL 6 = 70mL 7 = W 5mL 8 = W 10mL 9 = W 30mL 10 = W 55mL	S	2	0	10	[-]
2	Adapter tube length 1 = 0.5 m 2 = 1.0 m 3 = 1.5 m 4 = 2.0 m	A	1	1	4	[-]

Data packet legend

Syringe size = 5mL

Adapter tube length = 0.5m

STX	0	9	D	A	2	3	S	0	1	A	1	8	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA30 Acquisition of delay timer]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	30	30	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	0	B	F	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	On delay timer	,	4	0	9999	[0.001s]
2	Off delay timer	,	4	0	9999	[0.001s]

Data packet legend

On delay timer = 1.234s

Off delay timer = 1.234s

STX	0	E	D	A	3	0	,	1	2	3	4	,	1	2	3	4	B	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA31 Acquisition of DIS-monitoring function setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	31	31	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	1	B	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Allowable	,	4	100	1000	[0.1kPa] [0.0001bar]
2	Dispensing OFF time	,	4	50	9999	[0.001s]

Data packet legend

Allowable = 10.0kPa

Dispensing OFF time = 1.000s

STX	0	E	D	A	3	1	,	0	1	0	0	,	1	0	0	0	C	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA32 Acquisitions of channel name]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	32	32	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	2	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Channel name	,	14	14	14	[Character]

Data packet legend

Channel name = MUSASHI

STX	1	3	D	A	3	2	,	M	U	S	A	S	H	I	SP	SP	SP	SP	SP	SP	8	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	---	---	-----

[DA33 Acquisition of ECO mode setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	33	33	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	3	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	ECO mode 0 = OFF 1 = ON	,	1	0	1	[-]

Data packet legend

ECO mode = OFF

STX	0	6	D	A	3	3	,	0	5	3	ETX
-----	---	---	---	---	---	---	---	---	---	---	-----

[DA34 Σ function: Acquisition of Vol. corr. setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	34	34	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	4	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Vol. reduction corr. OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Alpha correction value	,	4	-200	+999	[-]
3	Delta correction	,	3	50	200	[%]

Data packet legend

Vol. reduction corr. = ON
Alpha correction value = 100
Delta correction = 110%

STX	0	F	D	A	3	4	,	1	,	+	1	0	0	,	1	1	0	9	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA35 Σ function: Acquisition of fluid-drip prevention setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	35	35	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	5	B	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Fluid-drip prevention OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Vacuum correction value	,	4	-500	+500	[0.01kPa] [0.0001bar]

Data packet legend

Fluid-drip prevention = ON
Vacuum correction value = -0.50kPa

STX	0	B	D	A	3	5	,	1	,	-	0	5	0	5	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

Fluid-drip prevention = ON
Vacuum correction value = 0.0100bar

STX	0	B	D	A	3	5	,	1	,	+	1	0	0	5	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA36 Σ function: Acquisition of remaining vol. alert setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	36	36	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	6	B	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Remaining vol. alert OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Remaining volume detection level	,	2	0	99	[%]
3	RSM count	,	2	1	10	[times]

Data packet legend

Remaining vol. alert = ON
 Remaining volume detection level = 10%
 RSM count = 5 times

STX	0	C	D	A	3	6	,	1	,	1	0	,	0	5	2	4	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA37 Σ function: Acquisition of remaining vol. corr. setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense channel	-	3	1	400	[CH]
2	Data packet specification	D	2	37	37	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	7	B	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Remaining vol. corr. OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Correction value	,	3	-99	+99	[%]

Data packet legend

Remaining vol. corr. = ON
 Correction value = -10%

STX	0	A	D	A	3	7	,	1	,	-	1	0	8	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA38 Acquisition of recipe]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Data packet specification	D	2	38	38	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	8	B	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Start CH.	,	3	1	400	[CH]
2	End CH.	,	3	Start CH.	400	[CH]

Data packet legend

Start CH. = 90CH
End CH. = 110CH

STX	0	C	D	A	3	8	,	0	9	0	,	1	1	0	1	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA39 Acquisition of auto-increment function setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Data packet specification	D	2	39	39	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	3	9	B	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Auto-increment Function OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Auto-increment Function dispensing mode 0 = TIMED mode 1 = MANUAL mode	,	1	0	1	[-]
3	ECO mode 0 = OFF 1 = ON	,	1	0	1	[-]
4	Switching Condition 0 = Total time 1 = Total count 2 = Pot life	,	1	0	2	[-]
5	Switching value	,	4	1	9999	[s]
6						[shot]
7						[s]

Data packet legend

Auto-increment Function = ON
DIS-mode = TIMED mode
ECO mode = OFF
Switching Condition = Total count
Switching value = 30shot

STX	1	1	D	A	3	9	,	1	,	0	,	0	,	1	,	0	0	3	0	4	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA40 Acquisition of recipe details]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Data packet specification	D	2	40	40	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	4	0	B	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe name	,	14	14	14	[Character]
2	Nozzle name	,	14	14	14	[Character]
3	Liquid material name	,	14	14	14	[Character]

Data packet legend

Recipe name = RECIPE
 Nozzle name = NOZZLE
 Liquid material name = LIQUID

STX	3	1	D	A	4	0	,	R	E	C	I	P	E	SP	SP	SP	SP	SP	SP	SP	SP	SP	SP	,	N	O	Z	Z	L	E	SP	SP	SP	SP	SP	SP	SP	SP	SP
,	L	I	Q	U	I	D	SP	SP	SP	SP	SP	SP	SP	SP	C	D	ETX																						

[DA41 Acquisition of temperature unit conditions]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Recipe number	-	3	1	100	[-]
2	Data packet specification	D	2	41	41	[-]

Request packet legend

STX	0	8	U	L	0	0	1	D	4	1	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Temperature controller 1 control OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Temperature controller 1 setting temperature	,	4	0	1000	[0.1°C]
3	Temperature controller 1 upper limit	,	4	0	1000	[0.1°C]
4	Temperature controller 1 lower limit	,	4	0	1000	[-0.1°C]
5	Temperature controller 2 Temperature unit control OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
6	Temperature controller 2 setting temperature	,	4	0	1000	[0.1°C]
7	Temperature controller 2 upper limit	,	4	0	1000	[0.1°C]
8	Temperature controller 2 lower limit	,	4	0	1000	[-0.1°C]
9	Temperature controller 3 Temperature unit control OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
10	Temperature controller 3 setting temperature	,	4	0	1000	[0.1°C]
11	Temperature controller 3 upper limit	,	4	0	1000	[0.1°C]
12	Temperature controller 3 lower limit	,	4	0	1000	[-0.1°C]
13	Temperature controller 4 Temperature unit control OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
14	Temperature controller 4 setting temperature	,	4	0	1000	[0.1°C]
15	Temperature controller 4 upper limit	,	4	0	1000	[0.1°C]
16	Temperature controller 4 lower limit	,	4	0	1000	[-0.1°C]
17	Temperature controller 5 Temperature unit control OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
18	Temperature controller 5 setting temperature	,	4	0	1000	[0.1°C]
19	Temperature controller 5 upper limit	,	4	0	1000	[0.1°C]
20	Temperature controller 5 lower limit	,	4	0	1000	[-0.1°C]

Data packet legend

With following setting

Temperature unit	Control OFF/ON	Setting temperature	Upper Limit	Lower Limit
Controller 1	ON	60.0°C	2.0°C	2.0°C
Controller 2	ON	55.0°C	1.0°C	1.0°C
Controller 3	OFF	25.0°C	1.0°C	1.0°C
Controller 4	OFF	25.0°C	1.0°C	1.0°C
Controller 5	OFF	25.0°C	1.0°C	1.0°C

STX	5	9	D	A	4	1	.	1	.	0	6	0	0	.	0	0	2	0	.	0	0	2	0	.	1	.	0	5	5	0	.	0	0	1	0	.	0	0	1	0
.	0	.	0	2	5	0	.	0	0	1	0	.	0	0	1	0	.	0	.	0	2	5	0	.	0	0	1	0	.	0	0	1	0							
.	0	.	0	2	5	0	.	0	0	1	0	.	0	0	1	0	D	5	ETX																					

[DA42 Acquisition of dispensing parameter correction function setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Select target 0 = Dispensing time 1 = Dispensing pressure 2 = Vacuum pressure	-	3	0	2	[-]
2	Data packet specification	D	2	42	42	[-]

Request packet legend

STX	0	8	U	L	0	0	0	D	4	2	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Correction method 0 = OFF 1 = Correction amount 2 = Correction ratio	,	1	0	2	[-]
2	Correction amount	,	5	-9999	+9999	[-]
				-5000	+5000	[0.1kPa] [0.001bar]
				-2000	+2000	[0.1kPa] [0.001bar]
				-500	+500	[0.01kPa] [0.0001bar]
3	Correction ratio	,	3	1	999	[%]

Data packet legend

Correction method = Correction amount
Correction amount = 10kPa
Correction ratio = 100%

STX	1	0	D	A	4	2	,	1	,	0	0	0	1	0	,	1	0	0	7	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA43 Acquisition of Interval feature setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	43	43	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	4	3	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Interval feature OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Link CH.	,	3	0	400	[CH]
3	Count	,	4	1	9999	[times]
4	Interval	,	7	10	9999999	[0.001s]
5	Repeat	,	4	1	9999	[times]
6	Wait time	,	7	10	9999999	[0.001s]
7	Interval mode 0 = SET 1 = MAN	,	1	0	1	[-]

Data packet legend

Interval feature = ON
 Link CH. = 0CH
 Count = 123 times
 Interval = 1.000s
 Repeat = 1234 times
 Wait time = 2.000s
 Interval mode = SET

STX	2	6	D	A	4	3	,	1	,	0	0	0	,	0	1	2	3	,	0	0	0	1	0	0	0
,	1	2	3	4	,	0	0	0	2	0	0	0	,	0	5	4	ETX								

[DA44 Acquisition of I/O signal setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	44	44	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	4	4	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Series 0 = ML-808GX 1 = SuperΣ	,	1	0	1	[-]
2	Output signal spec. 0 = In-dispensing 1 = Dispense complete	,	1	0	1	[-]
3	ENABLE function 0 = Inactive 1 = Active	,	1	0	1	[-]
4	On Error 0 = OFF 1 = ON	,	1	0	1	[-]

Data packet legend

Series = SuperΣ
Output signal spec. = Dispense complete
ENABLE function = Inactive
On Error = OFF

STX	0	C	D	A	4	4	,	1	,	1	,	0	,	0	2	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA45 Acquisition of configure setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	45	45	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	4	5	B	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Language 0 = English 1 = Japanese 2 = Chinese	,	1	0	2	[-]
2	P.UNIT 0 = kPa 1 = bar	,	1	0	1	[-]
3	Buzzer OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
4	T-ready Enable 0 = OFF 1 = ON	,	1	0	1	[-]
5	Warm-ready enable 0 = OFF 1 = ON	,	1	0	1	[-]
6	Pop-up auto OFF 0 = OFF 1 = ON	,	1	0	1	[-]
7	Immed. Inc. (Alnc) 0 = OFF 1 = ON	,	1	0	1	[-]
8	Control mode 0 = SPD 1 = STB	,	1	0	1	[-]
9	Sleep timer 0 = OFF (Manual) 1 = 5 minutes 2 = 10 minutes 3 = 30 minutes 4 = 1 hours 5 = 2 hours 6 = 3 hours 7 = 4 hours 8 = 5 hours	,	1	0	8	[-]

Data packet legend

Language	=	Japanese
P.UNIT	=	kPa
Buzzer OFF/ON	=	ON
T-ready Enable	=	OFF
Warm-ready enable	=	OFF
Pop-up auto OFF	=	OFF
Immed. Inc. (Alnc)	=	OFF
Control mode	=	STB
Sleep mode	=	OFF

STX	1	6	D	A	4	5	,	1	,	0	,	1	,	0	,	0	,	0	,	1	,	0	6	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA46 Acquisition of alert function setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	46	46	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	4	6	B	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Empty alarm 0 = OFF (disabled) 1 = ON (enabled)	,	1	0	1	[-]
2	Empty warning threshold	,	9	1	999999999	[shot]
3	Valve warning threshold	,	9	1	999999999	[shot]
4	RSM advance notice lv.	,	2	0	50	[%]

Data packet legend

Empty alarm = OFF
Empty warning threshold = 1000shot
Valve warning threshold = 50,000,000shot
RSM advance notice lv. = 10%

STX	1	D	D	A	4	6	,	0	,	0	0	0	0	0	1	0	0	0	,	0	5	0	0	0	0	0	0	0	,	1	0	F	5	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA48 Acquisition of password lock setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	48	48	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	4	8	B	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Password lock OFF/ON 0 = OFF 1 = ON	,	1	0	1	[-]
2	Password	,	Variable	0	14	[Character]

Data packet legend

Password lock = ON
Password = 1234

STX	0	B	D	A	4	7	,	1	,	1	2	3	4	4	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA49 Acquisition of input signal allocation setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	49	49	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	4	9	B	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	D-Sub3 PIN	,	2	0	18	[-]
2	D-Sub4 PIN	,	2	0	18	[-]
3	D-Sub5 PIN	,	2	0	18	[-]
4	D-Sub6 PIN	,	2	0	18	[-]
5	D-Sub7 PIN	,	2	0	18	[-]
6	D-Sub8 PIN	,	2	0	18	[-]
7	D-Sub15 PIN	,	2	0	18	[-]
8	D-Sub16 PIN	,	2	0	18	[-]
9	D-Sub17 PIN	,	2	0	18	[-]
10	D-Sub18 PIN	,	2	0	18	[-]
11	D-Sub19 PIN	,	2	0	18	[-]
12	D-Sub20 PIN	,	2	0	18	[-]

Data packet legend

D-Sub3 PIN	=	STEP signal	D-Sub15 PIN	=	CD80 signal
D-Sub4 PIN	=	CHS signal	D-Sub16 PIN	=	PRESET signal
D-Sub5 PIN	=	CD02 signal	D-Sub17 PIN	=	CD01 signal
D-Sub6 PIN	=	CD08 signal	D-Sub18 PIN	=	CD04 signal
D-Sub7 PIN	=	CD20 signal	D-Sub19 PIN	=	CD10 signal
D-Sub8 PIN	=	ENABLE signal	D-Sub20 PIN	=	CD40 signal

STX	2	8	D	A	4	9	,	0	1	,	0	9	,	1	1	,	1	3	,	1	5	,	0	0	
,	1	7	,	0	2	,	1	0	,	1	2	,	1	4	,	1	6	E	4	ETX					

[DA50 Acquisition of output signal allocation setting]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	50	50	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	0	B	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

• Series settings for ML-808GX

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	D-Sub9 PIN	,	2	0	15	[-]
2	D-Sub10 PIN	,	2	0	15	[-]
3	D-Sub12 PIN	,	2	0	15	[-]
4	D-Sub21 PIN	,	2	0	15	[-]
5	D-Sub22 PIN	,	2	0	15	[-]
6	D-Sub23 PIN	,	2	0	15	[-]

• Series settings for SuperΣ

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	D-Sub10 PIN	,	2	0	15	[-]
2	D-Sub22 PIN	,	2	0	15	[-]
3	D-Sub23 PIN	,	2	0	15	[-]

Data packet legend

Series settings for SuperΣ

D-Sub10 PIN = EXE1 signal

D-Sub22 PIN = RSM signal

D-Sub23 PIN = PSE1 signal

STX	0	D	D	A	5	0	,	1	1	,	0	3	,	1	3	F	5	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA51 Acquisition of time (year, month, day, hour, minute, second)]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	51	51	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	1	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Year	,	4	2000	2099	[Year]
2	Month	,	2	1	12	[Month]
3	Day	,	2	1	31	[Day]
4	Hour	,	2	0	23	[Hour]
5	Minute	,	2	0	59	[Minute]
6	Second	,	2	0	59	[Second]

Data packet legend

Year = 2023
Month = January
Day = 1
Hour = 12 hours
Minute = 0 minutes
Second = 0 seconds

STX	1	8	D	A	5	1	,	2	0	2	3	,	0	1	,	0	1	,	1	2	,	0	0	,	0	0	F	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA52 Acquisition of controller name]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	52	52	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	2	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Controller name	,	10	10	10	[Character]

Data packet legend

Controller name = DESPENSER1

STX	0	F	D	A	5	2	,	D	E	S	P	E	N	S	E	R	1	9	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA53 Acquisition of dispense status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	53	53	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	3	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispense status 0 = Dispense waiting 1 = In-dispensing	,	1	0	1	[-]
2	Dispense availability 0 = Dispense possible 1 = Dispense not possible	,	1	0	1	[-]
3	Error status	,	10	0	2147483647	[-]
4	Alarm status	.	10	0	2147483647	[-]

bit	Error status	bit	Error status
0	"Setting data load error"	10	"Temp. unit error: HB warning"
1	"Startup error"	11	"Temp. unit error: Timeout"
2	"Image data error"	12	"Temp. unit error: Setting error"
3	"RTC error"	13	"Internal error 1"
4	"Solenoid valve error"	14	"Internal error 2"
5	"Air supply pressure error"	15	"Internal error 3"
6	"Exhaust error"	16	"Internal error 4"
7	"Vacuum pressure error"	17	Reserved
8	"Temp. unit error: HS warning"	18	Reserved
9	"Temp. unit error: Input error"	19	"Internal error 7"

bit	Alarm status	bit	Alarm status
0	"Air supply pressure warning"	3	"Temp. unit alarm: Warning 1"
1	"Solenoid valve alarm"	4	"Temp. unit alarm: Warning 2"
2	"Empty alarm"	5	"DIS-monitoring alarm"
		6	"Σ (sigma) mode remaining amount alert"

Data packet legend

Dispense status	=	Dispense waiting
Dispense availability	=	Dispense possible
Error status	=	No error
Alarm status	=	0 = No alarm

[illegible]

[DA54 Acquisition of air supply pressure]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	54	54	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	4	B	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current air supply pressure	,	3	0	999	[0.1kPa] [0.001bar]

Data packet legend

Air Supply Pressure = 500kPa

STX	0	8	D	A	5	4	,	5	0	0	E	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA55 Acquisition of stopwatch value]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	55	55	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	5	B	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Stopwatch	,	7	1	9999999	[0.001s]

Data packet legend

Stopwatch = 1.234s

STX	0	C	D	A	5	5	,	0	0	0	1	2	3	4	1	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA56 Acquisition of Current recipe]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	56	56	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	6	B	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current recipe number	,	3	1	100	[-]

If it does not belong to a recipe, data is omitted and the response will only be " , (0x2C)" (comma).

Data packet legend

If it belongs to a recipe:

Recipe number = 3

STX	0	8	D	A	5	6	,	0	0	3	E	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

If it does not belong to a recipe:

STX	0	5	D	A	5	6	,	7	F	ETX
-----	---	---	---	---	---	---	---	---	---	-----

[DA57 Acquisition of Auto-increment status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	57	57	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	7	B	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Progress status 0 = Waiting 1 = Progressing 2 = Complete	,	1	0	2	[-]
2	Remaining volume	,	4	0	9999	[s]
	Total time					[shot]
	Total count					[s]
	Pot life					

If the function is OFF, data is omitted and the response will only be " , (0x2C)" (comma).

Data packet legend

If function is ON

Progress status = Waiting

Remaining volume = 1234shot

STX	0	B	D	A	5	7	,	0	,	1	2	3	4	4	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If function is OFF

STX	0	6	D	A	5	7	,	,	5	1	ETX
-----	---	---	---	---	---	---	---	---	---	---	-----

[DA58 Acquisition of DIS-monitoring test status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	58	58	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	8	B	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	DIS-monitoring test status 0 = Waiting 1 = Testing	,	1	0	1	[-]
2	DIS-monitoring test result 0 = Not performed 1 = NG 2 = OK	,	1	0	2	[-]

Data packet legend

DIS-monitoring test status = Waiting

DIS-monitoring test result = OK

STX	0	8	D	A	5	8	,	0	,	2	E	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA59 Acquisition of empty alarm state]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	59	59	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	5	9	B	5	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current remaining volume	,	3	0	100	[%]
2	Current detection count*	,	2	0	10	[times]

* If the automatic remaining volume alert function is OFF, data is omitted and the response will only be ", " (comma).

Data packet legend

Current remaining volume = 100%

Current detection count = 0 times

STX	D	A	5	9	,	1	0	0	,	0	0	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	-----

[DA60 Acquisition of temperature unit status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	60	60	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	0	B	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current temperature of temperature controller 1	,	4	0	9999	[0.1°C]
2	Current temperature of temperature controller 2	,	4	0	9999	[0.1°C]
3	Current temperature of temperature controller 3	,	4	0	9999	[0.1°C]
4	Current temperature of temperature controller 4	,	4	0	9999	[0.1°C]
5	Current temperature of temperature controller 5	,	4	0	9999	[0.1°C]

Data packet legend

Current temperature of temperature controller 1 = 25.0°C
 Current temperature of temperature controller 2 = 25.0°C
 Current temperature of temperature controller 3 = 25.0°C
 Current temperature of temperature controller 4 = 25.0°C
 Current temperature of temperature controller 5 = 25.0°C

STX	1	D	D	A	6	0	,	0	2	5	0	,	0	2	5	0	,	0	2	5	0	,	0	2	5	0	,	0	2	5	0	E	1	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA61 Acquisition of value after DIS-correction]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	61	61	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	1	B	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Dispensing time after DIS-correction	,	7	1	9999999	[0.001s]
2	Dispensing pressure after DIS-correction	,	4	300	5000	[0.1kPa]
						[0.001bar]
				50	2000	[0.1kPa]
						[0.001bar]
3	Vacuum pressure after DIS-correction	,	3	0	500	[-0.01kPa]
						[-0.0001bar]

If the DIS-correction correction method is OFF, data is omitted and the response will only be ", (0x2C)" (comma).

Data packet legend

Dispensing time after DIS-correction = 1.234s
 Dispensing pressure after DIS-correction = 200.0kPa
 Vacuum pressure after DIS-correction = -0.10kPa

STX	1	5	D	A	6	1	,	0	0	0	1	2	3	4	,	2	0	0	0	,	0	1	0	7	D	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA62 Acquisition of interval status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	62	62	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	2	B	B	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Current count	,	4	0	9999	[times]
2	Current repeat	,	4	0	9999	[times]

If the function is OFF, data is omitted and the response will only be ", " (comma).

Data packet legend

Current count = 3
Current repeat = 10

STX	0	E	D	A	6	2	,	0	0	0	3	,	0	0	1	0	C	2	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA63 Acquisition of diagnosis status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	63	63	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	3	B	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Diagnosis status 0 = Waiting 1 = Diagnosing	,	1	0	1	[-]
2	Dispensing pressure diagnosis result 0 = Errors 1 = No errors	,	1	0	1	[-]
3	Vacuum pressure diagnosis result 0 = Errors 1 = No errors	,	1	0	1	[-]
4	Solenoid valve diagnosis result 0 = Errors 1 = No errors	,	1	0	1	[-]

During diagnosis, diagnosis result data is omitted and the response will only be ", " (comma).

Data packet legend

Diagnosis status = Waiting
Dispensing pressure diagnosis result = No errors
Vacuum pressure diagnosis result = No errors
Solenoid valve diagnosis result = No errors

STX	0	C	D	A	6	3	,	0	,	1	,	1	,	1	2	C	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

Diagnosis status = Diagnosing

STX	0	9	D	A	6	3	,	0	,	,	,	C	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA64 Acquisition of temperature unit auto-tuning status]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	64	64	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	4	B	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Temperature controller 1 Auto-tuning status 0 = NG 1 = OK 2 = Not performed 3 = Performing	,	1	0	3	[-]
2	Temperature controller 2 Auto-tuning status 0 = NG 1 = OK 2 = Not performed 3 = Performing	,	1	0	3	[-]
3	Temperature controller 3 Auto-tuning status 0 = NG 1 = OK 2 = Not performed 3 = Performing	,	1	0	3	[-]
4	Temperature controller 4 Auto-tuning status 0 = NG 1 = OK 2 = Not performed 3 = Performing	,	1	0	3	[-]
5	Temperature controller 5 Auto-tuning status 0 = NG 1 = OK 2 = Not performed 3 = Performing	,	1	0	3	[-]

Data packet legend

Temperature controller 1 auto-tuning status = OK
 Temperature controller 2 auto-tuning status = Not performed
 Temperature controller 3 auto-tuning status = Not performed
 Temperature controller 4 auto-tuning status = Not performed
 Temperature controller 5 auto-tuning status = Not performed

STX	0	E	D	A	6	4	,	1	,	2	,	2	,	2	,	2	C	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA65 Acquisition of total error log number]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	65	65	[-]

Request packet legend

STX	0	8	U	L	0	0	0	D	6	5	B	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Total error log number	,	4	1	1000	[logs]

Data packet legend

Total error log number = 23

STX	0	9	D	A	6	5	,	0	0	2	3	B	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA66 Acquisition of error log (main)]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Log	-	3	0	999	[-]
2	Data packet specification	D	2	66	66	[-]

Request packet legend

STX	0	8	U	L	0	0	0	D	6	6	B	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Date of occurrence * The response is placed within double quotation marks.	,	19	2000/01/01 00:00:00	2099/12/31 23:59:59	[Character]
2	Error code * See Basic Operations for details.	,	3	1	999	[-]
3	Total counter	,	9	0	999999999	[shot]

If there is no specified error log, data is omitted and the response will only be " , (0x2C)" (comma).

Data packet legend

Date of occurrence = 2023/01/01 12:00:00
Error code = 006
Total counter = 2shot

STX	2	6	D	A	6	6	,	2	0	2	3	/	0	1	/	0	1	SP	1	2	:	0	0	:	0	0
,	0	0	6	,	0	0	0	0	0	0	0	0	2	3	D	ETX										

[DA67 Acquisition of error log (sub)]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Log	-	3	0	999	[-]
2	Data packet specification	D	2	67	67	[-]

Request packet legend

STX	0	8	U	L	0	0	0	D	6	7	B	6	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Error pattern	,	1	0	9	[-]
2	Related data 1	,	See below.			
3	Related data 2	,				
4	Related data 3	,				
5	Related data 4	,				
6	Related data 5	,				
7	Related data 6	,				

If there is no related data, data is omitted and the response will be " , (0x2C)" (comma).

If error pattern is "0"

Related data	Details	Minimum value	Maximum value	Unit
1 to 6	None	-	-	-

Data packet legend

STX	0	C	D	A	6	7	,	0	,	,	,	,	,	3	7	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If error pattern is "1"

Related data	Details	Digit number	Minimum value	Maximum value	Unit
1	Air Supply Pressure	3	0	999	[kPa] [0.01bar]
2	DIS-mode 0 = Normal / TIMED mode 1 = NORMAL / MANUAL mode: 2 = Σ / TIMED mode 3 = Σ / MANUAL mode 4 = Dummy shot mode	1	0	4	[-]
3	Dispensing time	7	10	9999999	[0.001s]
4	Dispensing pressure V5	4	300	5000	[0.1kPa] [0.001bar]
	Dispensing pressure V2	4	50	2000	[0.1kPa] [0.001bar]
5	Vacuum pressure	3	0	500	[-0.01kPa] [-0.00001bar]
6	Dispense status when error occurred 0 = Dispense waiting 1 = In-dispensing	1	0	1	[-]

Data packet legend

Air Supply Pressure	=	500kPa
DIS-mode	=	Normal / TIMED mode
Dispensing time	=	1.000s
Dispensing pressure	=	100.0kPa
Vacuum pressure	=	-0.10kPa
Dispense status when error occurred	=	Dispense waiting

STX	1	F	D	A	6	7	,	1	,	5	0	0	,	0	,	0	0	0	1	0	0	0	,	1	0	0	0	,	0	1	0	,	0	9	A	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If error pattern is "2"

Related data	Details	Digit number	Minimum value	Maximum value	Unit
1	Default mode 1 = Screen operation 2 = RS-232C communications	1	1	2	[-]
2 to 6	None	-	-	-	-

Data packet legend

Default mode = Screen operation

STX	0	D	D	A	6	7	,	2	,	1	,	,	,	,	0	3	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If error pattern is "3"

Related data	Details	Digit number	Minimum value	Maximum value	Unit
1	Temperature unit number	1	1	5	[-]
2	Current temperature	4	0	9999	[0.1°C]
3	Setting temperature	4	0	1000	[0.1°C]
4	Upper Limit	4	0	1000	[0.1°C]
5	Lower Limit	4	0	1000	[0.1°C]
6	None	-	-	-	-

Data packet legend

Temperature unit number = Temperature controller 1

Current temperature = 48.0°C

Setting temperature = 50.0°C

Upper Limit = 2.0°C

Lower Limit = 1.5°C

STX	1	D	D	A	6	7	,	3	,	1	,	0	4	8	0	,	0	5	0	0	,	0	0	2	0	,	0	0	1	5	,	E	8	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If error pattern is "4"

Related data	Details	Digit number	Minimum value	Maximum value	Unit
1	Air Supply Pressure	3	0	999	[kPa] [0.01bar]
2	Dispensing time	7	10	9999999	[0.001s]
3	Dispensing pressure	4	300	5000	[0.1kPa] [0.001bar]
					[0.1kPa] [0.001bar]
		4	50	2000	[0.001bar] [-0.01kPa]
					[-0.00001bar]
4	Vacuum pressure	3	0	500	[0.001bar]
5	User counter	9	0	999999999	[shot]
6	Pressure reached	4	0	9999	[0.1kPa] [0.001bar]

Data packet legend

Air Supply Pressure = 500kPa

Dispensing time = 1.000s

Dispensing pressure = 200kPa

Vacuum pressure = -0.00kPa

User counter = 200shot

Pressure reached = 180kPa

STX	2	A	D	A	6	7	,	4	,	5	0	0	,	0	0	0	1	0	0	0	,	2	0	0	0
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

,	0	0	0	,	0	0	0	0	0	0	2	0	0	,	1	8	0	0	8	0	ETX
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If error pattern is "5"

Related data	Details	Minimum value	Maximum value	Unit
1	Air Supply Pressure	0	999	[kPa] [bar]
2	Diagnosis result 0 = Failure 1 = Pass	0	1	[-]
3 to 6	None	-	-	-

Data packet legend

Air Supply Pressure = 500kPa
Diagnosis result = Pass

STX	1	0	D	A	6	7	,	5	,	5	0	0	,	1	,	,	,	7	E	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

If error pattern is "6"

Related data	Details	Minimum value	Maximum value	Unit
1	Error response	0	7	-
2 to 6	None	-	-	-

Data packet legend

STX	0	D	D	A	6	7	,	6	,	7	,	,	,	,	F	9	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

[DA68 Acquisition of model information]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	68	68	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	8	B	5	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Product model	V5 V2	14	S-SIGMA-CM4-CTR-V5 S-SIGMA-CM4-CTR-V2	S-SIGMA-CM4-CTR-V5 S-SIGMA-CM4-CTR-V2	[-]
2	Firmware1	,	10	SCM4-01.00	SCM4-99.99	[-]
3	Firmware2	,	9	0001.0000	FFFF.FFFF	[-]
4	Firmware3	,	9	0000.0000	FFFF.FFFF	[-]

Data packet legend

Product model = S-SIGMA-CM4-CTR-V5
Firmware1 = SCM4-01.00
Firmware2 = 01.00
Firmware3 = 00.00

STX	3	2	D	A	6	8	,	S	-	S	I	G	M	A	-	C	M	4	-	V	5	,	S	C	M	4	-	0	1	.	0	0	
	,	0	0	0	1	.	0	0	0	0	,	0	0	0	0	.	0	0	0	0	C	E	ETX										

[DA69 Acquisition of Firmware 4]

Request packet data 1 and 2

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Unused data*	-	3	0	0	[-]
2	Data packet specification	D	2	69	69	[-]

* Unused data. Always insert 3 digits of 0.

Request packet legend

STX	0	8	U	L	0	0	0	D	6	9	B	4	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	-----

Data packet data

No.	Details	Prefix	Digit number	Minimum value	Maximum value	Unit
1	Firmware4	,	10	CM4H-01.00	CM4H-99.99	[-]

Data packet legend

Firmware4 = CM4H-01.00

STX	0	F	D	A	6	9	,	C	M	4	H	-	0	1	.	0	0	4	2	ETX
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

Troubleshooting

If there are any problems, first read the following carefully and take appropriate action.
Contact our sales department if the problem is not resolved with the actions outlined below.

Symptoms	No.	Cause	Action
No response packet is received.	1	A cable is disconnected.	● Check the cable connection conditions.
	2	There is an error with the connection specifications.	● See P.3"RS-232C Communication Connection Specifications".
	3	There is an error with the communication speed.	● Adjust the communication speed to match the host device.
	4	There is an error with the communication specifications and connection signal.	● See P.3"RS-232C Communication Specifications".
	5	There is an error with the communication port.	● Check the communication port of the host device.
The response packet error type is "2," indicating a response packet error.	6	Unable to receive footer ("ETX(0x03)").	● Check the connection status.
	7	There is an error with the checksum.	● See P.64"Checksum Calculation" and check if there are any errors with the checksum calculation method.
	8	No header ("STX(0x02)").	● The request packet begins with the header ("STX(0x02)"). Check the request packet.
	9	There is an error with the command.	● See P.65"Request Packet List" and correct the command.
	10	The data is outside the allowable setting range.	● See the relevant command in P.73"Command" and correct the data.
	11	Not in receive status	● Tap the [←BACK] or [←] icon to display the Dispense screen. ● If in In-dispensing, wait until dispense end. (P.119)

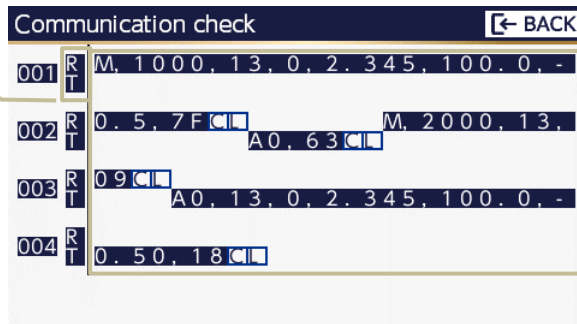
4 Checking the Transmission Data with the Controller (Communication check)

This device can check the transmission data with the host device.

See P.4 "Changing RS-232C Communication Settings" and move to the "RS-232C communication setting" screen. Then tap "Com. check" to display the "Communication check" screen.

R: This device receive status
T: This device transmit status

* Press and hold the screen deletes the history.



The transmission data history is displayed.

The following control code is omitted from being displayed.

Control Code	Data	Display
STX	0x02	S
ETX	0x03	E
EOT	0x04	O
ENQ	0x05	Q
ACK	0x06	A
LF	0x0a	L
CR	0x0d	C
CAN	0x18	N

ASCII Code Chart

		Top bit 4							
		0	1	2	3	4	5	6	7
Lower bit 4	0	NUL	DLE	SP	0	@	P	'	p
	1	SOH	DC1	!	1	A	Q	a	q
	2	STX	DC2	"	2	B	R	b	r
	3	ETX	DC3	#	3	C	S	c	s
	4	EOT	DC4	\$	4	D	T	d	t
	5	ENQ	NAK	%	5	E	U	e	u
	6	ACK	SYN	&	6	F	V	f	v
	7	BEL	ETB	'	7	G	W	g	w
	8	BS	CAN	(	8	H	X	h	x
	9	HT	EM	)	9	I	Y	i	y
	A	LF	SUB	*	:	J	Z	j	z
	B	VT	ESC	+	;	K	[	k	{
	C	FF	FS	,	<	L	\	l	
	D	CR	GS	-	=	M	]	m	}
	E	SO	RS	.	>	N	^	n	~
	F	SI	US	/	?	O	_	o	DEL

Communication Ethernet

Please check the Basics before reading this book.

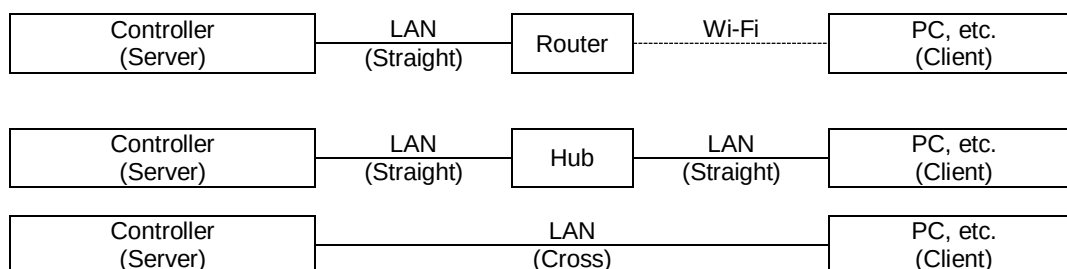
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Connection using Ethernet communications

1 Overview

This unit can be controlled via Ethernet from an Ethernet compatible device such as a PC. Ethernet-compatible devices can set and acquire commands such as channel step and preset, and dispense conditions under the HTTP protocol.



Two functions are provided using Ethernet communication.

Web server function : By operating a dedicated Web page displayed on a general-purpose Web browser (Microsoft® Edge, Google Chrome™, etc.), you can check the operating status and setting values of the controller, and you can set parameters.

Web API function : To set or get commands such as channel step/preset, and dispense conditions, Web API can be used. It allows you to develop your own control software easily.

* Use this function after connecting to the company network. If you need to connect to an external Internet, do not connect directly and be sure to take security measures such as using a firewall.

Ethernet communication specifications

The communication specifications used by this device are as follows.

Standard (LAN port)	RJ-45 10BASE-T/100BASE-TX
Com. protocol	TCP/IP (HTTP) *1
DHCP	Incompatible
Number of terminals that can be connected simultaneously	3
IP address	192.168.0.10 (default)
Subnet mask	255.255.255.0 (default)
Default gateway	192.168.0.1 (default)
Port number	1024, 1025, 1026
Supported character code	UTF-8
Read only	OFF (default)
Web page refresh interval (Web server function)	0 to 60 minutes (Default: 0 minutes (Web page does not refresh automatically))

*1 HTTPS is not supported.

Ethernet communication specifications

A commercially available LAN cable (RJ45, Category 3 or higher) can be used.

Access Rights

Two privileges are available, "operator" and "administrator".

- | | | |
|---------------|---|---|
| Operator | : | There are some restrictions on the display screen of the Web server and the available Web API.
It is possible to obtain the status of the controller and issue an operation command, but not possible to change the setting value. |
| Administrator | : | All screens of the web server can be displayed and operated.
Also, you can use all Web APIs. |

You will have operator privileges immediately after you connect the line. If you successfully log in as an administrator, you will have administrator privileges.

To log in as an administrator, you are required to have "administrator password".

For information on changing the setting of "administrator password", see P.6 "Administrator setting".

If "Read only" is set to ON, only operator privileges are available. Note that only setting import requests will be accepted (anything else returns a 403 error).

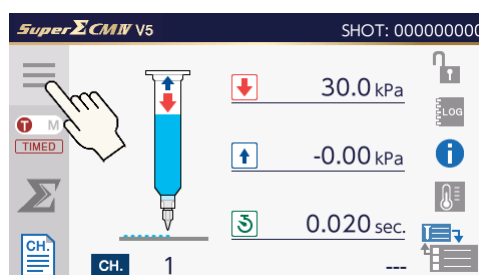
For information on changing the setting of "Read only", see P.7 "Read only".

Setting change of Ethernet communication

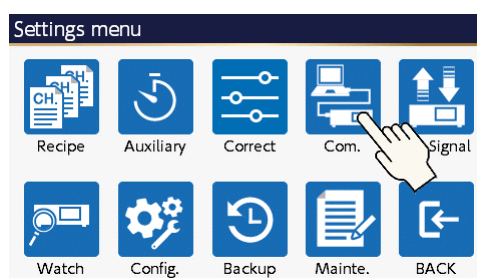
Item	Description
"Controller setting"	Edits and displays various settings for Ethernet communications. IP address/Subnet mask/Default gateway/MAC address
"Administrator setting"	Sets a password to login as an administrator. Displays the date and time of the last login.
"Read only"	When this is ON, restricts operations to only getting controller settings.
"Web page refresh interval"	Sets the interval for automatically updating the web page screen when using the web server.
"Com. check"	Allows you to check the connection status of Ethernet communication. For details See P. 47"6 Check the communication status with the controller (communication check)".

To change the Ethernet communication settings, move to the "Ethernet communication setting" screen by following the steps below.

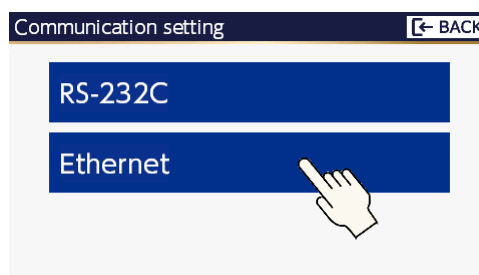
- 1 Tap the [☰] icon in Dispense Screen A or B.



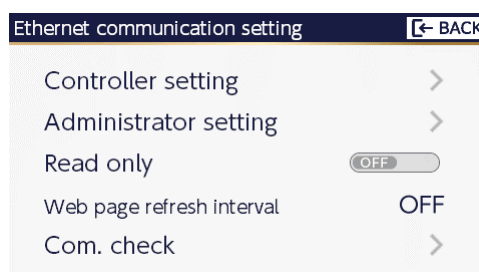
- 2 The "Settings menu" screen is displayed. Tap the [Settings icon] icon.



- 3 The "Communication setting" screen is displayed. Tap "Ethernet".



- 4 "Ethernet communication setting" screen appears. Set each item.



"Controller setting"

On the "Ethernet communication setting" screen, tap "Controller setting." "Controller setting" screen appears.

Controller setting		← BACK
IP address	192.168.0.10	
Subnet mask	255.255.255.0	
Default gateway	192.168.0.1	
MAC address	D4-61-37-CF-FF-FF	

- "IP address"
This is the IP address to set for the controller.
Tap to move to the edit screen. Set each of the four values separated by dots.
- "Subnet mask"
This is the subnet mask to set for the controller.
Tap to move to the edit screen. Set each of the four values separated by dots.
- "Default gateway"
This is the default gateway to set for the controller.
Tap to move to the edit screen. Set each of the four values separated by dots.
- "MAC address"
This is the MAC address that was set for the controller.
Displays only, cannot be edited.

*The changes of IP address, subnet mask and default gateway will take effect after reboot."

"Administrator setting"

On the "Ethernet" screen, tap "Administrator setting". "Administrator setting" screen appears.

Administrator setting		← BACK
Administrator password		>
Last log-in date	2023/10/15	
	15:00:00	

- "Administrator password"
This password is required for administrator login. Tap to move to the edit screen.
*This is different from the password setting on the "Configure" screen.
- "Last log-in date"
Displays the date and time of the last administrator login.

"Read only"

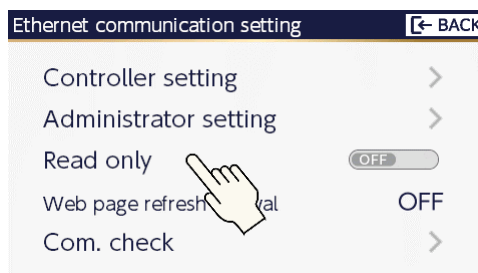
When set to ON, setting changes and operation commands via Ethernet communication are prohibited.

- Only operator privileges are available for all ports.
- The Web API that can be requested on this unit is limited to acquisition of settings only.
- The screen displayed on the web browser is restricted. In particular, the "Remote" screen is only displayed and it cannot be edited.

For details, see P.11 "Screen list".

See below for how to change the settings.

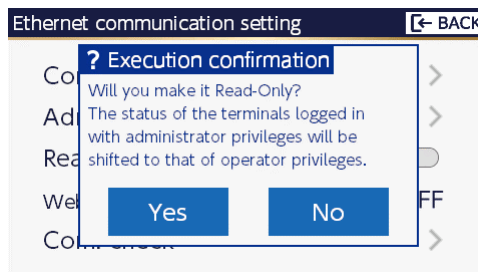
- 1 On the "Ethernet communication setting" screen, tap "Read only".



- 2 The setting change confirmation screen appears. Select [Yes] to change the settings.

*The figure on the right is for when "Read only" is set to ON.

*If you are connecting with an Ethernet-compatible device, changing "Read only" to ON will shift to operator privilege.



"Web page refresh interval"

This is the parameter related to the Web server function.

You can set the interval for automatically updating the web page.

On the "Ethernet communication setting" screen, tap "Web page refresh interval". The numeric input screen appears.

Default: 0 minutes (Web page does not refresh automatically)
Range: 0 to 60 min.

Number of clients that can be connected simultaneously

The number of clients that can use Ethernet communication simultaneously is 3 for each controller.
To send a URL, specify a port number for an IP address of the controller.

Available port numbers : 1024, 1025, 1026

Example:

IP address : 192.168.0.10
Port number : 1024

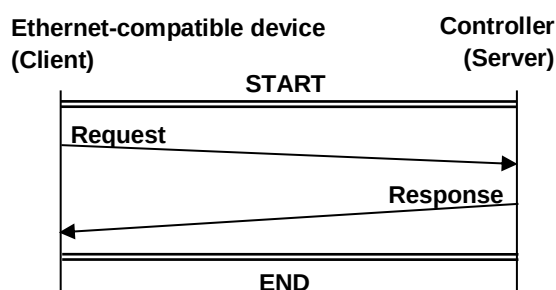
- To display the operation status confirmation screen on the web browser, go to <http://192.168.0.10:1024/>
- To get the current channel with Web API, go to <http://192.168.0.10:1024/v1/status/channel>

Communication sequence

Follow the steps below for Ethernet communication between this unit and an Ethernet-compatible device.

- 1 Send a request from an Ethernet-compatible device.
- 2 The controller that received the request sends a response to the Ethernet-compatible device, and the communication ends.

If there is no problem with the content of the request or the state of the controller, and the request is processed correctly, the status code will be 200 (OK).
On the other hand, if the request was not processed correctly, the status code will be a value of a 400 number.



Status code

When the controller receives a request, a 3-digit number (status code) is issued as a result of system processing.

Status code	Explanation of status code	Content
200	OK	Successfully processed.
400	Bad Request	The request was rejected due to reasons such as the controller not being in a wait status. If the state does not belong to any status code 401 or higher, the code will be this status code. Different messages are sent for each error cause. For details, See P.21 "List of reasons for rejections to requests".
401	Unauthorized	Administrator login failed. A request requiring administrator privileges was received while in the state of operator privileges.
403	Forbidden	A request other than acquisition of settings was received when "Read only" was set to ON.
404	Not Found	The page cannot be found because there is an error in the sent URL.
405	Method Not Allowed	The specified method does not correspond to the URL.
411	Length Required	"Content-Length" is not attached to the header part.
413	Payload Too Large	Posted data size is too large.
414	URI Too Long	URL is too long.

2 Web Server

Overview

You can access the controller from a web browser on an Ethernet-compatible device such as a PC. There is no need for special software, and you can check the operation status of the controller and acquire and edit setting values simply by entering URLs or operating buttons.

Supported browser

For the web server function, the web browsers whose operation has been confirmed by us are as follows.

Ethernet-compatible device	Web browser	Web browser version	Web browser compatible OS
PC	Microsoft® Edge	109.0.1518.55	Microsoft® Windows® 10
	Google Chrome™	88.0.4324.182	Microsoft® Windows® 10
	Mozilla Firefox®	108.0.2	Microsoft® Windows® 10
Tablet	Mobile Google Chrome™	88.0.4324.181	Android® 11

Screen Layout

The screen consists of four areas.

The screenshot displays the Super Σ CMIV web interface in a browser window. The address bar shows the URL 192.168.0.10:1024. The interface is divided into four main areas:

- (1) Header:** Contains the Super Σ CMIV logo and a user status bar showing "User: Operator" and "IP address: 192.168.0.100".
- (2) Main Table:** A table with columns: Status, Channel, Σ, Recipe, Auto. Inc., and Temp. unit. It lists various system parameters such as Air supply pressure, Controller time, ERROR, ALARM, Remain [%], User counter, Total counter, DIS. state, Auto. Inc. state, Interval feature, and Password lock.
- (3) Left Navigation Menu:** A vertical list of buttons for navigating between different sections: Op state, CH. list, Recipe list, CH. common, Log list, Backup, Maintenance, Model info., Remote, Admin only, and API test.
- (4) Bottom Control Panel:** A row of buttons for system control, including "CH. clear", "CH. list", "CH. edit", "CH. delete", "Σ", "0000", "INTERVAL", "INTERVAL OFF", "RECIPE", and a lightning bolt icon.

- (1) Banner
Click to display the operating status screen.
If the controller name is set, it will be displayed on the right side of the banner.

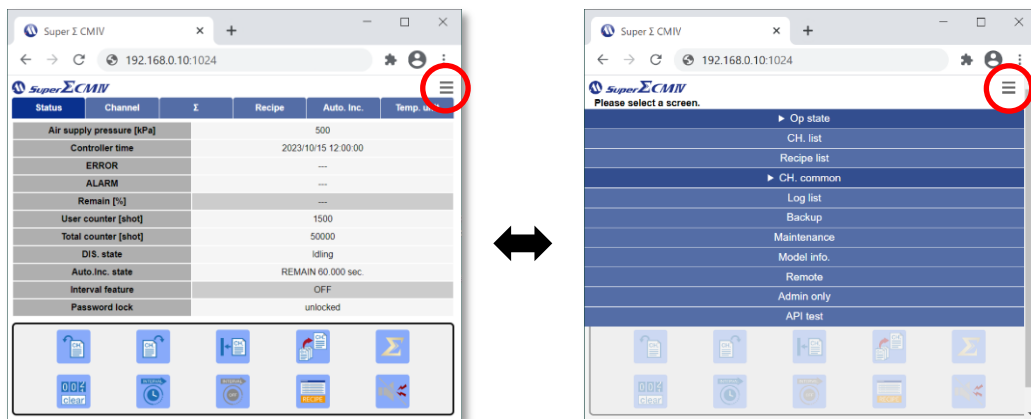


- (2) User information
The privilege and IP address of the Ethernet-compatible device which is being accessed are displayed.
- (3) Side bar
By pressing the button, you can move to each screen prepared for each purpose.
- (4) Main
Displays the operating status and setting data.
For each screen, refer to P. 11 "Screen explanation details".

- * Buttons that require administrator privileges to run are grayed out under operator privileges. When you press one of them, the administrator password input screen appears. Enter the administrator password and press the [Submit] button. If the password is correct, you will have administrator privileges and be able to view all screens.
For screens that require an administrator login, refer to P. 11 "Screen list".

The screen shows a sidebar on the left with options: Op state, CH. list, Recipe list, CH. common, and Log list. The main area has the text 'Enter the administrator password.' followed by a text input field containing 'password' and a blue 'Submit' button.

- * If the width of the screen is narrow, the user information and sidebar are hidden. You can switch the display of the screen list by pressing the red circle part in the figure below. You can move to each screen in the same way as the sidebar.



Screen list

The screen displayed on the web browser is as follows.

Screen	From the sidebar	Administrator only	When read only status	Detailed explanation
"Op state"	○	—	○	P. 11
"CH. list"	○	—	○	P. 13
"Recipe list"	○	—	○	P. 13
"CH. common"	○	○	—	P. 13
"Log list"	○	—	○	P. 14
"Backup"	○	△ *1	○	P. 14
"Maintenance"	○	—	—	P. 14
"Model information"	○	—	○	P. 15
"Remote"	○	△ *2	○ *3	P. 15
"Admin only"	○	○	—	P. 15
"API test"	○	○	—	P. 15
"Edit CH."	—	○	—	P. 16
"Edit recipe"	—	○	—	P. 17
"Add recipe"	—	○	—	P. 17
"Input signal alloc."	—	○	—	P. 18
"Output signal alloc."	—	○	—	P. 18
"Log details"	—	○	○	P. 18

*1 Administrator login is required to use the import function.

*2 Administrator login is required to operate the screen.

*3 Screen operation is not available.

Screen explanation details

This is an explanation of each screen displayed on the web browser.

Screens that can be accessed from the sidebar are indicated with ◇, and screens that can be accessed by other operations are indicated with ◆.

◇"Op state" screen

You can check the operating status of the controller and parameter settings related to the current channel. Information is categorized by tabs.

1 "Status"

Status	Channel	I	Recipe	Auto. Inc.	Temp. unit
Air supply pressure [kPa]				500	
Controller time				2023/10/15 12:00:00	
ERROR				---	
ALARM				---	
Remain [%]				---	
User counter [shot]				1500	
Total counter [shot]				50000	
DIS. state				Idling	
Auto. Inc. state				REMAIN: 60 000 sec	
Interval feature				OFF	
Password lock				unlocked	

Allows you to check controller statuses such as air supply pressure and error status.

This is displayed immediately after connecting to the web server or when the banner is clicked.

2 "Channel"

Status	Channel	I	Recipe	Auto. Inc.	Temp. unit
Current CH.				1	
CH. name					
DIS-mode				Timed	
DIS-time [sec.]				0.020	
DIS-press. [kPa]				30.0	
Vacuum [kPa]				-0.00	
Syringe size				5mL	
Tube length				1.0m	
Plunger				None	
ECO mode				OFF	
On delay [sec.]				0.000	
Off delay [sec.]				0.000	
DIS-monitoring				unchecked	
Dispensing OFF time [sec.]				0.500	
Allowable range [kPa]				100.0	

Displays the current channel and its settings.

[Edit]

Moves to the "Edit CH." screen.

*Administrator login is required.

3 "Σ"

Status	Channel	I	Recipe	Auto. Inc.	Temp. unit
Current CH.				1	
DIS-mode				I + Timed	
Volume reduction correction				ON	
Alpha correction				0	
Delta correction [%]				100	
Fluid-drip prevention				ON	
Vacuum correction [kPa]				0.00	
Remaining vol. alert				ON	
RSM level [%]				20	
RSM count				5	
Edit					

Displays the current channel and Σ function settings.

[Edit]

Moves to the "Edit CH." screen.

*Administrator login is required.

4 "Recipe"

Status	Channel	I	Recipe	Auto. Inc.	Temp. unit
Recipe no.				1	
Start CH.				1	
End CH.				15	
I mode CH.				Not include	
Recipe name				Sample01	
Liquid				Liquid	
Nozzle				Nozzle	
Auto-increment				Total time	
Temp. unit setting				OFF	
Edit					

If the current channel belongs to a recipe, displays the settings for that recipe.

[Edit]

Move to the "Edit recipe" screen.

*Administrator login is required.

5 "Auto-increment"

Status	Channel	I	Recipe	Auto. Inc.	Temp. unit
ON/OFF				ON	
DIS-mode				TIMED	
Switching condition				Total time	
Increment value [sec.]				60	
Edit					

If the current channel belongs to a recipe, displays the auto-increment setting for that recipe.

[Edit]

Move to the "Edit recipe" screen.

*Administrator login is required.

6 "Temp. unit"

Status	Channel	I	Recipe	Auto. Inc.	Temp. unit
ON/OFF	#1	#2	#3	#4	#5
Measured [°C]	ON	ON	OFF	OFF	OFF
Setting [°C]	49.8	ERROR	---	---	---
Upper limit [°C]	50.0	40.0	25.0	25.0	25.0
Lower limit [°C]	+1.5	+1.0	+1.0	+1.0	+1.0
	-1.5	-1.0	-1.0	-1.0	-1.0
Edit					

If the current channel belongs to a recipe, displays the temperature unit setting value and error status of that recipe.

[Edit]

Move to the "Edit recipe" screen.

*Administrator login is required.

Various icons are placed on the "Op state" screen to give commands to the controller when they are clicked.
*These icons will not be displayed when "Read only" is set to ON.



Icon	Function	Icon	Function	Icon	Function	Icon	Function
	Moves down one channel. Same as DOWN signal.		Increases the channel number by 1. Same as STEP signal.		Calls the default channel. Same as PRESET signal.		Calls the specified channel.
	Full sampling starts now.		Turns ON the interval feature.		Turns OFF the interval feature.		Clears the User counter.
	Calls the start channel of the specified recipe.		Stops the buzzer sound.				

◇ "CH. list" screen

CH	CH name	DIS-mode	DIS-time [sec.]	DIS-press. [kPa]	Vacuum [kPa]	ECO mode	Syringe size	Tube length	Plunger	+	-	+	-
1	test1	Timed	0.020	30.0	-0.00	OFF	10mL	1.0m	None				
2	test2	Timed	0.025	35.0	-0.00	OFF	10mL	1.0m	None				
3	test3	Timed	0.030	40.0	-0.00	OFF	10mL	1.0m	None				
4	test4	Timed	0.035	45.0	-0.00	OFF	10mL	1.0m	None				
5	test5	Timed	0.040	50.0	-0.00	OFF	10mL	1.0m	None				
6	test6	Timed	0.045	55.0	-0.00	OFF	10mL	1.0m	None				
7	test7	Timed	0.050	60.0	-0.00	OFF	10mL	1.0m	None				
8	sigma	Σ+Timed	0.500	100.0	-0.00	OFF	10mL	1.0m	None				
9		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
10		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
11		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
12		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
13		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
14		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
15		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
16		Timed	0.020	60.7	-0.00	OFF	10mL	1.0m	None				
17		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
18	test1	Timed	0.020	30.0	-0.00	OFF	10mL	1.0m	None				
19		Timed	0.001	30.0	-0.00	OFF	10mL	1.0m	None				
20		Timed	0.020	68.9	-0.00	OFF	10mL	1.0m	None				

You can check the setting data of all channels.

By clicking a data line you can move to the "Edit CH." screen of its channel.

*Administrator login is required.

*By clicking the header part, that column is folded up in the display. Clicking the header part with the + display enables that column to be unhidden.

◇ "Recipe list" screen

Recipe no.	Start CH	End CH	I mode CH	Recipe name	Liquid	Nozzle	Auto-increment	Temp. unit setting
1	1	15	Not include	Sample001	Liquid A	Nozzle A	Total count	ON
2	-	-	-	-	-	-	-	-
3	41	60	Not include	Sample002	Liquid B	Nozzle B	OFF	OFF
4	101	140	Include	Sample003	Liquid C	Nozzle C	OFF	ON
5	-	-	-	-	-	-	-	-
6	-	-	-	-	-	-	-	-
7	-	-	-	-	-	-	-	-
8	-	-	-	-	-	-	-	-
9	-	-	-	-	-	-	-	-
10	-	-	-	-	-	-	-	-
11	-	-	-	-	-	-	-	-
12	-	-	-	-	-	-	-	-
13	-	-	-	-	-	-	-	-
14	-	-	-	-	-	-	-	-
15	-	-	-	-	-	-	-	-
16	-	-	-	-	-	-	-	-
17	-	-	-	-	-	-	-	-
18	-	-	-	-	-	-	-	-
19	-	-	-	-	-	-	-	-
20	-	-	-	-	-	-	-	-

You can check the setting data of all recipes.

Colored rows represent predefined recipes.

By clicking one of them, you can move to the recipe edit screen for that recipe.

Uncolored rows represent undefined recipes.

By clicking one of them, you can move to the "Add recipe" screen.

*Administrator login is required.

*By clicking the header part, that column is folded up in the display. Clicking the header part with the + display enables that column to be unhidden.

◇ "CH. common" screen

You can view and change the configure setting parameters of the controller.

Parameters are categorized by tabs.

*Administrator login is required to access it.

*When you press the [Submit] button, the data will be sent and the data in the controller will be changed.

*When you press the [Read cur val.] button, the data will be received and the data in the controller will be displayed again.

1 "Configure"

Configure	Interval	DIS-correction	RS-232C	I/O signal
Language			English ☒ 日本語 ☐ 中文	
P.UNIT			kPa ☒ bar	
Empty alarm			☐	
Empty warning thresh. [shot]			100000	
Valve warning thresh. [shot]			80000000	
Buzzer			☒	
T-ready enable			☐	
Warm-ready enable			☐	
Pop-up auto OFF			☒	
Immed. Inc. (Ainc)			☐	
Control mode			☐ Speedy ☒ Stable	
Sleep timer			OFF v	
RSM advance notice lv. [%]			10	
Screen brightness			50	
Password			0000	
Submit Read cur val.				

You can check and set the parameters on the controller's "Configure" screen.

2 "Interval feature"

Configure	Interval	DIS-correction	RS-232C	I/O signal
ON/OFF			☒	
Link CH. [ch]			20	
Count [shot]			4	
Interval [sec.]			0.2	
Repeat			5	
Wait time [sec.]			20	
Mode			☐ MAN ☒ SET	
Submit Read cur val.				

You can check and set the parameters related to the interval feature.

3 "DIS-correction"

You can check and set parameters related to DIS-correction.

4 "RS-232C"

You can check and set parameters related to RS-232C communication.

5 "I/O signal"

You can check and set parameters related to I/O signals.
For input signal allocation and output signal allocation, click "jump" to move to the dedicated screen.

◇ "Log list" screen

Date of occurrence	Code	Message
2023/02/02 14:12:03	302	User counter clear
2023/02/02 14:11:53	202	Temp. unit error clear
2023/02/02 14:11:49	012	Temp. unit error: Timeout
2023/02/02 14:11:45	006	Air supply pressure error
2023/02/02 14:11:45	100	Air supply pressure warning
2023/02/02 14:11:24	012	Temp. unit error: Timeout
2023/02/02 14:11:09	006	Air supply pressure error
2023/02/02 14:11:09	100	Air supply pressure warning
2023/02/02 14:10:48	006	Air supply pressure error
2023/02/02 14:10:48	100	Air supply pressure warning
2023/02/02 14:10:28	006	Air supply pressure error
2023/02/02 14:10:28	100	Air supply pressure warning
2023/02/02 14:10:05	006	Air supply pressure error
2023/02/02 14:10:05	100	Air supply pressure warning
2023/02/02 14:09:48	006	Air supply pressure error
2023/02/02 14:09:48	100	Air supply pressure warning
2023/02/02 14:09:23	006	Air supply pressure error
2023/02/02 14:09:23	100	Air supply pressure warning
2023/02/02 14:09:06	006	Air supply pressure error
2023/02/02 14:09:06	100	Air supply pressure warning

Up to 20 items of log data recorded in the controller are displayed.

By clicking it, you can move to the "Log details" screen.

◇ "Backup" screen

Data can be imported and exported via JSON files.
Also, the log recorded in the controller can be acquired in TSV file format.

*Administrator login is required to use the import function.

◇ "Maintenance" screen

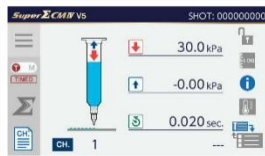
Sensor calibration, diagnostics, and counter clearing can be performed.

◇ "Model info." screen

Product model	S-SIGMA-CM4-V5
I/O specifications	NPN
Firmware1	SCM4-01.00
Firmware2	01.00
Firmware3	00.00
Firmware4	CM4H-01.00
Total counter [shot]	50000
User counter [shot]	1500
Controller time	2023/10/15 12:00:00

Displays model information such as the product model of the controller.

◇ "Remote" screen



The screen displayed on the controller can be operated remotely.

*Administrator login is required for remote operation.

◇ "Admin only" screen

Controller info.		All memory clear Parameter clear Log clear Time adjust Controller name
IP address	192.168.0.10	
Subnet mask	255.255.255.0	
Default Gateway	192.168.0.1	
MAC address	D4-61-37-CF-FF-FF	
Browser refresh interval [min.]	10	
Administrator info.		
Last log-in	2023/01/18 12:34:56	
Password	0000	

Displays information necessary for Ethernet communication.

*Administrator login is required to access it.

[All memory clear]

Restore the settings to factory defaults.

*Ethernet communication settings will not be initialized.

[Parameter clear]

Initializes all recipe and channel data.

[Log clear]

Clears the log information.

[Time adjust]

Sends the time information of the Ethernet-compatible device to the controller and updates the time of the controller.

[Controller name]

Sets the controller name.

◇ "API test" screen

API:

Data for POST:

Received Data:

```
{
  "status": "Ready",
  "alarm": 0,
  "alarm": 0
}
```

GET

POST

You can make HTTP requests to the controller. It can be used to check the operation of Web API.

*Administrator login is required to access it.

◆ "Edit CH." screen

- For Normal mode CH
 - Channel setting

Channel	Σ
CH	100
CH name	sample #5
DIS-mode	☒ TIMED ☐ MANUAL
DIS-time [sec.]	4.500
DIS-press. [kPa]	30.0
Vacuum [kPa]	0.05
Syringe size	10mL
Tube length	1.0m
Plunger	☒ None ☐ Use
ECO mode	☐
On delay [sec.]	0.500
Off delay [sec.]	0.500
Dispensing OFF time [sec.]	0.500
Allowable range [kPa]	100.0
DIS-monitoring	unchecked

Submit Read cur val. Set as cur CH.

• Σ Setting

Channel	Σ
Σ mode is OFF for this CH.	

You can view and update channel information.

*Administrator login is required to access it.
*Settings relating to the Σ function cannot be changed.

[Submit]

Sends the channel information described in the table to the controller.

[Read cur val.]

Reacquire the channel information being displayed.

[Set as cur CH]

Makes the displayed channel the current channel.

- For Σ mode CH
 - Channel setting

Channel	Σ
It is a Σ mode CH.	
CH	100
CH name	sample #5
On delay [sec.]	0.500
Off delay [sec.]	0.500
Dispensing OFF time [sec.]	0.500
Allowable range [kPa]	100.0
DIS-monitoring	unchecked

Submit Read cur val. Set as cur CH.

• Σ Setting

Channel	Σ
Volume reduction correction	
ON/OFF	☒
Alpha correction	0
Delta correction [%]	100
Fluid-drip prevention	
ON/OFF	☒
Vacuum correction [kPa]	0.00
Remaining vol. alert	
ON/OFF	☒
RSM level [%]	20
RSM count	5

Submit Read cur val. Set as cur CH.

You can view and update channel information.

*Administrator login is required to access it.
*DIS-condition settings and DIS-auxiliary settings cannot be edited.

[Submit]

Sends the channel information described in the table to the controller.

[Read cur val.]

Reacquire the channel information being displayed.

[Set as cur CH]

Makes the displayed channel the current channel.

◆ "Edit recipe" screen

● Recipe basic settings

Recipe	Auto. inc.	Temp. unit
Recipe no.		1
Start CH. [ch]		1
End CH. [ch]		15
I mode CH.		Not include
Recipe name		sample001
Liquid		Liquid A
Nozzle		Nozzle A

● Auto-increment function settings

Recipe	Auto. inc.	Temp. unit
ON/OFF	☒	
DIS-mode	☒ TIMED ☐ MANUAL	
ECO mode	☐	
Switching condition	☒ Total time ☐ Total count ☐ Pot life	
Total time [sec.]		60
Total count [shot]		9999
Pot life [sec.]		9999

Recipe	Auto. inc.	Temp. unit
Recipe no.		1
Start CH.		1
End CH.		15
I mode CH.		Not include

● Temperature unit setting

Recipe	Auto. inc.	Temp. unit
ON/OFF	☒	
Setting [°C]	50.0	
Upper limit [°C]	1.5	
Lower limit [°C]	1.5	

Recipe	Auto. inc.	Temp. unit
Recipe no.		1
Start CH.		1
End CH.		15
I mode CH.		Not include

You can view and update recipe information.

*Administrator login is required to access it.

*If the recipe range contains a Σ mode CH, the auto-increment function cannot be turned ON.

[Submit]

Sends the information set in the table to the controller.

[Read cur val.]

Reacquires the displayed recipe information.

[Del. this recipe]

Clears the displayed recipe.

◆ "Add recipe" screen

Please select a CH range.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140
141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160
161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220
221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260
261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280
281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320
321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340
341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360
361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380
381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400

Recipe no. 7
Start CH. 251
End CH. 255
Num. of CHs. 5

Explanatory notes
 The number of channels and setting in a recipe.
 The selected CH.
 CH of the registered recipe.
 New CH of the registered recipe.

By clicking an undefined recipe row in the table of the recipe list screen, this screen appears.

To select a recipe which is to be newly added, click the cell of the start channel and end channel of it.

[Submit]

Recipes can be defined in the channel range selected on the table. After clicking, the screen moves to the "Edit recipe" screen.

[Reset]

You can reset the selection status.

*Cells in the blue box indicate Σ mode CH.

◆ "Input signal alloc." screen

Allocation can be modified by a drag-and-drop operation.

Setup value				Candidate			
02	DIS			STEP			
03	TMS	04	CHS	05	CD02	06	CD08
07	CD20	08	ENABLE	15	CD80	16	TMD
17	CD01	18	CD04	19	CD10	20	CD40

Submit Read cur val.

By clicking "jump" in the input signal allocation on "I/O signal" tab on "CH. common" screen, this screen appears.

You can allocate the input signal of the D-Sub25 pin.

◆ "Output signal alloc." screen

Allocation can be modified by a drag-and-drop operation.

Setup value				Candidate			
09	END	10	EXE1	11	PON	12	NDVO
21	DSO	22	RSM	23	PSE1	24	DVO

Submit Read cur val.

By clicking "jump" in the output signal allocation on "I/O signal" tab on "CH. common" screen, this screen appears.

You can allocate the output signal of the D-Sub25 pin.

◆ "Log details" screen

Code	006
Message	Air supply pressure error
Date of occurrence	2023/01/18 12:34:56
Total counter [shot]	2204

Related data

Error pattern	Dispensing
Supply [kPa]	0
DIS-mode	Timed
DIS-time [sec.]	5.000
DIS-press. [kPa]	97.1
Vacuum [kPa]	-0.00
DIS-state	Idle

Next log Previous log

This is displayed when you click the table on the "Log list" screen.

Displays detailed log information for the row you clicked.

3 Web API

Overview

Describes Web API specification of this device.

By using Web API, it is possible to get and change the configuration parameters of the controller.

Except for some Web APIs, the data will be in JSON format.

HTTP method

The HTTP methods that can be used with this unit are only POST and GET.

[POST]

This is used when logging in and changing controller setting values.

Requests must be accompanied by string data, such as JSON data.

- When processed successfully
The status code will be 200 (OK).
- When processed abnormally
The status code will be a 400s number.
The error cause will be attached to the response in JSON format.

[GET]

This is used to issue an operation start command to the controller and to acquire setting values.

- When processed successfully
The status code will be 200 (OK).
When sending a setting value acquisition request, the request data is attached to the response in JSON format.
- When processed abnormally
The status code will be a 400s number.
The error cause will be attached to the response in JSON format.

List of Web APIs

Web API		Operation	HTTP method	URL	Error	
Login		Registration	POST	/v1/login	A	
Logout		Registration	GET	/v1/logout	-	
Data operation	General purpose	Registration	POST	/v1/import	B, C, E, F, X	
	All data	List acquisition	GET	/v1/export/data	B	
	Log	List acquisition	GET	/v1/export/log	B	
Screen		Acquisition	GET	/v1/screen	-	
Time		Registration (administrator)	POST	/v1/time	A, B, E, F	
		Acquisition	GET	/v1/time	-	
		Update	GET	/v1/current/channel/{id}	B, C, E, F, I	
Execution	To the start channel of the specified recipe	Update	GET	/v1/current/recipe/{id}	B, C, E, F, I, L	
	Dispensing stop	Execution	GET	/v1/current/dispense/0	E	
	Dispensing start	Execution	GET	/v1/current/dispense/1	B, C, D, E, F, R	
	Move up one channel	Execution	GET	/v1/current/step	B, C, E, F, I, O	
	Move down one channel	Execution	GET	/v1/current/down	B, C, E, F, I, O	
	Call the default channel	Execution	GET	/v1/current/preset	B, C, E, F, I	
	To TIMED mode	Execution	GET	/v1/current/timed	B, C, E, F, I, J, M, Q, R	
	To MANUAL mode	Execution	GET	/v1/current/manual	B, C, E, F, I, J, M, Q, R	
	Interval feature ON	Update	GET	/v1/current/interval/1	B, C, E, F	
	Interval feature OFF	Update	GET	/v1/current/interval/0	B, C, E, F, M	
	DIS-monitoring test	Execution	GET	/v1/current/test	B, C, D, E, F, R	
	Temperature unit auto-tuning	Execution	GET	/v1/current/autoTune/{id}/{onoff}	B, C, E, F	
	Buzzer cut	Execution	GET	/v1/current/buzzerCut	-	
	Solenoid valve alarm OFF	Execution	GET	/v1/current/warningOff	-	
	Full sampling	Execution	GET	/v1/current/sampling/full	B, C, D, E, F, M, Q, R	
	Empty sampling	Execution	GET	/v1/current/sampling/empty	B, C, D, E, F, M, P, R	
	Detailed sampling	Execution	GET	/v1/current/sampling/detail/{remain}	B, C, D, E, F, M, P, R	
	To dummy shot mode	Execution	GET	/v1/current/dummyShotMode/1	B, C, E, F, H, I, J, P, R	
	Cancel dummy shot mode	Execution	GET	/v1/current/dummyShotMode/0	B, C, E, F, H, P, R	
	Acquisition of settings	Dispense status, etc.	Acquisition	GET	/v1/status/main	-
		Current channel	Acquisition	GET	/v1/status/channel	-
		Current recipe	Acquisition	GET	/v1/status/recipe	-
		Occurring error	Acquisition	GET	/v1/status/error	-
		Active alarm	Acquisition	GET	/v1/status/alarm	-
		Total counter	Acquisition	GET	/v1/status/totalCounter	-
		User counter	Acquisition	GET	/v1/status/userCounter	-
		Air Supply Pressure	Acquisition	GET	/v1/status/supply	-
Auto-increment state		Acquisition	GET	/v1/status/autoInc	-	
Number of interval count, etc.		Acquisition	GET	/v1/status/interval	-	
Actual dispensing time		Acquisition	GET	/v1/status/stopWatch	-	
Status of temp. unit		Acquisition	GET	/v1/status/tempUnit	-	
Status of Σ mode		Acquisition	GET	/v1/status/sigma	-	
Remaining vol.		Acquisition	GET	/v1/status/remain	-	
D-SUB25 pin input status		Acquisition	GET	/v1/status/dsubio/in	-	
D-SUB25 pin output status	Acquisition	GET	/v1/status/dsubio/out	-		
Initialization	All data*	Execution (administrator)	GET	/v1/clear/allMemory	B, C, E, F	
	All channel data	Execution (administrator)	GET	/v1/clear/parameter	B, C, E, F	
	Log	Execution (administrator)	GET	/v1/clear/log	B, C, E, F	
	User counter	Execution	GET	/v1/clear/counter	B, C, E, F	
	Recipe	Execution	GET	/v1/clear/recipe/{id}	B, C, E, F, L	
	Σ mode	Execution	GET	/v1/clear/sigma/{id}	B, C, E, F, L, R	
Calibration	Sensor calibration	Execution	GET	/v1/calibrate/sensor/{option}	B, C, E, F, G	
Diagnosis	Dispensing pressure diagnosis	Execution	GET	/v1/diagnosis/dispense/{onoff}	B, C, D, E, F, G, R	
	Vacuum pressure diagnosis	Acquisition	GET	/v1/diagnosis/dispense/result	-	
		Execution	GET	/v1/diagnosis/vacuum/{onoff}	B, C, D, E, F, G, R	
		Acquisition	GET	/v1/diagnosis/vacuum/result	-	
	Valve diagnosis	Execution	GET	/v1/diagnosis/valve/{onoff}	B, C, D, E, F, G, R	
		Acquisition	GET	/v1/diagnosis/valve/result	-	
Model information		Acquisition	GET	/v1/info/machine/data	B	
Common to all channels	Interval feature setting	Acquisition	GET	/v1/info/common/interval/data	B	
		Acquisition of range value	GET	/v1/info/common/interval/range	B	
	DIS-correction setting	Acquisition	GET	/v1/info/common/correction/data	B	
		Acquisition of range value	GET	/v1/info/common/correction/range	B	
	RS-232C communication setting	Acquisition	GET	/v1/info/common/rs232c/data	B	
		Acquisition of range value	GET	/v1/info/common/rs232c/range	B	
	Ethernet communication setting	Acquisition	GET	/v1/info/common/ethernet/data	B	
		Acquisition of range value	GET	/v1/info/common/ethernet/range	B	
	External input/output signal setting	Acquisition	GET	/v1/info/common/dsubio/data	B	
		Acquisition of range value	GET	/v1/info/common/dsubio/range	B	
Configure setting	Acquisition	GET	/v1/info/common/configure/data	B		
	Acquisition of range value	GET	/v1/info/common/configure/range	B		
Recipe	Individual acquisition	GET	/v1/info/recipe/data/{id}	B		
	List acquisition	GET	/v1/info/recipe/data/all	B		
	Acquisition of range value	GET	/v1/info/recipe/range	B		
Channel	Individual acquisition	GET	/v1/info/channel/data/{id}	B		
	List acquisition	GET	/v1/info/channel/data/all	B		
	Acquisition of range value	GET	/v1/info/channel/range	B		

*Ethernet communication settings will not be initialized.

List of reasons for rejections to requests

If you send a request but it doesn't get processed and the status code is 400 (Bad Request), the following message will be attached to the response.

Id	"message"	Reasons for rejections to requests
A	"Format error."	There is a format error of the string attached to the request.
B	"Importing/exporting now."	The controller is displaying a screen related to the backup function.
C	"Dispensing now."	Dispensing now.
D	"Dispensing is restricted."	The dispense operation is disabled, such as when the ENABLE signal is OFF.
E	"Processing now."	A process such as diagnosis is in progress.
F	"Manual operating now."	A screen other than the dispense screen is displayed and you are directly operating the controller.
G	"Air supply pressure error has occurred."	An air supply pressure error has occurred.
H	"The current CH. is in manual mode"	The current CH is in manual mode.
I	"The current CH. is the link CH. Of the interval feature."	The interval feature is ON and link CH. is in use.
J	"The interval feature is enabled."	The interval feature is ON. (Link CH. is set to OFF.)
K	"The current CH. does not belong to a recipe."	The current CH. does not belong to a recipe.
L	"The specified recipe is not defined"	The specified recipe is undefined.
M	"The auto-increment feature is enabled."	The auto-increment function is enabled in the recipe where the current CH belongs.
N	"The specified temperature controller is disabled."	The specified temp. unit controller is turned OFF.
O	"The specified range will be exceeded once executed."	Executing the request will cause it to go out of the configurable range.
P	"The current CH. is in the normal mode."	The current CH. is in the normal mode.
Q	"The current CH. is in the sigma mode"	The current CH. is in the Σ mode.
R	"Internal communication error has occurred."	Internal communication error has occurred.
X	"There is a problem with importing data."	There is an error in the format or data of the imported data.

*Messages will be shown in English regardless of the language setting.

Usage example of API test screen

Web API can be tested on the API test screen.

GET request

(1)
Enter an Web API
starting with "/v1".

(2)
Send the request by
pressing the [GET]
button.

(3)
If there was data or an
error message with the
response, it will be
displayed here.

POST request

(1)
Enter an Web API
starting with "/v1".

(2)
Enter the data to be
sent.

(3)
Send the request by
pressing the [POST]
button.

(4)
If the response was
accompanied by an
error message, it will be
displayed here.

Web API details

Description example

◇ [HTTP method] [URL]	
[F u n c t i o n]	This is a functional overview of the Web API.
[Access right]	This is permission to use Web API.
[U R L d a t a]	Descriptions of values to be set on the URL, such as {id} and {onoff}.
[Transmission data]	For the POST method, it describes the format of the data to be sent.
[Received data]	For the GET method, it describes the format of the received data.
[Receive example]	For the GET method, this is an example of the data you receive.
[Error type]	This is the request reject reason code received when a 400 error occurs.

◆ Administrator login

◇ POST /v1/login

[F u n c t i o n]	Sends the administrator password. If successful, you will be able to execute Web APIs that require administrator privileges.
[Access right]	Administrator, operator
[Transmission data]	password=<Administrator password>
	If the administrator password is "0000": password=0000
[Error type]	A

◇ GET /v1/logout

[F u n c t i o n]	Returns to operator privileges.
[Access right]	Administrator, operator
[Error type]	N/A

◆ Data editing

◇ POST /v1/import

[F u n c t i o n]	Data settings
[Access right]	Administrator
[Transmission data]	See "4 JSON data format"
[Error type]	B, C, E, F, X

◇ GET /v1/export/data

[F u n c t i o n]	Acquisition of all data
[Access right]	Administrator, operator
[Received data]	See "4 JSON data format"
	*This is the same file that is sent to the USB memory when you tap "Export all data" on the export screen of the controller.
[Error type]	B

◇ GET /v1/export/log

[F u n c t i o n]	Acquisition of log data
[Access right]	Administrator, operator
[Received data]	Text data in TSV format
	*This is the same file that is sent to the USB memory when you tap "Export log data" on the export screen of the controller.
[Error type]	B

◆ Controller screen

◇ GET /v1/screen

[F u n c t i o n]	Acquires the screen displayed on the controller.
[Access right]	Administrator, operator
[Received data]	Image data in JPEG format
[Error type]	N/A

◆ Time

◇ POST /v1/time

[F u n c t i o n]	Time adjust of the controller's internal clock
[Access right]	Administrator
[Received data]	See "5 Time data format".
[Error type]	A, B, E, F

◇ GET /v1/time

[F u n c t i o n]	Time acquisition of the controller's internal clock
[Access right]	Administrator, operator
[Received data]	See "5 Time data format".
[Error type]	N/A

◆ Execution

◇ GET /v1/current/channel/{id}

[F u n c t i o n]	Move to specified channel
[Access right]	Administrator, operator
[U R L d a t a]	{id} : Channel number (1 to 400)
[Error type]	B, C, E, F, I

◇ GET /v1/current/recipe/{id}

[F u n c t i o n]	Moves to the start channel of the specified recipe
[Access right]	Administrator, operator
[U R L d a t a]	{id} : Recipe number (1 to 100)
[Error type]	B, C, E, F, I, L

◇ GET /v1/current/dispense/0

[F u n c t i o n]	Stops dispensing operation
[Access right]	Administrator, operator
[Error type]	E

◇ GET /v1/current/dispense/1

[F u n c t i o n]	Starts dispensing operation
[Access right]	Administrator, operator
[Error type]	B, C, D, E, F, R

◇ GET /v1/current/step

[F u n c t i o n]	Moves up one channel (same as STEP signal)
[Access right]	Administrator, operator
[Error type]	B, C, E, F, I, O

◇ **GET /v1/current/down**

[F u n c t i o n] Moves down one channel (same as DOWN signal)
[Access right] Administrator, operator
[Error type] B, C, E, F, I, O

◇ **GET /v1/current/preset**

[F u n c t i o n] Calls default channel (same as PRESET signal)
If channel belongs to a recipe:
Changes the channel to the start CH of the belonging recipe.
If channel does not belong to a recipe:
Changes the channel to 1CH.
[Access right] Administrator, operator
[Error type] B, C, E, F, I

◇ **GET /v1/current/timed**

[F u n c t i o n] Switches to timed mode
[Access right] Administrator, operator
[Error type] B, C, E, F, I, J, M, Q, R

◇ **GET /v1/current/manual**

[F u n c t i o n] Switches to manual mode
[Access right] Administrator, operator
[Error type] B, C, E, F, I, J, M, Q, R

◇ **GET /v1/current/interval/{onoff}**

[F u n c t i o n] ON/OFF switching of the interval feature
[Access right] Administrator, operator
[U R L d a t a] {onoff} : 0 = OFF, 1 = ON
[Error type] B, C, E, F, M

◇ **GET /v1/current/test**

[F u n c t i o n] Starts DIS-monitoring test
*You can check that the DIS-monitoring test is in progress at "/v1/status/main".
[Access right] Administrator, operator
[Error type] B, C, D, E, F, R

◇ **GET /v1/current/autoTune/{id}/{onoff}**

[F u n c t i o n] Starts Temp. unit auto-tuning.
* You can check that the auto-tuning is in progress at "/v1/status/main".
[Access right] Administrator, operator
[U R L d a t a] {id} : Temp. unit controller number (1 to 5)
{onoff} : 0 = OFF, 1 = ON
[Error type] B, C, E, F

◇ **GET /v1/current/buzzerCut**

[F u n c t i o n] Buzzer cut
[Access right] Administrator, operator
[Error type] N/A

◇ GET /v1/current/warningOff

[F u n c t i o n] Solenoid valve alarm OFF
[Access right] Administrator, operator
[Error type] N/A

◇ GET /v1/current/sampling/full

[F u n c t i o n] Starts full sampling
[Access right] Administrator, operator
[Error type] B, C, D, E, F, M, Q, R

◇ GET /v1/current/sampling/empty

[F u n c t i o n] Starts empty sampling
[Access right] Administrator, operator
[Error type] B, C, D, E, F, M, Q, R

◇ GET /v1/current/sampling/detail/{remain}

[F u n c t i o n] Starts detailed sampling
[Access right] Administrator, operator
[U R L d a t a] {remain} : User remaining vol.
[Error type] B, C, D, E, F, M, Q, R

◇ GET /v1/current/dummyShotMode/{onoff}

[F u n c t i o n] Switches dispensing mode of current CH to dummy shot mode.
*You can check that system is in Σ mode at "/v1/status/main".
[Access right] Administrator, operator
[U R L d a t a] {onoff} : 0 = OFF; Switches from dummy shot mode to Σ +TIMED mode
: 1= ON; Switches from Σ +TIMED mode to dummy shot mode
[Error type] B, C, E, F, H, P, R

◆ Acquisition of setting

◇ GET /v1/status/main

[F u n c t i o n] Acquisition of controller status, error, and alarm occurrence status
 [Access right] Administrator, operator
 [Received data] "status" : Controller status (message in the table below)
 "error" : 0 = No error
 1 = Error in progress
 "alarm" : 0 = No alarm
 1 = Alarm in progress
 [Receive example] In cases of in-dispensing, no error, alarm in progress
 {
 "status" : "Shot",
 "error" : 0,
 "alarm" : 1,
 }
 [Error type] N/A

Controller status list

message	Content
"Ready"	Ready state (waiting for dispense command)
"Not ready"	Dispense is not possible due to a reason other than the following conditions
"Shot"	In-dispensing
"End"	Dispense complete
"On interval shot"	Interval dispensing/ dispense completed
"On DIS-press. watch test"	In DIS-monitoring test
"On auto tuning"	In temp. unit auto-tuning
"On sensor calibration"	In sensor calibration
"On DIS-press. diagnosis"	In dispensing pressure diagnosis
"On vacuum diagnosis"	In vacuum pressure diagnosis
"On valve diagnosis"	In valve diagnosis
"On sigma sampling (Full)"	In full sampling
"On sigma sampling (Empty)"	In empty sampling
"On sigma sampling (Detail)"	In detailed sampling

◇ GET /v1/status/channel

[F u n c t i o n] Acquires current channel
 [Access right] Administrator, operator
 [Received data] "value" : Current channel (1 to 400)
 [Receive example] If the current channel is 200CH.
 {
 "value" : 200
 }
 [Error type] N/A

◇ GET /v1/status/recipe

[F u n c t i o n] Acquires current recipe ID
 [Access right] Administrator, operator
 [Received data] "value" : Recipe ID (1 to 100, null)
 [Receive example 1] If the current channel belongs to recipe 15
 {
 "value" : 15
 }
 [Receive example 2] If the current channel does not belong to a recipe
 {
 "value" : null
 }
 [Error type] N/A

◇ GET /v1/status/error

[Function] Acquires all occurring errors

[Access right] Administrator, operator

[Received data] "error" Enumerates the errors that are occurring in an array format.
For error details, refer to the table below.

[Receive example 1] If an air supply pressure error and a temp. unit error occur

```
{
  "error" : [
    {
      "code" : "006",
      "message" : "Air supply pressure error"
    },
    {
      "code" : "009",
      "message" : "Temp. unit error: HS warning"
    }
  ]
}
```

[Receive example 2] If no error is occurring

```
{
  "error":null
}
```

[Error type] N/A

Error list

"code"	"message"	Content
"001"	"Setting data load error"	Setting data cannot be loaded.
"002"	"Startup error"	Failed to start up the program.
"003"	"Image data error"	Image data cannot be loaded.
"004"	"RTC error"	Time information cannot be loaded.
"005"	"Solenoid valve error"	This error occurs when the solenoid valve unit has worn and is at increased risk of failure.
"006"	"Air supply pressure error"	An air supply pressure error has occurred.
"007"	"Exhaust error"	Exhaust error has occurred.
"008"	"Vacuum pressure error"	A vacuum pressure error has occurred.
"009"	"Temp. unit error: HS warning"	Internal parts error of temperature controller.
"010"	"Temp. unit error: Input error"	Temp. sensor connection error of temperature controller.
"011"	"Temp. unit error: HB warning"	Heater connection error of temperature controller.
"012"	"Temp. unit error: Timeout"	A connection error in the communication cable, or the power to the temp. unit controller is not turned ON.
"013"	"Temp. unit error: Setting error"	Settings cannot be written to the temp. unit controller.
"014"	"Internal error 1"	Internal error (connection) occurred.
"015"	"Internal error 2"	Internal error (connection) occurred.
"016"	"Internal error 3"	Internal error (connection) occurred.
"017"	"Internal error 4"	Version error has been detected.
"018"	(Reserved)	-
"019"	(Reserved)	-
"020"	"Internal error 7"	Internal error (communication) occurred.

◇ GET /v1/status/alarm

[F u n c t i o n] Acquires all occurring alarms
 [Access right] Administrator, operator
 [Received data] "alarm" Enumerates the occurring alarms in an array format.
 For the alarm contents, refer to the table below.

[Receive example 1] When an air supply pressure alarm and an empty alarm are occurring

```
{
  "alarm" : [
    {
      "code" : "100",
      "message" : "Air supply pressure warning"
    },
    {
      "code" : "102",
      "message" : "Empty alarm"
    }
  ]
}
```

[Receive example 2] When no alarm is occurring

```
{
  "alarm" : null
}
```

[Error type] N/A

Alarm list

"code"	"message"	Content
"100"	"Air supply pressure warning"	Air supply pressure is low.
"101"	"Solenoid valve alarm"	Solenoid valve replacement timing.
"102"	"Empty alarm"	Empty alarm has occurred.
"103"	"Temp. unit alarm: Warning 1"	Temp. unit alarm (Warning 1) has occurred.
"104"	"Temp. unit alarm: Warning 2"	Temp. unit alarm (Warning 2) has occurred.
"105"	"DIS-monitoring alarm"	Dispensing pressure is out of allowable range.
"016"	"Sigma mode remaining alarm"	Σ mode remaining volume alert has occurred.
"017"	(Reserved)	-

◇ GET /v1/status/totalCounter

[F u n c t i o n] Acquires total counter
 [Access right] Administrator, operator
 [Received data] "value" :0 ~ 999999999
 [Receive example] When the total counter is 1,000,000 shots

```
{
  "value" : 1000000
}
```

[Error type] N/A

◇ GET /v1/status/userCounter

[F u n c t i o n] Acquires user counter
 [Access right] Administrator, operator
 [Received data] "value" :0 ~ 999999999
 [Receive example] When the user counter is 1,000,000 shots

```
{
  "value" : 1000000
}
```

[Error type] N/A

◇ GET /v1/status/supply

[F u n c t i o n] Acquires current air supply pressure
 [Access right] Administrator, operator
 [Received data] "value" : 0 ~ 999
 [Receive example] When air supply pressure is 500 kPa

```
{
  "value" : 500
}
```

[Error type] N/A

*Only "kPa" is used as the pressure unit.

◇ GET /v1/status/autolnc

[F u n c t i o n] Acquires the progress of the auto-increment feature
 [Access right] Administrator, operator
 [Received data] "aiEnd" : 0 = in progress, 1 = completed
 "term" : Auto-increment switching condition
 "remain" : Remaining amount of auto-increment

	"term"	"remain"
	Character string	Numeric value
Total time	"total time"	0.000 ~ 9999.999
Total count	"total count"	0 ~ 9999
Pot life	"pot life"	0.000 ~ 9999.999

[Receive example 1] When the switching condition is the total count and the remaining number is 30 shots

```
{
  "aiEnd" : 0,
  "term" : "total count",
  "remain" : 30
}
```

[Receive example 2] When the switching condition is pot life and the auto increment is completed

```
{
  "aiEnd" : 1,
  "term" : "pot life",
  "remain" : 0.000
}
```

[Receive example 3] If the current channel does not belong to a recipe
 Or if the auto-increment feature is OFF

```
{
  "aiEnd" : null,
  "term" : null,
  "remain" : null
}
```

[Error type] N/A

◇ GET /v1/status/interval

[F u n c t i o n] Acquires the progress of the interval feature
 [Access right] Administrator, operator
 [Received data] "count" : Current count (0 to 9999)
 "repeat" : Current repeat (0 to 9999)

[Receive example 1] When the current count is 30 shots and the current repeat is 20

```
{
  "count" : 30,
  "repeat" : 20
}
```

[Receive example 2] When the interval feature is OFF

```
{
  "count" : null,
  "repeat" : null
}
```

[Error type] N/A

◇ GET /v1/status/stopWatch

[F u n c t i o n] Acquires actual dispensing time
[Access right] Administrator, operator
[Received data] "value" : 0.000 ~ 9999.999
[Receive example] When the actual dispensing time is 123.456 sec.
{
 "value" : 123.456
}
[Error type] N/A

◇ GET /v1/status/tempUnit

[F u n c t i o n] Acquires the operating status of the temp. unit controller.
[Access right] Administrator, operator
[Received data] "id" : 1 ~ 5
 "status" : "normal", "alarm", "error"
 "measured" : 0.0 ~ 999.9
[Receive example 1] Temp. unit 1 and 3 are used,
 temp. unit 1 is operating normally, current temperature is 30.0°C, temp. unit 3 is in
 error
 {
 "tempUnit": [
 {
 "id" : 1,
 "status" : "normal",
 "measured" : 30.0
 },
 {
 "id" : 3,
 "status" : "error",
 "measured" : null
 }
]
 }
[Receive example 2] The current recipe does not use a temp. unit controller.
 {
 "tempUnit" : null
 }
[Error type] N/A

◇ GET /v1/status/sigma

[F u n c t i o n] Acquires Σ mode
[Access right] Administrator, operator
[Received data] "sigmaMode" : 0 = Normal mode, 1 = Σ mode
 "shotMode" : 0 = TIMED mode, 1 = MANUAL mode
 "dummyShot" : 0 = Dummy shot mode OFF, 1 = Dummy shot mode ON
[Receive example] In Σ +TIMED mode, dummy shot mode OFF
 {
 "sigmaMode" : 1,
 "shotMode" : 0,
 "dummyShot" : 0
 }
[Error type] N/A

◇ GET /v1/status/remain

[F u n c t i o n] Acquires current remaining volume
[Access right] Administrator, operator
[Received data] "value" : 0 ~ 100
[Receive example] If remaining vol. is 80%
{
 "value" : 80
}
For Normal mode
{
 "value" : null
}
[Error type] N/A

◇ GET /v1/status/dsubio/in

[F u n c t i o n] Acquires D-SUB25 pin input status
[Access right] Administrator, operator
[Received data] "value" : 0 = OFF, 1 = ON
Pin numbers from the beginning 2, 3, 4, 5, 6, 7, 8, 15, 16, 17, 18,
19, 20
[Receive example] If the number 2, 4, and 17 pins are ON.
{
 "value" : [1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0]
}
[Error type] N/A

◇ GET /v1/status/dsubio/out

[F u n c t i o n] Acquires D-SUB25 pin output status
[Access right] Administrator, operator
[Received data] "value" : 0 = OFF, 1 = ON
Pin numbers from the beginning 9, 10, 11, 12, 21, 22, 23, 24
[Receive example] If the number 11, 21, and 24 pins are ON.
{
 "value" : [0, 0, 1, 0, 1, 0, 0, 1]
}
[Error type] N/A

◆ Initialization

◇ GET /v1/clear/allMemory

[F u n c t i o n] Resets all settings to default.
 *Ethernet communication settings will not be initialized.
[Access right] Administrator
[Error type] B, C, E, F

◇ GET /v1/clear/parameter

[F u n c t i o n] Resets the settings saved for each channel to their default values.
[Access right] Administrator
[Error type] B, C, E, F

◇ GET /v1/clear/log

[F u n c t i o n] Clears operation history (log).
[Access right] Administrator
[Error type] B, C, E, F

◇ GET /v1/clear/counter

[F u n c t i o n] Clears the User counter.
[Access right] Administrator, operator
[Error type] B, C, E, F

◇ GET /v1/clear/recipe/{id}

[F u n c t i o n] Clears recipe with specified ID.
[Access right] Administrator
[U R L d a t a] {id} :1 ~ 100
[Error type] B, C, E, F, L

◇ GET /v1/clear/sigma/{id}

[F u n c t i o n] Cancels the Σ mode of the channel with the specified ID.
[Access right] Administrator
[U R L d a t a] {id} :1 ~ 400
[Error type] B, C, E, F, L, R

◆ Calibration

◆ GET /v1/calibrate/sensor/{option}

[Function] Starts sensor calibration.

*You can confirm that the sensor calibration is being carried out by checking `/v1/status/main`.

[Access right] Administrator, operator

Process ID	Priority	Administrator	Operator	Process Description
[URL data]	{option}	:	0	= Starts sensor calibration for the case of no air supply pressure. * Make sure the air supply pressure is shut off before sending the request. The current air supply pressure will be set to zero if the calibration is executed while the air supply pressure is not cut off.
			1	= Starts sensor calibration for the case with air supply pressure.

[Error type] B, C, E, F, G

◆ Diagnosis

◇ GET /v1/diagnosis/{type}/{onoff}

[F u n c t i o n] Issues a diagnostic start or abort command.

*You can check that the diagnosis is in progress at `/v1/status/main`.

*Make sure that the adapter tube is disconnected before starting the diagnosis.

[Access right] Administrator, operator

[U R L d a t a]	{type}	:	dispense	=	Dispensing pressure diagnosis
			vacuum	=	Vacuum pressure diagnosis
			valve	=	Valve diagnosis
	{onoff}	:	0	=	Aborts diagnosis.
			1	=	Starts diagnosis.

[Error type] B, C, D, E, F, G, R

◇ GET /v1/diagnosis/{type}/result

```
[Function] Acquires diagnostic results.
```

[Access right]	Administrator, operator
----------------	-------------------------

[U R L d a t a]	{type}	:	dispense	=	Dispensing pressure diagnosis
			vacuum	=	Vacuum pressure diagnosis
			valve	=	Valve diagnosis
[Received data]	"value"	:	0	=	Failure
			1	=	Pass

[Receive example] Passed the diagnosis

```
{
  "value": 1
}
```

[Error type] N/A

◆ Model information

◇ GET /v1/info/machine/data

[Function]	Acquires model information.
[Access right]	Administrator, operator
[Received data]	See "4 JSON data format".
[Error type]	B

◆ Common channel settings

◇ GET /v1/info/common/{option}/data

[Function]	Acquires the channel common data.		
[Access right]	Administrator, operator		
[URL data]	{option}	:	interval Data related to interval feature
			correction Data related to DIS-correction
			rs232c Data related to RS-232C communication
			ethernet Data related to Ethernet communication
			dsubio Data related to external input/output signals
			configure Configure setting data
[Received data]	See "4 JSON data format".		
[Error type]	B		

◇ GET /v1/info/common/{option}/range

[Function]	Acquires the settable range of channel common data.		
[Access right]	Administrator, operator		
[URL data]	{option}	:	interval Data related to interval feature
			correction Data related to DIS-correction
			rs232c Data related to RS-232C communication
			ethernet Data related to Ethernet communication
			dsubio Data related to external input/output signals
			configure Configure setting data
[Received data]	See "4 JSON data format".		
[Error type]	B		

◆ Recipe settings

◇ GET /v1/info/recipe/data/{id}

[Function]	Acquires data for the specified recipe.
[Access right]	Administrator, operator
[URL data]	{id} :1 ~ 100
[Received data]	See "4 JSON data format".
[Error type]	B

◆ GET /v1/info/recipe/data/all

[Function]	Acquires data for all recipes.
[Access right]	Administrator, operator
[Received data]	See "4 JSON data format".
[Error type]	B

◇ GET /v1/info/recipe/range

[Function]	Acquires the settable range of recipe data.
[Access right]	Administrator, operator
[Received data]	See "4 JSON data format".
[Error type]	B

◆ Channel settings

◇ GET /v1/info/channel/data/{id}

[F u n c t i o n] Acquires the data of the specified channel.
[Access right] Administrator, operator
[U R L d a t a] {id} :1 ~ 400
[Received data] See "4 JSON data format".
[Error type] B

◇ GET /v1/info/channel/data/all

[F u n c t i o n] Acquires data of all channels.
[Access right] Administrator, operator
[Received data] See "4 JSON data format".
[Error type] B

◇ GET /v1/info/channel/range

[F u n c t i o n] Acquires the settable range of channel data.
[Access right] Administrator, operator
[Received data] See "4 JSON data format".
[Error type] B

4 JSON data format

The import and export functions use JSON files that conform to the JSON data format below.

Structure

- Enclose the whole data with "{" and "}".
- It consists of three parts: an object key, a data key, and a data value.
- The key is enclosed by "".
- There is a ":" after the key.
- Object keys are used to separate each category.
There are the following types of object keys.

Object key	Content	Remarks
"info"	Controller information	Acquisition only
"setting"	Common channel settings	There are Web APIs that can be obtained for each category.
"recipe"	Recipe settings	Arranged by recipe.
"ch"	Channel setting	Arranged by channel.

- "info" key and "setting" key are followed by ": {". "recipe" key and "ch" key are followed by ": [".
 - A single recipe or channel is enclosed in "{" and "}", and the group of them is enclosed in "[" and "]".
 - Each parameter is described by a set of data key and data value.
- Data values can be shown in the following data types:

Data type	Content	Example
Character string	Enclose it with "".	"machineInfo": "ML8X-02.00"
Numeric value (boolean value)	The value will be 0 or 1. 0 means OFF, 1 means ON.	"buzzerOnOff": 1
Numeric value (integer)	Integer only	"currentCh": 14
Numeric value (floating point)	Value with decimal point	"disPress": 50.0
Numeric array (integer)	Array of integers	"stdMenu": [0, 1, 2, 3]

- If the data is followed by an object or data, they must be separated by ",".
- If you request range acquisition, the reply will be in the following format.

```
{
  "min": minimum
value,
  "max":
maximum value
}
```

Notices

- Recipe numbers and channels (excluding link CH) start with 0.
Example: CH.15 data ⇒ ""no":14"
- For undefined recipes, set the start channel and end channel to 400.
- The unit of pressure value is only "kPa" regardless of the pressure unit setting.
- The unit of temperature is "°C".

JSON data example

```
{
  "info": {
    "machineInfo": "S-SIGMA-CM4-V5", "firmware1": "SCM4-01.00"
  },
  "setting": {
    "currentCh": 0, "stdMenu": [0, 1, 2, 3]
  },
  "recipe": [
    {
      "no": 0, "startCh": 0, "endCh": 19, "comment": "work001"
    },
    {
      "no": 1, "startCh": 20, "endCh": 39, "comment": "work002"
    }
  ],
  "ch": [
    {
      "no": 0, "disPress": 35.5, "chName": "sample001"
    },
    {
      "no": 1, "disPress": 50.0, "chName": "sample002"
    }
  ]
}
```

Data list

A list of data used in JSON files.

- * Keys that are not listed are data for controlling this unit. Do not edit.
- * For the correspondences of data values, refer to the instruction manual for Communication (RS-232C).

Controller information object

Object key "info"

Data acquisition URL /v1/info/machine/data

Range acquisition None

URL

Content	Data key	Data form	Minimum value	Maximum value	Unit	R/W
Model name	"machineInfo"	Character string	-	-	-	R
Firmware1	"firmware1"	Character string	-	-	-	R
Firmware2	"firmware2"	Character string	-	-	-	R
Firmware3	"firmware3"	Character string	-	-	-	R
Firmware4	"firmware4"	Character string	-	-	-	R
User counter	"userCnt"	Numeric value (integer)	0	999999999	-	R
Total counter	"totalCnt"	Numeric value (integer)	0	999999999	-	R
Operation time 1	"uptimePress"	Numeric value (integer)	0	999999999	-	R
Operation time 2	"uptimeVac"	Numeric value (integer)	0	999999999	-	R
I/O specifications	"ioSpec"	Character string	"NPN"	"PNP"	-	R
MAC address	"macAddress"	Character string	-	-	-	R

Channel common setting object

Object key "setting"

Category Interval feature setting

Data acquisition URL /v1/info/common/interval/data

Range acquisition /v1/info/common/interval/range

URL

Content	Data key	Data form	Minimum value	Maximum value	Unit	R/W
Interval feature	Function ON/OFF	Boolean value	0	1	-	R/W
	Count	Numeric value (integer)	1	9999	shot	R/W
	Interval	Numeric value (floating point)	0.010	9999.999	sec.	R/W
	Repeat	Numeric value (integer)	1	9999	-	R/W
	Wait time	Numeric value (floating point)	0.010	9999.999	sec.	R/W
	Interval mode	Numeric value (integer)	0	1	-	R/W
	Link CH.	Numeric value (integer)	0	400	CH	R/W

- * If the Link CH is 0, the channel link is OFF.

Category DIS-correction setting

Data acquisition URL /v1/info/common/correction/data

Range acquisition /v1/info/common/correction/range

URL

Content	Data key	Data form	Minimum value	Maximum value	Unit	R/W
DIS-correction	Dispensing time: Correction method	Numeric value (integer)	0	2	-	R/W
	Dispensing time: Amount	Numeric value (floating point)	-99.999	99.999	sec.	R/W
	Dispensing time: Ratio	Numeric value (integer)	1	999	%	R/W
	Dispensing pressure: Correction method	Numeric value (integer)	0	2	-	R/W
	Dispensing pressure: Amount (V2)	Numeric value (floating point)	-200.0	200.0	kPa	R/W
	Dispensing pressure: Amount (V5)	Numeric value (floating point)	-500.0	500.0	kPa	R/W
	Dispensing pressure: Ratio	Numeric value (integer)	1	999	%	R/W
	Vacuum pressure: Correction method	Numeric value (integer)	0	2	-	R/W
	Vacuum pressure: Amount	Numeric value (floating point)	-5.00	5.00	kPa	R/W
	Vacuum pressure: Ratio	Numeric value (integer)	1	999	%	R/W

Category RS-232C communication setting
 Data acquisition URL /v1/info/common/rs232c/data
 Range acquisition /v1/info/common/rs232c/range
 URL

Content			Data key	Data form	Minimum value	Maximum value	Unit	R/W
RS-232C setting	communication	Communication speed	"baudRate"	Numeric value (integer)	0	3	-	R/W
		Checksum ON/OFF	"checkSumOnOff"	Boolean value	0	1	-	R/W
		Com. protocol	"comFormat"	Numeric value (integer)	0	2	-	R/W

Category External input/output signal setting
 Data acquisition URL /v1/info/common/dsubio/data
 Range acquisition /v1/info/common/dsubio/range
 URL

Content		Data key	Data form	Minimum value	Maximum value	Unit	R/W
I/O signal setting	Series	"signalSeries"	Numeric value (integer)	0	1	-	R/W
	Output signal spec.	"signalSpec"	Numeric value (integer)	0	1	-	R/W
	ENABLE signal	"enableSignal"	Boolean value	0	1	-	R/W
	On Error	"outputOnError"	Boolean value	0	1	-	R/W
	Input signal allocation	"funcIn"	Numeric array (integer)	0	19	-	R/W
	Output signal allocation	"funcOut"	Numeric array (integer)	0	21	-	R/W

- * If the series setting is different from the controller setting value and JSON data without output signal allocation setting data is imported, the output signal allocation will be initialized.

Category Configure setting
 Data acquisition URL /v1/info/common/configure/data
 Range acquisition /v1/info/common/configure/range
 URL

Content		Data key	Data form	Minimum value	Maximum value	Unit	R/W
Current channel		"currentCh"	Numeric value (integer)	0	399	CH	R/W
Configure setting	Language	"language"	Numeric value (integer)	0	2	-	R/W
	Pressure unit	"pressUnit"	Numeric value (integer)	0	1	-	R/W
	Empty alarm	"rsmOnOff"	Boolean value	0	1	-	R/W
	Empty warning threshold	"remainCnt"	Numeric value (integer)	1	999999999	shot	R/W
	Valve warning threshold	"valveAlarmCnt"	Numeric value (integer)	1	999999999	shot	R/W
	Remaining volume detection notice	"rsmPreLv"	Numeric value (integer)	0	50	%	R/W
	Buzzer ON/OFF	"buzzerOnOff"	Boolean value	0	1	-	R/W
	T-ready Enable	"tempDisDisable"	Boolean value	0	1	-	R/W
	Warm-ready enable	"warmDisDisable"	Boolean value	0	1	-	R/W
	Pop-up auto OFF	"popAutoOff"	Boolean value	0	1	-	R/W
	Immed. Inc. (Alnc)	"chChangeInShot"	Boolean value	0	1	-	R/W
	Control mode	"valveControl"	Numeric value (integer)	0	1	-	R/W
	Sleep timer	"sleepTimer"	Numeric value (integer)	0	8	-	R/W
	Screen brightness	"backlightDuty"	Numeric value (integer)	5	100	%	R/W
	Password lock setting	Controller name	"machineName"	Character string	0	10	-
Password lock ON/OFF		"keyLockOnOff"	Boolean value	0	1	-	R/W
Password		"password"	Character string	0	14	-	R/W
Icon Allocation	Left side of the screen	"stdMenu"	Numeric array (integer)	0	11	-	R/W
	Right side of the screen	"stdSubMenu"	Numeric array (integer)	0	3	-	R/W
Time zone setting		"timeOffset"	Numeric value (integer)	0	37	-	R/W

- * For setting values for icon allocation, refer to the Basic instruction manual.
 * For time zone setting, see "5 Time data format".

Recipe settings object

Object key "recipe"

Data acquisition URL /v1/info/recipe/data/{id}
/v1/info/recipe/data/all

Range acquisition URL /v1/info/recipe/range

Content		Key	Data form	Minimum value	Maximum value	Unit	R/W
Recipe number		"no"	Numeric value (integer)	0	99	CH	R/W
Recipe settings	Start channel	"startCh"	Numeric value (integer)	0	400	CH	R/W
	End channel	"endCh"	Numeric value (integer)	0	400	CH	R/W
	Recipe name	"comment"	Character string	0	14	-	R/W
	Nozzle name	"nozzleName"	Character string	0	14	-	R/W
	Liquid material name	"liquidName"	Character string	0	14	-	R/W
Auto-increment setting	Function ON/OFF	"autoIncOnOff"	Boolean value	0	1	-	R/W
	DIS-mode	"aiShotMode"	Numeric value (integer)	0	1	-	R/W
	ECO mode	"aiAirEco"	Numeric value (integer)	0	1	-	R/W
	Switching Condition	"alncTerm"	Numeric value (integer)	0	2	-	R/W
	Switching value: Total time	"alncDataTime"	Numeric value (integer)	1	9999	sec.	R/W
	Switching value: Total count	"alncDataCnt"	Numeric value (integer)	1	9999	shot	R/W
	Switching value: Pot life	"alncDataLife"	Numeric value (integer)	1	9999	sec.	R/W
	Temp. unit 1: Control ON/OFF	"tempOnOff1"	Boolean value	0	1	-	R/W
	Temp. unit 1: Setting temperature	"tempSettingT1"	Numeric value (floating point)	0	100.0	°C	R/W
	Temp. unit 1: Upper limit	"tempUpperLimit1"	Numeric value (floating point)	0.0	100.0	°C	R/W
Temperature unit setting	Temp. unit 1: Lower limit	"tempLowerLimit1"	Numeric value (floating point)	-100.0	-0.0	°C	R/W
	Temp. unit 2: Control ON/OFF	"tempOnOff2"	Boolean value	0	1	-	R/W
	Temp. unit 2: Setting temperature	"tempSettingT2"	Numeric value (floating point)	0	100.0	°C	R/W
	Temp. unit 2: Upper limit	"tempUpperLimit2"	Numeric value (floating point)	0.0	100.0	°C	R/W
	Temp. unit 2: Lower limit	"tempLowerLimit2"	Numeric value (floating point)	-100.0	-0.0	°C	R/W
	Temp. unit 3: Control ON/OFF	"tempOnOff3"	Boolean value	0	1	-	R/W
	Temp. unit 3: Setting temperature	"tempSettingT3"	Numeric value (floating point)	0	100.0	°C	R/W
	Temp. unit 3: Upper limit	"tempUpperLimit3"	Numeric value (floating point)	0.0	100.0	°C	R/W
	Temp. unit 3: Lower limit	"tempLowerLimit3"	Numeric value (floating point)	-100.0	-0.0	°C	R/W
	Temp. unit 4: Control ON/OFF	"tempOnOff4"	Boolean value	0	1	-	R/W
	Temp. unit 4: Setting temperature	"tempSettingT4"	Numeric value (floating point)	0	100.0	°C	R/W
	Temp. unit 4: Upper limit	"tempUpperLimit4"	Numeric value (floating point)	0.0	100.0	°C	R/W
	Temp. unit 4: Lower limit	"tempLowerLimit4"	Numeric value (floating point)	-100.0	-0.0	°C	R/W
	Temp. unit 5: Control ON/OFF	"tempOnOff5"	Boolean value	0	1	-	R/W
	Temp. unit 5: Setting temperature	"tempSettingT5"	Numeric value (floating point)	0	100.0	°C	R/W
	Temp. unit 5: Upper limit	"tempUpperLimit5"	Numeric value (floating point)	0.0	100.0	°C	R/W
	Temp. unit 5: Lower limit	"tempLowerLimit5"	Numeric value (floating point)	-100.0	-0.0	°C	R/W

* When both the start and end channels are 400, it indicates that the recipe is undefined.

Channel configuration object

Object key "ch"

Data acquisition URL /v1/info/channel/data/{id}
/v1/info/channel/data/all

Range acquisition URL /v1/info/channel/range

Content		Key	Data form	Minimum value	Maximum value	Unit	R/W
Channel number		"no"	Numeric value (integer)	0	399	CH	R/W
Dispensing conditions	DIS-mode	"shotMode"	Numeric value (integer)	0	1	-	R/W
	Dispensing time	"disTime"	Numeric value (floating point)	0.010	9999.999	sec.	R/W
	Dispensing pressure (V2)	"disPress"	Numeric value (floating point)	5.0	200.0	kPa	R/W
	Dispensing pressure (V5)	"disPress"	Numeric value (floating point)	30.0	500.0	kPa	R/W
	Vacuum pressure	"disVacuum"	Numeric value (floating point)	0.00	5.00	-kPa	R/W
DIS-auxiliary setting	Syringe size	"syringeSize"	Numeric value (integer)	0		-	R/W
	Adapter tube length	"tubeLength"	Numeric value (floating point)	0.5	2.0	m	R/W
	Plunger	"usePlunger"	Numeric value (integer)	0	1	-	R/W
	ECO mode	"airEco"	Numeric value (integer)	0	1	-	R/W
Σ parameter	Σ mode	"sigmaMode"	Numeric value (integer)	0	1	-	R/W
	Dummy shot mode	"dummyShot"	Numeric value (integer)	0	1	-	R/W
	Vol. reduction corr. ON/ OFF	"volRedCorr"	Numeric value (integer)	0	1	-	R/W
	Alpha correction value	"corrAlpha"	Numeric value (integer)	-200	999	-	R/W
	Delta correction value	"corrDelta"	Numeric value (integer)	50	200	%	R/W
	Fluid-drip prevention ON/ OFF	"dropPrevent"	Numeric value (integer)	0	1	-	R/W
	Vacuum correction value	"corrVac"	Numeric value (floating point)	-5.00	5.00	kPa	R/W
	Remaining vol. alert ON/OFF	"rsmDetect"	Numeric value (integer)	0	1	-	R/W
	Remaining volume detection level	"rsmLevel"	Numeric value (integer)	0	99	%	R/W
	RSM count	"rsmCount"	Numeric value (integer)	1	10	shot	R/W
	Remaining vol. corr. ON/OFF	"rsmCorrOnOff"	Numeric value (integer)	0	1	-	R/W
	Correction value	"rsmCorr"	Numeric value (integer)	-99	99	%	R/W
Timer setting	On delay time	"onDelay"	Numeric value (floating point)	0.000	9.999	sec.	R/W
	Off delay time	"offDelay"	Numeric value (floating point)	0.000	9.999	sec.	R/W
DIS-monitoring function setting	Allowable	"watchPermit"	Numeric value (floating point)	10.0	100.0	kPa	R/W
	Dispensing OFF time	"watchMinOffTime"	Numeric value (floating point)	0.050	9.999	sec.	R/W
	Watch status	"watchResult"	Numeric value (integer)	0	2	-	R
Channel name		"chName"	Character string	0	14	-	R/W

Import error

Errors will be generated if there are syntax or data errors in the data to be imported.
Errors are output in JSON format.

If even one error occurs, the data will not be written.
Up to 50 errors will be detected, and then the import process will be interrupted immediately.
Some errors immediately abort the import process.

When an import is done from the backup screen with USB memory, an "ImportError.log" file is generated in the USB memory, and error messages are recorded in it.

Error list

Error code	Error name	Content	Suspend immediately
100	Syntax error	There is an error in the structure of the JSON file.	○
101	Key length error	The object key or data key exceeds 16 characters.	○
102	Data length error	Data exceeds 128 characters.	○
103	Data missing error	No data value.	○
110	Array error	Missing "no" key in recipe and channel arrays.	-
111	Array element count error	The array has too many (too few) elements.	-
120	Data type mismatch error	The data type of the data value is different from that was expected from the data key.	-
121	Numeric conversion error	Failed to get data value.	-
200	Out of range error	The data is out of range.	-
210	Allocation error 1	Data is duplicated in the allocated array.	-
211	Allocation error 2	The data is out of range in the allocated array.	-
220	Recipe error 1	The start channel is greater than the end channel.	-
221	Recipe error 2	The channel is commonly used for that of other recipes.	-
222	Recipe error 3	Σ mode CH is included in recipe with auto-increment function ON.	-
230	Σ error 1	Missing data necessary for Σ mode CH.	-
231	Σ error 2	There is data with an inappropriate combination in a Σ mode CH.	-
300	Undefined key	A key string not handled by the controller is used.	○

Error statement structure

Errors are output as array data bounded by the object key "error".
Each error information has the following data keys.
Some errors have irrelevant data keys, in which case the data will be null.

Data key	Content
"code"	Error code.
"message"	Error message.
"key"	Shows the key that caused the error.
"data"	Shows the data part character string that caused the error.
"order"	Shows which element in the array has a problem. *The ordinal number starting from zero will be shown. For recipes and channels, the data indicated by the "no" key of the data to be imported will be entered. In the case of an allocated array, the sequence enclosed in [] indicates the order.
"min"	Displays the minimum value that can be set. *In the case of error code 221, the recipe number of the duplicative recipe is displayed. The displayed number will be the smaller number.
"max"	Displays the maximum value that can be set. *In the case of error code 221, the recipe number of the duplicative recipe is displayed. The displayed number will be the larger number.

Error details

Description example

[Error code] [Error name]

[C o n t e n t] Error contents.
[K e y] A string or value to be inserted for each key.

100 Syntax error

[C o n t e n t] There is an error in the syntax of JSON data.
It may contain inappropriate delimiters.
[" c o d e "] 100
[" m e s s a g e "] "Parse error"
[" k e y "] null will be entered.
[" d a t a "] null will be entered.
[" o r d e r "] null will be entered.
[" m i n "] null will be entered.
[" m a x "] null will be entered.

101 Key error

[C o n t e n t] Object key or data key string is 16 or more
[" c o d e "] 101
[" m e s s a g e "] "Key length error"
[" k e y "] The string entered as the data key will be entered.
[" d a t a "] null will be entered.
[" o r d e r "] null will be entered.
[" m i n "] null will be entered.
[" m a x "] null will be entered.

102 Data length error

[C o n t e n t] The character string in the data section exceeded 128 characters.
[" c o d e "] 102
[" m e s s a g e "] "Data length error"
[" k e y "] The data key where an error occurred will be entered.
[" d a t a "] The character string entered as the data part character string will be entered.
[" o r d e r "] null will be entered.
[" m i n "] null will be entered.
[" m a x "] null will be entered.

103 Data missing error

[C o n t e n t] No data value.
[" c o d e "] 103
[" m e s s a g e "] "Data not found"
[" k e y "] The data key where an error occurred will be entered.
[" d a t a "] null will be entered.
[" o r d e r "] null will be entered.
[" m i n "] null will be entered.
[" m a x "] null will be entered.

110 Array error

[C o n t e n t]	There is no "no" key in the recipe or channel data arrays.
[" c o d e "]	110
[" m e s s a g e "]	"Array error"
[" k e y "]	"no" will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

111 Array element count error

[C o n t e n t]	The number of elements in [] was too few or too many.
[" c o d e "]	111
[" m e s s a g e "]	"Too few/many elements in array"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

200 Out of range error

[C o n t e n t]	The data value is out of range.
[" c o d e "]	200
[" m e s s a g e "]	"Out of range (min=<min>, max=<max>)"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	The character string entered as the data part character string will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	The minimum value that can be set will be entered.
[" m a x "]	The maximum value that can be set will be entered.

210 Data type mismatch error

[C o n t e n t]	The data type of the data value is different from that was expected from the data key.
[" c o d e "]	210
[" m e s s a g e "]	"Data type inconsistent"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	The character string entered as the data part character string will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

211 Numeric conversion error

[C o n t e n t]	Failed to get data value.
[" c o d e "]	211
[" m e s s a g e "]	"Convert error"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	The character string entered as the data part character string will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

210 Allocation error: duplicate

[C o n t e n t]	The same data is contained in the allocation data array.
[" c o d e "]	210
[" m e s s a g e "]	"Allocation error; Duplicated"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	The data value that caused the error will be entered.
[" o r d e r "]	The order of "data" in the array will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.
[T a r g e t k e y]	"stdMenu", "stdSubMenu", "funcIn", "funcOut"

211 Allocation error: out of range

[C o n t e n t]	The allocation data array contains out-of-range data.
[" c o d e "]	211
[" m e s s a g e "]	"Allocation error; Out of range (min=<min>, max=<max>)"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	The data value that caused the error will be entered.
[" o r d e r "]	The order of "data" in the array will be entered.
[" m i n "]	The minimum value that can be set will be entered.
[" m a x "]	The maximum value that can be set will be entered.
[T a r g e t k e y]	"stdMenu", "stdSubMenu", "funcIn", "funcOut"

220 Recipe error: out of sequence

[C o n t e n t]	The start channel of the recipe is larger than the end channel.
[" c o d e "]	220
[" m e s s a g e "]	"Recipe error; Wrong range (#<order>)"
[" k e y "]	null will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	The recipe number will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

221 Recipe error: duplicate

[C o n t e n t]	Different recipes contain the same channels.
[" c o d e "]	221
[" m e s s a g e "]	"Recipe error; Overwrapped (#<min> & #<max>)"
[" k e y "]	null will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	In the case that one channel is used in multiple recipes, the smaller recipe number is entered.
[" m a x "]	In the case that one channel is used in multiple recipes, the larger recipe number is entered.

222 Recipe error: Σ mode CH and Auto-increment Function

[C o n t e n t]	Σ mode CH is included in recipe with auto-increment function ON.
[" c o d e "]	222
[" m e s s a g e "]	"Recipe error; Sigma in Auto.Inc"
[" k e y "]	null will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	The recipe number will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

230 Σ error: Missing

[C o n t e n t]	Missing data necessary for Σ mode CH.
[" c o d e "]	230
[" m e s s a g e "]	"Sigma error; Incomplete data"
[" k e y "]	null will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	Channel number will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

231 Σ error: Inappropriate combination

[C o n t e n t]	There is data with an inappropriate combination in a Σ mode CH.
[" c o d e "]	231
[" m e s s a g e "]	"Sigma error; Bad combination"
[" k e y "]	null will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	Channel number will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

300 Undefined key error

[C o n t e n t]	An undefined key was used.
[" c o d e "]	300
[" m e s s a g e "]	"Undefined key"
[" k e y "]	The data key where an error occurred will be entered.
[" d a t a "]	null will be entered.
[" o r d e r "]	null will be entered.
[" m i n "]	null will be entered.
[" m a x "]	null will be entered.

5 Time data format

The time data of the controller in Web API uses date and time notation conforming to ISO8601 format.

Notation

	Notation	Example: Japan time 2023, September 20, 18:00
For Coordinated Universal Time (UTC)	YYYY-MM-DDThh:mm:ssZ	2023-09-20T18:00:00Z
For other time zones	YYYY-MM-DDThh:mm:ss+XX:XX	2023-09-20T09:00:00+09:00

In the above, the abbreviations are: YYYY = year, MM = month, DD = day, hh = hour, mm = minute, ss = second, +XX:XX = time zone.

Select your time zone from the list below.

id	Time zone	id	Time zone	id	Time zone	id	Time zone
0	-12:00	10	-03:30	20	+04:30	30	+09:30
1	-11:00	11	-03:00	21	+05:00	31	+10:00
2	-10:00	12	-02:00	22	+05:30	32	+10:30
3	-09:30	13	-01:00	23	+05:45	33	+11:00
4	-09:00	14	Z	24	+06:00	34	+12:00
5	-08:00	15	+01:00	25	+06:30	35	+12:45
6	-07:00	16	+02:00	26	+07:00	36	+13:00
7	-06:00	17	+03:00	27	+08:00	37	+14:00
8	-05:00	18	+03:30	28	+08:45		
9	-04:00	19	+04:00	29	+09:00		

* "id" is the setting value in JSON data "timeOffset".

If you send a "/v1/time" request on the API test screen, it will look like the figure below.

GET request

API: /v1/time

Data for POST:

Received Data:
2023-02-10T05:00:00+08:00

GET POST

POST request

API: /v1/time

Data for POST:
2023-02-10T05:00:00+08:00

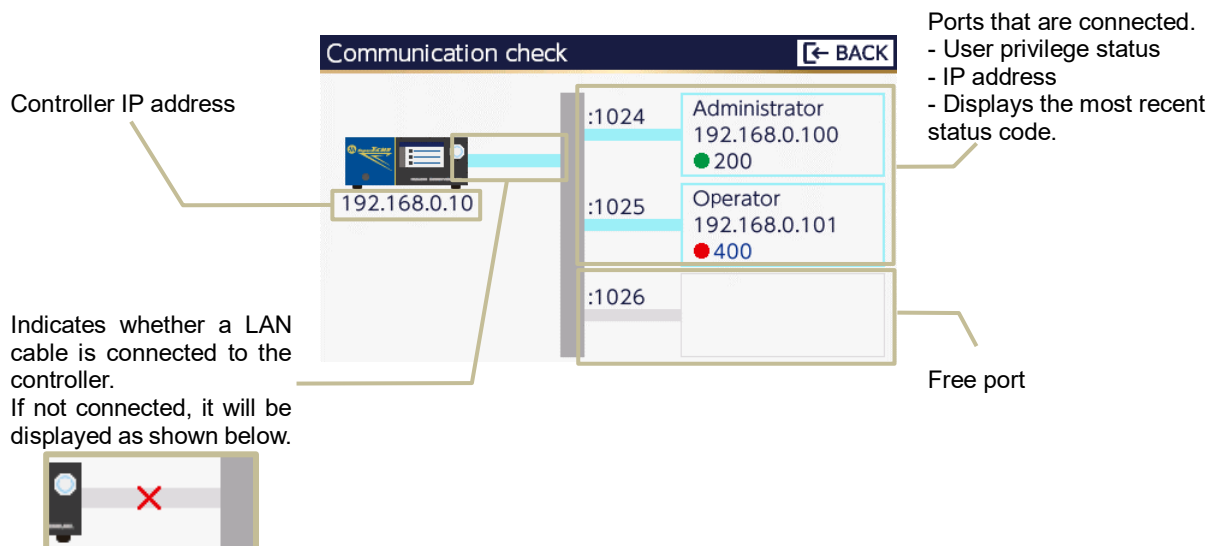
Received Data:

GET POST

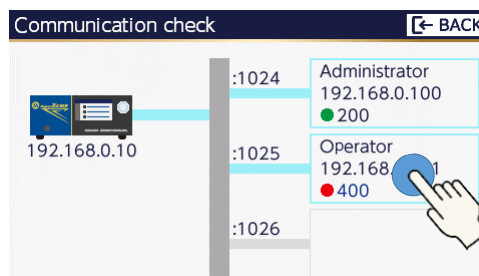
6 Check the communication status with the controller (communication check)

This unit can check the data sent and received with the host device.

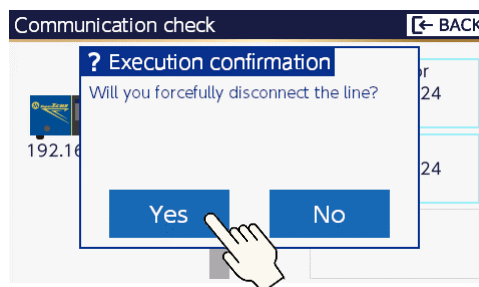
See P.5 "Setting change of Ethernet communication" and move to the "Ethernet communication setting" screen. Then tap "Com. check" to display the "Communication check" screen.



If you press and hold a connected port, a confirmation screen will appear asking whether to forcibly disconnect the line.



Tap [Yes] to disconnect the line and return to the state of free port.



7 Troubleshooting

If any trouble occurs, please read the following carefully and take appropriate action.

If you cannot solve the trouble with the following solutions, please contact our sales department.

For contact information, please see the back cover of this manual.

Web server function

Symptoms	No.	Cause	Action
The screen is not displayed. (Not updated)	1	Connecting now.	● Please wait.
	2	A LAN cable is disconnected.	● Check the connection status of the LAN cable.
	3	There is an error in the IP address.	● Check if the IP address set in the controller and the IP address entered in the URL match.
	4	Incorrect port number.	● Make sure the port number is one of 1024, 1025, 1026.
	5	Being accessed by another client.	● Please access from another port. ● Forcibly disconnect unused lines from the communication check screen on the controller.
	6	Being filtered by a proxy server.	● Depending on your network environment, you may need to exclude the connection with the device.
Cannot log in as an administrator.	1	Incorrect administrator password.	● Check your administrator password.
	2	It is in the read-only status.	● Turn OFF read-only. For Configuration Procedures, see P.7"Read only".
Icon is not displayed.	1	The web browser has failed to acquire an image file.	● Refresh the screen by pressing the F5 key, for example.
Button does not respond.	1	Acquisition of files required for web browser display has failed.	● Refresh the screen by pressing the F5 key, for example.

Web API function

Symptoms	No.	Cause	Action
No response.	1	Connecting now.	● Please wait.
	2	A LAN cable is disconnected.	● Check the connection status of the LAN cable.
	3	There is an error in the IP address.	● Check if the IP address set in the controller and the IP address entered in the URL match.
	4	Incorrect port number.	● Make sure the port number is one of 1024, 1025, 1026.
	5	Being accessed by another client.	● Please access from another port. ● Forcibly disconnect the line from the communication check screen on the controller.
	6	Being filtered by a proxy server.	● Depending on your network environment, you may need to exclude the connection with the device.
Cannot log in as an administrator.	1	Incorrect administrator password.	● Check your administrator password.
	2	There is a format error of the data to be posted.	● Check P.22"Web API details".
401 error (Unauthorized) has occurred.	1	Incorrect administrator password. ("POST /v1/login")	● Check your administrator password.
	2	There is a format error of the data to be posted. ("POST /v1/login", "POST /v1/time")	● Check P.22"Web API details".
	3	A Web API that requires administrator privileges is requested with operator privileges. ("POST /v1/import", "Get /v1/clear/allMemory", etc.)	● Log in as an administrator. For Web APIs that require administrator privileges, check P.20"List of Web APIs".
400, 402 and later errors occur.	1	There is an error in the URL or posted data. Or, the unit is in a situation where the request cannot be executed.	● Check P.8"Status code".



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